

This is revision **65** to **Quick Reference Handbook Vol. 1 , CSP A-022-080.**

To bring this manual up to date, remove old pages and insert revised pages as follows:

REMOVE:		INSERT:	
Chapter Section Subject	Page	Chapter Section Subject	Page
Title Page / Disclaimer	All	Title Page / Disclaimer	All
00-02	00-02-1 to 00-02-2	00-02	00-02-1 to 00-02-2
		00-04	00-04-1 to 00-04-4
SUPPLEMENTS	SUPP 1-1 to SUPP 1-2	SUPPLEMENTS	SUPP 1-1 to SUPP 1-2
EICAS Message List	EICAS MESSAGE LIST-4 to EICAS MESSAGE LIST-7	EICAS Message List	EICAS MESSAGE LIST-4 to EICAS MESSAGE LIST-7

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



Model CL-600-2B19

QUICK REFERENCE HANDBOOK

Volume 1

CSP A-022-080

This handbook contains suggested procedures only for quick reference. These procedures in no way supersede current procedures outlined in the approved Airplane Flight Manual and any revisions thereto. In case of conflict, the Airplane Flight Manual takes precedence.

BOMBARDIER INC.
BOMBARDIER AEROSPACE COMMERCIAL AIRCRAFT
CUSTOMER SUPPORT

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Rev No.	Description of Revision	Incorp By	Date
REV 59	Introduces changes to the QRH to reflect Airplane Flight Manual Revision 59. • Harmonization to the other programs, Manual Changes Request (MCRs) and incorporation of all previous Temporary Revisions (TRs). • Relocation of historical data for Record of Revisions (i.e. covering REV 1 thru REV 58) and Record of Temporary Revisions (i.e. covering QRH TR RJ/1 thru TR RJ/167) to the Bombardier Aerospace online website.	BCSG	21 Mar 2012
REV 60	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 60.	BCSG	30 Nov 2012
REV 61	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 62.	BCSG	01 Oct 2014
REV 62	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 63.	BCSG	13 Feb 2015
REV 63	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 65.	BCSG	11 Jan 2016
REV 64	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 66.	BCSG	22 Apr 2016

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Rev No.	Description of Revision	Incorp By	Date
REV 64A	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 66A.	BCSG	18 Nov 2016
REV 65	Introduces changes to the QRH Volume 1 to reflect Airplane Flight Manual Revision 67.		

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Record the date you insert each temporary revision in your manual.

Revision 59 is a major revision that incorporates all outstanding temporary revisions. A complete listing of all issued temporary revisions may be obtained from Bombardier Aerospace Customer Service.

TR No.	Subject	Incorp. By	Date

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

**Vol. 1****00-04-1**

Rev. 65, Jan 13/2017

Page	Date	Page	Date
Title Page / All*	Jan 13/2017	LIMITS 2 -1	Mar 21/2012
Disclaimer		-2	Mar 21/2012
00-02 -1 *	Jan 13/2017	LIMITS 3 -1	Mar 21/2012
-2 *	Jan 13/2017	-2	Mar 21/2012
		-3	Mar 21/2012
00-03 -1	Mar 21/2012	-4	Apr 22/2016
-2	Mar 21/2012	-5	Mar 21/2012
		-6	Nov 18/2016
00-04 -1 *	Jan 13/2017	-7	Nov 18/2016
-2 *	Jan 13/2017	-8	Mar 21/2012
-3 *	Jan 13/2017	-9	Nov 18/2016
-4 *	Jan 13/2017	-10	Mar 21/2012
SUPP-00 -1	Feb 13/2015	LIMITS 4 -1	Mar 21/2012
-2	Mar 21/2012	-2	Apr 22/2016
		-3	Oct 01/2014
SUPP 1 -1 *	Jan 13/2017	-4	Mar 21/2012
-2 *	Jan 13/2017	-5	Mar 21/2012
		-6	Mar 21/2012
SUPP 2 -1	Feb 13/2015	-7	Nov 30/2012
-2	Feb 13/2015	-8	Nov 30/2012
		-9	Mar 21/2012
SUPP 3 -1	Mar 21/2012	-10	Mar 21/2012
-2	Mar 21/2012	-11	Mar 21/2012
		-12	Mar 21/2012
SUPP 4 -1	Mar 21/2012	-13	Mar 21/2012
-2	Mar 21/2012	-14	Mar 21/2012
-3	Mar 21/2012	LIMITS 5 -1	Mar 21/2012
-4	Mar 21/2012	-2	Mar 21/2012
		-3	Mar 21/2012
SUPP 5 -1	Nov 30/2012	-4	Mar 21/2012
-2	Nov 30/2012	LIMITS 6 -1	Mar 21/2012
-3	Oct 01/2014	-2	Mar 21/2012
-4	Oct 01/2014		
LIMITS-00 -1	Apr 22/2016	LIMITS 7 -1	Oct 01/2014
-2	Apr 22/2016	-2	Mar 21/2012
-3	Oct 01/2014	-3	Mar 21/2012
-4	Oct 01/2014	-4	Nov 30/2012
		-5	Feb 13/2015
LIMITS 1 -1	Mar 21/2012	-6	Oct 01/2014
-2	Mar 21/2012	-7	Oct 01/2014
		-8	Nov 30/2012

* The asterisk indicates pages changed, added or deleted.

**QUICK REFERENCE
HANDBOOK
CSP A-022-080****LIST OF EFFECTIVE PAGES**

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Vol. 1

00-04-2

Rev. 65, Jan 13/2017

Page	Date	Page	Date
LIMITS 8 -1	Mar 21/2012	-3	Mar 21/2012
-2	Mar 21/2012	-4	Mar 21/2012
-3	Oct 01/2014	-5	Mar 21/2012
-4	Oct 01/2014	-6	Mar 21/2012
-5	Oct 01/2014	-7	Mar 21/2012
-6	Oct 01/2014	-8	Mar 21/2012
		-9	Mar 21/2012
LIMITS 9 -1	Apr 22/2016	-10	Mar 21/2012
-2	Apr 22/2016	-11	Mar 21/2012
		-12	Mar 21/2012
LIMITS 10 -1	Mar 21/2012	PERF 4 -1	Mar 21/2012
-2	Mar 21/2012	-2	Mar 21/2012
		-3	Mar 21/2012
PERF-00 -1	Mar 21/2012	-4	Mar 21/2012
-2	Mar 21/2012	-5	Mar 21/2012
-3	Mar 21/2012	-6	Mar 21/2012
-4	Mar 21/2012	-7	Mar 21/2012
-5	Mar 21/2012	-8	Mar 21/2012
-6	Mar 21/2012		
-7	Mar 21/2012	PERF 5 -1	Mar 21/2012
-8	Mar 21/2012	-2	Mar 21/2012
		-3	Mar 21/2012
PERF 1 -1	Mar 21/2012	-4	Mar 21/2012
-2	Mar 21/2012	-5	Mar 21/2012
-3	Mar 21/2012	-6	Mar 21/2012
-4	Mar 21/2012	-7	Mar 21/2012
-5	Mar 21/2012	-8	Mar 21/2012
-6	Mar 21/2012	-9	Mar 21/2012
-7	Mar 21/2012	-10	Mar 21/2012
-8	Mar 21/2012	-11	Mar 21/2012
-9	Mar 21/2012	-12	Mar 21/2012
-10	Mar 21/2012	-13	Mar 21/2012
-11	Mar 21/2012	-14	Mar 21/2012
-12	Mar 21/2012	-15	Mar 21/2012
-13	Mar 21/2012	-16	Mar 21/2012
-14	Mar 21/2012	-17	Mar 21/2012
-15	Mar 21/2012	-18	Mar 21/2012
-16	Mar 21/2012	-19	Mar 21/2012
		-20	Mar 21/2012
PERF 2 -1	Mar 21/2012	-21	Mar 21/2012
-2	Mar 21/2012	-22	Mar 21/2012
-3	Mar 21/2012	-23	Mar 21/2012
-4	Mar 21/2012	-24	Mar 21/2012
		-25	Mar 21/2012
PERF 3 -1	Mar 21/2012	-26	Mar 21/2012
-2	Mar 21/2012		

* The asterisk indicates pages changed, added or deleted.

QUICK REFERENCE
HANDBOOK
CSP A-022-080

LIST OF EFFECTIVE PAGES

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Vol. 1

00-04-3

Rev. 65, Jan 13/2017

Page	Date	Page	Date
PERF 6 -1	Mar 21/2012	MISC -1	Mar 21/2012
-2	Mar 21/2012	-2	Mar 21/2012
-3	Mar 21/2012		
-4	Mar 21/2012	PLAN-00 -1	Mar 21/2012
-5	Mar 21/2012	-2	Mar 21/2012
-6	Mar 21/2012		
-7	Mar 21/2012	PLAN -1	Mar 21/2012
-8	Mar 21/2012	-2	Mar 21/2012
-9	Mar 21/2012		
-10	Mar 21/2012	INTERMIX -1	Mar 21/2012
-11	Mar 21/2012	TOC	
-12	Mar 21/2012	-2	Mar 21/2012
-13	Mar 21/2012	-3	Mar 21/2012
-14	Mar 21/2012	-4	Mar 21/2012
-15	Mar 21/2012		
-16	Mar 21/2012	INTERMIX 1 -1	Mar 21/2012
-17	Mar 21/2012	-2	Mar 21/2012
-18	Mar 21/2012	-3	Mar 21/2012
-19	Mar 21/2012	-4	Mar 21/2012
-20	Mar 21/2012		
-21	Mar 21/2012	INTERMIX 2 -1	Mar 21/2012
-22	Mar 21/2012	-2	Mar 21/2012
-23	Mar 21/2012	-3	Mar 21/2012
-24	Mar 21/2012	-4	Mar 21/2012
-25	Mar 21/2012	-5	Mar 21/2012
-26	Mar 21/2012	-6	Mar 21/2012
-27	Mar 21/2012	-7	Mar 21/2012
-28	Mar 21/2012	-8	Mar 21/2012
-29	Mar 21/2012	-9	Mar 21/2012
-30	Mar 21/2012	-10	Mar 21/2012
-31	Mar 21/2012	-11	Mar 21/2012
-32	Mar 21/2012	-12	Mar 21/2012
-33	Mar 21/2012	-13	Mar 21/2012
-34	Mar 21/2012	-14	Mar 21/2012
-35	Mar 21/2012	-15	Mar 21/2012
-36	Mar 21/2012	-16	Mar 21/2012
		-17	Mar 21/2012
PERF 7 -1	Mar 21/2012	-18	Mar 21/2012
-2	Mar 21/2012		
-3	Mar 21/2012	INTERMIX 3 -1	Mar 21/2012
-4	Mar 21/2012	-2	Mar 21/2012
-5	Mar 21/2012	-3	Mar 21/2012
-6	Mar 21/2012	-4	Mar 21/2012
-7	Mar 21/2012		
-8	Mar 21/2012		
MISC-00 -1	Mar 21/2012		
-2	Mar 21/2012		

* The asterisk indicates pages changed, added or deleted.

QUICK REFERENCE
HANDBOOK
CSP A-022-080

LIST OF EFFECTIVE PAGES

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Page	Date	Page	Date
INTERMIX 4 -1	Mar 21/2012	-7 *	Jan 13/2017
-2	Mar 21/2012	-8	Feb 13/2015
-3	Mar 21/2012	-9	Apr 22/2016
-4	Mar 21/2012	-10	Feb 13/2015
		-11	Feb 13/2015
INTERMIX 5 -1	Mar 21/2012	-12	Feb 13/2015
-2	Mar 21/2012	-13	Feb 13/2015
-3	Mar 21/2012	-14	Feb 13/2015
-4	Mar 21/2012	-15	Feb 13/2015
-5	Mar 21/2012	-16	Feb 13/2015
-6	Mar 21/2012		
-7	Mar 21/2012		
-8	Mar 21/2012		
INTERMIX 6 -1	Mar 21/2012		
-2	Mar 21/2012		
-3	Mar 21/2012		
-4	Mar 21/2012		
-5	Mar 21/2012		
-6	Mar 21/2012		
-7	Mar 21/2012		
-8	Mar 21/2012		
-9	Mar 21/2012		
-10	Mar 21/2012		
-11	Mar 21/2012		
-12	Mar 21/2012		
-13	Mar 21/2012		
-14	Mar 21/2012		
-15	Mar 21/2012		
-16	Mar 21/2012		
-17	Mar 21/2012		
-18	Mar 21/2012		
-19	Mar 21/2012		
-20	Mar 21/2012		
EICAS -1	Feb 13/2015		
MSG TOC			
-2	Mar 21/2012		
EICAS -1	Feb 13/2015		
MESSAGE			
LIST			
-2	Apr 22/2016		
-3	Feb 13/2015		
-4 *	Jan 13/2017		
-5 *	Jan 13/2017		
-6 *	Jan 13/2017		

* The asterisk indicates pages changed, added or deleted.

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
-----------------	-------------

GENERAL

AIRPLANE OPTION CODES SUPP 1-1

ENGINE STARTING

ENGINE STARTING PROCEDURES SUPP 2-1
 General Notes, Cautions and Warnings SUPP 2-1
 Engine Start – External Air SUPP 2-1
 Engine Start – Cross Bleed SUPP 2-1
 Engine Start – Battery / External Air SUPP 2-2

AIR-CONDITIONING AND PRESSURIZATION

UNPRESSURIZED TAKE-OFF AND LANDING
PROCEDURE SUPP 3-1
 Before Take-off SUPP 3-1
 After Take-off SUPP 3-1
 Landing SUPP 3-2

REVERSE TAXI OPERATIONS

LIMITATIONS SUPP 4-1
NORMAL PROCEDURES SUPP 4-1
 Reverse Taxi Procedures SUPP 4-2

COLD WEATHER OPERATION

DEICING / ANTI-ICING SUPP 5-1
DEICING FLUID HOLDOVER TIMES SUPP 5-3

LIST OF ILLUSTRATIONS

REVERSE TAXI OPERATIONS

Reverse Taxi Operations SUPP 4-1

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

CONTENTS
SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

AIRPLANE OPTION CODES

Relevant pages of the QRH contain the following airplane option codes adjacent to the applicable text.

In cases where two or more option codes apply to the same text, and one of the options is the basic configuration, the option code <Type Spec> will appear, paired with other applicable option code(s). Example: <Type Spec> and <0004>.

Absence of an option code means that the data are applicable to all.

Airplane Option Codes		
Option Code	Customer Request	Description
<JAA>	CR 00-211	Data applicable to Joint Aviation Authorities registered airplanes only.
<0001>	CR 28-320	Metric Fuel Indication
<0002>	CR 00-008	Performance – Extended Range: 51000 lb MTOW
<0004>	CR 00-014	Performance – Long Range: 23995 kg MTOW
<0005>	CR 72-302	Power Plant - CRJ200 (CF34-3B1 Engine Basic Performance) <FAA> coded pages comply with FAR 25-109 at Amendment 25-41.
<0006>	CR 00-312	Performance – Additional 8-degree Flap setting
<0008>	CR 23-330	VHF Communication – Third Radio, 25 kHz Channel Spacing
<0010>	CR 23-350	Selcal
<0011>	CR 23-340	Single HF Radio
<0012>	CR 23-332	VHF Communication – Dual, 8.33 kHz Channel Spacing
<0013>	CR 25-359	Cargo Compartment – Automatic Temperature Control (40 Minute Diversion Capability)
<0014>	CR 23-334	VHF Communication – Third Radio, 8.33 kHz Channel Spacing
<0019>	CR 31-310	Data Concentrator Unit – Third DCU
<0020>	CR 33-310	Lights – Tail Logo Floodlights
<0021>	CR 33-303	Lights – Red Beacon Lights

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Airplane Option Codes		
Option Code	Customer Request	Description
<0024>	CR 34-332 CR 34-334 CR 34-336	Flight Management System (FMS) – Dual Installation
<0025>	CR 34-320	Inertial Reference System (IRS) – Dual Installation
<0026>	CR 34-380	Head-up Guidance System (HGS) – Category IIIa Capability
<0027>	CR 34-352	Global Positioning System (GPS) – Dual Sensors
<0039>	CR 31-320	EICAS 2000 Avionics Systems Update
<0040>	CR 34-328	Enhanced Ground Proximity Warning System (EGPWS)
<0045>	CR 34-360	Altimeter – Second Radio Altimeter
<0055>	CR 24-003	Electrical Power – Alternate Source for Cargo, Boarding and Maintenance Lights
<0071>	CR 35-006	Oxygen – Passenger, 22-Minute Chemical Oxygen Generators
<0096>	CR 34-312	Transponder – Mode S, Elementary Surveillance

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

GENERAL NOTES, CAUTIONS AND WARNINGS

NOTE

- 1. Accomplish all the normal checklists up to and including the “Cleared to start” checklist before performing the following supplementary procedures for engine start.
- 2. If an external source is required, it should be capable of supplying 45 psi manifold pressure at the initiation of start, however, lower pressure units, dependent upon altitude and flow, have been proven satisfactory.

ENGINE START – EXTERNAL AIR

- (1) APU LCV Close
- (2) EngineSTART

———— **END** ————

ENGINE START – CROSS BLEED

- (1) APU LCV Close
- (2) 10TH STAGE bleed valve (on operating engine)Open
- (3) ECS pageSelect and monitor



Beware of increased jet blast when increasing thrust.

- (4) Thrust lever (for operating engine)Advance as required to achieve a minimum of 60 psi bleed air pressure (approximately 85% N₂).
- (5) L or R ENGSTART

NOTE

If thrust lever was advanced to achieve 60 psi, reduce to idle after starter cutout.

———— **END** ————

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



ENGINE START – BATTERY / EXTERNAL AIR

- (1) FUEL, L BOOST PUMP ON
- (2) Ignition BARM
- (3) L ENG STARTSTART

———— **END** ————

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

UNPRESSURIZED TAKE-OFF AND LANDING PROCEDURE

The procedures outlined below are recommended when performing a no-engine bleed take-off and landing, with the APU inoperative or selected off for performance reasons.

When the engine bleeds cannot be used for take-off (due to hot ambient temperature – short runway combination, etc.) and the APU is inoperative, the following procedures are recommended:

Before Take-off

- (1) PRESS CONT switchAUTO
- (2) EMER DEPRESS ON
- (3) L and R PACK switches OFF
- (4) 10TH STAGE, ISOL Closed
- (5) 10TH STAGE, L and R BLEED
AIRCLOSED
- (6) RAM AIR switch OPEN

———— **END** ————

At a safe altitude, preferably between 1500 and 3000 feet above field elevation:

After Take-off

- (1) RAM AIR switch Close
- (2) 10TH STAGE, L and R BLEED
AIR OPEN
- (3) L and R PACK switchesOn
- (4) EMER DEPRESSOff

NOTE

If an engine failure occurs, delay turning on the bleeds until after obstacle clearance is assured.

———— **END** ————

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Landing

The following procedures are recommended if engine bleeds are not to be used during landing, with the APU inoperative or selected OFF for performance reasons.

When below 10000 feet AGL:

(1) PRESS CONT switchAUTO

At the turn to final approach:

(2) L and R PACK switches OFF

(3) 10TH STAGE, L and R BLEED
AIRCLOSED

(4) RAM AIR switch OPEN

(5) Landing ElevationSET to 3000 feet above
landing field elevation

NOTE

1. Avoid high rates of descent for passenger comfort.
2. After landing, ensure that the airplane is completely depressurized prior to opening any airplane doors.

END

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

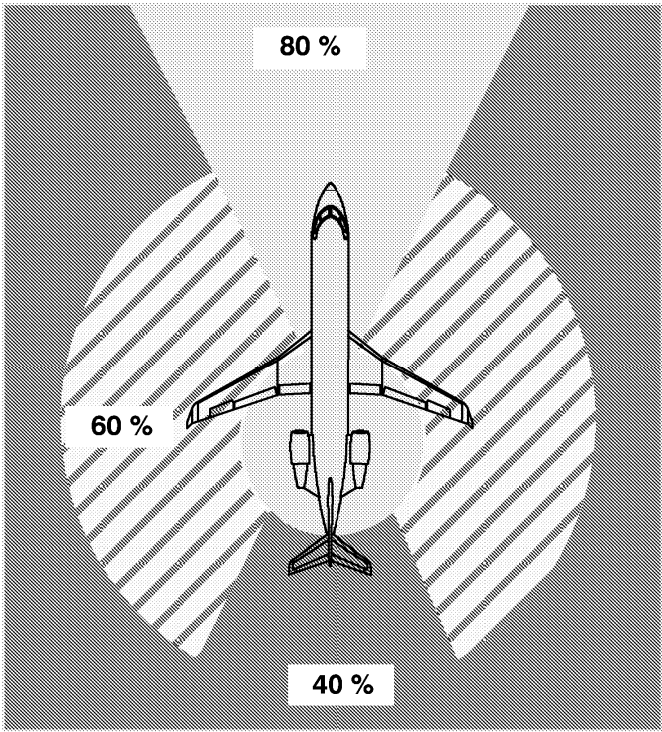
ABNORM

INDEX

LIMITATIONS

- Both engines’ thrust reversers must be operational.
- Fan speed must not exceed 80% N₁ reverse thrust.
- The application of brakes while the airplane is backing-up is prohibited.
- Reverse taxi operation is prohibited when ice, slush or snow is on ramp surface or during periods of heavy rain.
- Wind and fan speed limitations as detailed in the following illustration must be observed.

MAXIMUM REVERSE % N ₁	WIND CONDITION
80	Headwind / Any speed OR Any wind direction / <10 knots
60	Crosswind / >10 knots; < 40 knots
40	Crosswind / ≥ 40 knots OR Tailwind / ≥ 10 knots



NORMAL PROCEDURES

It is assumed that prior to the performance of the following procedures, the Pilot-In-Command (PIC) has ascertained that the prevailing conditions are suitable for the safe conduct of reverse taxi operations.

REVERSE TAXI PROCEDURES

After Engine Start:

- (1) Communication Establish
 - (2) External power / air connections Removed
 - (3) Chocks Removed
 - (4) Cleared-to-taxi signal Received from ground crew
 - (5) Thrust levers IDLE, then deploy thrust
reversers.
 - (6) Reverse thrust INCREASE gradually and
smoothly until the airplane
moves.
- Fan speed (% N_1) must not exceed the limits established in this supplement.

NOTE

1. Backing-up using thrust reversers may not always be possible. The capability to back-up is adversely affected by increasing airplane weight, aft centre of gravity, and/or ramp gradient.
2. Backing-up using thrust reversers can be consistently accomplished on a clean, level ramp, with airplane weights of up to 23133 kg (51000 lb), providing that cross-winds do not limit the fan speed to below 80% N_1 .
3. Reverse taxi using fan speeds in excess of 60% N_1 may result in noise and vibration levels which may not be acceptable for passenger comfort.

- (7) Rearward speedMaintain by modulating the thrust, as required.
- Reverse taxi rate must not exceed that of a normal walk.



Do not use brakes to stop the rearward movement of the airplane. If it is necessary to stop the airplane for any reason, the pilot must move the thrust levers out of reverse and stop the airplane with forward thrust.

Reverse Taxi During Turns:

- (1) Direction of turn Confirmed with ground crew.
 - (2) Nose wheel steering tiller Set to commence turn, after signal from ground crew has been received.
- Avoid large nosewheel deflections.

Termination of Reverse Taxi:

- (1) Thrust reversers Stow and apply forward thrust.



Do not use brakes to stop the rearward movement of the airplane.

- (2) Normal taxi Commence

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

CONTENTS
SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

DEICING / ANTI-ICING

Before spraying, if engines / APU operating:

- (1) Thrust leversIDLE
- (2) APUSee below
 - One / two step de-icing - APU as required
 - Gantry de-icing - APU shutdown
- (3) L and R PACK OFF
- (4) ANTI-ICE, WING All OFF
- (5) ANTI-ICE, LH and RH COWL All OFF
- (6) BleedsAll CLOSED
- (7) WipersOFF / PARK
- (8) Stabilizer trim Set for take-off
- (9) Flaps As required
- (10) AirplaneHead into wind, if possible
- (11) De-icing operatorAdvised
 - Engine and APU must not ingest de-icing fluid.
 - Avoid direct spray on the following:
 - Windshields, and all windows;
 - Air data probes and sensors / ports and AOA vanes;
 - Antennas;
 - Intakes and exhausts, vents and drains;
 - Winglets, wheels and brakes.
 - Type II, Type III and IV fluid MUST NEVER be applied to the following:
 - Windshields and side windows;
 - Air data probes and sensors;
 - Control surface cavities;
 - Engine and APU air intakes;
 - Lower side of radome;
 - Engine cowlings and thrust reverser components; and
 - NACA vent scoops.
 - Application of fluid should follow the sequence as follows:
 - Horizontal stabilizer,
 - Vertical stabilizer,
 - Top of fuselage
 - Sides of fuselage,
 - Wings.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

After spraying:

(12) Critical areas Inspect
to confirm complete de-icing

- Flight controls – free movement;
- Wings – upper / lower surfaces, pay particular attention to the leading edges
- Vertical and horizontal stabilizer – leading edges, upper / lower surfaces and side panels;
- Flaps, flap tracks and flap drives;
- Ailerons, elevators, rudder, spoilers and spoilerons;
- Air data probes / sensor, AOA vanes;
- Antennas;
- Fuel drains and NACA vents;
- Engine and APU intakes, APU exhaust;
- Landing gear and landing gear bays;
- Windshields, windows, door sills and surrounds.



1. Under no circumstances can an airplane that has been anti-iced, receive another coat of Type II / Type III / Type IV fluid on top of the existing film.
2. If holdover time is exceeded, surfaces must be deiced with a hot de-icing mixture of Type I fluid and water, before another application of Type II, Type III, or Type IV fluid is made.

When engines are running:

(13) Engine thrust Increase to 60% N_1 for
20 seconds to clear residual
fluid

(14) BLEED AIR, L and R 14TH
STAGE OPEN

(15) ANTI-ICE , WING As required

(16) ANTI-ICE , LH and RH COWL As required

NOTE

Wait at least 1 minute before opening the 10th stage bleed air valves and 3 minutes before turning the packs on to avoid contaminating the air-conditioning system with de-icing/anti-icing fluid. Contamination will cause objectionable fumes (causing throat irritation) and odours to enter the airplane.

- (17) BLEED AIR, L and R 10TH STAGE OPEN
- (18) L and R PACK On
- (19) Wipers As required
- (20) Take-off configuration Check

DEICING FLUID HOLDOVER TIMES

Guidelines for holdover times and de-icing fluid application are revised every year according to the Transport Canada TP 14052 and associated holdover time guideline document. The current documents are available at your country authorities website.

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS

PAGE

INTRODUCTION

INTRODUCTION LIMITS 1-1

KINDS OF AIRPLANE OPERATION LIMITS 1-1

CREW HEADSETS LIMITS 1-1

STRUCTURAL WEIGHT

STRUCTURAL WEIGHT LIMITATION LIMITS 2-1

OPERATING LIMITATIONS

ALTITUDE AND TEMPERATURE OPERATING LIMITS LIMITS 3-1

Take-off, Landing and Operating Limits LIMITS 3-1

TAKE-OFF LIMITS 3-1

Rotation Rate and Pitch Attitude LIMITS 3-1

Pitch Trim LIMITS 3-1

Take-off Pitch Target LIMITS 3-1

OPERATION IN ICING CONDITIONS LIMITS 3-2

Cowl Anti-ice System LIMITS 3-2

Wing Anti-ice System LIMITS 3-3

Thrust Settings LIMITS 3-5

Super-cooled Large Droplet (SLD) Icing LIMITS 3-6

COLD WEATHER OPERATIONS LIMITS 3-6

ENHANCED TAKE-OFF PROCEDURES AND WINTER OPERATIONS TRAINING LIMITS 3-8

RUNWAY SLOPES LIMITS 3-8

TAILWIND CONDITIONS LIMITS 3-8

MINIMUM FLIGHT CREW LIMITS 3-8

CARGO LIMITS 3-9

OPERATIONS USING QFE ALTIMETER SETTINGS LIMITS 3-9

POWER PLANT

ENGINES LIMITS 4-1

Engine Types LIMITS 4-1

Engine Indications LIMITS 4-1

Engine Warm-up LIMITS 4-2

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



LIMITS-00-2

Rev. 64, Apr 22/2016

CONTENTS

PAGE

ENGINE OPERATING LIMITS	LIMITS 4-2
Engine Operating Limits Table	LIMITS 4-2
Airplane Cold Soak	LIMITS 4-3
Oil Temperature	LIMITS 4-3
Oil Pressure	LIMITS 4-4
CONTINUOUS ENGINE IGNITION	LIMITS 4-4
AUTOMATIC PERFORMANCE RESERVE (APR)	LIMITS 4-4
STARTER CRANKING LIMITS (GROUND AND AIR)	LIMITS 4-4
Engine Start	LIMITS 4-4
Associated Conditions	LIMITS 4-4
Dry Motoring Cycle	LIMITS 4-5
ENGINE RELIGHT	LIMITS 4-5
FUEL	LIMITS 4-5
Fuel Load	LIMITS 4-5
Fuel Temperature	LIMITS 4-6
Fuel Grades	LIMITS 4-8
Fuel Additives	LIMITS 4-8
Fuel Cross-flow	LIMITS 4-10
OIL	LIMITS 4-10
Oil Grades	LIMITS 4-10
Oil Consumption	LIMITS 4-10
Engine Oil Level	LIMITS 4-10
Oil Replenishment System	LIMITS 4-11
AUXILIARY POWER UNIT (APU)	LIMITS 4-11
Type	LIMITS 4-11
Maximum RPM	LIMITS 4-11
Maximum EGT	LIMITS 4-11
Starting	LIMITS 4-11
APU Bleed Air	LIMITS 4-12
APU Generator	LIMITS 4-13
APU Indications	LIMITS 4-13

OPERATING SPEEDS

MAXIMUM OPERATING SPEED AND MACH NUMBER	LIMITS 5-1
DESIGN MANEUVERING SPEED	LIMITS 5-1

CONTENTS	PAGE
MAXIMUM FLAPS OPERATING SPEED	LIMITS 5-1
MAXIMUM FLAPS EXTENDED SPEED	LIMITS 5-2
MAXIMUM LANDING GEAR OPERATING SPEED	LIMITS 5-3
MAXIMUM LANDING GEAR EXTENDED SPEED	LIMITS 5-3
TIRE LIMIT SPEED	LIMITS 5-3
MAXIMUM AIRSPEED FOR ADG OPERATION	LIMITS 5-3
TURBULENCE PENETRATION SPEED	LIMITS 5-4
MINIMUM OPERATING LIMIT SPEED	LIMITS 5-4
WINDSHIELD WIPER (IF INSTALLED)	LIMITS 5-4
MANEUVERING LOADS	
MANEUVERING LIMIT LOAD FACTORS	LIMITS 6-1
SIDE-SLIP MANEUVERS	LIMITS 6-1
SYSTEM LIMITATIONS	
AIR-CONDITIONING AND PRESSURIZATION	LIMITS 7-1
AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS)	LIMITS 7-1
BLEED AIR SYSTEMS	LIMITS 7-2
ELECTRICAL SYSTEMS	LIMITS 7-2
Permissible Loads on AC System	LIMITS 7-2
Permissible Loads on DC Systems	LIMITS 7-2
Circuit Breaker Reset (In Flight)	LIMITS 7-2
FLIGHT CONTROLS – LIFT/DRAG DEVICES	LIMITS 7-3
Flaps	LIMITS 7-3
Flight Spoilers	LIMITS 7-3
Mach Trim	LIMITS 7-3
Ailerons	LIMITS 7-4
STALL PROTECTION SYSTEM	LIMITS 7-4
THRUST REVERSERS	LIMITS 7-5
I NOSEWHEEL STEERING SYSTEM	LIMITS 7-5

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



CONTENTS

PAGE

TAXI LIGHTS	LIMITS 7-5
WHEEL BRAKE COOLING LIMITATIONS	LIMITS 7-5
TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM (TCAS)	LIMITS 7-6
INTEGRATED STANDBY INSTRUMENT (ISI)	LIMITS 7-6
CONFIGURATION DEVIATION LIST (CDL)	LIMITS 7-6
ENHANCED GROUND PROXIMITY WARNING SYSTEM (EGPWS) <0040>	LIMITS 7-6
HYDRAULIC 3B PUMP	LIMITS 7-7
AIR TRAFFIC CONTROL TRANSPONDER, ENHANCED MODE S <0096>	LIMITS 7-8

NAVIGATION SYSTEM LIMITATIONS

FLIGHT MANAGEMENT SYSTEM (FMS) <0024> or <0049> or <0050>	LIMITS 8-1
Operating Data	LIMITS 8-1
FMS-4200 Operating Limitations	LIMITS 8-1
GLOBAL POSITIONING SYSTEM (GPS) <0027> or <0047>	LIMITS 8-4
General	LIMITS 8-4
RNP-10 Operations <0024> and <0027>	LIMITS 8-5

HEAD-UP GUIDANCE SYSTEM <0026>

HEAD-UP GUIDANCE SYSTEM LIMITATIONS	LIMITS 9-1
--	------------

REQUIRED EQUIPMENT LIST

Required Equipment List	LIMITS 10-1
Category II Required Equipment List	LIMITS 10-1
Category IIIA Required Equipment List <0026>	LIMITS 10-1
BRNAV Required Equipment List	LIMITS 10-2

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

INTRODUCTION

GENERAL

The Quick Reference Handbook Manual (QRH) contains limitations data for the operation of the following airplane model, manufactured by Bombardier Inc., Montreal, Quebec, Canada:

CL-600-2B19

Observance of the limitations included in this chapter is mandatory.

In the event of a disagreement between the QRH and the AFM, the AFM takes precedence.

KINDS OF AIRPLANE OPERATION

The airplane is certified in the transport category for day and night operations, in the following conditions when the equipment and instruments required by the airworthiness and operating regulations are approved, installed and in an operable condition:

- VFR and IFR
- Flight in icing conditions

The airplane is certified for ditching when the safety equipment specified by the applicable regulations is installed.

Effectivity:

- Airplanes 7004 thru 7006 and 7008 thru 7014 **not incorporating** Service Bulletin SB 601R-23-004, Communications – Audio Integrating – Installation of the Audio Electronic Unit; and
- Airplanes 7003 thru 7046 **not incorporating** Service Bulletin SB 601R-34-013, Navigation – Independent Position Determining – TCAS Aural Warning Level Adjustment.

CREW HEADSETS

All flight compartment crew members must wear headsets at all times.

The headsets approved for use are as follows:

- Sennheiser HME–1410 KA, or
- Telex Airman 750.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

STRUCTURAL WEIGHT

STRUCTURAL WEIGHT LIMITATION

Weight	kg	lb	Airplane Option Code
Maximum Ramp Weight (MRW)	23247	51250	<0002>
	24154	53250	<0004>
Maximum Take-Off Weight (MTOW)	23133	51000	<0002>
	23995	52900	<0004>
Maximum Landing Weight (MLW)	21319	47000	
Maximum Zero Fuel Weight (MZFW)	19958	44000	
Minimum flight weight	13608	30000	

NOTE

The Maximum Take-off Weight (MTOW) and/or Maximum Landing Weight (MLW) may be further limited due to performance considerations.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

OPERATING LIMITATIONS

ALTITUDE AND TEMPERATURE OPERATING LIMITS

Take-off, Landing and Operating Limits

Maximum airport pressure altitude for take-off and landing is 10000 feet.

Maximum operating altitude is 41000 feet.

Maximum ambient air temperature approved for take-off and landing is ISA + 35°C.

Minimum ambient temperature approved for take-off is -40°C (-40°F).

TAKE-OFF

Rotation Rate and Pitch Attitude

WARNING

Excessive rotation rates (exceeding 3 degrees per second) or over-rotations may lead to high pitch attitudes and angles of attack being attained while the aircraft is near the ground. This can reduce stall margins significantly, resulting in stick shaker/pusher activation and potentially loss of control. Pilots must rotate smoothly towards the target pitch attitude then transition to speed control.

Pitch Trim

WARNING

Failure to set the pitch trim appropriate to the computed centre of gravity may result in excessive rotation rate at take-off.

- Pitch trim must be set according to the airplane’s computed centre of gravity.

Effectivity:

- Airplanes **not incorporating** the -904 or the -037 Flight Control Computer:

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Take-off Pitch Target

- The initial target for rotation is 10 degrees.
- If the flight director is used for take-off, set the pitch target for 10 degrees. (Refer to Flight Crew Operating Manual, Volume 2 (CSP A-013): SUPPLEMENTARY PROCEDURES – Automatic Flight Control System – TAKE-OFF).
- Take-off performance data in the Airplane Flight Manual (CSP A-012), Chapter 6 remains applicable. (Refer to Airplane Flight Manual, (CSP A-012): – PERFORMANCE – Take-off Performance.)

OPERATION IN ICING CONDITIONS

Cowl Anti-ice System

(1) Ground Operations

The cowl anti-ice system must be ON when the OAT is 10°C (50°F) or below and visible moisture in any form is present (such as fog with visibility of 1500 metres [one mile] or less, rain, snow, sleet and ice crystals).

The cowl anti-ice system must also be ON when the OAT is 10°C (50°F) or below when operating on runways, ramps, or taxiways where surface snow, ice, standing water, or slush is present.

(2) Flight Operations

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is –40°C (–40°F) or below.

The cowl anti-ice system must be ON:

- when in icing conditions, or
- when ICE is annunciated by the ice detection system.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Wing Anti-ice System

(1) Ground Operations

NOTE

Icing conditions exist on the ground when the OAT is 5°C (41°F) or below and:

- visible moisture in any form (such as clouds, fog or mist), is present below 400 feet AGL, or
- the runway is wet or contaminated, or
- in the presence of any precipitation (such as rain, snow, sleet or ice crystals).

Effectivity:

- Commencing 1 December 2008:
- Airplanes 7003 thru 8076, 8082, 8086, 8090 thru 8092, 8096 and 8097 **not incorporating** the following Service Bulletin:
 - SB A601R-30-032, Ice and Rain Protection – Wing Anti-Ice System – Inspection of the Wing Anti-Ice Piccolo Tubes.

Take-off in icing conditions, which would require the use of the wing anti-ice system, is prohibited.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



LIMITS 3-4

Rev. 64, Apr 22/2016

- The wing anti-ice system must be selected ON during final taxi prior to take-off if the OAT is 5°C (41°F) or below, unless Type II, Type III or Type IV anti-icing fluids have been applied. During single engine taxi operations, final taxi prior to take-off is defined as that period after the second engine is started.

NOTE

1. If wing anti-ice is selected ON during taxi, **L WING A/I** or **R WING A/I** caution messages may be posted but must be verified out and **WING A/ICE ON** advisory message posted prior to take-off. If wing anti-ice is not required for take-off, it should be selected OFF just prior to take-off.
2. To prevent wing contamination from reverse jet blast, operating the thrust reversers during taxi operations on wet and contaminated surfaces should be avoided.

The wing anti-ice system must be selected and confirmed ON for take-off when, the OAT is 5°C (41°F) or below and:

- visible moisture in any form (such as clouds, fog or mist), is present below 400 feet AGL, or
- the runway is wet or contaminated, or
- in the presence of any precipitation (such as rain, snow, sleet or ice crystals).

NOTE

If the wing anti-ice system is selected ON for take-off, the cowl anti-ice system must also be selected ON.

- When Type II, Type III or Type IV anti-icing fluids have been applied, the wing anti-ice system must only be selected and confirmed ON just prior to thrust increase for take-off.
- Refer to Flight Crew Operating Manual, Volume 2 (CSP A-013): SUPPLEMENTARY PROCEDURES – Cold Weather Operations. – Phase of Flight Procedures.

(2) Flight Operations

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is –40°C (–40°F) or below.

The wing anti-ice system must be ON:

- when ICE is annunciated by the ice detection system, or
- when in icing conditions and the airspeed is less than 230 KIAS.

Effectivity:

- Commencing 1 December 2008:
- Airplanes 7003 thru 8076, 8082, 8086, 8090 thru 8092, 8096 and 8097 **not incorporating** the following Service Bulletin:
 - SB A601R-30-032, Ice and Rain Protection – Wing Anti-Ice System – Inspection of the Wing Anti-Ice Piccolo Tubes.
- Continued flight in conditions requiring the use of wing anti-ice is prohibited.
- If the wing anti-ice system was selected ON in flight, leave icing conditions, and
 - If the visible portion of the wings can be confirmed, from the cockpit, to be free of ice and the **TAT** for approach and landing is greater than 10°C, accomplish a normal approach and landing.
 - If the visible portion of the wings cannot be confirmed, from the cockpit, to be free of ice and the **TAT** for approach and landing is less than or equal to 10°C, accomplish ABNORMAL PROCEDURES, Ice and Rain Protection, Wing Anti-Ice Piccolo Duct Damage Suspected Procedure.

Thrust Settings

For take-offs in icing conditions, from 0°C to 5°C OAT and at pressure altitudes of 2000 feet and below, the take-off thrust settings and associated airplane performance for 10th stage bleeds closed and engine cowl anti-icing on can be used with engine cowl and wing anti-icing bleeds switched on.

Super-cooled Large Droplet (SLD) Icing

Continued operation in areas where Super-cooled Large Droplet (SLD) icing conditions exist is prohibited.

SLD icing conditions are indicated by ice accretion on the flight compartment side windows:

- The wing anti-icing system must be ON in SLD icing conditions.
- The cowl anti-icing system must be ON in SLD icing conditions.
- Leave icing conditions when side window icing occurs.

COLD WEATHER OPERATIONS

WARNING

Even small amounts of frost, ice, snow, or slush on the wing leading edges and forward upper wing surface may adversely change the stall speeds, stall characteristics and the protection provided by the stall protection system, which may result in loss of control on take-off.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Take-off is prohibited with frost, ice, snow or slush adhering to any critical surface (wings, horizontal stabilizer, vertical stabilizer, control surfaces and engine inlets).

Although the upper fuselage is not defined as a critical surface, it must nonetheless be de-iced to remove contamination, other than allowable frost, anytime the wing and tail surfaces require de-icing.

NOTE

1. Take-off is permitted with frost adhering to:
 - the upper surface of the fuselage; and/or
 - the underside of the wing, that is caused by cold soaked fuel,in accordance with the instructions provided in the Flight Crew Operating Manual, Volume 2 (CSP A-013): SUPPLEMENTARY PROCEDURES – Cold Weather Operations – PRE-FLIGHT PREPARATION – External Safety Inspection.
2. Comprehensive procedures for operating in cold weather are provided in the Flight Crew Operating Manual, Volume 2 (CSP A-013): SUPPLEMENTARY PROCEDURES – Cold Weather Operations.

In addition to a visual check, a tactile check of the wing leading edge, wing forward upper surface and wing rear upper surface is required during the External Walkaround inspection to determine that the wing is free from frost, ice, snow or slush when:

- the Outside Air Temperature (OAT) is 5°C (41°F) or less, or
- the wing fuel temperature is 0°C (32°F) or less, or
- the atmospheric conditions have been conducive to frost formation.

NOTE

Ice and frost may continue to adhere to wing surfaces for some time even at OAT above 5°C (41°F).

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

ENHANCED TAKE-OFF PROCEDURES AND WINTER OPERATIONS TRAINING

The limitations in this paragraph are effective commencing 1 November, 2008.

No take-off shall be conducted where the OAT is 5°C (41°F) or below, unless the pilot-in-command has successfully completed specific training, within the preceding 12 calendar months, for take-off procedures, ground icing conditions and cold weather operations.

No take-off shall be conducted by a pilot where the OAT is 5°C (41°F) or below, unless that pilot has successfully completed the specific training, within the preceding 12 calendar months, for take-off procedures, ground icing conditions and cold weather operations.

If neither of the preceding requirements has been complied with, the wing anti-ice system must be selected ON for take-off, just prior to thrust increase for take-off, if the OAT is 5°C (41°F) or below.

Completion of the following Bombardier Aerospace course will meet the intent of this training requirement:

- Bombardier Aerospace Enhancement to Take-Off Operational Safety Margins Training.

RUNWAY SLOPES

The maximum runway slopes approved for take-off and landing are:

- +2% (uphill)
- -2% (downhill)

TAILWIND CONDITIONS

The maximum tailwind component approved for take-off and landing is 10 knots.

MINIMUM FLIGHT CREW

The minimum flight crew is one pilot and one copilot.

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CARGO

Flight must be within 40 minutes of a suitable airport, if cargo is carried in the cargo compartment. <0013> or <0059>

OPERATIONS USING QFE ALTIMETER SETTINGS

Airplane operations using QFE altimeter settings must be conducted in accordance with the Airplane Flight Manual, Chapter 7, Supplement 22, Operations Using QFE Altimeter Settings.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

POWER PLANT

ENGINES

Engine Types

Type: General Electric CF34-3B1, quantity two. <0005> or <0068>

Engine Indications

The engine limit display markings on EICAS must be used to determine compliance with the maximum/minimum limits and precautionary ranges. If EICAS markings show more conservative limits than those specified in the Engine Indications table, the limit markings on the EICAS must be used.

CF34-3B1 <0005> or <0068>			
Engine Indications			
Indication	Red (Maximum/ Minimum Limits)	Amber (Caution Range)	Green (Normal Operating Range)
N ₁ % RPM	98.6	–	0 to 98.5
N ₂ % RPM: Wing anti-ice on	99.3	0 to 77.9	78 to 99.2
N ₂ % RPM: Wing anti-ice off	99.3	–	0 to 99.2
ITT °C: Normal Take-off	900 (first 2 min.)	–	0 to 900
	884 (next 3 min.)		0 to 884
ITT °C: Go-around or APR Thrust	928 (first 2 min.)	–	0 to 928
	900 (next 3 min.)		0 to 900
ITT °C: Maximum Continuous Thrust (MCT)	900* 874 to 900† 900†	874 to 900*	0 to 874
OIL TEMP °C	163	150 to 162	–40 to 149

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CF34-3B1 <0005> or <0068>			
Engine Indications			
Indication	Red (Maximum/ Minimum Limits)	Amber (Caution Range)	Green (Normal Operating Range)
OIL PRESS psi	0 to 25	116 to 156	26 to 115
* During MCT, amber band and red tick mark appear if ITT is greater than 874°C. † During MCT, amber band becomes a red band if ITT remains above 874°C for more than 30 seconds or if ITT is greater than 900°C. Red tick mark stays at 900°C.			

Engine Warm-up

The engine must remain at IDLE until oil pressure reaches normal operating range. During all starts, do not exceed 75% N₁ for two minutes after start, or until all operating indications are in the normal range, whichever is longer.

ENGINE OPERATING LIMITS

Engine Operating Limits Table

CF 34-3B1 <0005> or <0068>			
Condition	Core RPM N ₂ %	Fan RPM N ₁ %	ITT °C
Start	20	–	900
Idle	56.5 to 68.0†	–	–
Acceleration	–	–	900
Maximum Continuous	99.2	98.6	874
Normal Take-off	98.3	96.2	884 (5 min.)* 900 (2 min. out of 5 total transient)*

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	LIMITS 4-3
	Rev. 61, Oct 01/2014

CF 34–3B1 <0005> or <0068>			
Condition	Core RPM N ₂ %	Fan RPM N ₁ %	ITT °C
APR Thrust	99.4	98.6	900 (5 min.)* 928 (2 min. out of 5 total transient)*
* Transient limits † Engine to engine N ₂ split at ground idle power should not be greater than 2% N ₂ . If N ₂ is 57% or less with an OAT of –20°C or warmer, do not accelerate above idle.			

NOTE

1. The take-off, go-around and maximum continuous thrust N₁ values for the CF34-3B1 engine are presented on the appropriate engine thrust setting charts contained in this manual. <0005> or <0068>
2. If above 40000 feet one air-conditioning unit or cowl anti-ice must be selected on for each engine.

Airplane Cold Soak

Before the first flight of a day, when the airplane is cold-soaked at an ambient temperature of –30°C (–22°F) or below for more than 8 hours:

- The engines must be motored for 60 seconds and the fan rotation must be verified before an engine start is initiated.
- Thrust reversers must be actuated until the deploy and stow cycles are less than 5 seconds.

The minimum ambient temperature for starting a cold soaked APU on the ground is –40°C (–40°F).

Oil Temperature

Min for Start	-40°C
Max Cont	+155°C
Maximum Permissible	+163°C (15 Minutes)

QUICK REFERENCE HANDBOOK CSP A-022-080	LIMITATIONS
---	--------------------

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

	LIMITS 4-4
	Rev. 59, Mar 21/2012

Oil Pressure

Idle	25 psi minimum
Take-off	45 psi minimum
Max Cont	115 psi maximum
Max Transient (Cold Start)	156 psi maximum

CONTINUOUS ENGINE IGNITION

Continuous engine ignition must be used during the following:

- Take-offs and landings on contaminated runways;
- Take-offs with high cross-wind components;
- Flight through moderate or heavier intensity rain;
- Flight through moderate or heavier intensity turbulence;
- Flight in the vicinity of thunderstorms.

AUTOMATIC PERFORMANCE RESERVE (APR)

If take-off performance is predicated upon the use of APR, the APR system must be verified operative prior to take-off.

STARTER CRANKING LIMITS (GROUND AND AIR)

Engine Start

The starter must not be used if indicated N₂ rpm exceeds 55%.

Start	Maximum Time On	Followed By
1	1 minute	10 seconds off
2	1 minute	10 seconds off
3	1 minute	5 minutes off
4 and subs	1 minute	5 minutes off

Associated Conditions

At initiation of thrust lever movement from SHUT OFF to IDLE:

- ITT must be 120°C or less for all ground starts.
- ITT must be 90°C or less for starts. all in-flight

QUICK REFERENCE HANDBOOK CSP A-022-080	LIMITATIONS
---	--------------------

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Dry Motoring Cycle

The starter must not be used if indicated N₂ rpm exceeds 55%.
The dry motoring cycle may be used for:

- Engine ground starts
- Engine air starts
- Motoring of cold soaked engines

Start	Maximum Time On	Followed By
1	90 seconds	5 minutes off
2 and subs	30 seconds	5 minutes off
Thrust levers must be at SHUT OFF		

ENGINE RELIGHT

Engine starting in flight is only permitted within the envelope:

Relight Type	Envelope
Windmilling	Altitude from 21000 feet to 15000 feet: Speed 300 KIAS to V _{MO} /M _{MO} and from 12 to 55% N ₂ .
	Altitude less than 15000 feet: Speed from 300 KIAS to V _{MO} /M _{MO} and from 9 to 55% N ₂ .
Starter-assisted Cross Bleed (> 60 psi)	Altitude from 21000 feet to 15000 feet: Speed from 180 KIAS up to V _{MO} and from 0 to 55% N ₂ .
	Altitude less than 15000 feet: Speed from V _{REF} up to V _{MO} and from 0 to 55% N ₂ .

FUEL

Fuel Load

The maximum permissible fuel imbalance between the contents of the main left tank and the main right tank is 363 kg (800 lb).
Fuel remaining in a tank when the appropriate fuel quantity indicator reads zero is not usable.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



LIMITS 4-6

Rev. 59, Mar 21/2012

The fuel quantities that follow are based on 6.8 lb per US gal and 2.2046 lb per kg conversion factors. The maximum usable fuel load for each fuel tank is as follows:

	Pressure Refueling	Gravity Refueling
Left main tank	2159 kg (4760 lb)	2036 kg (4488 lb)
Right main tank	2159 kg (4760 lb)	2036 kg (4488 lb)
Centre tank	2267 kg (4998 lb)	2237 kg (4930 lb)
Total	6585 kg (14518 lb)	6309 kg (13906 lb)

Take-off with a fuel load in excess of 227 kg (500 lb) in the centre tank is not permitted unless:

- each main wing tank is above 1996 kg (4400 lb); or
- each main wing tank is above 907 kg (2000 lb): and
 - the allowable zero fuel weight is reduced by the weight of the fuel in the centre tank in excess of 227 kg (500 lb); and
 - the Centre of Gravity (CG) in this configuration is verified to be within the allowable CG envelope as calculated from the Weight and Balance Manual (WBM) (CSP A-041).

The minimum fuel quantity for go-around is 202 kg (450 lb) per wing (with the airplane level) and assuming a maximum airplane climb attitude of 10 degrees nose up.

Fuel Temperature

Take-off with engine fuel temperature indications below 5°C (41°F) is prohibited.

Take-off with bulk fuel temperature indications below the limits stated is prohibited.

During flight, bulk fuel temperature must remain above the applicable bulk fuel freezing point.

Fuel Type	Bulk Fuel Take-off Limit – Minimum Temperature	Bulk Fuel Freezing Point
ASTM D1655 JET A	–30°C	–40°C
ASTM D1655 JET A-1	–37°C	–47°C
MIL-DTL-5624 JP-5	–36°C	–46°C
MIL-DTL-83133 JP-8	–37°C	–47°C

QUICK REFERENCE
HANDBOOK
CSP A-022-080

LIMITATIONS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	LIMITS 4-7
	Rev. 60, Nov 30/2012

Fuel Type	Bulk Fuel Take-off Limit – Minimum Temperature	Bulk Fuel Freezing Point
GB 6537-2006 No. 3 Jet	–37°C	–47°C
CIS TS-1	–43°C	See NOTE
CIS RT	–40°C	
Ukrainian TS-1	–42°C	–52°C
Ukrainian RT	–42°C	–52°C

NOTE

- 1. Russian / CIS TS-1 and RT fuels with a freezing point of not higher than –50°C are approved for use wherein the ground level OAT is not below –30°C during the 24 hours before departure.
- 2. Russian / CIS fuel with a freezing point of not higher than –55°C (RT) and –60°C (TS-1) for use in low temperature regions are available at operator’s request.
- 3. The principle grade available in Russia (and members of the CIS) is TS-1 (written as TC-1 in Russian cyrillic script).

Use of wide-cut fuel is prohibited except for non-revenue ferry flights, with the following fuel temperature limitations:

Fuel Type	Bulk Fuel Take-off Limit Minimum Temperature	Bulk Fuel Maximum Operating Temperature	Bulk Fuel Freezing Point
ASTM D6615 Jet B	–40°C	+45°C	–50°C
MIL-DTL-5624 JP-4	–48°C	+45°C	–58°C

Powered cross-flow is prohibited when using wide-cut fuel and the APU is to be operated.

QUICK REFERENCE HANDBOOK CSP A-022-080	LIMITATIONS
---	--------------------

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Fuel Grades

Fuels conforming to any of the following specifications are approved for use. Mixing of fuels is permitted.

Canadian	American	British	Chinese	CIS/ Ukrainian	NATO
CGSB-3. 23	ASTM D1655 JET A				
	ASTM D1655 JET A-1	DEF STAN 91-91	GB 6537- 2006 No. 3 Jet	RT TS-1☆	F-35
CGSB-3. 24	MIL-DTL- 83133 JP-8	DEF STAN 91-87			F-34
	MIL-DTL- 5624 JP-5	DEF STAN 91-86			F-44
CIS/Ukrainian Fuels Inspection Requirement ☆ Engine fuel system components must be inspected in compliance with SB 601R-73-009, CF34-3A1 and CF34-3B1 — Russian Fuel Fleet Leader Sampling Program.					

Fuel Additives

- Alcohol-based anti-icing additives of Fuel System Icing Inhibitors (FSII) type Fluid I and Fluid I-M should not be used continuously. Russian / CIS fuel containing (or pre-mixed with) alcohol-based additives of anti-icing Fuel System Icing Inhibitor (FSII) type Fluid I and Fluid I-M should not be used continuously. Contact engine manufacturer for details of impact on a reduced component inspection program if continuous use has been determined.
- Anti-icing additives Diethylene Glycol Monomethyl Ether (DiEGME) per MIL-DTL-85470:

GEAE Fuel Specification D50TF2 Class and Fuel Description	Additive Maximum Concentration
Class A: Aviation Kerosene (Jet A)	0.15%
Class B: Wide-cut Distillate (Jet B, JP-4)	0.15%
Class C: Low Freeze Kerosene (Jet A-1, JP-8, No. 3 Jet)	0.15%

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	LIMITS 4-9
	Rev. 59, Mar 21/2012

GEAE Fuel Specification D50TF2 Class and Fuel Description	Additive Maximum Concentration
Class D: High Flash Kerosene (JP-5)	0.15%
Class E: Low Freeze Kerosene (CIS RT)	0.20%

- Kathon FP 1.5 biocide additive at a concentration of not more than 100 parts per million (ppm) by weight for the initial dose, to prevent the growth of micro-organisms. Contact the additive manufacturer for specific use instructions. The maintenance dose should be as per the additive manufacturer’s instructions and should not exceed 50 ppm.
- SOHIO Biobor JF biocide additive at a concentration of not more than 270 ppm (20 ppm elemental boron) by weight for the initial dose, to prevent the growth of micro-organisms. Contact the additive manufacturer for specific use instructions. The maintenance dose should be as per the additive manufacturer’s instructions and should not exceed 135 ppm.
- Anti-static additive STADIS 450 at a concentration of 3 g/m³ for the initial dose. Maximum dosage should not exceed 5 g/m³.
- Fuel system additive SIGBOL may be used in concentrations less than 0.0005% by weight. This additive should not be used continuously. Contact engine manufacturer for details of impact on the component inspection program.
- Corrosion inhibitors listed in the table that follows are approved by the concentrations indicated. It is recommended that corrosion inhibitors be blended with the fuel to provide lubricity. The corrosion inhibitor must be added after water removal and downstream of any clay filters (these processes remove the inhibitor).

Additive	Maximum Concentration
Apollo PRI-19	23 g/m ³
Octel DCI-4A	23 g/m ³
Octel DCI-6A	6 g/m ³
Hitec E-580	23 g/m ³
Nalco 5403	23 g/m ³
Nalco 5405	11 g/m ³

QUICK REFERENCE HANDBOOK CSP A-022-080	LIMITATIONS
---	--------------------

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Fuel Cross-flow

Powered cross-flow and gravity cross-flow must be off for take-off.

OIL

Oil Grades

NOTE

Mixing of different types of oils is prohibited.

Component	MIL-L-23699 (Type II)	MIL-L-7808 (Type I)	Castrol 4000
Engines	√	√	√
Air Turbine Starter	√	√	√
Integrated Drive Generator	√	√	√
Auxiliary Power Unit	√	√	√
Air Cycle Machines	√	√	√

Oil Consumption

Maximum oil consumption, on each engine, is 189 cubic centimetres per hour (6.4 US ounces per hour / 0.05 US gallon per hour).

Engine Oil Level

Engine oil levels must be checked as follows:

- The maximum duration without engine oil servicing is 16 operating hours.

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- For airplane operations in excess of 16 operating hours (without engine oil tank servicing), the engine oil level must be checked from the flight compartment ENGINE OIL LEVEL panel within 3 minutes to 2 hours after every engine shutdown.

NOTE

The engine oil level panel is used with the engines not running. Erroneous indications will occur if utilized during any other condition, as stated above.

Oil Replenishment System

If use of the oil replenishment system is required, then the engine(s) should be replenished within 15 minutes to 2 hours after engine shutdown.

Maximum refill allowable is 1890 cubic centilitres (2 US quarts) without dry motoring the engine.

If the oil system has to be replenished to maximum capacity and the replenishment period has been exceeded, the engine(s) must be dry motored.

AUXILIARY POWER UNIT (APU)

Type

Garret GTCP-36-150RJ

Maximum RPM

107%

Maximum EGT

743°C

Starting

Minimum ambient temperature for starting a cold soaked APU on the ground is -40°C.

The following APU start cycles are permitted:

- Using airplane batteries on the ground or for normal in-flight start:

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



LIMITS 4-12

Rev. 59, Mar 21/2012

Start	Maximum Time On	Followed By
1	30 seconds	–
2	30 seconds	20 minutes off
3	30 seconds	–
4	30 seconds	40 minutes off

- Using ground power:

Start	Maximum Time On	Followed By
1	15 seconds	–
2	15 seconds	20 minutes off
3	15 seconds	–
4	15 seconds	40 minutes off

- If, in either case a successful start is not obtained, a further start must not be attempted for a period of at least 40 minutes.

NOTE

It is recommended that a 2-minute delay be observed between cranking attempts to allow for cooling of starter and start contactor for APU drainage.

Maximum EGT:

- 974°C (not to be exceeded under any operating conditions).

APU Bleed Air

APU Bleed Air Limitations	
System/Condition	Limitation
Air-conditioning	Bleed air extraction from the APU is not permitted above 15000 feet.

	LIMITS 4-13
	Rev. 59, Mar 21/2012

APU Bleed Air Limitations	
System/Condition	Limitation
Engine-start during ground operations	No bleed air extraction limitation. Each engine may be started using the APU as a bleed air source. If both engines are to be started using the APU bleed air, then the operating engine's thrust must not exceed 70% N ₂ .
Engine-start during flight	During single engine operations, APU bleed air extraction for an engine start is not permitted. During double engine failure conditions, APU bleed air extraction for engine starts is permitted. However, APU bleed air extraction is limited to 13000 feet and below.

APU Generator

The maximum permissible load on the APU generator in flight is 30 kVA.

APU Indications

The APU limit display markings on the EICAS must be used to determine compliance with the maximum/minimum limits and precautionary ranges. If EICAS markings show more conservative limits than those specified in the APU Indications table, the limit markings on the EICAS should be used.

APU Indications			
Indication	Red (Maximum/ Minimum Limits)	Amber (Caution Range)	Green (Normal Operating Range)
APU EGT °C	Greater than 743	713 to 742	0 to 712
APU RPM %	Greater than 107	101 to 106	0 to 100

QUICK REFERENCE HANDBOOK CSP A-022-080	LIMITATIONS
---	--------------------

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

OPERATING SPEEDS

MAXIMUM OPERATING SPEED AND MACH NUMBER

Maximum operating limit speeds, must not be deliberately exceeded in any regime of flight (climb, cruise or descent), unless a higher speed is specifically authorized for flight test or training operations.

DESIGN MANEUVERING SPEED

Full application of rudder and aileron controls as well as maneuvers that involve angles of attack near the stall, must be confined to speeds below V_A .



Avoid rapid and large alternating control inputs, especially in combination with large changes in pitch, roll, or yaw (e.g., large side slip angles) as they may cause structural failure at any speed, including below V_A .

Effectivity:

- Airplanes **incorporating** the following Service Bulletin:
- SB 601R-11-090, Placards and Markings – Flight Compartment – Installation of the New Maximum Flaps Operating Speed Placard.

MAXIMUM FLAPS OPERATING SPEED

The maximum speeds at which it is safe to extend the flaps are:

- FLAPS 0 to 8: 200 KIAS
- FLAPS 8 to 20: 200 KIAS
- FLAPS 20 to 30: up to V_{FE} for FLAPS 30
- FLAPS 30 to 45: up to V_{FE} for FLAPS 45

The maximum speed at which it is safe to retract the flaps is V_{FE} for the initial flaps position.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

MAXIMUM FLAPS EXTENDED SPEED

The maximum speeds at which the airplane may be flown with the flaps extended are:

- FLAPS 8: 230 KIAS
- FLAPS 20: 230 KIAS
- FLAPS 30: 185 KIAS
- FLAPS 45: 170 KIAS

Effectivity:

- Airplanes 7003 thru 7319 **not incorporating** the following Service Bulletin:
- SB 601R-11-071, Placards and Markings, Reduced V_{FE} , or
- SB 601R-11-076, Placards and Markings, Reduced V_{FE} , or
- SB 601R-11-080, Placards and Markings – Flight Compartment – Installation of the New Airspeed Limitation Placard and Removal of the Flap Inspection Placard.

The maximum speeds at which the airplane may be flown with the flaps extended are:

- FLAPS 8: 230 KIAS
- FLAPS 20: 230 KIAS
- FLAPS 30: 196 KIAS
- FLAPS 45: 191 KIAS

Effectivity:

- Airplanes 7003 thru 7319 **incorporating** the following Service Bulletin:
- SB 601R-11-071, Placards and Markings, Reduced V_{FE} , or
- SB 601R-11-076, Placards and Markings, Reduced V_{FE} , or

The maximum speeds at which the airplane may be flown with the flaps extended are:

- FLAPS 8: 215 KIAS
- FLAPS 20: 215 KIAS
- FLAPS 30: 185 KIAS
- FLAPS 45: 170 KIAS

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Effectivity:

Airplanes 7320 thru 7903 **not incorporating** the following Service Bulletin:

SB 601R-11-080, Placards and Markings – Flight Compartment – Installation of the New Airspeed Limitation Placard and Removal of the Flap Inspection Placard.

The maximum speeds at which the airplane may be flown with the flaps extended are:

FLAPS 8:

215 KIAS

FLAPS 20:

215 KIAS

FLAPS 30:

185 KIAS

FLAPS 45:

170 KIAS

MAXIMUM LANDING GEAR OPERATING SPEED

The maximum airspeed at which it is safe to extend the landing gear is 250 KIAS.

The maximum airspeed at which it is safe to retract the landing gear is 200 KIAS.

MAXIMUM LANDING GEAR EXTENDED SPEED

The maximum airspeed at which the airplane may be flown with the landing gear extended and locked is 250 KIAS.

TIRE LIMIT SPEED

The tire limit speed is 182 knots ground speed.

MAXIMUM AIRSPEED FOR ADG OPERATION

Effectivity:

Airplanes 7002 thru 7304:

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

The maximum airspeeds for continuous operation of the air-driven generator are as follows:

- Following automatic or manual deployment, speeds up to 250 KIAS are permitted for the remainder of the flight.
- When operationally necessary (e.g. windmill airstarts) speeds up to 330 KIAS are permitted for a period of twelve (12) minutes, and speeds between 331 and 335 KIAS are permitted for a period of four (4) minutes.
- For flight test or pilot training deployment, the maximum airspeed is 215 KIAS.

TURBULENCE PENETRATION SPEED

Maximum airspeed for turbulence penetration is 280 KIAS or 0.75 Mach, whichever is lower.

MINIMUM OPERATING LIMIT SPEED

Intentional speed reduction below the onset of stall warning, as defined by stick shaker operation, is prohibited unless a lower speed is specifically authorized for flight test or training operations.

WINDSHIELD WIPER (IF INSTALLED)

The windshield wiper must not be operated above 220 KIAS.
If the windshield wiper has failed in a non-parked position, the airplane speed must not exceed 220 KIAS.

MANEUVERING LOADS

MANEUVERING LIMIT LOAD FACTORS

These load factors limit the permissible angles of bank in turns and the severity of pull-up and push-over maneuvers:

- Flaps up: –1.0 g to 2.5 g
- Flaps down: 0.0 g to 2.0 g

SIDE-SLIP MANEUVERS

Avoid unnecessary and large side-slip maneuvers during low-speed, high-altitude cruise.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SYSTEM LIMITATIONS

AIR-CONDITIONING AND PRESSURIZATION

The maximum positive differential pressure is 8.7 psi.

The maximum negative differential pressure is –0.5 psi.

During ground maneuvering, the pressure differential must not exceed 0.1 psi.

During initial landing (at touchdown), the maximum differential pressure must not exceed 1.0 psi.

The airplane must be completely depressurized prior to opening any of the airplane doors.

The cabin pressurization system must not be operated to –1500 feet when the system is in manual mode.

AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS)

Operation with the autopilot engaged is prohibited at altitudes below 600 feet AGL, except when performing the following:

- For visual and non-precision approaches, the minimum autopilot use height is 400 feet AGL.

Effectivity:

- Airplanes **incorporating** the –404 or the –604 Flight Control Computer:
- For precision approaches, the minimum autopilot use height is 60 feet AGL. <JAA>
- Use of IAS mode (flight director or autopilot coupled) during approach is not permitted.
- APPR mode VOR approaches (flight director or autopilot coupled) are not permitted.

Effectivity:

- Airplanes **incorporating** the –704 or the –037 Flight Control Computer:
- For precision approaches, the minimum autopilot use height is 80 feet AGL. <JAA>
- Use of DES submode (flight director or autopilot coupled) during approach is not permitted.

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



LIMITS 7-2

Rev. 59, Mar 21/2012

BLEED AIR SYSTEMS

The bleed air 10th stage valves must be closed for take-off and landing if the engine cowl and/or wing anti-ice systems have been selected on.

ELECTRICAL SYSTEMS

Permissible Loads on AC System

Individual AC generator loading must not exceed the following values:

Altitude (Feet)	Load Limitation (kVA)	
	Main Generator (Each)	APU Generator
0 – 35000	30	30
35001 to 37000	25	30
37001 and above	25	0

Permissible Loads on DC Systems

(1) Ground Operations

To protect the flight compartment CRT displays, the maximum permissible time for ground operations with DC power only is 5 minutes.

(2) Flight Operations

The maximum permissible continuous load on each TRU is 100 amp.

Circuit Breaker Reset (In Flight)

A circuit breaker must not be reset or cycled (i.e., opened or closed) unless doing so is consistent with explicit procedures specified in this AFM or unless, in the judgement of the Pilot-In-Command (PIC), that resetting or cycling of the circuit breaker is necessary for the safe completion of the flight.

FLIGHT CONTROLS – LIFT/DRAG DEVICES

Flaps

En-route use of flaps is prohibited.

Flight with flaps extended at altitudes above 15000 feet is prohibited.

Effectivity:

- Airplanes 7904 and subsequent; and
- Airplanes **incorporating** the following Service Bulletin:
 - SB 601R-11-080, Placards and Markings – Flight Compartment – Installation of the New Airspeed Limitation Placard and Removal of the Flap Inspection Placard.
- Except for touch-and-go training, take-off flaps must be selected from the 0 detent.
- Touch-and-go training is not permitted in cross-winds greater than 14 knots.

Flight Spoilers

Flight spoilers must not be extended in flight below an altitude of 300 feet AGL.

To ensure adequate maneuver margins, flight spoilers must not be extended in flight at airspeeds below the recommended approach speed plus 17 KIAS (refer to Chapter 6, PERFORMANCE – Landing Performance).

Effectivity:

- Airplanes 7003 thru 7112 **not incorporating** the following Service Bulletin:
 - SB 601R-27-053, Replacement of the Horizontal Stabilizer Trim Control Unit.

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Mach Trim

Use of Mach trim is prohibited.

If the autopilot is disengaged or inoperative, speed must be restricted to 250 KIAS (0.7 M).

NOTE

Pending incorporation of Alert Service Bulletin A601R-27-054, Flight Controls – Horizontal Stabilizer Control System - Deactivation of the Mach Trim Engage / Disengage Switch/light, the MACH TRIM caution message will be displayed on EICAS and the Mach Trim Engage / Disengage Switch "INOP" light will be illuminated (on). The EICAS message can be scrolled out of view prior to take-off, but the switch "INOP" light will remain illuminated (on).

Effectivity:

- Airplanes that **do not comply** with Transport Canada Airworthiness Directive CF-2002-35R2, Aileron Stiffness, Part C – Termination of Control Checks:

Ailerons

When departing from a wet or contaminated runway:

- at 10000 feet MSL, or when the **SAT** is less than or equal to 0°C, whichever occurs later;
- the ailerons must be operated manually every 5000 feet until the top of climb.

STALL PROTECTION SYSTEM

- The Stall Protection System test must be accomplished before the first flight of the day. (Refer to Flight Crew Operating Manual, Volume 2 (CSP A-013): NORMAL PROCEDURES – Prior to Start – ORIGINATING CHECK – STALL Test as per FCOM TR RJ/179 or later applicable revision.)
- Both stall protection system switches must remain on for all phases of flight.

THRUST REVERSERS

Thrust reversers are approved for ground use only.

The thrust reversers are intended for use during full stop landings. Do not attempt a go-around maneuver after deployment of the thrust reversers.

Take-off with any of the following thrust reverser lights on, icons or EICAS messages displayed is prohibited:

- THRUST REVERSER UNLK light,
- REV icon on N₁ gauge,
- **L REV UNLOCKED** or **R REV UNLOCKED** caution message.

During landing, application of the reverse thrust is permitted as follows:

- From touchdown speed to 60 KIAS - maximum N₁,
- Below 60 KIAS, reverse thrust must be reduced to idle.

During pre-flight check of the thrust reversers with the airplane stationary, reverse thrust must be limited to reverse idle.

NOSEWHEEL STEERING SYSTEM

Towbarless towing is prohibited, unless the operation is performed in compliance with the Aircraft Maintenance Manual (AMM) (CSP A-001) towbarless towing requirements.

TAXI LIGHTS

The taxi lights must be switched OFF whenever the airplane is stationary in excess of 10 minutes.

WHEEL BRAKE COOLING LIMITATIONS

- Brake cooling times (established in accordance with PERFORMANCE – Take-off Performance – MAXIMUM ALLOWABLE BRAKE TEMPERATURE FOR TAKE-OFF) must be observed between a landing or a Rejected Take-Off (RTO) and a subsequent take-off, to ensure that sufficient brake energy is available to bring the airplane to a complete stop, if the subsequent take-off is rejected.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM (TCAS)

The use of **TCAS** must comply with the appropriate national operational regulations. <JAA>

Effectivity:

- Airplanes 7891 and subsequent, or
- Airplanes 7003 thru 7890 **incorporating** the following Service Bulletin:
- SB 601R-34-139, Navigation – Integrated Standby Instruments System – Replace the Standby Attitude Indicator (SAI) and the Standby Altimeter/Indicated Airspeed (SA/IAS) Indicator with the Integrated Standby Instrument (ISI), or
- SB 601R-34-137, Navigation – Integrated Standby Instrument System (ISIS) – Installation of the Thales Standby Instrument (ISI).

INTEGRATED STANDBY INSTRUMENT (ISI)

Flight Operations

- When NAV 1 is tuned to a valid ILS frequency, the ISI will display localizer and glide slope deviation while on the backcourse approach. Use of the ISI localizer and backcourse information is prohibited during backcourse approaches.

CONFIGURATION DEVIATION LIST (CDL)

If the airplane is to be operated with certain secondary airframe and/or any nacelle parts missing, operation must be in accordance with the limitations specified in the basic Airplane Flight Manual, and as amended by the Configuration Deviation List (Appendix 1).

ENHANCED GROUND PROXIMITY WARNING SYSTEM (**EGPWS**)

<0040>

Airplane navigation must not be predicated upon the use of the terrain display.

To avoid giving unwanted alerts, the terrain awareness alerting and display functions must be inhibited on take-off, approach or landing within 15 nm of an airport not included in the EGPWS airport database.

The terrain data base, terrain displays and alerting system do not account for man-made obstructions, except for all known man-made obstructions in Canada, the United States and Mexico.



Effectivity:

- Airplanes **equipped** with GPS:<0027> or <0047>

1. If GPS is inoperative (identified by NO GPS RAIM, GPS NOT AVAILABLE or GPS – FMS DISAGREE message on the FMS CDU), the EGPWS may fail to alert when a terrain threat exists, during ambient surface temperatures below ISA, during non-standard temperature lapse rates or if incorrect barometric settings are set on the altimeter. <0040>

Effectivity:

- Airplanes **incorporating** the following Service Bulletin:
- SB A601R-29-031, Hydraulic Power – Hydraulic System No. 3 – Deactivation and Removal of the Accumulator Assembly, and
- Airplanes **not incorporating** the following Service Bulletin:
- SB 601R-29-034, Hydraulic Power – Hydraulic System No. 3 – Installation in a New Location of the New Accumulator and the Nitrogen Pressure Gauge of Hydraulic System No. 3.

HYDRAULIC 3B PUMP

The Hydraulic 3B pump switch must be selected ON instead of AUTO for all phases of flight.

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



LIMITS 7-8

Rev. 60, Nov 30/2012

AIR TRAFFIC CONTROL TRANSPONDER, ENHANCED MODE S
<0096>

The installed Mode S system satisfies the data requirements of ICAO Doc. 7030/4 “Regional Supplementary Procedures for SSR Mode S Enhanced Surveillance in Designated European Airspace”. The following table represents system capability to transmit data parameters:

ATC Transponder, Enhanced Mode S Transmission Capability	
Parameter	Available / Not Available
Magnetic Heading <0025>	Available (See NOTE)
Indicated Airspeed	Available
Mach Number	Available
Vertical Rate	Inertial rate of climb/descent is available. <0025>
Roll Angle	Available
Track Angle Rate <0025>	Available
True Track Angle <0024> or <0049> or <0050>	Available
Ground Speed <0024> or <0049> or <0050>	Available
Selected Altitude	Available
Barometric Pressure Setting	Available

NOTE

MAG heading must be selected for display on the Primary Flight Display. If TRUE heading is selected, the Enhanced Mode S Transponder will downlink true heading. <0025>

NAVIGATION SYSTEM LIMITATIONS

FLIGHT MANAGEMENT SYSTEM (FMS) <0024> or <0049> or <0050>

Operating Data <0024> or <0049> or <0050>

The Flight Management System (FMS) must be operated in accordance with the latest edition of the following:

- Airplane Flight Manual, and
- Flight Management System Pilot Guide that follows:

Publication Number	Applicability	
523-0778363	FMS-4200	<FMC-002, FMC-006, FMC-011, and FMC-013>

FMS-4200 Operating Limitations <0024> or <0049> or <0050>

(1) FMC-002, FMC-006, FMC-011, and FMC-013

- Software Program Version
 - The FMS is approved for use only with the software program version given in the table that follows:

Software Program Version (SCID)	Applicability
829-7739-008	<FMC-002>
829-7739-012	<FMC-006>
832-4119-012	<FMC-011>
829-7739-028	<FMC-013>

- AC 20-130A
 - The FMS meets the enroute, terminal and non-precision approach lateral performance / accuracy criteria of AC 20-130A.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



LIMITS 8-2

Rev. 59, Mar 21/2012

- RNP-10 Operations <0024> and <0025>
 - The dual FMS installation, when used in conjunction with the dual Inertial Reference System (IRS), meets the requirements of FAA Order 8400.12A, Rev. A, dated January 2, 1998 - Required Navigation Performance (RNP-10) Operational Approval, as a primary means of navigation for flights up to 6.2 hours in duration, without updating. The determination of flight duration starts when the system is placed in the navigation mode.

NOTE

The demonstration of performance with the above criteria does not constitute approval to conduct RNP-10 operations.

- AC 90-100A US Terminal and Enroute Area Navigation (RNAV) Operations
 - When equipped with an operating FMS and GPS, the aircraft meets the functional and the ± 1 nm accuracy requirements of AC 90-100A US Terminal and Enroute Area Navigation (RNAV) Operations.
 - AC 90 – 100A operations must not be conducted unless all of the required equipment specified below is operational

AC 90–100A Required Equipment List

Equipment	Requirements for AC 90–100A
Flight Management Computer (2) <0024>	One (1) must be operational.
FMS Control Display Unit (2) <0024>	One (1) must be operational.
Global Positioning System (2) <0027>	One (1) must be operational.

NOTE

Compliance with the standard noted above does not constitute an operational approval.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- TGL-10 PRNAV Operations
 - The FMS installation meets the airworthiness certification requirements of JAA Temporary Guidance Leaflet No. 10, Airworthiness and Operational Approval for Precision RNAV Operations in Designated European Airspace.
 - Precision RNAV operations must not be conducted unless:
 - all of the required equipment specified in the PRNAV Required Equipment List table is operational,

PRNAV Required Equipment List	
Equipment	Requirements for PRNAV
Flight Management Computer	One (1) must be operational.
FMS Control Display Unit	One (1) must be operational and controlling an operational FMC.
VHF NAV, DME, GPS	One (1) VHF NAV and one (1) DME must be operational, or one (1) GPS is operational.

- the following messages are not displayed on any PFD or CDU:
 - FMS DR
 - IRS ONLY <0025>
 - VOR/DME DIST >75 NM
 - FMS NAV INVALID
- ILS, LOC, LOC-BC, LDA, SDF and MLS approaches using the FMS are prohibited. The FMS does not provide automatic transition to ILS.
- The FMS is approved for use only during enroute, terminal and non-precision approach phases of flight.
- FMS thrust, range, fuel management and altitude/speed capability information is advisory only.
- IFR enroute and terminal navigation is prohibited unless the pilot verifies the currency of the database and verifies waypoints for accuracy by reference to current publications.
- Instrument approaches must be accomplished with instrument approach procedures that are retrieved from the FMS database. The FMS must incorporate the current update cycle. The pilot must verify approach waypoints for accuracy by reference to current publications.



LIMITS 8-4

Rev. 61, Oct 01/2014

- The area navigation system used for IFR Class 1 navigation meets the performance/accuracy criteria of AC 20-130A for enroute and terminal area navigation.
- The approach must not be continued past the final approach fix if an FMS “NO APPR” message is displayed on the PFD.
- The VNAV system information is not temperature compensated. FMS generated altitudes and V-paths are not corrected for non-ISA conditions.
- The performance database must be verified to be current and valid for the appropriate engine model installed on the airplane:

Engine Model	Performance Database Part Number	Applicability
CF34-3B1 (Basic) <0005> and <0039>	832-4172-103	<0039>

- The FMS calculated thrust setting must not be used if the pressure altitude is greater than 36000 feet. <0039>
- Editing of the runway extension waypoint distance during a synchronized visual approach is prohibited.
(FMC-002 only) <0024>

GLOBAL POSITIONING SYSTEM (GPS) <0027> or <0047>

General

Other approved navigation equipment appropriate to the route of flight (enroute and terminal) must be installed and operating.

The GPS may only be used for approach guidance if the reference coordinate data system for the instrument approach is WGS-84 or NAD-83.

	LIMITS 8-5
	Rev. 61, Oct 01/2014

RNP-10 Operations <0024> and <0027>

The dual GPS installation, when used in conjunction with the Prediction Program 832-3443-005 or later applicable revision, and with two operational FMS-4200 systems, has been found to comply with the requirements of MNPS operations and as a primary means of navigation in oceanic and remote airspace operations. Likewise, this compliance is considered as meeting RNP-10 requirements, without time limitations.

NOTE

The demonstration of performance with these criteria does not constitute approval to conduct MNPS, oceanic/remote airspace or RNP-10 operations.

QUICK REFERENCE HANDBOOK CSP A-022-080	LIMITATIONS
--	-------------

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

	LIMITS 8-6
	Rev. 61, Oct 01/2014

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QUICK REFERENCE HANDBOOK CSP A-022-080	LIMITATIONS
---	--------------------

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

HEAD-UP GUIDANCE SYSTEM <0026>

HEAD-UP GUIDANCE SYSTEM LIMITATIONS

Effectivity:

- Airplanes **equipped** with the (-503) HGS Computer:

NOTE

Data provided herein for AFM chapters 1 through 6 cover use of HGS in operations not less than Category I minima.

The following table identifies the combinations of mode and flight phase of the HGS which have been certified.

- An "X" indicates the combination is prohibited from use;
- A "✓" indicates no restriction unless accompanied by a limiting footnote;
- A footnote indicates an additional reference or a conditional authorization.

Limitations on the Use of HGS <JAA>

	Flight Phases					
Selected HGS Mode	Take-off	Climb	Cruise	Descent	Approach	Landing and Roll-out
PRI	✓ [1], [2]	✓	✓	✓	✓ [6]	✓ [12], [13]
VMC [7]	X	X	✓	✓	✓ [3]	✓ [12], [13]
FD	X	X	X	X	X	X
AI	X	X	X	X	✓ [3], [4], [9]	✓ [12], [13]
AII	X	X	X	X	✓ [3], [4], [9], [10]	✓ [12], [13]

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



LIMITS 9-2

Rev. 64, Apr 22/2016

Limitations on the Use of HGS <JAA>

Selected HGS Mode	Flight Phases						Landing and Roll-out
	Take-off	Climb	Cruise	Descent	Approach		
AIII	X	X	X	X	✓ [3], [5], [8], [9], [11]		✓ [8],[12]

General

- The system must be operated in accordance with the HGS Pilot Guide P/N 9701-0971 Dated May 1999 or later applicable edition.

Excessive or Unusual Attitudes

- In the event of loss of situation awareness or an upset leading to the occurrence of an unusual attitude, attempted recovery with HGS is prohibited. Refer to the PFDs.

Low Visibility Take-off

- The HGS take-off mode has been shown to meet the requirements of FAR-AWO Subpart 4. Airplanes equipped with HGS may be permitted by the operators' local authority to take-off in an RVR of 75 metres or more.

- [1] Rolling take-offs requiring localizer information are prohibited.
- [2] Take-offs using back course localizer information are prohibited.
- [3] Non-precision approaches are prohibited.
- [4] Glide slope angles outside the range of – 2.5 to –3.5 degrees are prohibited.
- [5] Glide slope angles outside the range of – 2.5 to –3.0 degrees are prohibited.
- [6] Except for go-around, use of PRI mode below 800 feet AGL is prohibited.
- [7] Use of VMC mode in IMC is prohibited.
- [8] Use of flaps other than 45 is prohibited.
- [9] Use of any A mode to monitor autopilot-coupled approaches is prohibited.
- [10] Additional limitations and procedures for AII mode are covered in Supplement 4.
- [11] Additional limitations and procedures for AIII mode are covered in Supplement 8.
- [12] Landing with the combiner deployed is prohibited if the standard airplane sun visor is not stowed.
- [13] The landing and roll-out must be done visually, however, it is permissible for the HGS combiner to be left deployed and for the mode symbology to be displayed.

REQUIRED EQUIPMENT LIST

REQUIRED EQUIPMENT LIST

Category II Required Equipment List

An ILS approach to Category II minima must not be commenced or continued unless all required airborne equipment, specified in the following Category II Required Equipment List and their ground installations, are operating satisfactorily.

Category II Required Equipment List	
Equipment	Prior to Approach
VHF NAV 1 and 2	Both must be operational.
VHF COM 1 and 2	Both must be operational.
PFD 1 and 2	One (1) PFD available and operational for each side.
Stab Channel 1 and 2	One (1) channel must be operational.
Radio Altimeter's	Both must be operational, with display on both sides.
IRS 1 and 2	Both must be operational.
ADC 1 and 2	Both must be operational.
FD 1 and 2	Both must be operational.
EFIS Comparator Monitors	Must be operational.
AFCS PITCH TRIM	Must be operational.
Autopilot	Must be operational.
Hydraulics (3)	All systems must be operational.
Electrics	Two (2) generators on and sharing load, i.e., two (2) main generators or one (1) main generator and APU generator on.

Category IIIA Required Equipment List <0026>

The following airplane equipment is required to be operative.

Category IIIA Required Equipment List	
Equipment	Quantity
Head-up Guidance System	One
VHF Navigation Receiver	Two

QUICK REFERENCE HANDBOOK CSP A-022-080	LIMITATIONS
---	--------------------

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Category IIIA Required Equipment List	
Equipment	Quantity
VHF Communication Receiver	Two
Primary Flight Display	Two
Stabilizer Channel	One
Radio Altimeter	Two
Inertial Reference System	Two
Air Data Computer	Two
Generator	Two
Flight Directors	Two
Yaw Dampers	One
Ground proximity Warning System	One

BRNAV Required Equipment List

The following equipment is required to be operative.

BRNAV Required Equipment List	
Equipment	Quantity
VOR	One
DME	One
FMC	One, same side
FMC CDU	One, same side
FMS Database	Current
MFD FMS MAP Mode	One (on PF side)
EFIS Control Panel	One (on PF side)

BRNAV shall not be continued if an FMS “CHECK POSITION” message is displayed.

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
-----------------	-------------

TAKE-OFF DATA

TAKE-OFF	PERF 1-1
Stabilizer Trim Setting	PERF 1-1
Normal Take-off Thrust Setting	PERF 1-2
Speedbook	PERF 1-6
Take-off Weight Reduction with APR OFF	PERF 1-16

CLIMB

CLIMB	PERF 2-1
Maximum Climb Thrust Setting	PERF 2-1

IN-FLIGHT

IN-FLIGHT	PERF 3-1
Maximum Continuous Thrust Setting	PERF 3-1
Maximum Cruise Thrust Setting	PERF 3-4
Minimum Drag Speed	PERF 3-7
Altitude Capability	PERF 3-9
Best Rate of Climb (Best ROC)	PERF 3-12

APPROACH AND LANDING

APPROACH AND LANDING	PERF 4-1
Go-Around Thrust Setting	PERF 4-1
Actual Landing Distance – FLAPS 45	PERF 4-4
Actual Landing Distance – Flaps Other Than 45	PERF 4-4
Brake Cooling Requirements	PERF 4-5

REDUCED THRUST TAKE-OFF

REDUCED THRUST TAKE-OFF	PERF 5-1
Reduced Thrust Take-off Setting	PERF 5-1

WET AND CONTAMINATED RUNWAY

WET AND CONTAMINATED RUNWAY	PERF 6-1
Take-off Weight and V ₁ Reduction	PERF 6-1
Actual Landing Distance – FLAPS 45	PERF 6-15
Actual Landing Distance – Flaps Other Than 45	PERF 6-19
Actual Landing Distance – Wet (Advisory Data Only)	PERF 6-31

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS

PAGE

ANTI-SKID

ANTI-SKID INOPERATIVE – DRY / WET

RUNWAY TAKE-OFF PERF 7-1

Take-off Weight and V₁ Reduction PERF 7-1

Actual Landing Distance – FLAPS 45 PERF 7-7

Actual Landing Distance – Flaps Other

Than 45 PERF 7-8

LIST OF ILLUSTRATIONS

TAKE-OFF DATA

Stabilizer Trim Setting for FLAPS 8 or 20

Take-off PERF 1-1

Normal Take-off Thrust Setting, %N₁ Zero

Engine Bleeds <0005> PERF 1-2

Normal Take-off Thrust Setting, %N₁ 10th

Stage Bleeds On – PACKs On <0005>..... PERF 1-3

Normal Take-off Thrust Setting, %N₁ Cowl

Anti-ice On <0005> PERF 1-4

Speedcard – Legend <0006> and <Metric> PERF 1-6

Speedcard – 15400 kg <TC> or <JAA> and <0005> and

<0006> and <Metric> PERF 1-7

Speedcard – 16000 kg <TC> or <JAA> and <0005> and

<0006> and <Metric> PERF 1-8

Speedcard – 17000 kg <TC> or <JAA> and <0005> and

<0006> and <Metric> PERF 1-9

Speedcard – 18000 kg <TC> or <JAA> and <0005> and

<0006> and <Metric> PERF 1-10

Speedcard – 19000 kg <TC> or <JAA> and <0005> and

<0006> and <Metric> PERF 1-11

Speedcard – 20000 kg <TC> or <JAA> and <0005> and

<0006> and <Metric> PERF 1-12

Speedcard – 21000 kg <TC> or <JAA> and <0005> and

<0006> and <Metric> PERF 1-13

Speedcard – 22000 kg <TC> or <JAA> and <0005> and

<0006> and <Metric> PERF 1-14

Speedcard – 23000 kg <TC> or <JAA> and <0005> and

<0006> and <Metric> PERF 1-15

Speedcard – 24000 kg <TC> or <JAA> and <0005> and

<0006> and <Metric> PERF 1-16

Take-off Weight Reduction with APR Off <0005>

and <Metric> PERF 1-16

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
-----------------	-------------

CLIMB

Maximum Climb Thrust Settings, %N ₁ PACKs	
On	PERF 2-1
Maximum Climb Thrust Settings, %N ₁ PACKs	
On; Cowl Anti-ice On	PERF 2-2
Maximum Climb Thrust Settings, %N ₁ PACKs	
On; Wing and Cowl Anti-ice On	PERF 2-3

IN-FLIGHT

Maximum Continuous Thrust Setting, %N ₁	
PACKs On <0005>	PERF 3-1
Maximum Continuous Thrust Setting, %N ₁	
PACKs On; Cowl Anti-ice On <0005>	PERF 3-2
Maximum Continuous Thrust Setting, %N ₁	
PACKs On; Wing and Cowl Anti-ice On <0005>	PERF 3-3
Maximum Cruise Thrust Settings, %N ₁ PACKs	
On	PERF 3-4
Maximum Cruise Thrust Settings, %N ₁ PACKs	
On; Cowl Anti-ice On	PERF 3-5
Maximum Cruise Thrust Settings, %N ₁ PACKs	
On; Wing and Cowl Anti-ice On	PERF 3-6
Minimum Drag Speed <Metric>	PERF 3-7
Altitude Capability 0.70 M <Metric>	PERF 3-9
Altitude Capability 0.74 M <Metric>	PERF 3-10
Altitude Capability 0.77 M <Metric>	PERF 3-11
Best Rate of Climb (KCAS) <Metric>	PERF 3-12

APPROACH AND LANDING

Go-around Thrust Settings, %N ₁ Zero Engine	
Bleed <0005>	PERF 4-1
Go-around Thrust Settings, %N ₁ 10th Stage	
Bleeds On – PACKs On <0005>	PERF 4-2
Go-around Thrust Settings, %N ₁ Cowl Anti-ice	
On <0005>	PERF 4-3
Go-around Thrust Settings, %N ₁ Wing and	
Cowl Anti-ice On <0005>	PERF 4-3
Landing Data – Landing Gear Down, FLAPS 45	
Landing Speed (V _{REF}) and Actual Landing	
Distance <Metric>	PERF 4-4
Landing Data – Actual Landing Distance	
Factors When Landing with Flaps Other	
Than 45	PERF 4-4
V _{1MBE} , FLAPS 8 <0006> and <Metric>	PERF 4-5
V _{1MBE} , FLAPS 20 <Metric>	PERF 4-5

CONTENTS
SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
-----------------	-------------

REDUCED THRUST TAKE-OFF

Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude –1000 Feet <0005>	PERF 5–2
Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude 0 Feet (Sea Level) <0005>	PERF 5–3
Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude 1000 Feet <0005>	PERF 5–4
Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude 2000 Feet <0005>	PERF 5–5
Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude 3000 Feet <0005>	PERF 5–6
Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude 4000 Feet <0005>	PERF 5–7
Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude 5000 Feet <0005>	PERF 5–8
Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude 6000 Feet <0005>	PERF 5–9
Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude 7000 Feet <0005>	PERF 5–10
Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude 8000 Feet <0005>	PERF 5–11
Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude 9000 Feet <0005>	PERF 5–12
Reduced Thrust Take-off Setting, %N ₁ , Zero Engine Bleed – Pressure Altitude 10000 Feet <0005>	PERF 5–13
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude –1000 Feet <0005>	PERF 5–14
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude 0 Feet (Sea Level) <0005>	PERF 5–15
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude 1000 Feet <0005>	PERF 5–16

CONTENTS
SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude 2000 Feet <0005>	PERF 5–17
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude 3000 Feet <0005>	PERF 5–18
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude 4000 Feet <0005>	PERF 5–19
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude 5000 Feet <0005>	PERF 5–20
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude 6000 Feet <0005>	PERF 5–21
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude 7000 Feet <0005>	PERF 5–22
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude 8000 Feet <0005>	PERF 5–23
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude 9000 Feet <0005>	PERF 5–24
Reduced Thrust Take-off Setting, %N ₁ , One PACK On Per Engine – Pressure Altitude 10000 Feet <0005>	PERF 5–25

WET AND CONTAMINATED RUNWAY

Take-off Weight and V ₁ Reduction – FLAPS 8 (Wet Runway) <0005> and <0006> and <Metric>	PERF 6–3
Take-off Weight and V ₁ Reduction – FLAPS 8 (3.2 mm Water or Equivalent) <0005> and <0006> and <Metric>	PERF 6–4
Take-off Weight and V ₁ Reduction – FLAPS 8 (6.4 mm Water or Equivalent) <0005> and <0006> and <Metric>	PERF 6–5
Take-off Weight and V ₁ Reduction – FLAPS 8 Compacted Snow <0005> and <0006> and <Metric>	PERF 6–7
Take-off Weight and V ₁ Reduction – FLAPS 8 Ice <0005> and <0006> and <Metric>	PERF 6–8
Take-off Weight and V ₁ Reduction – FLAPS 20 (Wet Runway) <0005> and <Metric>	PERF 6–9
Take-off Weight and V ₁ Reduction – FLAPS 20 (3.2 mm Water or Equivalent) <0005> and <Metric>	PERF 6–10

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
Take-off Weight and V ₁ Reduction – FLAPS 20 (6.4 mm Water or Equivalent) <0005> and <Metric>	PERF 6–11
Take-off Weight and V ₁ Reduction – FLAPS 20 Compacted Snow <0005> and <Metric>	PERF 6–13
Take-off Weight and V ₁ Reduction – FLAPS 20 Ice <0005> and <Metric>	PERF 6–14
Actual Landing Distance, FLAPS 45, Both Thrust Reversers Operative, (Standing Water / Slush) <0005> or <0068> and <Metric>	PERF 6–15
Actual Landing Distance, FLAPS 45, Both Thrust Reversers Operative, Dry Snow or Wet Snow <0005> or <0068> and <Metric>	PERF 6–16
Actual Landing Distance, FLAPS 45, Both Thrust Reversers Operative, Compacted Snow <0005> or <0068> and <Metric>	PERF 6–17
Actual Landing Distance, FLAPS 45, Both Thrust Reversers Operative, Ice <0005> or <0068> and <Metric>	PERF 6–18
Actual Landing Distance, FLAPS 0, Both Thrust Reversers Operative, (Standing Water / Slush / Dry Snow or Wet Snow) <0005> or <0068> and <Metric>	PERF 6–19
Actual Landing Distance, FLAPS 8, Both Thrust Reversers Operative, (Standing Water / Slush / Dry Snow or Wet Snow) <0005> or <0068> and <0006> and <Metric>	PERF 6–20
Actual Landing Distance, FLAPS 20, Both Thrust Reversers Operative, (Standing Water / Slush / Dry Snow or Wet Snow) <0005> or <0068> and <Metric>	PERF 6–21
Actual Landing Distance, FLAPS 30, Both Thrust Reversers Operative, (Standing Water / Slush / Dry Snow or Wet Snow) <0005> or <0068> and <Metric>	PERF 6–22
Actual Landing Distance, FLAPS 0, Both Thrust Reversers Operative, (Compacted Snow) <0005> or <0068> and <Metric>	PERF 6–23
Actual Landing Distance, FLAPS 8, Both Thrust Reversers Operative, (Compacted Snow) <0005> or <0068> and <0006> and <Metric>	PERF 6–24
Actual Landing Distance, FLAPS 20, Both Thrust Reversers Operative, (Compacted Snow) <0005> or <0068> and <Metric>	PERF 6–25
Actual Landing Distance, FLAPS 30, Both Thrust Reversers Operative, (Compacted Snow) <0005> or <0068> and <Metric>	PERF 6–26

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS

PAGE

Actual Landing Distance, FLAPS 0, Both Thrust
Reversers Operative, (Ice) <0005> or <0068> and
<Metric> PERF 6–27

Actual Landing Distance, FLAPS 8, Both Thrust
Reversers Operative, (Ice) <0005> or <0068> and
<0006> and <Metric> PERF 6–28

Actual Landing Distance, FLAPS 20, Both
Thrust Reversers Operative, (Ice) <0005> or <0068>
and <Metric> PERF 6–29

Actual Landing Distance, FLAPS 30, Both
Thrust Reversers Operative, (Ice) <0005> or <0068>
and <Metric> PERF 6–30

Actual Landing Distance, FLAPS 0, Both Thrust
Reversers Operative, (Wet) Advisory Data
Only <0005> or <0068> and <Metric> PERF 6–31

Actual Landing Distance, FLAPS 8, Both Thrust
Reversers Operative, (Wet) Advisory Data
Only <0005> or <0068> and <0006> and <Metric> PERF 6–32

Actual Landing Distance, FLAPS 20, Both
Thrust Reversers Operative, (Wet) Advisory
Data Only <0005> or <0068> and <Metric> PERF 6–33

Actual Landing Distance, FLAPS 30, Both
Thrust Reversers Operative, (Wet) Advisory
Data Only <0005> or <0068> and <Metric> PERF 6–34

Actual Landing Distance, FLAPS 45, Both
Thrust Reversers Operative, (Wet) Advisory
Data Only <0005> or <0068> and <Metric> PERF 6–35

ANTI-SKID

Anti-Skid Inoperative – Dry Runway Take-off
Data – Take-off Weight and V₁ Reduction –
FLAPS 8 <0006> and <Metric> PERF 7–3

Anti-Skid Inoperative – Dry Runway Take-off
Data – Take-off Weight and V₁ Reduction –
FLAPS 20 <Metric> PERF 7–4

Anti-Skid Inoperative – Wet Runway Take-off
Data – Take-off Weight and V₁ Reduction –
FLAPS 8 <0006> and <Metric> PERF 7–5

Anti-Skid Inoperative – Wet Runway Take-off
Data – Take-off Weight and V₁ Reduction –
FLAPS 20 <Metric> PERF 7–6

Anti-Skid Inoperative – Landing Speed and
Actual Landing Distance (Landing Gear Down –
FLAPS 45) <Metric> PERF 7–7

Anti-Skid Inoperative – Landing Distance
Increase Factor for Flap Failures PERF 7–8

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

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CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

TAKE-OFF

Stabilizer Trim Setting

Stabilizer Trim Setting for FLAPS 8 or 20 Take-off

C.G. %MAC	All Weights
9	8.2
10	8.0
12	7.7
14	7.4
16	7.1
18	6.7
20	6.4
22	6.1
24	5.8
26	5.4
28	5.1
30	4.8
32	4.5
34	4.2
35	4.0

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Normal Take-off Thrust Setting

Normal Take-off Thrust Setting, %N₁ Zero Engine Bleeds <0005>

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING ZERO ENGINE BLEED CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	80.9	81.3	82.1	82.9	83.9	84.9	85.9
-35	-31	81.8	82.1	83.0	83.8	84.8	85.8	86.8
-30	-22	82.6	83.0	83.8	84.7	85.6	86.6	87.7
-25	-13	83.4	83.8	84.7	85.5	86.5	87.5	88.6
-20	-4	84.2	84.7	85.5	86.4	87.4	88.4	89.5
-15	5	85.1	85.5	86.3	87.2	88.2	89.3	90.3
-10	14	85.9	86.3	87.2	88.0	89.1	90.1	91.2
-5	23	86.7	87.1	88.0	88.9	89.9	91.0	92.0
0	32	87.5	87.9	88.8	89.7	90.7	91.8	92.9
5	41	88.3	88.7	89.6	90.5	91.5	92.6	93.3
10	50	89.0	89.5	90.4	91.3	92.4	92.7	92.7
15	59	89.8	90.3	91.1	92.1	92.2	92.2	92.1
20	68	90.6	91.0	91.7	91.7	91.7	91.7	91.6
25	77	91.3	91.3	91.3	91.3	91.3	91.2	91.0
30	86	90.9	90.9	90.9	90.8	90.7	90.5	90.3
35	95	90.2	90.3	90.3	90.1	90.0	89.8	89.6
40	104	89.4	89.4	89.5	89.4	89.2		
45	113	88.6	88.6	88.6	88.7			
50	122	87.7	87.7	87.8				

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Normal Take-off Thrust Setting, %N₁ 10th Stage Bleeds On –
PACKs On <0005>

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING 10th STAGE BLEEDS ON – PACKS ON CF34–3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	–1000	0	2000	4000	6000	8000	10000
–40	–40	80.9	81.3	82.1	82.9	83.9	84.9	85.9
–35	–31	81.7	82.1	82.9	83.8	84.7	85.7	86.8
–30	–22	82.5	82.9	83.7	84.6	85.6	86.6	87.7
–25	–13	83.3	83.7	84.6	85.5	86.4	87.5	88.5
–20	–4	84.1	84.5	85.4	86.3	87.3	88.3	89.4
–15	5	84.9	85.4	86.2	87.1	88.1	89.2	90.3
–10	14	85.7	86.2	87.0	88.0	89.0	90.1	91.2
–5	23	86.5	87.0	87.8	88.8	89.8	90.9	92.0
0	32	87.3	87.8	88.7	89.6	90.7	91.8	92.6
5	41	88.1	88.6	89.5	90.5	91.5	92.2	92.4
10	50	88.9	89.4	90.3	91.3	91.7	91.9	91.8
15	59	89.7	90.2	91.1	91.4	91.4	91.3	91.3
20	68	90.5	90.8	91.0	91.0	90.9	90.8	90.7
25	77	90.6	90.6	90.6	90.5	90.4	90.2	90.0
30	86	90.1	90.2	90.1	89.9	89.8	89.5	89.2
35	95	89.4	89.5	89.5	89.3	89.0	88.8	88.4
40	104	88.6	88.5	88.6	88.5	88.2		
45	113	87.7	87.6	87.7	87.6			
50	122	86.8	86.8	86.8				

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Normal Take-off Thrust Setting, %N₁ Cowl Anti-ice On <0005>

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING COWL ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	80.9	81.3	82.1	82.9	83.9	84.9	85.9
-35	-31	81.7	82.1	82.9	83.8	84.7	85.7	86.8
-30	-22	82.5	82.9	83.7	84.6	85.6	86.6	87.7
-25	-13	83.3	83.7	84.6	85.4	86.4	87.5	88.5
-20	-4	84.1	84.5	85.4	86.3	87.3	88.3	89.4
-15	5	84.9	85.3	86.2	87.1	88.1	89.2	90.3
-10	14	85.7	86.1	87.0	87.9	89.0	90.0	91.2
-5	23	86.5	87.0	87.8	88.8	89.8	90.9	92.0
0	32	87.3	87.8	88.7	89.6	90.7	91.8	92.6
5	41	88.1	88.6	89.5	90.4	91.5	92.2	92.4
10	50	88.9	89.4	90.3	91.3	91.7	91.9	91.8

cf34-3b1_basic_nto_cowl_qrh_n1.ps

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Normal Take-off Thrust Setting, %N₁ Wing and Cowl Anti-ice
On <0005>

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING WING AND COWL ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	80.9	81.3	82.1	82.9	83.9	84.9	85.9
-35	-31	81.7	82.1	82.9	83.8	84.7	85.7	86.8
-30	-22	82.5	82.9	83.7	84.6	85.6	86.6	87.6
-25	-13	83.3	83.7	84.6	85.4	86.4	87.4	88.5
-20	-4	84.1	84.5	85.4	86.3	87.3	88.3	88.7
-15	5	84.9	85.3	86.2	87.1	87.8	88.3	88.7
-10	14	85.7	86.1	86.8	87.2	87.8	88.3	88.7
-5	23	86.2	86.6	86.8	87.2	87.8	88.3	88.7
0	32	86.2	86.6	86.8	87.2	87.8	88.3	88.7
5	41	86.2	86.6	86.8	87.2	87.8	88.3	88.6
10	50	86.2	86.6	86.8	87.2	87.8	87.9	87.9

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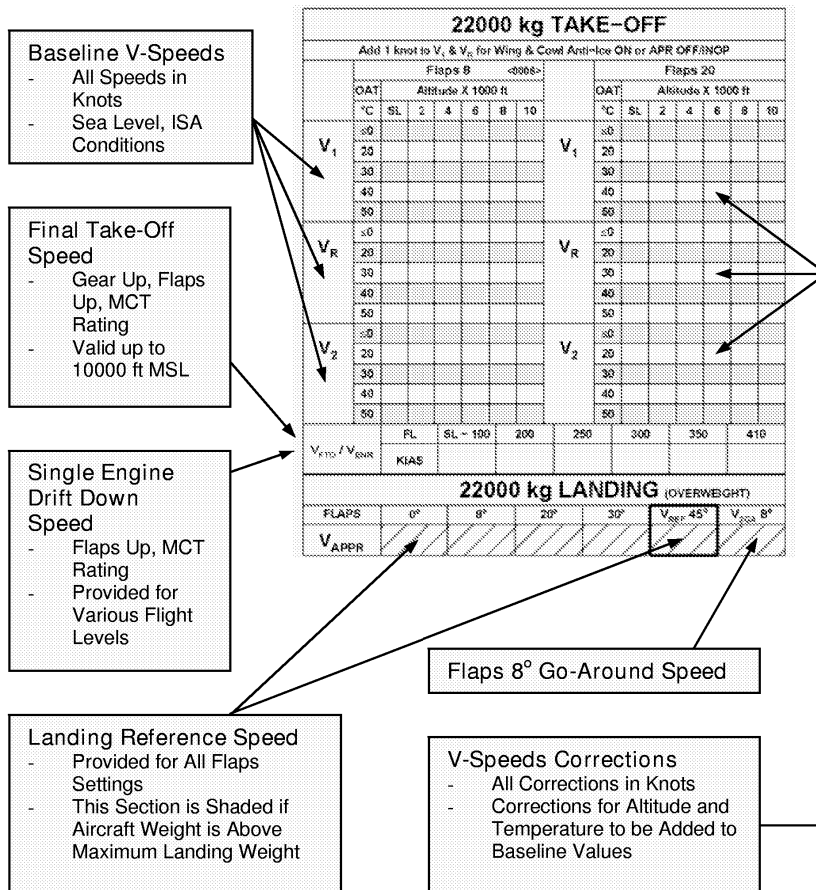
NOTE

In icing conditions, between 0°C and +5°C, at pressure altitudes of 2000 feet and below, the take-off thrust settings with cowl anti-icing on may be used when taking-off with the wing and cowl anti-icing on.



Speedbook

Speedcard – Legend <0006> and <Metric>



Notes

- V_1 and V_R Data are for a Balanced Field Length
- V_1 and V_R Data are Based on 0 Runway Slope and 0 Wind
- V_1 Must Not Be Less Than V_{1MCG} Nor Greater Than V_{1MBE}
- Speed Increment with Packs On Engine Bleed is Negligible
- -- : Data is not available for interpolation

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Speedcard – 15400 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

15400 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 108	≤0	0	2	3	5	6	8	V ₁ 98	≤0	0	2	3	5	7	8
	20	0	2	4	6	8	10		20	0	2	4	7	9	11
	30	2	4	6	8	10	12		30	2	4	7	9	11	13
	40	4	6	8	10	12	13		40	4	7	9	11	13	14
	50	6	9	10	12	14	15		50	7	9	11	13	15	17
V _R 113	≤0	0	1	2	3	4	5	V _R 105	≤0	0	1	2	4	5	6
	20	0	1	3	4	5	7		20	0	2	3	5	6	7
	30	1	2	4	5	6	8		30	1	3	4	6	7	8
	40	2	3	5	6	8	9		40	3	4	6	7	8	9
	50	3	5	6	8	9	10		50	4	6	7	8	9	10
V ₂ 127	≤0	0	0	0	0	0	0	V ₂ 117	≤0	0	0	0	0	0	0
	20	0	0	0	0	0	0		20	0	0	0	0	0	0
	30	0	0	0	0	0	0		30	0	0	0	0	0	0
	40	0	0	0	0	0	0		40	0	0	0	0	0	0
	50	0	0	0	0	0	0		50	0	0	0	0	0	0
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		152		152		159		167		175		186	
15400 kg LANDING															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		150		138		132		128		120		127			

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 16000 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

16000 kg TAKE-OFF																	
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP																	
	Flaps 8 <0006>									Flaps 20							
	OAT	Altitude X 1000 ft								OAT	Altitude X 1000 ft						
	°C	SL	2	4	6	8	10	°C		SL	2	4	6	8	10		
V ₁ 112	≤0	-1	1	3	4	6	7	V ₁ 101	≤0	0	2	4	5	7	8		
	20	0	1	3	6	8	9		20	0	2	5	7	9	11		
	30	1	3	5	7	9	11		30	2	4	7	9	11	13		
	40	3	5	8	9	11	12		40	4	7	9	11	13	15		
	50	6	8	10	11	13	14		50	7	9	11	13	15	16		
V _R 116	≤0	-1	1	2	3	4	5	V _R 108	≤0	0	1	2	3	5	5		
	20	0	1	2	4	5	7		20	0	2	3	4	6	7		
	30	1	2	4	5	6	8		30	1	3	4	5	7	8		
	40	2	3	5	6	8	9		40	3	4	5	7	8	9		
	50	3	5	6	7	9	10		50	4	5	7	8	9	9		
V ₂ 129	≤0	0	0	0	0	0	0	V ₂ 119	≤0	0	0	0	0	0	0		
	20	0	0	0	0	0	0		20	0	0	0	0	0	0		
	30	0	0	0	0	0	0		30	0	0	0	0	0	0		
	40	0	0	0	0	0	0		40	0	0	0	0	0	0		
	50	0	0	0	0	0	0		50	0	0	0	0	0	0		
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410			
		KIAS		154		154		162		170		179		190			
16000 kg LANDING																	
FLAPS	0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°						
V _{APPR}	152		140		134		130		122		129						

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 17000 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

17000 kg TAKE-OFF																	
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP																	
	Flaps 8 <0006>									Flaps 20							
	OAT	Altitude X 1000 ft								OAT	Altitude X 1000 ft						
	°C	SL	2	4	6	8	10	°C		SL	2	4	6	8	10		
V ₁ 117	≤0	0	1	3	4	6	7		V ₁ 106	≤0	0	2	4	5	7	8	
	20	0	1	4	6	8	9			20	1	2	5	7	9	11	
	30	1	3	6	8	9	11			30	2	4	7	9	11	12	
	40	4	6	8	9	11	12			40	5	7	9	11	13	15	
	50	6	8	9	11	12	13			50	7	9	11	13	15	15	
V _R 121	≤0	-1	0	1	3	4	5		V _R 113	≤0	0	1	2	3	4	5	
	20	0	1	2	4	5	6			20	0	1	3	4	5	6	
	30	0	2	3	5	6	7			30	1	2	4	5	6	7	
	40	2	3	5	6	7	8			40	2	4	5	6	7	8	
	50	3	5	6	7	8	9			50	4	5	6	7	8	8	
V ₂ 133	≤0	0	0	0	0	0	0		V ₂ 123	≤0	0	0	0	0	0	0	
	20	0	0	0	0	0	0			20	0	0	0	0	0	0	
	30	0	0	0	0	0	0			30	0	0	0	0	0	0	
	40	0	0	0	0	0	0			40	0	0	0	0	0	0	
	50	0	0	0	0	0	0			50	0	0	0	0	0	0	
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410			
		KIAS		158		159		167		176		184		196			
17000 kg LANDING																	
FLAPS	0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°						
V _{APPR}	156		144		138		134		126		133						

<CRJ200 QRH takeoff basic met Jun09 17000.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 18000 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

18000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 122	≤0	0	1	3	5	6	7	V ₁ 112	≤0	0	1	3	4	6	7
	20	0	2	4	6	7	9		20	0	1	4	6	8	9
	30	1	4	6	7	9	10		30	1	3	5	7	9	12
	40	4	6	7	9	10	12		40	3	5	7	9	12	13
	50	6	8	9	11	12	12		50	6	8	9	12	13	13
V _R 125	≤0	0	1	2	3	4	5	V _R 117	≤0	0	1	2	3	4	5
	20	0	1	3	4	6	7		20	0	2	3	4	5	6
	30	1	3	4	5	7	8		30	1	3	4	5	6	7
	40	2	4	5	6	8	9		40	3	4	5	6	7	8
	50	4	5	6	8	9	9		50	4	5	6	7	8	8
V ₂ 137	≤0	0	0	0	0	0	0	V ₂ 126	≤0	0	0	0	0	1	1
	20	0	0	0	0	0	0		20	0	0	0	0	1	1
	30	0	0	0	0	0	0		30	0	0	0	0	1	1
	40	0	0	0	0	0	0		40	0	0	0	0	1	1
	50	0	0	0	0	0	0		50	0	0	0	0	1	1
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		163		164		172		181		190		202	
18000 kg LANDING															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		160		148		142		138		130		137			

<CRJ200 QRH takeoff basic met Jun09 18000.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 19000 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

19000 kg TAKE-OFF																	
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP																	
	Flaps 8 <0006>									Flaps 20							
	OAT	Altitude X 1000 ft								OAT	Altitude X 1000 ft						
	°C	SL	2	4	6	8	10	°C		SL	2	4	6	8	10		
V ₁ 127	≤0	0	2	3	4	5	7	V ₁ 116	≤0	0	2	3	5	6	8		
	20	0	2	4	6	7	9		20	0	2	4	6	8	11		
	30	1	4	5	7	8	10		30	2	4	6	8	10	13		
	40	4	5	7	9	10	11		40	4	6	8	11	12	13		
	50	6	7	9	10	11	12		50	7	8	11	12	12	---		
V _R 130	≤0	-1	1	2	3	4	5	V _R 122	≤0	-1	1	2	2	3	5		
	20	0	1	2	4	5	6		20	0	1	2	3	4	6		
	30	1	2	4	5	6	7		30	1	2	3	4	5	7		
	40	2	3	5	6	7	8		40	2	3	4	5	6	7		
	50	3	5	6	7	8	9		50	3	4	5	6	6	---		
V ₂ 141	≤0	0	0	0	0	0	0	V ₂ 130	≤0	0	0	0	0	0	1		
	20	0	0	0	0	0	0		20	0	0	0	0	0	1		
	30	0	0	0	0	0	0		30	0	0	0	0	0	1		
	40	0	0	0	0	0	0		40	0	0	0	0	0	1		
	50	0	0	0	0	0	0		50	0	0	0	0	0	---		
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410			
		KIAS		168		169		177		186		196		208			
19000 kg LANDING																	
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°					
V _{APPR}		163		151		145		141		133		141					

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 20000 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

20000 kg TAKE-OFF														
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP														
	Flaps 8 													

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 21000 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

21000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 -0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 137	≤0	0	1	2	3	4	6	V ₁ 125	≤0	0	2	3	5	7	10
	20	0	1	3	4	6	8		20	0	2	4	6	10	12
	30	1	3	4	6	7	9		30	2	4	6	9	11	12
	40	3	4	6	7	8	9		40	4	6	9	10	12	--
	50	5	6	7	8	9	9		50	6	9	9	10	--	--
V _R 138	≤0	0	1	2	3	4	6	V _R 130	≤0	0	1	1	2	5	6
	20	1	2	3	4	5	7		20	0	1	2	3	6	7
	30	1	3	4	5	6	8		30	1	2	3	4	6	7
	40	3	4	5	6	7	8		40	2	3	4	5	7	--
	50	4	5	6	7	8	8		50	3	4	4	5	--	--
V ₂ 148	≤0	0	0	0	0	0	0	V ₂ 136	≤0	0	0	1	1	2	3
	20	0	0	0	0	0	0		20	0	0	1	1	2	3
	30	0	0	0	0	0	0		30	0	0	1	1	2	3
	40	0	0	0	0	0	0		40	0	0	1	1	2	--
	50	0	0	0	0	0	0		50	0	0	1	1	--	--
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		176		177		186		196		206		220	
21000 kg LANDING															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		170		158		152		148		140		148			

<CRJ200_QRH_takeoff_basic_met_Jun09_21000.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 22000 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

22000 kg TAKE-OFF																	
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP																	
	Flaps 8 <0006>									Flaps 20							
	OAT	Altitude X 1000 ft								OAT	Altitude X 1000 ft						
	°C	SL	2	4	6	8	10	°C		SL	2	4	6	8	10		
V ₁ 141	≤0	0	1	2	3	5	6	V ₁ 130	≤0	0	1	3	4	8	10		
	20	0	1	3	4	6	8		20	0	1	3	6	10	11		
	30	1	3	4	6	7	9		30	1	3	5	8	10	11		
	40	3	4	6	7	8	9		40	3	6	7	8	10	---		
	50	5	6	7	8	8	---		50	6	7	8	---	---	---		
V _R 143	≤0	-1	0	2	2	3	5	V _R 134	≤0	-1	0	1	2	5	6		
	20	0	1	2	3	4	6		20	0	1	2	3	6	7		
	30	1	2	3	4	5	7		30	0	1	2	4	6	7		
	40	2	3	4	5	6	7		40	1	2	3	4	6	---		
	50	3	4	5	6	6	---		50	2	3	4	---	---	---		
V ₂ 152	≤0	0	0	0	0	0	0	V ₂ 140	≤0	0	0	0	0	2	3		
	20	0	0	0	0	0	0		20	0	0	0	0	2	3		
	30	0	0	0	0	0	0		30	0	0	0	0	2	3		
	40	0	0	0	0	0	0		40	0	0	0	0	2	---		
	50	0	0	0	0	0	---		50	0	0	0	---	---	---		
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410			
		KIAS		180		182		191		201		211		226			
22000 kg LANDING (OVERWEIGHT)																	
FLAPS	0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°						
V _{APPR}	173		161		155		151		143		152						

<CRJ200_QRH_takeoff_basic_met_Jun09_22000.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 23000 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

23000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 -0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 145	≤0	0	1	2	3	5	7	V ₁ 134	≤0	0	1	3	6	9	11
	20	0	1	3	4	6	9		20	0	1	3	8	10	12
	30	1	3	4	6	8	9		30	1	3	6	9	10	--
	40	3	4	6	7	8	9		40	3	6	7	9	--	--
	50	5	6	7	7	--	--		50	6	7	7	--	--	--
V _R 147	≤0	-1	1	1	2	4	6	V _R 137	≤0	0	1	2	4	6	8
	20	0	1	2	3	5	7		20	0	1	2	5	7	9
	30	1	2	3	4	6	7		30	1	2	3	6	7	--
	40	2	3	4	5	6	7		40	2	3	4	6	--	--
	50	3	4	5	5	--	--		50	3	4	4	--	--	--
V ₂ 155	≤0	0	0	0	0	0	1	V ₂ 143	≤0	0	0	0	2	3	4
	20	0	0	0	0	0	1		20	0	0	0	2	3	4
	30	0	0	0	0	0	1		30	0	0	0	2	3	--
	40	0	0	0	0	0	1		40	0	0	0	2	--	--
	50	0	0	0	0	--	--		50	0	0	0	--	--	--
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		184		186		195		206		216		232	
23000 kg LANDING (OVERWEIGHT)															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		176		164		158		154		146		155			

<CRJ200_QRH_takeoff_basic_met_Jun09_23000.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 24000 kg <TC> or <JAA> and <0005> and <0006> and <Metric>

24000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 149	≤0	0	1	2	3	5	7	V ₁ 138	≤0	0	1	4	7	9	13
	20	0	1	3	4	7	9		20	0	1	5	8	9	--
	30	1	3	4	6	8	9		30	1	3	7	8	--	--
	40	3	4	6	7	8	--		40	4	5	7	--	--	--
	50	4	6	6	7	--	--		50	5	6	--	--	--	--
V _R 151	≤0	-1	0	1	2	4	6	V _R 141	≤0	0	1	3	4	6	10
	20	0	1	2	3	5	7		20	0	1	3	5	6	--
	30	0	2	3	4	6	7		30	1	2	4	5	--	--
	40	2	3	4	5	6	--		40	2	2	4	--	--	--
	50	3	4	4	5	--	--		50	2	3	--	--	--	--
V ₂ 158	≤0	0	0	0	0	1	2	V ₂ 146	≤0	0	0	1	2	3	6
	20	0	0	0	0	1	2		20	0	0	1	2	3	--
	30	0	0	0	0	1	2		30	0	0	1	2	--	--
	40	0	0	0	0	1	--		40	0	0	1	--	--	--
	50	0	0	0	0	--	--		50	0	0	--	--	--	--
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		188		190		200		210		222		237	
24000 kg LANDING (OVERWEIGHT)															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		180		168		162		158		150		158			

<CRJ200_QRH_takeoff_basic_met_Jun09_24000.ps>

Take-off Weight Reduction with APR OFF

Take-off Weight Reduction with APR Off <0005> and <Metric>

Weight (Kg)	Weight Reduction (Kg)
16000	1350
18000	1400
20000	1450
22000	1600
24000	1650
26000	1700
28000	1700

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CLIMB

CLIMB

Maximum Climb Thrust Setting

Maximum Climb Thrust Settings, %N₁ PACKs On

DISPLAYED FAN SPEED										
MAXIMUM CLIMB THRUST SETTING PACKS ON CF34-3B1										%N1
OAT		PRESSURE ALTITUDE (Feet)								
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000	36000	41000
-70	-94								93.7	92.5
-65	-85							91.9	94.9	93.5
-60	-76						90.2	93.0	96.1	94.6
-55	-67						91.2	94.1	97.0	95.1
-50	-58					89.2	92.2	95.1	96.4	94.3
-45	-49				87.8	90.2	93.3	96.1	95.8	93.7
-40	-40	80.2	82.9	85.9	88.7	91.2	94.3	95.9	95.0	93.1
-35	-31	81.1	83.8	86.8	89.7	92.1	95.2	95.1	94.3	92.4
-30	-22	81.9	84.7	87.7	90.6	93.1	94.6	94.6	93.7	
-25	-13	82.7	85.6	88.6	91.5	94.0	94.3	93.9	92.7	
-20	-4	83.6	86.4	89.5	92.4	93.6	93.5	93.0		
-15	5	84.4	87.3	90.4	93.3	92.8	92.7	91.9		
-10	14	85.2	88.1	91.2	92.6	91.9	91.8	90.9		
-5	23	86.0	88.9	92.1	91.8	91.0	90.8			
0	32	86.8	89.8	91.3	90.8	90.1	90.0			
5	41	87.6	90.6	90.5	90.1	89.3				
10	50	88.4	89.9	89.7	89.3	88.5				
15	59	89.1	89.1	89.0	88.5					
20	68	88.4	88.4	88.2	87.7					
25	77	87.6	87.6	87.3						
30	86	86.7	86.7	86.4						
35	95	85.9	85.9	85.6						
40	104	85.1	85.1							
45	113	84.4	84.3							
50	122	83.6								

cf34-3b1_mcl_acu_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Maximum Climb Thrust Settings, %N₁ PACKs On; Cowl Anti-ice On

DISPLAYED FAN SPEED	
MAXIMUM CLIMB THRUST SETTING PACKS ON, COWL ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)							
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000	33000
-70	-94								92.7
-65	-85							91.9	93.8
-60	-76						90.2	93.0	94.9
-55	-67						91.2	94.1	95.4
-50	-58					89.2	92.2	95.1	95.4
-45	-49				87.8	90.2	93.3	95.2	94.8
-40	-40	80.2	82.9	85.9	88.7	91.2	94.1	94.4	94.0
-35	-31	81.1	83.8	86.8	89.7	92.1	94.1	93.8	93.6
-30	-22	81.9	84.7	87.7	90.6	93.1	93.6	93.3	92.5
-25	-13	82.7	85.6	88.6	91.5	93.1	93.0	92.2	91.5
-20	-4	83.6	86.4	89.5	92.4	92.5	92.1	91.1	90.2
-15	5	84.4	87.3	90.4	92.4	91.6	91.2	90.0	
-10	14	85.2	88.1	91.2	91.6	90.6	90.1	89.0	
-5	23	86.0	88.9	91.2	90.6	89.6	89.3		
0	32	86.8	89.8	90.4	89.7	88.8	88.3		
5	41	87.6	89.8	89.5	88.9	87.9			
10	50	88.4	89.0	88.7	88.1	87.1			

cf34-3b1_mcl_acu_cowl_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Maximum Climb Thrust Settings, %N₁ PACKs On; Wing and Cowl Anti-ice On

DISPLAYED FAN SPEED	
MAXIMUM CLIMB THRUST SETTING PACKS ON, COWL AND WING ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)							
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000	33000
-70	-94								83.5
-65	-85							92.0	84.5
-60	-76						90.2	92.0	85.4
-55	-67						91.2	92.0	86.4
-50	-58					89.2	91.5	92.0	87.3
-45	-49				87.8	90.2	91.5	92.0	88.3
-40	-40	80.2	82.9	85.9	88.7	90.4	91.5	91.4	89.3
-35	-31	81.1	83.8	86.8	89.7	90.4	91.5	90.3	89.2
-30	-22	81.9	84.7	87.7	89.8	90.4	90.7	89.2	88.1
-25	-13	82.7	85.6	88.6	89.8	90.4	89.7	88.1	87.2
-20	-4	83.6	86.4	89.0	89.8	89.5	88.5	87.3	86.2
-15	5	84.4	87.3	89.0	89.8	88.4	87.4	86.3	
-10	14	85.2	87.8	89.0	88.9	87.5	86.2	85.6	
-5	23	86.0	87.8	89.0	88.0	86.4	85.4		
0	32	86.5	87.8	88.1	87.0	85.4	84.5		
5	41	86.5	87.8	87.2	86.0	84.3			
10	50	86.5	87.0	86.2	84.9	83.1			

cf34-3b1_mcl_acu_cowl_wing_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

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CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

IN-FLIGHT

IN-FLIGHT

Maximum Continuous Thrust Setting

Maximum Continuous Thrust Setting, %N₁ PACKs On <0005>

DISPLAYED FAN SPEED	
MAXIMUM CONTINUOUS THRUST SETTING PACK ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)									
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000	36000	41000	
-70	-94								89.8	90.8	
-65	-85							88.9	90.8	91.9	
-60	-76						88.7	89.9	91.9	93.0	
-55	-67						89.6	90.9	93.0	94.0	
-50	-58					89.2	90.5	92.0	94.0	95.0	
-45	-49				88.5	90.1	91.5	93.0	95.0	94.3	
-40	-40	82.2	84.4	86.6	89.4	91.1	92.5	94.0	94.5	93.7	
-35	-31	83.0	85.2	87.5	90.3	92.1	93.4	95.0	93.8	93.2	
-30	-22	83.8	86.1	88.4	91.2	93.0	94.3	94.9	93.3		
-25	-13	84.7	87.0	89.3	92.2	94.0	95.2	94.3	92.4		
-20	-4	85.5	87.8	90.2	93.1	94.9	94.8	93.6			
-15	5	86.3	88.7	91.1	94.0	95.1	94.2	92.8			
-10	14	87.1	89.5	92.0	94.9	94.5	93.5	91.8			
-5	23	88.0	90.4	92.8	94.7	93.7	92.7				
0	32	88.8	91.2	93.7	93.9	93.0	91.7				
5	41	89.6	92.1	93.4	93.2	92.3					
10	50	90.5	92.9	92.9	92.6	91.5					
15	59	91.3	92.7	92.4	91.8						
20	68	92.1	92.1	91.6	91.0						
25	77	91.8	91.5	90.9							
30	86	91.1	90.7	89.9							
35	95	90.3	89.9	89.0							
40	104	89.5	88.9								
45	113	88.6	88.0								
50	122	87.6									

cf34-3b1_basic_mct_acu_qrh_n1.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Maximum Continuous Thrust Setting, %N₁ PACKs On; Cowl
Anti-ice On <0005>

DISPLAYED FAN SPEED	
MAXIMUM CONTINUOUS THRUST SETTING PACK ON, COWL ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000
-65	-85							88.7
-60	-76						88.4	89.7
-55	-67						89.3	90.7
-50	-58					89.2	90.2	91.8
-45	-49				88.5	90.1	91.2	92.8
-40	-40	82.2	84.4	86.6	89.4	91.1	92.1	93.8
-35	-31	83.0	85.2	87.5	90.3	92.0	93.1	94.0
-30	-22	83.8	86.1	88.4	91.2	93.0	94.0	93.4
-25	-13	84.7	86.9	89.3	92.2	94.0	94.1	92.6
-20	-4	85.5	87.8	90.1	93.1	94.4	93.5	91.7
-15	5	86.3	88.6	91.0	94.0	93.9	92.6	90.6
-10	14	87.1	89.5	91.9	94.1	93.1	91.8	89.6
-5	23	88.0	90.4	92.8	93.5	92.3	90.8	
0	32	88.8	91.2	92.9	92.7	91.6	89.9	
5	41	89.6	92.1	92.5	92.1	90.8		
10	50	90.4	92.4	91.9	91.4	90.0		

cf34-3b1_basic_mct_acu_cowl_qrh_n1.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Maximum Continuous Thrust Setting, %N₁ PACKs On; Wing and Cowl Anti-ice On <0005>

DISPLAYED FAN SPEED	
MAXIMUM CONTINUOUS THRUST SETTING PACK ON, COWL AND WING ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000
-65	-85							87.5
-60	-76						87.5	88.5
-55	-67						88.5	89.6
-50	-58					88.7	89.6	90.7
-45	-49				88.4	89.8	90.8	91.6
-40	-40	82.2	84.4	86.6	89.4	90.2	90.8	91.6
-35	-31	83.0	85.2	87.5	89.7	90.2	90.8	91.0
-30	-22	83.8	86.1	88.4	89.7	90.2	90.8	90.0
-25	-13	84.7	86.9	88.9	89.7	90.2	90.2	89.0
-20	-4	85.5	87.8	88.9	89.7	90.2	89.4	88.0
-15	5	86.3	88.4	88.9	89.7	89.6	88.6	87.0
-10	14	87.1	88.4	88.9	89.7	89.0	87.8	86.1
-5	23	87.5	88.4	88.9	89.2	88.3	86.7	
0	32	87.5	88.4	88.9	88.6	87.5	85.7	
5	41	87.5	88.4	88.4	87.9	86.6		
10	50	87.5	88.4	87.7	87.0	85.4		

cf34-3b1_basic_mct_acu_cowl_wing_qrh_n1.ps

<0005>

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Maximum Cruise Thrust Setting

Maximum Cruise Thrust Settings, %N₁ PACKs On

DISPLAYED FAN SPEED	
MAXIMUM CRUISE THRUST SETTING PACKS ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)									
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000	36000	41000	
-70	-94								92.8	91.0	
-65	-85							91.1	93.9	92.1	
-60	-76						89.9	92.1	95.0	93.1	
-55	-67						90.9	93.2	95.7	93.7	
-50	-58					88.9	91.9	94.3	94.9	93.0	
-45	-49				87.3	89.9	92.9	95.3	94.3	92.5	
-40	-40	80.3	82.9	85.5	88.3	90.9	93.9	94.8	93.8	91.6	
-35	-31	81.2	83.8	86.4	89.2	91.9	94.9	94.5	92.8	90.5	
-30	-22	82.0	84.7	87.3	90.1	92.8	94.4	93.6	91.7		
-25	-13	82.9	85.6	88.2	91.0	93.8	93.5	92.7	90.7		
-20	-4	83.7	86.4	89.1	92.0	93.0	92.5	91.7			
-15	5	84.5	87.3	90.0	92.9	92.1	91.6	90.8			
-10	14	85.3	88.1	90.9	92.0	91.3	90.9	90.0			
-5	23	86.1	88.9	91.7	91.3	90.6	90.2				
0	32	86.9	89.8	91.1	90.6	89.8	89.5				
5	41	87.7	90.6	90.4	89.9	89.2					
10	50	88.5	89.9	89.7	89.2	88.5					
15	59	89.3	89.2	89.0	88.6						
20	68	88.6	88.5	88.4	87.9						
25	77	87.9	87.8	87.7							
30	86	87.1	87.1	87.0							
35	95	86.4	86.4	86.3							
40	104	85.7	85.7								
45	113	85.0	85.0								
50	122	84.2									

cf34-3b1_mcr_acu_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Maximum Cruise Thrust Settings, %N₁ PACKs On;
Cowl Anti-ice On

DISPLAYED FAN SPEED	
MAXIMUM CRUISE THRUST SETTING PACKS ON, COWL ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)							
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000	33000
-70	-94								91.2
-65	-85							91.1	92.3
-60	-76						89.9	92.1	93.4
-55	-67						90.9	93.2	94.0
-50	-58					88.9	91.9	94.2	93.9
-45	-49				87.3	89.9	92.9	94.2	93.2
-40	-40	80.3	82.9	85.5	88.3	90.9	93.9	93.9	92.7
-35	-31	81.2	83.8	86.4	89.2	91.9	94.0	93.1	91.7
-30	-22	82.0	84.7	87.3	90.1	92.8	93.2	92.1	90.6
-25	-13	82.9	85.6	88.2	91.0	92.8	92.2	91.0	89.4
-20	-4	83.7	86.4	89.1	92.0	91.9	91.3	90.1	88.4
-15	5	84.5	87.3	90.0	92.0	91.0	90.5	89.2	
-10	14	85.3	88.1	90.9	91.1	90.3	89.7	88.3	
-5	23	86.1	88.9	90.9	90.4	89.5	88.9		
0	32	86.9	89.8	90.3	89.6	88.7	88.3		
5	41	87.7	89.8	89.5	88.8	88.0			
10	50	88.5	89.1	88.8	88.2	87.3			

cf34-3b1_mcr_acu_cowl_qrh_n1.ps

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Maximum Cruise Thrust Settings, %N₁ PACKs On; Wing and Cowl Anti-ice On

DISPLAYED FAN SPEED	
MAXIMUM CRUISE THRUST SETTING PACKS ON, COWL & WING ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)							
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000	33000
-70	-94								83.5
-65	-85							91.1	84.5
-60	-76						89.9	91.6	85.4
-55	-67						90.9	91.6	86.4
-50	-58					88.9	91.3	91.6	87.3
-45	-49				87.3	89.9	91.3	91.6	88.3
-40	-40	80.3	82.9	85.5	88.3	90.2	91.3	90.7	88.9
-35	-31	81.2	83.8	86.4	89.2	90.2	91.3	89.5	87.8
-30	-22	82.0	84.7	87.3	89.7	90.2	90.2	88.3	86.8
-25	-13	82.9	85.6	88.2	89.7	90.2	89.2	87.3	85.9
-20	-4	83.7	86.4	89.0	89.7	89.4	88.4	86.5	85.1
-15	5	84.5	87.3	89.0	89.7	88.5	87.4	85.8	
-10	14	85.3	88.0	89.0	89.0	87.6	86.6	85.0	
-5	23	86.1	88.0	89.0	88.1	86.7	85.6		
0	32	86.7	88.0	88.3	87.2	85.9	84.6		
5	41	86.7	88.0	87.4	86.4	85.0			
10	50	86.7	87.2	86.7	85.6	84.2			

cf34-3b1_mcr_acu_cowl_wing_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Minimum Drag Speed

Minimum Drag Speed <Metric>

Weight (kg)	Lower Flight Levels								
	≤100	150	200	250	270	290	300	310	320
	Minimum Drag Speed (Mach)								
	Minimum Drag Speed (KIAS)								
15400	– 156	– 157	– 161	– 164	– 166	– 168	0.459 169	0.470 169	0.481 169
16000	– 159	– 161	– 164	– 168	– 170	0.457 172	0.468 172	0.479 172	0.491 173
17000	– 164	– 167	– 170	– 175	0.451 177	0.472 177	0.483 178	0.494 178	0.507 179
18000	– 169	– 173	– 176	– 181	0.464 182	0.486 183	0.497 183	0.510 184	0.523 184
19000	– 175	– 178	– 182	0.456 187	0.477 187	0.500 188	0.512 189	0.525 190	0.539 190
20000	– 180	– 184	– 188	0.468 192	0.490 193	0.514 194	0.527 195	0.541 195	0.555 196
21000	– 186	– 189	– 194	0.480 197	0.503 198	0.528 199	0.542 200	0.556 201	0.571 202
22000	– 191	– 194	– 200	0.492 202	0.515 203	0.542 205	0.556 206	0.571 207	0.585 208
23000	– 196	– 199	0.451 205	0.503 207	0.528 208	0.556 210	0.571 212	0.585 212	0.596 212
24000	– 200	– 205	0.461 210	0.515 212	0.541 214	0.570 216	0.584 217	0.595 216	0.606 216

Mach numbers that may be too low to be posted on the primary flight display are replaced by a – character in the table

<CRJ200_QRH_VMD_met_lower_Apr09.ps>

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Weight (kg)	Higher Flight Levels								
	330	340	350	360	370	380	390	400	410
	Minimum Drag Speed (Mach)								
	Minimum Drag Speed (KIAS)								
15400	0.493 169	0.505 170	0.519 171	0.533 171	0.547 172	0.563 173	0.579 174	0.591 174	0.603 174
16000	0.503 173	0.516 174	0.530 175	0.544 175	0.560 176	0.576 178	0.589 178	0.600 177	0.612 177
17000	0.520 179	0.534 180	0.548 181	0.564 182	0.580 183	0.592 183	0.603 182	0.615 182	0.630 182
18000	0.537 185	0.551 186	0.567 187	0.583 189	0.594 188	0.605 187	0.618 187	0.633 187	0.648 188
19000	0.554 191	0.569 193	0.584 193	0.595 193	0.607 192	0.620 192	0.635 192	0.650 193	0.667 194
20000	0.571 198	0.585 198	0.596 198	0.608 197	0.621 197	0.636 197	0.652 198	0.668 199	
21000	0.585 203	0.596 202	0.608 202	0.621 202	0.636 202	0.652 203	0.668 204		
22000	0.596 207	0.608 207	0.620 206	0.635 207	0.651 207	0.668 208			
23000	0.607 211	0.620 211	0.634 211	0.650 212	0.667 213	0.684 214			
24000	0.618 215	0.633 216	0.648 216	0.665 217	0.682 218				

No data are provided where the aircraft is limited by maximum cruise thrust or by 1.3 G buffet margin

<CRJ200_QRH_VMD_met_higher_Apr09.ps>

Altitude Capability

Altitude Capability 0.70 M <Metric>

Altitude Capability – 0.70 M*						
Weight At Altitude (kg)	500 fpm Rate of Climb			300 fpm Rate of Climb		
	Deviation from ISA (°C)					
	10 and below	15	20	10 and below	15	20
	Pressure Altitude (feet)					
15400	40400	39300	38200	41000	40700	39500
16000	39800	38800	37700	41000	40200	39000
17000	38900	37900	36800	40200	39300	38200
18000	38000	37100	36000	39300	38400	37400
19000	37200	36200	32000	38500	37600	36600
20000	36400	35300	31100	37700	36800	35900
21000	35700	31800	30300	36900	36000	34400
22000	34400	31000	29500	36200	34900	31700
23000	31400	30100	28600	35200	33600	30900
24000	30600	29300	27700	34000	31500	30100
	If airplane CG is forward from the nominal 25% MAC position, increase airplane gross weight by 50 kg for every 1% MAC forward of nominal CG position.					

<CRJ200_QRH_altcap_070_met_Apr09.ps>

*NOTE

The data are assuming anti-ice off and 25% MAC CG.

Altitude capability values ensure that the following considerations are met:

- 1. Rate of climb available is not less than minimum specified value, for a climb speed schedule of 250 KCAS/0.70M.
- 2. Specified Mach Number can be maintained at 0.70M with not more than maximum cruise thrust.
- 3. Specified Mach Number is not less than minimum drag speed.

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Altitude Capability 0.74 M <Metric>

Altitude Capability – 0.74 M*						
Weight At Altitude (kg)	500 fpm Rate of Climb			300 fpm Rate of Climb		
	Deviation from ISA (°C)					
	10 and below	15	20	10 and below	15	20
	Pressure Altitude (feet)					
15400	38900	37700	27800	40400	39200	37800
16000	38400	37200	27500	39900	38700	37300
17000	37600	36400	27000	39000	37900	36600
18000	36900	35600	26500	38200	37100	36000
19000	36100	27800	26000	37500	36300	27800
20000	35000	27300	25400	36700	35400	27300
21000	33200	26700	24800	36000	33700	26700
22000	27700	26100	--	34800	28000	--
23000	27100	25500	--	33300	27400	--
24000	26500	24800	--	31800	26900	--
	If airplane CG is forward from the nominal 25% MAC position, increase airplane gross weight by 50 kg for every 1% MAC forward of nominal CG position.					

*NOTE

<CRJ200_QRH_altcap_074_met_Apr09.ps>

The data are assuming anti-ice off and 25% MAC CG.

Altitude capability values ensure that the following considerations are met:

1. Rate of climb available is not less than minimum specified value,
for a climb speed schedule of 290 KCAS/0.74M.
2. Specified Mach Number can be maintained at 0.74M with not more
than maximum cruise thrust.
3. Specified Mach Number is not less than minimum drag speed.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Altitude Capability 0.77 M <Metric>

Altitude Capability – 0.77 M*						
Weight At Altitude (kg)	500 fpm Rate of Climb			300 fpm Rate of Climb		
	Deviation from ISA (°C)					
	10 and below	15	20	10 and below	15	20
	Pressure Altitude (feet)					
15400	30900	23500	21100	38900	25100	22600
16000	25500	23300	20900	38500	24900	22500
17000	29100	22900	--	37700	24600	--
18000	24700	22500	--	37000	24200	--
19000	24200	22100	--	36300	23900	--
20000	23800	21600	--	35200	23500	--
21000	23300	--	--	25200	--	--
22000	22800	--	--	24800	--	--
23000	22400	--	--	24300	--	--
24000	21900	--	--	23900	--	--
	If airplane CG is forward from the nominal 25% MAC position, increase airplane gross weight by 50 kg for every 1% MAC forward of nominal CG position.					

<CRJ200_QRH_altcap_077_met_Apr09.ps>

*NOTE

The data are assuming anti-ice off and 25% MAC CG.

Altitude capability values ensure that the following considerations are met:

1.

Rate of climb available is not less than minimum specified value,
for a climb speed schedule of 320 KCAS/0.77M.
2.

Specified Mach Number can be maintained at 0.77M with not more
than maximum cruise thrust.
3.

Specified Mach Number is not less than minimum drag speed.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Best Rate of Climb (Best ROC)

Best Rate of Climb (KCAS) <Metric>

This speed is recommended when maximum rate of climb is desired in order to reach an altitude on minimum climb time. The recommended maximum rate of climb speed is related to flaps up, all engines operating and is a function of aircraft all-up weight and altitudes as shown in the table below.

CRJ200 Best Rate of Climb (KCAS)

Alt (ft)	A/C All-Up Weight (kg)											
	14000	15000	16000	17000	18000	19000	20000	21000	22000	23000	24000	25000
5000	227	230	231	232	232	234	235	236	237	240	240	242
10000	220	221	224	224	224	225	227	231	231	232	235	238
15000	211	212	214	214	218	222	222	223	227	232	232	236
20000	199	202	205	208	209	212	215	218	221	224	227	231
25000	195	197	200	204	207	210	214	217	221	224	226	229
27000	193	196	198	202	206	209	211	215	218	221	224	227
29000	191	193	197	199	203	207	209	213	216	220	224	226
31000	196	197	198	200	202	206	208	211	215	218	220	224
33000	191	194	198	201	205	208	208	211	215	220	224	228
35000	190	192	195	199	199	203	209	214	219	224	-	-
37000	182	187	190	190	196	203	209	214	217	-	-	-
39000	180	180	182	186	192	199	-	-	-	-	-	-
41000	180	180	180	184	-	-	-	-	-	-	-	-

Please note that no extrapolation of the data in the table is permitted and that the altitude is not to exceed those presented in the table for a given all-up gross weight/altitude condition.

crj200_bestROCm.ps

NOTE: When following the speeds indicated above, it is possible that the climb N1 specified in the FMS, QRH or FPCCM might not be attainable even at the upper physical limit of the thrust lever position. However the aircraft will achieve its best ROC even though the N1 is not reached.

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

APPROACH AND LANDING

Go-Around Thrust Setting

The following thrust setting tables are given for a go-around with either both engines operating or one engine inoperative.

Go-around Thrust Settings, %N₁ Zero Engine Bleed <0005>

DISPLAYED FAN SPEED	
GO AROUND THRUST SETTING ZERO ENGINE BLEED CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	82.4	83.1	84.0	84.8	85.8	86.8	87.8
-35	-31	83.3	83.9	84.8	85.7	86.7	87.7	88.7
-30	-22	84.2	84.8	85.6	86.5	87.5	88.6	89.6
-25	-13	85.0	85.6	86.5	87.4	88.4	89.5	90.5
-20	-4	85.9	86.4	87.3	88.3	89.3	90.3	91.4
-15	5	86.7	87.3	88.2	89.1	90.1	91.2	92.3
-10	14	87.5	88.1	89.0	90.0	91.0	92.1	93.2
-5	23	88.3	88.9	89.9	90.8	91.9	93.0	94.1
0	32	89.2	89.8	90.7	91.7	92.8	93.9	95.0
5	41	90.0	90.6	91.6	92.6	93.6	94.7	95.4
10	50	90.8	91.4	92.4	93.4	94.5	94.8	94.8
15	59	91.5	92.3	93.2	94.3	94.3	94.3	94.2
20	68	92.3	93.1	93.8	93.8	93.8	93.7	93.6
25	77	93.1	93.4	93.4	93.3	93.3	93.2	93.0
30	86	92.9	93.0	93.0	92.8	92.7	92.5	92.3
35	95	92.3	92.4	92.4	92.2	92.0	91.8	91.7
40	104	91.5	91.5	91.5	91.5	91.3		
45	113	90.6	90.6	90.7	90.7			
50	122	89.9	89.9	89.9				

cf34-3b1_basic_apr_zero_qrh_n1.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Go-around Thrust Settings, %N₁ 10th Stage Bleeds On – PACKs On <0005>

DISPLAYED FAN SPEED	
GO AROUND THRUST SETTING 10th STAGE BLEEDS ON – PACKS ON CF34–3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	–1000	0	2000	4000	6000	8000	10000
–40	–40	82.4	83.1	84.0	84.8	85.8	86.8	87.8
–35	–31	83.3	84.0	84.8	85.7	86.7	87.7	88.7
–30	–22	84.2	84.8	85.6	86.6	87.6	88.6	89.6
–25	–13	85.0	85.6	86.5	87.4	88.4	89.5	90.5
–20	–4	85.9	86.5	87.3	88.3	89.3	90.4	91.4
–15	5	86.7	87.3	88.2	89.1	90.1	91.3	92.3
–10	14	87.5	88.1	89.0	90.0	91.0	92.2	93.2
–5	23	88.3	88.9	89.9	90.8	91.9	93.0	94.1
0	32	89.2	89.8	90.7	91.7	92.8	93.9	94.3
5	41	90.0	90.6	91.6	92.5	93.6	94.0	94.1
10	50	90.8	91.4	92.4	93.3	93.6	93.7	93.6
15	59	91.5	92.3	93.0	93.3	93.3	93.2	93.1
20	68	92.3	92.7	92.9	92.8	92.7	92.7	92.4
25	77	92.5	92.5	92.4	92.3	92.2	92.0	91.7
30	86	92.1	92.1	92.0	91.8	91.6	91.3	91.0
35	95	91.4	91.4	91.3	91.1	90.8	90.5	90.3
40	104	90.5	90.4	90.4	90.3	90.0		
45	113	89.6	89.5	89.5	89.5			
50	122	88.7	88.7	88.6				

cf34–3b1_basic_apr_acu_qrh_n1.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Go-around Thrust Settings, %N₁ Cowl Anti-ice On <0005>

DISPLAYED FAN SPEED	
GO AROUND THRUST SETTING COWL ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	82.4	83.1	83.9	84.8	85.8	86.8	87.8
-35	-31	83.3	83.9	84.8	85.7	86.7	87.7	88.7
-30	-22	84.2	84.7	85.6	86.5	87.5	88.5	89.6
-25	-13	85.0	85.6	86.5	87.4	88.4	89.4	90.5
-20	-4	85.9	86.4	87.3	88.2	89.3	90.3	91.4
-15	5	86.7	87.3	88.2	89.1	90.1	91.2	92.3
-10	14	87.5	88.1	89.0	90.0	91.0	92.1	93.2
-5	23	88.3	88.9	89.9	90.8	91.9	93.0	94.1
0	32	89.2	89.7	90.7	91.7	92.7	93.9	94.6
5	41	90.0	90.6	91.5	92.5	93.6	94.2	94.4
10	50	90.8	91.4	92.4	93.4	93.9	93.9	93.9

cf34-3b1_basic_apr_cowl_qrh_n1.ps<0005>

Go-around Thrust Settings, %N₁ Wing and Cowl Anti-ice On <0005>

DISPLAYED FAN SPEED	
GO AROUND THRUST SETTING WING AND COWL ANTI-ICE ON CF34-3B1	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	82.4	83.1	83.9	84.8	85.8	86.8	87.8
-35	-31	83.3	83.9	84.8	85.7	86.7	87.6	88.7
-30	-22	84.2	84.8	85.6	86.5	87.5	88.5	89.6
-25	-13	85.0	85.6	86.5	87.4	88.4	89.4	90.5
-20	-4	85.9	86.4	87.3	88.3	89.3	90.3	90.6
-15	5	86.7	87.2	88.2	89.1	90.1	90.3	90.6
-10	14	87.5	88.1	89.0	89.8	90.1	90.3	90.6
-5	23	88.3	88.9	89.6	89.8	90.1	90.3	90.6
0	32	89.1	89.2	89.6	89.8	90.1	90.3	90.6
5	41	89.1	89.2	89.6	89.8	90.1	90.3	90.4
10	50	89.1	89.2	89.6	89.8	90.1	90.0	89.9

cf34-3b1_basic_apr_cowl_wing_qrh_n1.ps<0005>

Actual Landing Distance – FLAPS 45

Landing Data – Landing Gear Down, FLAPS 45 Landing Speed
(V_{REF}) and Actual Landing Distance <Metric>

Weight (Kg)	V _{REF} * (KIAS)	Pressure Altitude (Feet x 1,000)					
		0	2	4	6	8	10
		Actual Landing Distance* (Meters)					
15400	120	704	733	764	799	836	878
16000	122	723	753	785	821	860	903
17000	126	754	786	819	857	898	944
18000	130	785	818	854	894	937	987
19000	133	816	850	888	930	977	1029
20000	137	846	883	923	968	1017	1072
21000	140	877	915	957	1005	1057	1118
22000	144	907	948	993	1043	1099	1170
23000	147	939	983	1031	1086	1149	1229
24000	151	975	1022	1074	1133	1208	1296

★ V_{REF} = Speed at 50' screen height (KIAS), Flaps 45°
※ Actual Landing Distance = Dry Landing Field Length x 0.6
Wet Landing Field Length = Dry Landing Field Length x 1.15

Actual Landing Distance – Flaps Other Than 45

Landing Data – Actual Landing Distance Factors When
Landing with Flaps Other Than 45

Flaps Setting	Distance (No Reversers)	Distance (W/ Reversers)
0°	Dist. x 1.70 (70%)	Dist. x 1.65 (65%)
8°	Dist. x 1.55 (55%)	Dist. x 1.50 (50%)
20°	Dist. x 1.25 (25%)	Dist. x 1.22 (22%)
30°	Dist. x 1.20 (20%)	Dist. x 1.18 (18%)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Brake Cooling Requirements

V_{1MBE} , FLAPS 8 <0006> and <Metric>

V _{1MBE} , Flaps 8° (Sea Level, 0 Wind, 0 Runway Slope)								
OAT		Airplane Weight (Kg x 1000)						
°C	°F	24	23	22	21	20	19	18
-40	-40	192	196	200	204	208	212	216
-30	-22	187	191	195	199	203	207	211
-20	-4	182	186	190	194	198	202	206
-10	14	178	182	186	190	194	198	202
0	32	173	177	181	185	189	193	197
10	50	169	173	177	181	185	189	193
20	68	166	170	174	178	182	186	190
30	86	163	167	171	175	179	183	187
40	104	161	165	169	173	177	181	185
50	122	159	163	167	171	175	179	183

V_{1MBE} , FLAPS 20 <Metric>

V _{1MBE} , Flaps 20° (Sea Level, 0 Wind, 0 Runway Slope)								
OAT		Airplane Weight (Kg x 1000)						
°C	°F	24	23	22	21	20	19	18
-40	-40	195	199	203	207	211	215	219
-30	-22	190	194	198	202	206	210	214
-20	-4	185	189	193	197	201	205	209
-10	14	181	185	189	193	197	201	205
0	32	176	180	184	188	192	196	200
10	50	172	176	180	184	188	192	196
20	68	169	173	177	181	185	189	193
30	86	166	170	174	178	182	186	190
40	104	164	168	172	176	180	184	188
50	122	162	166	170	174	178	182	186

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



PERF 4-6

Rev. 59, Mar 21/2012

- At weights less than 18000 kg, use V_{1MBE} for 18000 kg.
- Reduce V_{1MBE} by 3 KIAS per 1000 feet altitude above sea level.

Wind Correction:

Reduce V_{1MBE} by 1.5 KIAS/1 kt tailwind

Increase V_{1MBE} by 0.3 KIAS/1 kt headwind

Rwy Slope Correction:

Reduce V_{1MBE} by 2.0 KIAS/1% downhill

Increase V_{1MBE} by 2.0 KIAS/1% uphill

BTMS Operative:

A minimum brake cooling time of fifteen (15) minutes must be observed between a landing stop or a rejected take-off and the subsequent take-off.

Brake temperatures, as obtained from the BTMS, must be stable or decreasing before take-off. The temperature of the warmest brake at the start of the take-off roll must not exceed the BTMS units as shown in the following table:

V_1 / V_{1MBE}	BTMS Units
≤ 0.85	06
> 0.85	05

BTMS Inoperative:

Following a rejected take-off with a brakes application speed of not more than 80 KIAS, or a landing at a weight not exceeding the maximum permissible quick turn-around landing weight, the brake cooling times shown in the following table must be observed.

V_1 / V_{1MBE}	Min. Brake Cooling Time (Minutes)
< 0.90	10
≤ 0.92	15
≤ 0.94	20
≤ 0.96	28
≤ 0.98	36
≤ 1.00	45

Following a rejected take-off with a brakes application speed greater than 80 KIAS, or a landing at a weight exceeding the maximum permissible quick turn-around landing weight, the cooling times shown in the following table must be observed, followed by an inspection of the main wheels, brakes and tires before the next take-off:

	PERF 4-7
	Rev. 59, Mar 21/2012

V₁/ V_{1MBE}	Min. Brake Cooling Time (Minutes)
< 0.85	40
≤0.90	60
≤0.95	90
≤1.00	140

NOTE

The maintenance inspection should be carried out in accordance with the Aircraft Maintenance manual, Chapter 05-51-27, HIGH ENERGY STOP (HES)/REJECTED TAKE-OFF (RTO) – INSPECTION CHECK.

BTMS Operative – Training and Non-Passenger Operation:

The minimum brake cooling time after a landing and the maximum allowable brake temperature for take-off specified in this section must be observed to ensure sufficient brake energy capability in the event of a rejected take-off.

A minimum brake cooling time of five (5) minutes shall be observed between a landing stop and the subsequent take-off. At the end of this period, the temperature of the warmest brake shall not exceed 04 BTMS units.

NOTE

- 1. If the above conditions are not met, this procedure no longer applies. Use the existing instructions in PERF 04 (i.e., minimum brake cooling time of fifteen (15) minutes).
- 2. The temperature indications may continue to increase after the five (5) minute period but they will remain within limits.
- 3. To minimize brake temperature increase upon landing, use of maximum reverse thrust, ground lift dumping and minimum practical wheel braking, is recommended.

During climb, retract the landing gear only when all four (4) BTMS temperature indications are decreasing and are lower than 06 BTMS units.

QUICK REFERENCE HANDBOOK CSP A-022-080	APPROACH AND LANDING
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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

During flight, if any of the four (4) BTMS temperature indications increases to a value higher than 06 BTMS units, extend the landing gear or leave the landing gear extended, as applicable, to cool the brakes. The BTMS temperature indications shall be decreasing and lower than 06 BTMS units prior to landing.

Whether the landing gear is up or down during flight, operations shall be conducted in accordance with the performance data in the Airplane Flight Manual, Chapter 7, Supplements: SUPPLEMENT 5 – Flight With Landing Gear Down.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

REDUCED THRUST TAKE-OFF

Reduced Thrust Take-off Setting

To interpolate between pressure altitude, actual ambient temperature or assumed temperature, when using the reduced thrust take-off tables, the following steps must be followed:

- The pressure altitude to be used in determining the reduced %N₁ setting, for altitudes below 4000 feet, must be rounded up to the next highest altitude. For altitudes above 4000 feet, the %N₁ must be rounded down to the next lowest pressure altitude.
- The assumed temperature to be used in determining the reduced %N₁ setting must be rounded down to the next lowest assumed temperature.
- The actual ambient temperature to be used in determining the reduced %N₁ setting must be rounded up to the next highest actual ambient temperature.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude –1000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = –1000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	25	30	31	35	40	45	50
–40	80.9	80.9	80.9	80.9	80.9	80.9	80.9
–35	81.8	81.8	81.8	81.8	81.8	81.8	81.8
–30	82.6	82.6	82.6	82.6	82.6	82.6	82.6
–25	83.4	83.4	83.4	83.4	83.4	83.4	83.4
–20	84.2	84.2	84.2	84.2	84.2	84.2	84.2
–15	85.1	85.0	85.0	85.0	85.0	85.0	85.0
–10	85.9	85.0	85.0	85.0	85.0	85.0	85.0
–5	86.7	85.5	85.0	85.0	85.0	85.0	85.0
0	87.5	86.3	85.8	85.0	85.0	85.0	85.0
5	88.3	87.1	86.6	85.7	85.0	85.0	85.0
10	89.0	87.8	87.4	86.5	85.0	85.0	85.0
15	89.8	88.6	88.1	87.2	85.7	85.0	85.0
20	90.6	89.4	88.9	88.0	86.5	85.0	85.0
25		90.1	89.7	88.7	87.2	85.7	85.0
30			90.4	89.5	88.0	86.5	85.0
31				89.8	88.2	86.7	85.3
35					88.7	87.2	85.7
40						87.9	86.4
45							87.1
50							

flex_n1_3b1_zero_–1000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude 0 Feet (Sea Level) <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = SEA LEVEL	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	23	25	30	35	40	45	50
–40	81.3	81.3	81.3	81.3	81.3	81.3	81.3
–35	82.1	82.1	82.1	82.1	82.1	82.1	82.1
–30	83.0	83.0	83.0	83.0	83.0	83.0	83.0
–25	83.8	83.8	83.8	83.8	83.8	83.8	83.8
–20	84.7	84.7	84.7	84.7	84.7	84.7	84.7
–15	85.5	85.0	85.0	85.0	85.0	85.0	85.0
–10	86.3	85.8	85.0	85.0	85.0	85.0	85.0
–5	87.1	86.6	85.5	85.0	85.0	85.0	85.0
0	87.9	87.4	86.3	85.0	85.0	85.0	85.0
5	88.7	88.2	87.1	85.8	85.0	85.0	85.0
10	89.5	89.0	87.8	86.6	85.0	85.0	85.0
15	90.3	89.8	88.6	87.3	85.8	85.0	85.0
20	91.0	90.6	89.4	88.1	86.5	85.0	85.0
23		91.0	89.8	88.5	86.9	85.4	85.0
25			90.1	88.8	87.2	85.7	85.0
30				89.6	88.0	86.5	85.0
35					88.7	87.2	85.7
40						87.9	86.4
45							87.0
50							

flex_n1_3b1_zero_0_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude 1000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 1000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	21	25	30	35	40	45	48
–40	81.7	81.7	81.7	81.7	81.7	81.7	81.7
–35	82.5	82.5	82.5	82.5	82.5	82.5	82.5
–30	83.4	83.4	83.4	83.4	83.4	83.4	83.4
–25	84.2	84.2	84.2	84.2	84.2	84.2	84.2
–20	85.1	85.0	85.0	85.0	85.0	85.0	85.0
–15	85.9	85.0	85.0	85.0	85.0	85.0	85.0
–10	86.7	85.8	85.0	85.0	85.0	85.0	85.0
–5	87.5	86.6	85.5	85.0	85.0	85.0	85.0
0	88.3	87.4	86.3	85.0	85.0	85.0	85.0
5	89.1	88.2	87.0	85.7	85.0	85.0	85.0
10	89.9	89.0	87.8	86.5	85.0	85.0	85.0
15	90.7	89.8	88.6	87.3	85.8	85.0	85.0
20	91.5	90.5	89.4	88.0	86.5	85.0	85.0
21		90.7	89.5	88.2	86.7	85.2	85.0
25			90.1	88.8	87.3	85.8	85.0
30				89.5	88.0	86.5	85.6
35					88.7	87.2	86.3
40						87.9	87.0
45							87.7
48							

flex_n1_3b1_zero_1000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude 2000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 2000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)							
(°C)	19	20	25	30	35	40	45	46
–40	82.1	82.1	82.1	82.1	82.1	82.1	82.1	82.1
–35	83.0	83.0	83.0	83.0	83.0	83.0	83.0	83.0
–30	83.8	83.8	83.8	83.8	83.8	83.8	83.8	83.8
–25	84.7	84.7	84.7	84.7	84.7	84.7	84.7	84.7
–20	85.5	85.2	85.0	85.0	85.0	85.0	85.0	85.0
–15	86.3	86.0	85.0	85.0	85.0	85.0	85.0	85.0
–10	87.2	86.9	85.8	85.0	85.0	85.0	85.0	85.0
–5	88.0	87.7	86.6	85.5	85.0	85.0	85.0	85.0
0	88.8	88.5	87.4	86.3	85.0	85.0	85.0	85.0
5	89.6	89.3	88.2	87.1	85.8	85.0	85.0	85.0
10	90.4	90.1	89.0	87.8	86.6	85.1	85.0	85.0
15	91.1	90.9	89.8	88.6	87.3	85.8	85.0	85.0
19		91.5	90.4	89.2	87.9	86.4	85.0	85.0
20			90.6	89.4	88.1	86.6	85.1	85.0
25				90.1	88.8	87.3	85.8	85.5
30					89.6	88.0	86.5	86.2
35						88.7	87.2	86.9
40							87.9	87.6
45								88.3
46								

flex_n1_3b1_zero_2000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude 3000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 3000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	17	20	25	30	35	40	44
–40	82.5	82.5	82.5	82.5	82.5	82.5	82.5
–35	83.4	83.4	83.4	83.4	83.4	83.4	83.4
–30	84.2	84.2	84.2	84.2	84.2	84.2	84.2
–25	85.1	85.0	85.0	85.0	85.0	85.0	85.0
–20	85.9	85.2	85.0	85.0	85.0	85.0	85.0
–15	86.8	86.1	85.0	85.0	85.0	85.0	85.0
–10	87.6	86.9	85.8	85.0	85.0	85.0	85.0
–5	88.4	87.7	86.6	85.4	85.0	85.0	85.0
0	89.2	88.5	87.4	86.2	85.0	85.0	85.0
5	90.0	89.3	88.2	87.0	85.7	85.0	85.0
10	90.8	90.1	89.0	87.8	86.5	85.0	85.0
15	91.6	90.9	89.8	88.6	87.2	85.8	85.0
17		91.3	90.1	88.9	87.5	86.1	85.0
20			90.5	89.3	88.0	86.5	85.4
25				90.1	88.7	87.2	86.1
30					89.5	88.0	86.8
35						88.7	87.5
40							88.2
44							

flex_n1_3b1_zero_3000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude 4000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 4000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	15	20	25	30	35	40	42
–40	82.9	82.9	82.9	82.9	82.9	82.9	82.9
–35	83.8	83.8	83.8	83.8	83.8	83.8	83.8
–30	84.7	84.7	84.7	84.7	84.7	84.7	84.7
–25	85.5	85.0	85.0	85.0	85.0	85.0	85.0
–20	86.4	85.2	85.0	85.0	85.0	85.0	85.0
–15	87.2	86.1	85.0	85.0	85.0	85.0	85.0
–10	88.0	86.9	85.8	85.0	85.0	85.0	85.0
–5	88.9	87.7	86.6	85.4	85.0	85.0	85.0
0	89.7	88.5	87.4	86.2	85.0	85.0	85.0
5	90.5	89.3	88.2	87.0	85.6	85.0	85.0
10	91.3	90.1	89.0	87.7	86.4	85.0	85.0
15		90.9	89.7	88.5	87.2	85.8	85.2
20			90.5	89.3	87.9	86.5	85.9
25				90.0	88.7	87.2	86.7
30					89.4	88.0	87.4
35						88.7	88.1
40							88.8
42							

flex_n1_3b1_zero_4000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude 5000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 5000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	13	15	20	25	30	35	40
–40	83.4	83.4	83.4	83.4	83.4	83.4	83.4
–35	84.3	84.3	84.3	84.3	84.3	84.3	84.3
–30	85.2	85.0	85.0	85.0	85.0	85.0	85.0
–25	86.0	85.5	85.0	85.0	85.0	85.0	85.0
–20	86.9	86.4	85.2	85.0	85.0	85.0	85.0
–15	87.7	87.2	86.1	85.0	85.0	85.0	85.0
–10	88.5	88.1	86.9	85.8	85.0	85.0	85.0
–5	89.4	88.9	87.7	86.6	85.4	85.0	85.0
0	90.2	89.7	88.5	87.4	86.2	85.0	85.0
5	91.0	90.6	89.3	88.2	86.9	85.6	85.0
10	91.8	91.4	90.1	89.0	87.7	86.3	85.0
13		91.8	90.6	89.4	88.2	86.8	85.4
15			90.9	89.7	88.5	87.1	85.7
20				90.5	89.3	87.8	86.4
25					90.0	88.6	87.1
30						89.3	87.9
35							88.6
40							

flex_n1_3b1_zero_5000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude 6000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 6000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	11	15	20	25	30	35	38
–40	83.9	83.9	83.9	83.9	83.9	83.9	83.9
–35	84.8	84.8	84.8	84.8	84.8	84.8	84.8
–30	85.6	85.0	85.0	85.0	85.0	85.0	85.0
–25	86.5	85.5	85.0	85.0	85.0	85.0	85.0
–20	87.4	86.4	85.2	85.0	85.0	85.0	85.0
–15	88.2	87.2	86.0	85.0	85.0	85.0	85.0
–10	89.1	88.1	86.9	85.7	85.0	85.0	85.0
–5	89.9	88.9	87.7	86.6	85.3	85.0	85.0
0	90.7	89.7	88.5	87.4	86.1	85.0	85.0
5	91.5	90.5	89.3	88.2	86.9	85.5	85.0
10	92.4	91.3	90.1	88.9	87.6	86.2	85.4
11		91.5	90.3	89.1	87.8	86.4	85.5
15			90.9	89.7	88.4	87.0	86.1
20				90.5	89.2	87.7	86.9
25					89.9	88.5	87.6
30						89.2	88.3
35							89.1
38							

flex_n1_3b1_zero_6000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude 7000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 7000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	9	15	20	25	30	35	36
–40	84.4	84.4	84.4	84.4	84.4	84.4	84.4
–35	85.3	85.0	85.0	85.0	85.0	85.0	85.0
–30	86.1	85.0	85.0	85.0	85.0	85.0	85.0
–25	87.0	85.5	85.0	85.0	85.0	85.0	85.0
–20	87.9	86.4	85.2	85.0	85.0	85.0	85.0
–15	88.7	87.2	86.1	85.0	85.0	85.0	85.0
–10	89.6	88.1	86.9	85.7	85.0	85.0	85.0
–5	90.4	88.9	87.7	86.5	85.2	85.0	85.0
0	91.3	89.7	88.5	87.3	86.0	85.0	85.0
5	92.1	90.5	89.3	88.1	86.8	85.4	85.1
9		91.2	90.0	88.7	87.4	86.0	85.7
10		91.4	90.1	88.9	87.5	86.1	85.8
15			90.9	89.7	88.3	86.9	86.6
20				90.5	89.1	87.6	87.3
25					89.8	88.4	88.1
30						89.1	88.8
35							89.6
36							

flex_n1_3b1_zero_7000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude 8000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 8000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	7	10	15	20	25	30	34
–40	84.9	84.9	84.9	84.9	84.9	84.9	84.9
–35	85.8	85.0	85.0	85.0	85.0	85.0	85.0
–30	86.6	85.9	85.0	85.0	85.0	85.0	85.0
–25	87.5	86.8	85.5	85.0	85.0	85.0	85.0
–20	88.4	87.7	86.4	85.2	85.0	85.0	85.0
–15	89.3	88.5	87.2	86.0	85.0	85.0	85.0
–10	90.1	89.4	88.1	86.9	85.6	85.0	85.0
–5	91.0	90.2	88.9	87.7	86.4	85.1	85.0
0	91.8	91.0	89.7	88.5	87.3	85.9	85.0
5	92.6	91.9	90.5	89.3	88.0	86.7	85.6
7		92.2	90.9	89.6	88.4	87.0	85.9
10			91.4	90.1	88.8	87.5	86.3
15				90.9	89.6	88.2	87.1
20					90.4	89.0	87.8
25						89.7	88.6
30							89.3
34							

flex_n1_3b1_zero_8000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude 9000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 9000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	5	10	15	20	25	30	32
–40	85.4	85.0	85.0	85.0	85.0	85.0	85.0
–35	86.3	85.0	85.0	85.0	85.0	85.0	85.0
–30	87.2	85.9	85.0	85.0	85.0	85.0	85.0
–25	88.1	86.8	85.5	85.0	85.0	85.0	85.0
–20	88.9	87.6	86.4	85.2	85.0	85.0	85.0
–15	89.8	88.5	87.2	86.0	85.0	85.0	85.0
–10	90.7	89.4	88.1	86.8	85.5	85.0	85.0
–5	91.5	90.2	88.9	87.6	86.3	85.0	85.0
0	92.3	91.0	89.7	88.5	87.1	85.8	85.2
5		91.9	90.5	89.3	87.9	86.6	86.0
10			91.4	90.1	88.7	87.3	86.8
15				90.9	89.5	88.1	87.6
20					90.3	88.9	88.3
25						89.6	89.1
30							89.8
32							

flex_n1_3b1_zero_9000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, Zero Engine Bleed –
Pressure Altitude 10000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 10000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	3	5	10	15	20	25	30
–40	85.9	85.4	85.0	85.0	85.0	85.0	85.0
–35	86.8	86.3	85.0	85.0	85.0	85.0	85.0
–30	87.7	87.2	85.9	85.0	85.0	85.0	85.0
–25	88.6	88.1	86.8	85.5	85.0	85.0	85.0
–20	89.5	89.0	87.6	86.4	85.1	85.0	85.0
–15	90.3	89.8	88.5	87.2	85.9	85.0	85.0
–10	91.2	90.7	89.4	88.0	86.8	85.4	85.0
–5	92.0	91.6	90.2	88.9	87.6	86.3	85.0
0	92.9	92.4	91.0	89.7	88.4	87.1	85.7
3		92.9	91.5	90.2	88.9	87.5	86.2
5			91.9	90.5	89.2	87.8	86.5
10				91.3	90.0	88.6	87.3
15					90.8	89.4	88.0
20						90.2	88.8
25							89.5
30							

flex_n1_3b1_zero_10000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude –1000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = –1000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	20	25	26	30	31	35	40	45	50
–40	80.9	80.9	80.9	80.9	80.9	80.9	80.9	80.9	80.9
–35	81.7	81.7	81.7	81.7	81.7	81.7	81.7	81.7	81.7
–30	82.5	82.5	82.5	82.5	82.5	82.5	82.5	82.5	82.5
–25	83.3	83.3	83.3	83.3	83.3	83.3	83.3	83.3	83.3
–20	84.1	84.1	84.1	84.1	84.1	84.1	84.1	84.1	84.1
–15	84.9	84.9	84.9	84.9	84.9	84.9	84.9	84.9	84.9
–10	85.7	85.1	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–5	86.5	85.9	85.5	85.0	85.0	85.0	85.0	85.0	85.0
0	87.3	86.7	86.3	85.6	85.1	85.0	85.0	85.0	85.0
5	88.1	87.5	87.1	86.3	85.9	85.0	85.0	85.0	85.0
10	88.9	88.3	87.9	87.1	86.6	85.7	85.0	85.0	85.0
15	89.7	89.0	88.7	87.9	87.4	86.5	85.0	85.0	85.0
20		89.8	89.4	88.7	88.2	87.3	85.7	85.0	85.0
25			90.2	89.4	88.9	88.0	86.4	85.0	85.0
26				89.7	89.2	88.3	86.7	85.2	85.0
30					89.6	88.7	87.1	85.6	85.0
31						89.0	87.4	85.9	85.0
35							87.9	86.3	85.0
40								87.0	85.5
45									86.2
50									

flex_n1_3b1_acu_-1000_qrh.ps

<0005>

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude 0 Feet (Sea Level) <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = SEA LEVEL	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	19	20	23	25	30	35	40	45	50
–40	81.3	81.3	81.3	81.3	81.3	81.3	81.3	81.3	81.3
–35	82.1	82.1	82.1	82.1	82.1	82.1	82.1	82.1	82.1
–30	82.9	82.9	82.9	82.9	82.9	82.9	82.9	82.9	82.9
–25	83.7	83.7	83.7	83.7	83.7	83.7	83.7	83.7	83.7
–20	84.5	84.5	84.5	84.5	84.5	84.5	84.5	84.5	84.5
–15	85.4	85.2	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–10	86.2	86.0	85.6	85.1	85.0	85.0	85.0	85.0	85.0
–5	87.0	86.8	86.4	85.9	85.0	85.0	85.0	85.0	85.0
0	87.8	87.6	87.2	86.7	85.6	85.0	85.0	85.0	85.0
5	88.6	88.4	88.0	87.5	86.4	85.0	85.0	85.0	85.0
10	89.4	89.2	88.8	88.3	87.1	85.8	85.0	85.0	85.0
15	90.2	90.0	89.5	89.1	87.9	86.5	85.0	85.0	85.0
19		90.6	90.2	89.7	88.5	87.1	85.5	85.0	85.0
20			90.3	89.9	88.7	87.3	85.7	85.0	85.0
23				90.3	89.1	87.7	86.1	85.0	85.0
25					89.4	88.0	86.4	85.0	85.0
30						88.7	87.1	85.6	85.0
35							87.8	86.3	85.0
40								87.0	85.4
45									86.1
50									

flex_n1_3b1_acu_0_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude 1000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 1000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	17	20	21	25	30	35	40	45	48
–40	81.7	81.7	81.7	81.7	81.7	81.7	81.7	81.7	81.7
–35	82.5	82.5	82.5	82.5	82.5	82.5	82.5	82.5	82.5
–30	83.3	83.3	83.3	83.3	83.3	83.3	83.3	83.3	83.3
–25	84.1	84.1	84.1	84.1	84.1	84.1	84.1	84.1	84.1
–20	85.0	85.0	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–15	85.8	85.3	85.2	85.0	85.0	85.0	85.0	85.0	85.0
–10	86.6	86.1	86.0	85.1	85.0	85.0	85.0	85.0	85.0
–5	87.4	87.0	86.8	85.9	85.0	85.0	85.0	85.0	85.0
0	88.2	87.8	87.6	86.7	85.5	85.0	85.0	85.0	85.0
5	89.0	88.6	88.4	87.5	86.3	85.0	85.0	85.0	85.0
10	89.8	89.4	89.2	88.3	87.1	85.7	85.0	85.0	85.0
15	90.7	90.1	90.0	89.1	87.8	86.5	85.0	85.0	85.0
17		90.5	90.3	89.4	88.1	86.8	85.2	85.0	85.0
20			90.8	89.8	88.6	87.2	85.7	85.0	85.0
21				90.0	88.7	87.4	85.8	85.0	85.0
25					89.3	87.9	86.4	85.0	85.0
30						88.7	87.1	85.6	85.0
35							87.8	86.3	85.4
40								87.0	86.1
45									86.7
48									

flex_n1_3b1_acu_1000_qrh.ps

<0005>

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude 2000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 2000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	15	19	20	25	30	35	40	45	46
–40	82.1	82.1	82.1	82.1	82.1	82.1	82.1	82.1	82.1
–35	82.9	82.9	82.9	82.9	82.9	82.9	82.9	82.9	82.9
–30	83.7	83.7	83.7	83.7	83.7	83.7	83.7	83.7	83.7
–25	84.6	84.6	84.6	84.6	84.6	84.6	84.6	84.6	84.6
–20	85.4	85.0	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–15	86.2	85.6	85.4	85.0	85.0	85.0	85.0	85.0	85.0
–10	87.0	86.4	86.2	85.1	85.0	85.0	85.0	85.0	85.0
–5	87.8	87.2	87.0	85.9	85.0	85.0	85.0	85.0	85.0
0	88.7	88.1	87.8	86.7	85.5	85.0	85.0	85.0	85.0
5	89.5	88.9	88.6	87.5	86.3	85.0	85.0	85.0	85.0
10	90.3	89.7	89.4	88.3	87.1	85.7	85.0	85.0	85.0
15		90.4	90.2	89.0	87.8	86.5	85.0	85.0	85.0
19			90.8	89.7	88.4	87.1	85.5	85.0	85.0
20				89.8	88.6	87.3	85.7	85.0	85.0
25					89.3	88.0	86.4	85.0	85.0
30						88.7	87.1	85.6	85.3
35							87.8	86.3	86.0
40								87.0	86.7
45									87.4
46									

flex_n1_3b1_acu_2000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude 3000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 3000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	13	15	17	20	25	30	35	40	44
–40	82.5	82.5	82.5	82.5	82.5	82.5	82.5	82.5	82.5
–35	83.3	83.3	83.3	83.3	83.3	83.3	83.3	83.3	83.3
–30	84.2	84.2	84.2	84.2	84.2	84.2	84.2	84.2	84.2
–25	85.0	85.0	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–20	85.8	85.5	85.2	85.0	85.0	85.0	85.0	85.0	85.0
–15	86.7	86.3	86.0	85.4	85.0	85.0	85.0	85.0	85.0
–10	87.5	87.2	86.9	86.2	85.1	85.0	85.0	85.0	85.0
–5	88.3	88.0	87.7	87.0	85.9	85.0	85.0	85.0	85.0
0	89.1	88.8	88.5	87.8	86.7	85.4	85.0	85.0	85.0
5	90.0	89.6	89.3	88.6	87.5	86.2	85.0	85.0	85.0
10	90.8	90.4	90.1	89.4	88.2	87.0	85.6	85.0	85.0
13		90.9	90.6	89.9	88.7	87.5	86.1	85.0	85.0
15			90.9	90.2	89.0	87.8	86.4	85.0	85.0
17				90.5	89.3	88.1	86.7	85.2	85.0
20					89.8	88.5	87.1	85.6	85.0
25						89.3	87.9	86.4	85.1
30							88.6	87.1	85.9
35								87.8	86.6
40									87.3
44									

flex_n1_3b1_acu_3000_qrh.ps<0005>

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude 4000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 4000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)							
(°C)	11	15	20	25	30	35	40	42
–40	82.9	82.9	82.9	82.9	82.9	82.9	82.9	82.9
–35	83.8	83.8	83.8	83.8	83.8	83.8	83.8	83.8
–30	84.6	84.6	84.6	84.6	84.6	84.6	84.6	84.6
–25	85.5	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–20	86.3	85.7	85.0	85.0	85.0	85.0	85.0	85.0
–15	87.1	86.5	85.4	85.0	85.0	85.0	85.0	85.0
–10	88.0	87.3	86.2	85.0	85.0	85.0	85.0	85.0
–5	88.8	88.2	87.0	85.8	85.0	85.0	85.0	85.0
0	89.6	89.0	87.8	86.6	85.4	85.0	85.0	85.0
5	90.5	89.8	88.6	87.4	86.1	85.0	85.0	85.0
10	91.3	90.6	89.4	88.2	86.9	85.6	85.0	85.0
11		90.7	89.6	88.4	87.1	85.7	85.0	85.0
15			90.2	89.0	87.7	86.3	85.0	85.0
20				89.7	88.4	87.1	85.6	85.0
25					89.2	87.8	86.3	85.7
30						88.5	87.0	86.4
35							87.8	87.1
40								87.8
42								

flex_n1_3b1_acu_4000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude 5000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 5000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)							
(°C)	9	13	15	20	25	30	35	40
–40	83.4	83.4	83.4	83.4	83.4	83.4	83.4	83.4
–35	84.2	84.2	84.2	84.2	84.2	84.2	84.2	84.2
–30	85.1	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–25	85.9	85.2	85.0	85.0	85.0	85.0	85.0	85.0
–20	86.8	86.1	85.7	85.0	85.0	85.0	85.0	85.0
–15	87.6	86.9	86.5	85.3	85.0	85.0	85.0	85.0
–10	88.5	87.8	87.3	86.2	85.0	85.0	85.0	85.0
–5	89.3	88.6	88.2	87.0	85.8	85.0	85.0	85.0
0	90.2	89.4	89.0	87.8	86.6	85.3	85.0	85.0
5	91.0	90.3	89.8	88.6	87.4	86.1	85.0	85.0
9		90.9	90.4	89.2	88.0	86.7	85.3	85.0
10		91.1	90.6	89.4	88.2	86.9	85.4	85.0
13			91.1	89.8	88.6	87.3	85.9	85.0
15				90.2	88.9	87.6	86.2	85.0
20					89.7	88.4	86.9	85.4
25						89.1	87.7	86.2
30							88.4	86.9
35								87.6
40								

flex_n1_3b1_acu_5000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude 6000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 6000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	7	10	11	15	20	25	30	35	38
–40	83.9	83.9	83.9	83.9	83.9	83.9	83.9	83.9	83.9
–35	84.7	84.7	84.7	84.7	84.7	84.7	84.7	84.7	84.7
–30	85.6	85.0	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–25	86.4	85.8	85.7	85.0	85.0	85.0	85.0	85.0	85.0
–20	87.3	86.7	86.5	85.6	85.0	85.0	85.0	85.0	85.0
–15	88.1	87.6	87.4	86.5	85.3	85.0	85.0	85.0	85.0
–10	89.0	88.4	88.2	87.3	86.1	85.0	85.0	85.0	85.0
–5	89.8	89.2	89.1	88.2	86.9	85.8	85.0	85.0	85.0
0	90.7	90.1	89.9	89.0	87.7	86.6	85.2	85.0	85.0
5	91.5	90.9	90.7	89.8	88.5	87.3	86.0	85.0	85.0
7		91.2	91.0	90.1	88.9	87.7	86.3	85.0	85.0
10			91.5	90.6	89.3	88.1	86.8	85.3	85.0
11				90.7	89.5	88.3	86.9	85.4	85.0
15					90.1	88.9	87.5	86.0	85.2
20						89.7	88.3	86.8	85.9
25							89.0	87.5	86.6
30								88.3	87.4
35									88.1
38									

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<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude 7000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 7000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	5	9	10	15	20	25	30	35	36
–40	84.4	84.4	84.4	84.4	84.4	84.4	84.4	84.4	84.4
–35	85.2	85.0	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–30	86.1	85.3	85.1	85.0	85.0	85.0	85.0	85.0	85.0
–25	86.9	86.2	86.0	85.0	85.0	85.0	85.0	85.0	85.0
–20	87.8	87.1	86.9	85.6	85.0	85.0	85.0	85.0	85.0
–15	88.7	87.9	87.7	86.5	85.3	85.0	85.0	85.0	85.0
–10	89.5	88.8	88.6	87.3	86.1	85.0	85.0	85.0	85.0
–5	90.4	89.6	89.4	88.1	86.9	85.7	85.0	85.0	85.0
0	91.2	90.5	90.2	89.0	87.7	86.5	85.1	85.0	85.0
5		91.3	91.1	89.8	88.5	87.3	85.9	85.0	85.0
9			91.7	90.4	89.2	87.9	86.5	85.0	85.0
10				90.6	89.3	88.1	86.6	85.2	85.0
15					90.1	88.8	87.4	85.9	85.6
20						89.6	88.1	86.7	86.4
25							88.9	87.4	87.1
30								88.1	87.8
35									88.6
36									

flex_n1_3b1_acu_7000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude 8000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 8000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	3	5	7	10	15	20	25	30	34
–40	84.9	84.9	84.9	84.9	84.9	84.9	84.9	84.9	84.9
–35	85.7	85.3	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–30	86.6	86.2	85.9	85.1	85.0	85.0	85.0	85.0	85.0
–25	87.5	87.1	86.8	86.0	85.0	85.0	85.0	85.0	85.0
–20	88.3	87.9	87.6	86.9	85.6	85.0	85.0	85.0	85.0
–15	89.2	88.8	88.5	87.7	86.5	85.2	85.0	85.0	85.0
–10	90.1	89.7	89.3	88.6	87.3	86.1	85.0	85.0	85.0
–5	90.9	90.5	90.2	89.4	88.1	86.9	85.6	85.0	85.0
0	91.8	91.4	91.0	90.2	88.9	87.7	86.4	85.0	85.0
3		91.9	91.5	90.7	89.4	88.2	86.9	85.4	85.0
5			91.9	91.1	89.7	88.5	87.2	85.7	85.0
7				91.4	90.1	88.8	87.5	86.0	85.0
10					90.5	89.3	87.9	86.5	85.4
15						90.1	88.7	87.3	86.1
20							89.5	88.0	86.9
25								88.8	87.6
30									88.3
34									

flex_n1_3b1_acu_8000_qrh.ps

<0005>

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude 9000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 9000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)							
(°C)	1	5	10	15	20	25	30	32
–40	85.4	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–35	86.3	85.5	85.0	85.0	85.0	85.0	85.0	85.0
–30	87.1	86.4	85.1	85.0	85.0	85.0	85.0	85.0
–25	88.0	87.2	86.0	85.0	85.0	85.0	85.0	85.0
–20	88.9	88.1	86.8	85.6	85.0	85.0	85.0	85.0
–15	89.7	89.0	87.7	86.4	85.2	85.0	85.0	85.0
–10	90.6	89.8	88.5	87.3	86.0	85.0	85.0	85.0
–5	91.5	90.7	89.4	88.1	86.8	85.5	85.0	85.0
0	92.3	91.5	90.2	88.9	87.6	86.2	85.0	85.0
1		91.7	90.4	89.1	87.8	86.4	85.0	85.0
5			91.0	89.7	88.4	87.0	85.6	85.0
10				90.5	89.2	87.8	86.4	85.8
15					90.0	88.6	87.1	86.5
20						89.3	87.9	87.3
25							88.6	88.0
30								88.8
32								

flex_n1_3b1_acu_9000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁, One PACK On Per Engine – Pressure Altitude 10000 Feet <0005>

DISPLAYED FAN SPEED	
CF34–3B1 (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 10000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	–2	0	3	5	10	15	20	25	30
–40	85.9	85.5	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–35	86.8	86.4	85.9	85.5	85.0	85.0	85.0	85.0	85.0
–30	87.7	87.3	86.8	86.4	85.1	85.0	85.0	85.0	85.0
–25	88.5	88.2	87.7	87.3	86.0	85.0	85.0	85.0	85.0
–20	89.4	89.1	88.6	88.1	86.8	85.5	85.0	85.0	85.0
–15	90.3	90.0	89.5	89.0	87.7	86.4	85.1	85.0	85.0
–10	91.2	90.8	90.3	89.9	88.5	87.2	85.9	85.0	85.0
–5	92.0	91.7	91.2	90.7	89.4	88.0	86.7	85.3	85.0
–2		92.2	91.7	91.2	89.9	88.5	87.2	85.8	85.0
0			92.0	91.5	90.2	88.9	87.5	86.1	85.0
3				92.0	90.7	89.4	88.0	86.6	85.1
5					91.0	89.7	88.3	86.9	85.5
10						90.5	89.1	87.7	86.2
15							89.9	88.5	87.0
20								89.2	87.7
25									88.5
30									

flex_n1_3b1_acu_10000_qrh.ps

<0005>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

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CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

WET AND CONTAMINATED RUNWAY

Take-off Weight and V₁ Reduction

The following correction tables are predicated upon dispatch with both thrust reversers operative. For dispatch with one thrust reverser inoperative, additional corrections must be applied as indicated at the bottom of the tables. Separate tables must be used for FLAPS 8 or FLAPS 20 take-offs.

- (1) Calculation of Maximum Allowable Take-off Weight
- (a) Assuming a **dry runway**, determine the take-off weight limited by field length or obstacle clearance.

(b) Using the weight obtained from (a) and the airport pressure altitude, determine the weight correction from the appropriate table depending on the runway surface condition or type of contaminant. If applicable, apply further weight correction for one thrust reverser inoperative.

(c) Reduce the weight from (a) by the weight correction from (b).

(d) Determine the weight limited by climb requirements and the structural weight limit. The maximum allowable take-off weight is the most restrictive of those two limits and the weight from (c).
- (2) Calculation of V₁
- (a) Assuming a **dry runway**, determine V₁ using the actual weight, as determined in step (1) above.

(b) Using the actual weight and the airport pressure altitude, determine the V₁ correction from the appropriate table depending on the runway surface condition or type of contaminant. If applicable, apply further V₁ correction for one thrust reverser inoperative.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

(c) The V_1 to be set is the value from (a) reduced by the correction from (b).

NOTE

- 1. V_1 must not be less than V_{1MCG} nor greater than V_{1MBE} . If the V_1 obtained after applying the correction is lower than V_{1MCG} then set $V_1 = V_{1MCG}$
- 2. In the tables, a dash implies that the weight and/or the V_1 reduction cannot be applied for this runway surface condition or type of contaminant.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Take-off Weight and V₁ Reduction – FLAPS 8 (Wet Runway) <0005> and <0006> and <Metric>

Gross Weight (kg)	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)					
16000	150 10	200 10	200 9	250 9	250 9	250 8
17000	200 10	250 9	250 9	300 9	300 8	350 8
18000	250 10	250 9	300 9	300 9	350 8	350 7
19000	300 10	300 9	350 9	350 8	400 8	400 7
20000	300 9	350 9	400 9	400 8	450 7	450 7
21000	350 9	400 9	400 8	450 8	450 7	450 6
22000	400 9	400 8	450 8	450 7	450 7	450 6
23000	400 8	450 8	450 7	450 7	450 6	450 5
24000	400 8	450 7	450 7	450 6	450 5	500 4
25000	400 7	450 7	450 6	450 5	450 4	450 4
26000	400 7	400 6	450 5	450 4	450 4	350 3
27000	350 6	400 6	400 4	400 4	400 3	300 2
28000	350 6	400 5	400 4	400 4	300 3	200 2

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE

$\Delta W = -350 \text{ kg}$

$\Delta V_1 = -2 \text{ KIAS}$

cf34_3A1_to_wet_8_red_met.ps

Take-off Weight and V₁ Reduction – FLAPS 8 (3.2 mm Water or Equivalent) <0005> and <0006> and <Metric>

Gross Weight (kg)	3.2 mm WATER 3.8 mm SLUSH 6.4 mm WET SNOW 24.1 mm DRY SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)					
16000	15	14	13	12	12	11
17000	15	14	12	11	11	10
18000	14	13	12	11	10	9
19000	2150 13	2300 12	2450 11	2550 10	2650 9	2750 9
20000	1900 13	2000 12	2150 11	2250 10	2350 9	2400 8
21000	2100 12	2250 11	2400 10	2450 9	2500 9	2500 8
22000	2300 12	2450 11	2550 10	2600 9	2550 9	2450 8
23000	2500 11	2600 11	2650 10	2600 9	2450 8	2250 8
24000	2650 11	2650 10	2650 10	2550 9	2400 8	2200 8
25000	2750 10	2700 10	2650 9	2550 9	2450 8	2300 8
26000	2850 –	2700 –	2500 –	2400 –	2300 –	2200 –
27000	2800 –	2650 –	2500 –	2350 –	2250 –	2150 –
28000	2750 –	2600 –	2400 –	2300 –	2200 –	2100 –

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE

ΔW = –800 kg

ΔV₁ = –4 KIAS

For W < 21500 kg, refer to AFM

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Take-off Weight and V₁ Reduction – FLAPS 8 (6.4 mm Water or Equivalent) <0005> and <0006> and <Metric>

TAKE-OFF WEIGHT AND V₁ REDUCTION – FLAPS 8°

Gross Weight (kg)	6.4 mm WATER 7.8 mm SLUSH 12.7 mm WET SNOW 48.3 mm DRY SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)					
16000	8	8	7	7	6	6
17000	7	7	7	6	6	6
18000	6	6	6	6	5	5
19000	2450 6	2700 5	2950 5	3050 5	3000 5	3000 5
20000	2500 5	2800 5	3150 5	3250 4	3100 4	2950 4
21000	2750 5	3100 5	3400 5	3450 4	3250 4	3000 4
22000	3050 5	3250 5	3500 5	3550 4	3350 4	3150 4
23000	3250 5	3500 5	3700 5	3700 4	3500 4	3300 4
24000	3500 5	3650 5	3800 5	3750 4	3500 4	3250 4
25000	3650 5	3700 5	3700 5	3600 5	3450 4	3250 4
26000	3800 –	3750 –	3700 –	3600 –	3500 –	3350 –
27000	3750 –	3600 –	3450 –	3300 –	3200 –	3100 –
28000	3600 –	3450 –	3300 –	3150 –	3050 –	2950 –

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE

ΔW = -900 kg ΔV₁ = -2 KIAS

For W < 21000 kg, refer to AFM

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Take-off Weight and V₁ Reduction – FLAPS 8 (12.7 mm Water or Equivalent) <0005> and <0006> and <Metric>

TAKE-OFF WEIGHT REDUCTION – FLAPS 8°

Gross Weight (kg)	12.7 mm WATER 12.7 mm SLUSH 25.4 mm WET SNOW 76.2 mm DRY SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg)					
20000	4350	4400	4450	4500	4550	4600
21000	4450	4600	4700	4750	4750	4750
22000	4700	4800	4900	4950	4850	4800
23000	4950	5000	5050	5000	4850	4700
24000	5150	5100	5100	5000	4800	4650
25000	5350	5250	5200	5050	4950	4800
26000	5550	5400	5250	5100	5000	4900
27000	5600	5450	5250	5150	5050	4950
28000	5500	5350	5200	5050	4950	4850

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE
ΔW = -1100 kg ΔV₁ = not applicable
For W < 21500 kg, refer to AFM

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NOTE

The above table is based on V1/VR = 1.
No correction to V1 is required.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Take-off Weight and V₁ Reduction – FLAPS 8 Compacted
Snow <0005> and <0006> and <Metric>

Gross Weight (kg)	COMPACTED SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)					
16000	13	13	12	12	11	11
17000	13	12	12	11	11	10
18000	650 13	700 12	700 11	750 11	800 10	800 10
19000	650 12	700 12	750 11	750 11	800 10	800 9
20000	700 12	750 11	800 11	850 10	850 10	850 9
21000	800 12	800 11	850 11	850 10	850 9	850 9
22000	850 11	850 11	850 10	850 10	800 9	750 9
23000	900 11	900 10	900 10	850 9	700 9	600 8
24000	900 10	850 10	850 9	750 9	550 8	400 7
25000	900 10	900 9	850 9	750 8	550 7	400 7
26000	900 -	850 -	850 -	750 -	550 -	400 -
27000	800 -	750 -	700 -	600 -	450 -	250 -
28000	750 -	700 -	650 -	550 -	400 -	250 -

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE

$\Delta W = -500 \text{ kg}$ $\Delta V_1 = -4 \text{ KIAS}$

For W < 19000 kg, refer to AFM

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Take-off Weight and V₁ Reduction – FLAPS 8 Ice <0005> and <0006>
and <Metric>

Gross Weight (kg)	ICE					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)					
16000	41	42	41	40	38	36
17000	42	42	42	40	38	35
18000	42	42	42	40	38	35
19000	41	42	41	40	37	32
20000	37	37	36	33	28	4300 22
21000	27	27	26	23	19	2500 14
22000	26	26	24	3600 22	3000 17	2400 12
23000	4700 24	3700 24	3000 23	2500 20	2250 16	2200 11
24000	2800 23	2700 23	2600 22	2450 19	2300 15	2100 10
25000	2900 22	2750 22	2550 20	2350 18	2200 14	2050 9
26000	2950	2750	2550	2350	2150	2000
27000	2900	2700	2450	2300	2100	1950
28000	2750	2550	2400	2200	2050	1900

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE

Dispatch is prohibited

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Take-off Weight and V₁ Reduction – FLAPS 20 (Wet Runway) <0005> and <Metric>

Gross Weight (kg)	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)					
16000	– –	– –	300 8	350 7	350 7	400 7
17000	300 8	350 8	400 8	400 7	450 7	450 6
18000	350 8	400 8	450 7	450 7	450 7	450 6
19000	400 8	450 7	500 7	500 7	500 6	450 6
20000	450 8	500 7	500 7	500 7	550 6	450 6
21000	500 8	550 7	550 7	550 6	450 6	500 5
22000	550 7	550 7	600 7	550 6	500 6	450 5
23000	550 7	550 7	600 6	450 6	500 5	250 4
24000	550 7	600 6	450 6	450 5	300 4	100 3
25000	550 6	550 6	450 5	350 5	100 3	0 2
26000	550 6	350 6	400 5	150 4	0 3	0 2
27000	450 6	400 5	200 4	0 3	0 2	– –
28000	300 5	300 4	50 3	0 2	– –	– –

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE

ΔW = –350 kg ΔV₁ = –2 KIAS

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Take-off Weight and V₁ Reduction – FLAPS 20 (3.2 mm Water or Equivalent) <0005> and <Metric>

TAKE-OFF WEIGHT AND V₁ REDUCTION – FLAPS 20°

Gross Weight (kg)	3.2 mm WATER 3.8 mm SLUSH 6.4 mm WET SNOW 24.1 mm DRY SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)					
16000	14	13	12	11	10	9
17000	13	12	11	10	9	8
18000	12	11	10	10	9	8
19000	2150 12	2300 11	2450 10	2550 9	2650 8	2750 8
20000	1900 11	2000 10	2150 9	2250 9	2350 8	2400 7
21000	2100 11	2250 10	2400 9	2450 8	2500 8	2500 7
22000	2300 10	2450 9	2550 9	2600 8	2550 8	2450 7
23000	2500 10	2600 9	2650 8	2600 8	2450 8	2250 7
24000	2650 9	2650 9	2650 8	2550 8	2400 8	2200 7
25000	2750 9	2700 8	2650 8	2550 8	2450 8	2300 7
26000	2850 –	2700 –	2500 –	2400 –	2300 –	2200 –
27000	2800 –	2650 –	2500 –	2350 –	2250 –	2150 –
28000	2750 –	2600 –	2400 –	2300 –	2200 –	2100 –

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE
ΔW = -800 kg ΔV₁ = -4 KIAS
For W < 21500 kg, refer to AFM

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Take-off Weight and V₁ Reduction – FLAPS 20 (6.4 mm Water or Equivalent) <0005> and <Metric>

TAKE-OFF WEIGHT AND V₁ REDUCTION – FLAPS 20°

Gross Weight (kg)	6.4 mm WATER 7.8 mm SLUSH 12.7 mm WET SNOW 48.3 mm DRY SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)					
16000	7	6	6	6	6	5
17000	6	6	6	5	5	5
18000	5	5	5	5	5	4
19000	2450 5	2700 4	2950 4	3050 4	3000 4	3000 4
20000	2500 4	2800 4	3150 3	3250 3	3100 3	2950 3
21000	2750 3	3100 3	3400 3	3450 2	3250 2	3000 2
22000	3050 3	3250 2	3500 2	3550 2	3350 2	3150 1
23000	3250 2	3500 2	3700 2	3700 1	3500 1	3300 1
24000	3500 2	3650 1	3800 1	3750 1	3500 1	3250 0
25000	3650 1	3700 1	3700 1	3600 0	3450 0	3250 0
26000	3800 –	3750 –	3700 –	3600 –	3500 –	3350 –
27000	3750 –	3600 –	3450 –	3300 –	3200 –	3100 –
28000	3600 –	3450 –	3300 –	3150 –	3050 –	2950 –

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE
ΔW = –900 kg ΔV₁ = –2 KIAS
For W < 21000 kg, refer to AFM

slush_250_20_SI_Oct10.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Take-off Weight and V₁ Reduction – FLAPS 20 (12.7 mm Water or Equivalent) <0005> and <Metric>

TAKE-OFF WEIGHT REDUCTION – FLAPS 20°						
Gross Weight (kg)	12.7 mm WATER 12.7 mm SLUSH 25.4 mm WET SNOW 76.2 mm DRY SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg)					
20000	4350	4400	4450	4500	4550	4600
21000	4450	4600	4700	4750	4750	4750
22000	4700	4800	4900	4950	4850	4800
23000	4950	5000	5050	5000	4850	4700
24000	5150	5100	5100	5000	4800	4650
25000	5350	5250	5200	5050	4950	4800
26000	5550	5400	5250	5100	5000	4900
27000	5600	5450	5250	5150	5050	4950
28000	5500	5350	5200	5050	4950	4850

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE
ΔW = -1100 kg ΔV₁ = not applicable
For W < 21500 kg, refer to AFM

slush_500_20_SI_Oct10.ps

NOTE

The above table is based on V₁/VR = 1.
No correction to V₁ is required.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Take-off Weight and V₁ Reduction – FLAPS 20 Compacted
Snow <0005> and <Metric>

Gross Weight (kg)	COMPACTED SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)					
16000	- 11	- 10	- 10	- 9	- 9	- 8
17000	- 11	- 10	- 10	- 9	- 9	- 8
18000	650 10	700 10	700 9	750 9	800 8	800 8
19000	650 10	700 9	750 9	750 8	800 8	800 7
20000	700 10	750 9	800 9	850 8	850 7	850 7
21000	800 9	800 9	850 8	850 8	850 7	850 6
22000	850 9	850 8	850 8	850 7	800 7	750 6
23000	900 8	900 8	900 7	850 7	700 6	600 6
24000	900 8	850 7	850 7	750 6	550 5	400 5
25000	900 8	900 7	850 6	750 5	550 5	400 4
26000	900 -	850 -	850 -	750 -	550 -	400 -
27000	800 -	750 -	700 -	600 -	450 -	250 -
28000	750 -	700 -	650 -	550 -	400 -	250 -

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE

$\Delta W = -500 \text{ kg}$

$\Delta V_1 = -4 \text{ KIAS}$

For W < 19000 kg, refer to AFM

snow_20_SI.pro

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Take-off Weight and V₁ Reduction – FLAPS 20 Ice <0005> and
<Metric>

Gross Weight (kg)	ICE					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)					
16000	40	41	41	40	38	36
17000	40	40	40	39	37	35
18000	38	39	38	37	35	31
19000	35	35	34	32	29	26
20000	30	30	29	27	23	4300 18
21000	25	25	24	22	18	2500 14
22000	24	24	23	3600 20	3000 16	2400 11
23000	4700 22	3700 22	3000 21	2500 18	2250 14	2200 9
24000	2800 21	2700 21	2600 19	2450 16	2300 12	2100 7
25000	2900 19	2750 19	2550 18	2350 15	2200 11	2050 5
26000	2950 –	2750 –	2550 –	2350 –	2150 –	2000 –
27000	2900 –	2700 –	2450 –	2300 –	2100 –	1950 –
28000	2750 –	2550 –	2400 –	2200 –	2050 –	1900 –

DISPATCH WITH ONE THRUST REVERSER INOPERATIVE

Dispatch is prohibited

ice_20_Sl_Oct10.ps

Actual Landing Distance – FLAPS 45

Actual Landing Distance, FLAPS 45, Both Thrust Reversers
Operative, (Standing Water / Slush) <0005> or <0068> and <Metric>

Gross Weight (kg)	STANDING WATER / SLUSH					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1255	1320	1395	1470	1545	1625
16000	1295	1365	1440	1515	1600	1685
17000	1365	1440	1515	1600	1685	1775
18000	1435	1510	1595	1680	1770	1865
19000	1505	1585	1675	1760	1860	1960
20000	1575	1660	1750	1845	1945	2055
21000	1645	1730	1830	1925	2035	2155
22000	1715	1805	1905	2010	2125	2260
23000	1780	1880	1985	2095	2220	2370
24000	1855	1955	2065	2185	2325	2480
	Distance: + 50 m / kt of tail wind, –10 m / kt of head wind + 25 m / kt above V _{REF} + 290 m when one thrust reverser inoperative + 800 m when two thrust reversers inoperative					

qrh_main_F45_Sl.out

qrh_rj_ld_flap_45_red_vref_water_2tr_Sl_Sep10.ps

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 45, Both Thrust Reversers
Operative, Dry Snow or Wet Snow <0005> or <0068> and <Metric>

Gross Weight (kg)	DRY SNOW OR WET SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1260	1320	1385	1450	1520	1595
16000	1295	1360	1425	1495	1565	1645
17000	1355	1425	1495	1565	1645	1725
18000	1415	1485	1560	1635	1720	1805
19000	1475	1550	1630	1710	1795	1885
20000	1535	1615	1695	1780	1870	1965
21000	1595	1675	1760	1850	1945	2055
22000	1655	1740	1830	1925	2025	2165
23000	1715	1805	1900	1995	2120	2270
24000	1780	1870	1965	2075	2220	2380
	Distance: + 50 m / kt of tail wind, -10 m / kt of head wind + 30 m / kt above V _{REF} + 240 m when one thrust reverser inoperative + 610 m when two thrust reversers inoperative					

qrh_main_F45_SI.out

qrh_rj_ld_flap_45_red_vref_lsnw_2tr_SI_Sep10.ps

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 45, Both Thrust Reversers
Operative, Compacted Snow <0005> or <0068> and <Metric>

Gross Weight (kg)	COMPACTED SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1105	1155	1210	1265	1330	1395
16000	1135	1190	1245	1305	1365	1435
17000	1185	1240	1300	1365	1430	1500
18000	1240	1295	1360	1425	1495	1570
19000	1290	1350	1415	1485	1560	1635
20000	1340	1405	1470	1545	1620	1705
21000	1390	1455	1530	1605	1685	1775
22000	1440	1510	1585	1665	1750	1850
23000	1490	1565	1640	1725	1815	1925
24000	1540	1615	1700	1785	1885	2000
	Distance: + 35 m / kt of tail wind, -10 m / kt of head wind + 25 m / kt above V _{REF} + 125 m when one thrust reverser inoperative + 305 m when two thrust reversers inoperative					

qrh_main_F45_SI.out

qrh_rj_ld_flap_45_red_vref_csnow_2tr_SI_Sep10.ps

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 45, Both Thrust Reversers
Operative, Ice <0005> or <0068> and <Metric>

Gross Weight (kg)	ICE					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1640	1730	1825	1925	2025	2130
16000	1690	1785	1885	1985	2090	2200
17000	1775	1875	1985	2090	2205	2320
18000	1865	1970	2085	2195	2315	2440
19000	1950	2060	2180	2300	2430	2560
20000	2035	2155	2280	2410	2545	2680
21000	2125	2245	2380	2515	2655	2805
22000	2210	2340	2480	2620	2770	2940
23000	2300	2435	2580	2725	2890	3070
24000	2385	2525	2680	2835	3015	3205
	Distance: + 65 m / kt of tail wind, -15 m / kt of head wind + 25 m / kt above V _{REF} + 690 m when one thrust reverser inoperative + 2160 m when two thrust reversers inoperative					

qrh_main_F45_SI.out

qrh_rj_ld_flap_45_red_vref_ice_2tr_SI_Sep10.ps

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance – Flaps Other Than 45

Actual Landing Distance, FLAPS 0, Both Thrust Reversers
Operative, (Standing Water / Slush / Dry Snow or Wet
Snow) <0005> or <0068> and <Metric>

Gross Weight (kg)	STANDING WATER / SLUSH / DRY SNOW OR WET SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1725	1815	1915	2015	2125	2245
16000	1775	1870	1970	2075	2195	2320
17000	1860	1960	2065	2180	2305	2440
18000	1940	2050	2165	2285	2420	2560
19000	2030	2140	2260	2395	2535	2685
20000	2115	2235	2365	2500	2650	2810
21000	2200	2325	2465	2610	2770	2945
22000	2290	2425	2565	2720	2890	3085
23000	2380	2520	2670	2830	3015	3230
24000	2470	2615	2775	2945	3150	3375
	Distance: + 55 m / kt of tail wind, –20 m / kt of head wind + 30 m / kt above V _{REF} + 535 m when one thrust reverser inoperative + 1585 m when two thrust reversers inoperative					

qrh_main_F00_Sl.out

qrh_rj_ld_flap_00_red_vref_water_2tr_Sl_Sep10.ps

NOTE

Should only be used during flap failure to
determine required ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 8, Both Thrust Reversers
Operative, (Standing Water / Slush / Dry Snow or Wet
Snow) <0005> or <0068> and <0006> and <Metric>

Gross Weight (kg)	STANDING WATER / SLUSH / DRY SNOW OR WET SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1560	1645	1735	1825	1920	2025
16000	1610	1695	1785	1880	1985	2090
17000	1690	1780	1875	1975	2085	2200
18000	1770	1860	1965	2075	2190	2315
19000	1845	1945	2055	2170	2300	2430
20000	1925	2035	2145	2270	2405	2550
21000	2010	2120	2245	2370	2515	2670
22000	2090	2205	2335	2475	2625	2800
23000	2170	2295	2430	2575	2745	2935
24000	2255	2385	2530	2680	2865	3070
	Distance: + 55 m / kt of tail wind, -20 m / kt of head wind + 25 m / kt above V _{REF} + 460 m when one thrust reverser inoperative + 1355 m when two thrust reversers inoperative					

qrh_main_F08_SI.outqrh_rj_ld_flap_08_red_vref_water_2tr_SI_Sep10.ps

NOTE

Should only be used during flap failure to
determine required ALD.

Actual Landing Distance, FLAPS 20, Both Thrust Reversers
Operative, (Standing Water / Slush / Dry Snow or Wet
Snow) <0005> or <0068> and <Metric>

Gross Weight (kg)	STANDING WATER / SLUSH / DRY SNOW OR WET SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1470	1545	1630	1715	1810	1905
16000	1515	1595	1685	1770	1865	1965
17000	1590	1680	1765	1860	1965	2070
18000	1670	1760	1855	1950	2060	2180
19000	1745	1840	1940	2045	2160	2285
20000	1820	1920	2025	2140	2265	2395
21000	1900	2000	2115	2235	2365	2510
22000	1975	2085	2205	2330	2470	2630
23000	2050	2170	2295	2430	2580	2760
24000	2135	2255	2385	2530	2700	2890
	Distance: + 55 m / kt of tail wind, -20 m / kt of head wind + 25 m / kt above V _{REF} + 410 m when one thrust reverser inoperative + 1180 m when two thrust reversers inoperative					

qrh_main_F20_SI.out

qrh_rj_ld_flap_20_red_vref_water_2tr_SI_Sep10.ps

NOTE
Should only be used during flap failure to
determine required ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 30, Both Thrust Reversers
Operative, (Standing Water / Slush / Dry Snow or Wet
Snow) <0005> or <0068> and <Metric>

Gross Weight (kg)	STANDING WATER / SLUSH / DRY SNOW OR WET SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1395	1470	1550	1630	1720	1810
16000	1440	1515	1600	1685	1770	1870
17000	1515	1595	1680	1770	1865	1965
18000	1590	1675	1765	1860	1960	2065
19000	1660	1750	1845	1945	2055	2170
20000	1735	1830	1930	2035	2150	2275
21000	1810	1905	2015	2125	2250	2380
22000	1880	1985	2095	2215	2345	2500
23000	1960	2065	2185	2310	2450	2615
24000	2030	2145	2270	2405	2565	2740
	Distance: + 50 m / kt of tail wind, -20 m / kt of head wind + 25 m / kt above V _{REF} + 360 m when one thrust reverser inoperative + 1015 m when two thrust reversers inoperative					

qrh_main_F30_SI.out

qrh_rj_ld_flap_30_red_vref_water_2tr_SI_Sep10.ps

NOTE
Should only be used during flap failure to
determine required ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 0, Both Thrust Reversers
Operative, (Compacted Snow) <0005> or <0068> and <Metric>

Gross Weight (kg)	COMPACTED SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1460	1525	1600	1675	1760	1845
16000	1495	1565	1640	1720	1805	1895
17000	1555	1625	1705	1790	1880	1970
18000	1610	1690	1770	1860	1950	2050
19000	1670	1750	1835	1925	2025	2125
20000	1725	1810	1900	1995	2095	2205
21000	1785	1870	1965	2065	2170	2285
22000	1840	1935	2030	2130	2240	2370
23000	1900	1995	2095	2200	2320	2455
24000	1955	2055	2155	2265	2395	2540
	Distance: + 40 m / kt of tail wind, – 15 m / kt of head wind + 20 m / kt above V _{REF} + 185 m when one thrust reverser inoperative + 445 m when two thrust reversers inoperative					

qrh_main_F00_SI.out

qrh_rj_ld_flap_00_red_vref_csnow_2tr_SI_Sep10.ps

NOTE
Should only be used during flap failure to
determine required ALD.

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 8, Both Thrust Reversers
Operative, (Compacted Snow) <0005> or <0068> and <0006> and <Metric>

Gross Weight (kg)	COMPACTED SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1335	1395	1465	1535	1610	1685
16000	1365	1430	1500	1575	1650	1730
17000	1425	1490	1565	1640	1720	1805
18000	1480	1550	1625	1705	1790	1880
19000	1535	1610	1690	1770	1860	1955
20000	1590	1670	1750	1840	1930	2030
21000	1645	1725	1815	1905	2000	2105
22000	1700	1785	1875	1970	2070	2190
23000	1760	1845	1935	2035	2145	2270
24000	1815	1900	2000	2100	2220	2355
	Distance: + 35 m / kt of tail wind, -15 m / kt of head wind + 20 m / kt above V _{REF} + 165 m when one thrust reverser inoperative + 405 m when two thrust reversers inoperative					

qrh_main_F08_SI.out

qrh_rj_ld_flap_08_red_vref_csnow_2tr_SI_Sep10.ps

NOTE
Should only be used during flap failure to
determine required ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 20, Both Thrust Reversers
Operative, (Compacted Snow) <0005> or <0068> and <Metric>

Gross Weight (kg)	COMPACTED SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1270	1330	1395	1460	1530	1605
16000	1305	1365	1430	1500	1575	1650
17000	1360	1425	1495	1565	1645	1725
18000	1415	1480	1555	1630	1710	1795
19000	1470	1540	1615	1695	1780	1870
20000	1525	1600	1675	1760	1850	1945
21000	1580	1655	1735	1825	1915	2020
22000	1635	1715	1800	1890	1985	2100
23000	1685	1770	1860	1955	2060	2180
24000	1740	1825	1920	2015	2135	2265
	Distance: + 35 m / kt of tail wind, -15 m / kt of head wind + 20 m / kt above V _{REF} + 160 m when one thrust reverser inoperative + 385 m when two thrust reversers inoperative					

qrh_main_F20_SI.out

qrh_rj_ld_flap_20_red_vref_csnow_2tr_SI_Sep10.ps

NOTE
Should only be used during flap failure to
determine required ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 30, Both Thrust Reversers
Operative, (Compacted Snow) <0005> or <0068> and <Metric>

Gross Weight (kg)	COMPACTED SNOW					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1220	1275	1335	1400	1470	1540
16000	1250	1310	1375	1440	1510	1585
17000	1305	1370	1435	1500	1575	1655
18000	1360	1425	1495	1565	1645	1725
19000	1415	1480	1555	1630	1710	1795
20000	1465	1535	1615	1690	1780	1870
21000	1520	1595	1670	1755	1845	1940
22000	1575	1650	1730	1820	1910	2020
23000	1625	1705	1790	1880	1980	2100
24000	1680	1760	1850	1945	2055	2180
	Distance: + 35 m / kt of tail wind, –15 m / kt of head wind + 25 m / kt above V _{REF} + 145 m when one thrust reverser inoperative + 355 m when two thrust reversers inoperative					

qrh_main_F30_SI.out

qrh_rj_ld_flap_30_red_vref_csnow_2tr_SI_Sep10.ps

NOTE
Should only be used during flap failure to
determine required ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 0, Both Thrust Reversers
Operative, (Ice) <0005> or <0068> and <Metric>

Gross Weight (kg)	ICE					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	2170	2295	2430	2565	2710	2855
16000	2235	2365	2500	2645	2790	2945
17000	2340	2480	2625	2775	2935	3095
18000	2450	2590	2745	2905	3075	3245
19000	2555	2705	2870	3035	3215	3400
20000	2665	2820	2990	3165	3355	3550
21000	2770	2935	3115	3300	3495	3705
22000	2880	3055	3240	3430	3635	3865
23000	2990	3170	3365	3565	3785	4030
24000	3100	3285	3485	3695	3940	4200
	Distance: + 70 m / kt of tail wind, -25 m / kt of head wind + 30 m / kt above V _{REF} + 1055 m when one thrust reverser inoperative + 3560 m when two thrust reversers inoperative					

qrh_main_F00_SI.out

qrh_rj_ld_flap_00_red_vref_ice_2tr_SI_Sep10.ps

NOTE
Should only be used during flap failure to
determine required ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 8, Both Thrust Reversers
Operative, (Ice) <0005> or <0068> and <0006> and <Metric>

Gross Weight (kg)	ICE					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1995	2110	2230	2355	2485	2620
16000	2055	2170	2300	2430	2565	2700
17000	2155	2280	2415	2550	2695	2845
18000	2255	2390	2530	2675	2830	2985
19000	2360	2495	2645	2800	2960	3130
20000	2460	2605	2765	2925	3095	3270
21000	2560	2715	2880	3045	3230	3420
22000	2665	2825	2995	3175	3360	3575
23000	2770	2935	3115	3300	3500	3730
24000	2870	3045	3230	3425	3650	3890
	Distance: + 70 m / kt of tail wind, -25 m / kt of head wind + 30 m / kt above V _{REF} + 975 m when one thrust reverser inoperative + 3280 m when two thrust reversers inoperative					

qrh_main_F08_SI.out

qrh_rj_ld_flap_08_red_vref_ice_2tr_SI_Sep10.ps

NOTE

Should only be used during flap failure to
determine required ALD.

Actual Landing Distance, FLAPS 20, Both Thrust Reversers
Operative, (Ice) <0005> or <0068> and <Metric>

Gross Weight (kg)	ICE					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1885	1990	2105	2220	2345	2470
16000	1940	2055	2175	2290	2420	2550
17000	2040	2155	2285	2410	2545	2685
18000	2135	2260	2395	2530	2670	2820
19000	2235	2365	2505	2645	2800	2955
20000	2330	2465	2615	2765	2925	3090
21000	2425	2570	2725	2885	3055	3230
22000	2525	2675	2840	3005	3180	3380
23000	2625	2780	2950	3125	3315	3530
24000	2725	2885	3065	3245	3455	3680
	Distance: + 70 m / kt of tail wind, -25 m / kt of head wind + 30 m / kt above V _{REF} + 890 m when one thrust reverser inoperative + 2950 m when two thrust reversers inoperative					

qrh_main_F20_SI.out

qrh_rj_id_flap_20_red_vref_ice_2tr_SI_Sep10.ps

NOTE
Should only be used during flap failure to
determine required ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 30, Both Thrust Reversers
Operative, (Ice) <0005> or <0068> and <Metric>

Gross Weight (kg)	ICE					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Contaminated Runway (meters)					
15400	1795	1895	2005	2110	2225	2340
16000	1850	1955	2065	2180	2300	2420
17000	1940	2050	2170	2290	2420	2550
18000	2035	2150	2275	2405	2540	2675
19000	2125	2250	2380	2515	2660	2805
20000	2220	2350	2490	2630	2780	2935
21000	2315	2450	2595	2745	2905	3070
22000	2405	2550	2700	2860	3025	3210
23000	2500	2650	2810	2975	3155	3355
24000	2595	2750	2915	3085	3285	3500
	Distance: + 65 m / kt of tail wind, -15 m / kt of head wind + 25 m / kt above V _{REF} + 805 m when one thrust reverser inoperative + 2600 m when two thrust reversers inoperative					

qrh_main_F30_SI.out

qrh_rj_ld_flap_30_red_vref_ice_2tr_SI_Sep10.ps

NOTE
Should only be used during flap failure to
determine required ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance – Wet (Advisory Data Only)

Actual Landing Distance, FLAPS 0, Both Thrust Reversers
Operative, (Wet) Advisory Data Only <0005> or <0068> and <Metric>

Gross Weight (kg)	WET					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Wet Runway (meters)					
15400	1555	1640	1735	1840	1955	2075
16000	1600	1690	1790	1900	2015	2140
17000	1680	1775	1880	1995	2120	2255
18000	1755	1860	1970	2090	2225	2365
19000	1835	1945	2065	2190	2330	2480
20000	1915	2030	2155	2290	2435	2595
21000	1995	2115	2250	2390	2545	2715
22000	2080	2205	2340	2490	2655	2845
23000	2160	2290	2435	2595	2770	2980
24000	2240	2380	2535	2695	2895	3115
	Distance: + 60 m / kt of tail wind, –20 m / kt of head wind + 30 m / kt above V _{REF} + 340 m when one thrust reverser inoperative + 920 m when two thrust reversers inoperative					

qrh_main_F00_SI.out

qrh_rj_ld_flap_00_red_vref_wet_2tr_SI_Sep10.ps

NOTE

Should only be used along with abnormal or
emergency procedures to determine required
ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 8, Both Thrust Reversers
Operative, (Wet) Advisory Data Only <0005> or <0068> and <0006> and
<Metric>

Gross Weight (kg)	WET					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Wet Runway (meters)					
15400	1390	1465	1550	1640	1735	1845
16000	1430	1510	1600	1690	1795	1905
17000	1505	1590	1680	1780	1890	2010
18000	1575	1665	1765	1870	1990	2115
19000	1650	1745	1850	1965	2090	2225
20000	1725	1825	1935	2055	2190	2330
21000	1800	1905	2025	2150	2290	2445
22000	1875	1985	2110	2245	2390	2565
23000	1950	2070	2200	2340	2500	2690
24000	2030	2155	2290	2440	2615	2820
	Distance: + 55 m / kt of tail wind, –20 m / kt of head wind + 30 m / kt above V _{REF} + 290 m when one thrust reverser inoperative + 775 m when two thrust reversers inoperative					

qrh_main_F08_Sl.out

qrh_rj_ld_flap_08_red_vref_wet_2tr_Sl_Sep10.ps

NOTE
Should only be used along with abnormal or
emergency procedures to determine required
ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 20, Both Thrust Reversers
Operative, (Wet) Advisory Data Only <0005> or <0068> and <Metric>

Gross Weight (kg)	WET					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Wet Runway (meters)					
15400	1155	1220	1295	1370	1455	1550
16000	1190	1260	1335	1420	1510	1605
17000	1255	1330	1410	1500	1595	1700
18000	1320	1400	1485	1580	1680	1795
19000	1385	1470	1560	1660	1770	1890
20000	1455	1540	1640	1745	1860	1990
21000	1520	1615	1720	1830	1955	2090
22000	1590	1690	1800	1915	2045	2205
23000	1660	1765	1880	2005	2145	2315
24000	1730	1840	1960	2095	2255	2435
	Distance: + 50 m / kt of tail wind, -20 m / kt of head wind + 30 m / kt above V _{REF} + 260 m when one thrust reverser inoperative + 685 m when two thrust reversers inoperative					

qrh_main_F20_SI.out

qrh_rj_ld_flap_20_red_vref_wet_2tr_SI_Sep10.ps

NOTE
Should only be used along with abnormal or
emergency procedures to determine required
ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Actual Landing Distance, FLAPS 30, Both Thrust Reversers
Operative, (Wet) Advisory Data Only <0005> or <0068> and <Metric>

Gross Weight (kg)	WET					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Wet Runway (meters)					
15400	1095	1155	1225	1295	1375	1465
16000	1130	1195	1265	1340	1425	1515
17000	1190	1260	1335	1420	1505	1605
18000	1255	1325	1410	1495	1590	1695
19000	1315	1395	1480	1575	1675	1785
20000	1380	1465	1555	1655	1760	1880
21000	1445	1530	1630	1735	1850	1975
22000	1510	1600	1705	1815	1935	2085
23000	1575	1675	1780	1900	2035	2190
24000	1640	1745	1860	1985	2135	2305
	Distance: + 45 m / kt of tail wind, -15 m / kt of head wind + 25 m / kt above V _{REF} + 230 m when one thrust reverser inoperative + 590 m when two thrust reversers inoperative					

qrh_main_F30_SI.out

qrh_rj_ld_flap_30_red_vref_wet_2tr_SI_Sep10.ps

NOTE
Should only be used along with abnormal or
emergency procedures to determine required
ALD.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Actual Landing Distance, FLAPS 45, Both Thrust Reversers
Operative, (Wet) Advisory Data Only <0005> or <0068> and <Metric>

Gross Weight (kg)	WET					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Wet Runway (meters)					
15400	925	975	1030	1085	1150	1220
16000	955	1010	1065	1125	1190	1260
17000	1005	1060	1120	1185	1255	1335
18000	1060	1115	1180	1250	1325	1410
19000	1110	1175	1240	1315	1395	1485
20000	1165	1230	1300	1380	1465	1560
21000	1215	1285	1365	1445	1540	1640
22000	1270	1345	1425	1515	1615	1730
23000	1325	1405	1490	1585	1690	1820
24000	1380	1465	1555	1655	1780	1915
	Distance: + 40 m / kt of tail wind, – 10 m / kt of head wind + 25 m / kt above V _{REF} + 180 m when one thrust reverser inoperative + 455 m when two thrust reversers inoperative					

qrh_main_F45_SI.out

qrh_rj_ld_flap_45_red_vref_wet_2tr_SI_Sep10.ps

NOTE
Should only be used along with abnormal or
emergency procedures to determine required
ALD.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

ANTI-SKID

ANTI-SKID INOPERATIVE – DRY / WET RUNWAY TAKE-OFF

Take-off Weight and V₁ Reduction

The following correction tables are predicated for dispatch with one channel of the anti-skid system inoperative. Separate table must be used for FLAPS 8 or FLAPS 20 take-offs.

- (1) Calculation of the Maximum Allowable Take-off Weight
 - (a) Assuming a **normal, anti-skid operative take-off**, determine the take-off weight limited by field length or obstacle clearance on a **dry runway**.
 - (b) Using the weight obtained from (a) and the airport pressure altitude, determine the weight correction by using the proper Anti-Skid Inoperative Data: Dry Runway Take-off Data for a dry runway, or Wet Runway Take-off Data for a wet runway (wet runway data based on both thrust reversers operative).
 - (c) Reduce the weight from (a) by the weight correction from (b).
 - (d) Determine the weight limited by climb requirements and the structural weight limit. The maximum allowable take-off weight is the most restrictive of those two limits and the weight from (c).
- (2) Calculation of V₁
 - (a) Assuming a **normal, anti-skid operative take-off**, determine V₁ for a **dry runway** using the actual weight, as determined in step (1) above.
 - (b) Using the actual weight and the airport pressure altitude, determine the V₁ correction by using the proper Anti-Skid Inoperative Data: Dry Runway Take-off Data for a dry runway, or Wet Runway Take-off Data for a wet runway (wet runway data based on both thrust reversers operative).

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

(c) The V_1 to be set is the value from (a) reduced by the correction from (b).

NOTE

- 1. V_1 must not be less than V_{1MCG} nor greater than V_{1MBE} . If the V_1 obtained after applying the correction is lower than V_{1MCG} then set $V_1 = V_{1MCG}$.
- 2. In the tables, a dash implies that the weight and/or the V_1 reduction cannot be applied for this specific condition.

(3) V_{1MBE} Reduction

- (a) Determine V_{1MBE} for the actual take-off weight using the V_{1MBE} tables found in the brake cooling section in PERF-04.
- (b) Reduce V_{1MBE} obtained from (a) by 42 knots.

NOTE

If V_1 is greater than V_{1MBE} obtained after applying the V_{1MBE} reduction, reduce the actual take-off weight and repeat steps (2) and (3) until V_1 is less than V_{1MBE} . If V_1 is always greater than V_{1MBE} , take-off is not permitted.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Anti-Skid Inoperative – Dry Runway Take-off Data – Take-off Weight and V₁ Reduction – FLAPS 8 <0006> and <Metric>

Gross Weight (kg)	Pressure Altitude (feet)		
	SL	5000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)		
16000	1600 23	1650 19	1600 17
18000	1850 21	1750 17	1800 15
20000	1900 19	1900 16	1950 13
22000	2100 17	2100 14	1900 11
24000	2200 15	2100 12	1850 8
26000	2200 13	2000 10	1800 6

If V₁ reduction is obtained from the shaded area above, and if adjusted V₁ is less than V_{1MCG} (96 KIAS) :

- (a) Set V₁ = 96 KIAS;
- (b) Observe the V_{1MCG} field length limited take-off weight from the table below.

Take-Off Weight Limited by Field Length (kg) (Zero Wind; Zero Runway Slope)			
Runway Length (metres)	Pressure Altitude (feet)		
	SL	5000	10000
1500	–	–	–
1750	18000	–	–
2000	18000	17000	–
≥ 2500	18000	17000	16000



PERF 7-4

Rev. 59, Mar 21/2012

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Anti-Skid Inoperative – Dry Runway Take-off Data – Take-off
Weight and V_1 Reduction – FLAPS 20 <Metric>

Gross Weight (kg)	Pressure Altitude (feet)		
	SL	5000	10000
	Weight Reduction (kg) V_1 Reduction (KIAS)		
16000	1600 21	1600 18	1650 16
18000	1750 19	1800 16	1800 14
20000	1950 18	1950 15	1900 11
22000	2150 16	2050 13	1750 9
24000	2200 14	2000 11	1350 6
26000	2300 12	1800 8	1600 3

If V_1 reduction is obtained from the shaded area above, and if adjusted V_1 is less than V_{1MCG} (96 KIAS) :

- (a) Set V_1 = 96 KIAS;
- (b) Observe the V_{1MCG} field length limited take-off weight from the table below.

Take-Off Weight Limited by Field Length (kg) (Zero Wind; Zero Runway Slope)			
Runway Length (metres)	Pressure Altitude (feet)		
	SL	5000	10000
1500	–	–	–
1750	19000	–	–
2000	19000	16000	–
≥ 2500	19000	18000	17000

Anti-Skid Inoperative – Wet Runway Take-off Data – Take-off Weight and V₁ Reduction – FLAPS 8 <0006> and <Metric>

Gross Weight (kg)	Pressure Altitude (feet)		
	SL	5000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)		
16000	1200 26	1350 24	1400 22
18000	1450 26	1550 23	1650 20
20000	1650 25	1750 22	1900 18
22000	1850 23	2000 20	1950 16
24000	2050 21	2100 18	2000 12
26000	2150 19	2100 15	2050 10

If V₁ reduction is obtained from the shaded area above, and if adjusted V₁ is less than V_{1MCG} (96 KIAS) :

- (a) Set V₁ = 96 KIAS;
- (b) Observe the V_{1MCG} field length limited take-off weight from the table below.

Take-Off Weight Limited by Field Length (kg) (Zero Wind; Zero Runway Slope)			
Runway Length (metres)	Pressure Altitude (feet)		
	SL	5000	10000
1500	–	–	–
1750	16000	–	–
2000	19000	15000	–
≥ 2500	19000	17000	16000



PERF 7-6

Rev. 59, Mar 21/2012

Anti-Skid Inoperative – Wet Runway Take-off Data – Take-off
Weight and V₁ Reduction – FLAPS 20 <Metric>

Gross Weight (kg)	Pressure Altitude (feet)		
	SL	5000	10000
	Weight Reduction (kg) V ₁ Reduction (KIAS)		
16000	1200 24	1400 22	1500 19
18000	1500 23	1650 21	1800 18
20000	1800 22	1900 19	1950 16
22000	2050 21	2100 18	1850 14
24000	2200 19	2100 16	1450 10
26000	2300 18	2000 13	1750 6

If V₁ reduction is obtained from the shaded area above, and if adjusted V₁ is less than V_{1MCG} (96 KIAS) :

- (a) Set V₁ = 96 KIAS;
- (b) Observe the V_{1MCG} field length limited take-off weight from the table below.

Take-Off Weight Limited by Field Length (kg) (Zero Wind; Zero Runway Slope)			
Runway Length (metres)	Pressure Altitude (feet)		
	SL	5000	10000
1500	–	–	–
1750	17000	–	–
2000	20000	16000	–
≥ 2500	20000	19000	16000

Actual Landing Distance – FLAPS 45

Anti-Skid Inoperative – Landing Speed and Actual Landing Distance (Landing Gear Down – FLAPS 45) <Metric>

Weight (kg)	V _{REF} [★] (KIAS)	Pressure Altitude (Feet x 1,000)					
		0	2	4	6	8	10
		Actual Landing Distance [※] (Meters)					
15400	120	1257	1313	1375	1451	1537	1639
16000	122	1294	1353	1417	1496	1585	1689
17000	126	1356	1419	1486	1570	1664	1774
18000	130	1418	1485	1557	1645	1744	1860
19000	133	1480	1551	1627	1721	1825	1947
20000	137	1543	1618	1698	1797	1907	2035
21000	140	1605	1684	1770	1874	1990	2128
22000	144	1668	1751	1843	1952	2074	2230
23000	147	1732	1820	1916	2031	2165	2335
24000	151	1796	1888	1990	2110	2263	2443

★ V_{REF} = Speed at 50' screen height (KIAS), Flaps 45°

※ Actual Landing Distance = Dry Landing Field Length x 0.6

Wet Landing Field Length = Dry Landing Field Length x 1.15

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Anti-Skid Inoperative – Landing Data – Wet Landing Landing
Field Length (Wet), Advisory Data Only – FLAPS 45 <Metric>

Gross Weight (kg)	WET					
	Pressure Altitude (feet)					
	SL	2000	4000	6000	8000	10000
	Actual Landing Distance for Wet Runway (meters)					
15400	1805	1900	2005	2120	2240	2375
16000	1865	1965	2075	2190	2320	2460
17000	1960	2070	2185	2315	2450	2600
18000	2065	2175	2300	2435	2585	2745
19000	2165	2285	2420	2565	2720	2890
20000	2265	2395	2535	2690	2855	3040
21000	2370	2505	2655	2820	3000	3200
22000	2475	2620	2780	2955	3145	3375
23000	2580	2735	2905	3085	3300	3550
24000	2690	2850	3030	3225	3465	3735
	Distance: + 40 m / kt of tail wind, –10 m / kt of head wind + 25 m / kt above V _{REF} + 180 m when one thrust reverser inoperative + 455 m when two thrust reversers inoperative					

qrh_main_F45_SI.out

qrh_rj_ld_flap_45_red_vref_wet_2tr_no_ASkd_SI_Sep10.ps

NOTE

Should only be used along with abnormal or
emergency procedures to determine required ALD.

Actual Landing Distance – Flaps Other Than 45
Anti-Skid Inoperative – Landing Distance Increase Factor for
Flap Failures

LANDING DISTANCE INCREASE FACTOR FOR FLAP FAILURES			
Landing Field Length – Flaps 45			
Flaps Setting (Degrees)	Δ V _{REF} (KTS)	Actual Landing Distance Factor (Without Thrust Reversers)	Actual Landing Distance Factor (With Thrust Reversers)
0	30	1.65 (65%)	1.35 (35%)
8	18	1.50 (50%)	1.20 (20%)
20	12	1.30 (30%)	1.05 (5%)
30	8	1.20 (20%)	1.00 (0%)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
----------	------

MISCELLANEOUS

MISCELLANEOUS	MISC –1
Miscellaneous Data	MISC –1

LIST OF ILLUSTRATIONS

MISCELLANEOUS

Wind Component Conversion Table	MISC –1
Wind Component Conversion Chart	MISC –2

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

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CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

MISCELLANEOUS

MISCELLANEOUS

Miscellaneous Data

Wind Component Conversion Table

Angle (DEG)	Wind velocity (knots)									
	5	10	15	20	25	30	35	40	50	60
0	5	10	15	20	25	30	35	40	50	60
	0	0	0	0	0	0	0	0	0	0
10	5	10	15	20	25	30	34	39	49	59
	1	2	3	3	4	5	6	7	9	10
20	5	9	14	19	23	28	33	38	47	56
	2	3	5	7	9	10	12	14	17	21
30	4	9	13	17	22	26	30	35	43	52
	2	5	7	10	12	15	17	20	25	30
40	4	8	11	15	19	23	27	31	38	46
	3	6	10	13	16	19	22	26	32	39
50	3	6	10	13	16	19	22	26	32	39
	4	8	11	15	19	23	27	31	38	46
60	3	5	8	10	13	15	18	20	25	30
	4	9	13	17	22	26	30	35	43	52
70	2	3	5	7	9	10	12	14	17	21
	5	9	14	19	23	28	33	38	47	56
80	1	2	3	3	4	5	6	7	9	10
	5	10	15	20	25	30	34	39	49	59
90	0	0	0	0	0	0	0	0	0	0
	5	10	15	20	25	30	35	40	50	60
100	-1	-2	-3	-3	-4	-5	-6	-7	-9	-10
	5	10	15	20	25	30	34	39	49	59
110	-2	-3	-5	-7	-9	-10	-12	-14	-17	-21
	5	9	14	19	23	28	33	38	47	56
120	-2	-5	-7	-10	-12	-15	-17	-20	-25	-30
	4	9	13	17	22	26	30	35	43	52
130	-3	-6	-10	-13	-16	-19	-22	-26	-32	-39
	4	8	11	15	19	23	27	31	38	46
140	-4	-8	-11	-15	-19	-23	-27	-31	-38	-46
	3	6	10	13	16	19	22	26	32	39
150	-4	-9	-13	-17	-22	-26	-30	-35	-43	-52
	3	5	8	10	13	15	18	20	25	30
160	-5	-9	-14	-19	-23	-28	-33	-38	-47	-56
	2	3	5	7	9	10	12	14	17	21
170	-5	-10	-15	-20	-25	-30	-34	-39	-49	-59
	1	2	3	3	4	5	6	7	9	10
180	-5	-10	-15	-20	-25	-30	-35	-40	-50	-60
	0	0	0	0	0	0	0	0	0	0
Upper values (shaded): Components parallel to the runway Lower values: Crosswind components Angle measured between wind and runway										

wind_table.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



MISC-2

Rev. 59, Mar 21/2012

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

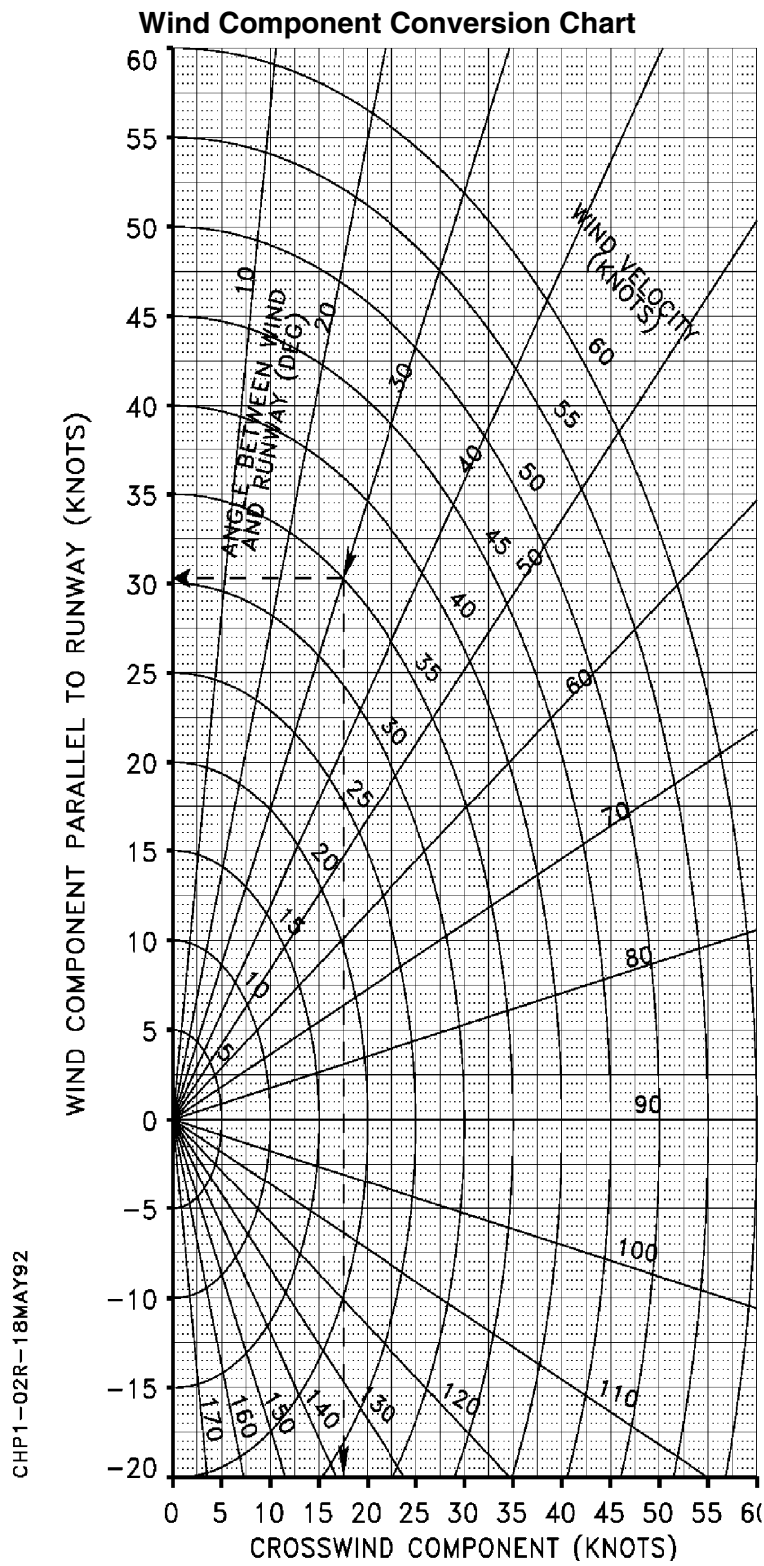
EICAS

EMER

IMM

ABNORM

INDEX



CONTENTS	PAGE
-----------------	-------------

FLIGHT PLANNING

OZONE TABLES	PLAN –1
Maximum Altitude	PLAN –1
Maximum Flight Duration at Altitude	PLAN –1

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

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CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

FLIGHT PLANNING

OZONE TABLES

Maximum Altitude

NOTE

Two checks are required to determine the cabin ozone concentration flight profile limitation. The first check determines the maximum allowable altitude while the second check determines the maximum flight duration at altitude. The most limiting of these checks should then be used.

The maximum allowable altitudes in accordance with FAR 25-832 (a) (1), are presented in the following tables. It is permitted to interpolate linearly between months or altitudes.

Maximum Flight Duration at Altitude

NOTE

Two checks are required to determine the cabin ozone concentration flight profile limitation. The first check determines the maximum allowable altitude while the second check determines the maximum flight duration at altitude. The most limiting of these checks should then be used.

The maximum allowable flight duration at altitude in accordance with FAR 25.832 (a) (2), are presented in the following tables.
It is permitted to interpolate linearly between months, latitudes or altitudes. There are no flight duration limitations below FL 350.

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
-----------------	-------------

GENERAL	
INTRODUCTION	INTERMIX 1-1
SUPPLEMENTARY PROCEDURES	INTERMIX 1-1
LIMITATIONS	INTERMIX 1-2
NORMAL PROCEDURES	INTERMIX 1-3
TAKE-OFF PERFORMANCE	
INTRODUCTION	INTERMIX 2-1
Normal Take-off Thrust Setting	INTERMIX 2-2
Maximum Continuous Thrust Setting	INTERMIX 2-5
Speedbook	INTERMIX 2-8
MAXIMUM CLIMB THRUST SETTINGS	
Maximum Climb Thrust Setting	INTERMIX 3-1
MAXIMUM CRUISE THRUST SETTINGS	
Maximum Cruise Thrust Setting	INTERMIX 4-1
APPROACH AND LANDING DATA	
Go-around Thrust Setting	INTERMIX 5-1
Actual Landing Distance – FLAPS 45	INTERMIX 5-4
Actual Landing Distance – Flaps Other than 45	INTERMIX 5-4
Brake Cooling Requirements	INTERMIX 5-5
REDUCED THRUST TAKE-OFF	
Reduced Thrust Take-off	INTERMIX 6-1

LIST OF ILLUSTRATIONS

TAKE-OFF PERFORMANCE	
Normal Take-off Thrust Setting, %N ₁ (Zero Engine Bleed)	INTERMIX 2-2
Normal Take-off Thrust Setting, %N ₁ (10th Stage Bleeds On – PACKs On)	INTERMIX 2-3
Normal Take-off Thrust Setting, %N ₁ (Cowl Anti-ice On)	INTERMIX 2-4
Normal Take-off Thrust Setting, %N ₁ (Wing and Cowl Anti-ice On)	INTERMIX 2-4
Maximum Continuous Thrust Setting, %N ₁ (PACK On)	INTERMIX 2-5
Maximum Continuous Thrust Setting, %N ₁ (PACK On; Cowl Anti-ice On)	INTERMIX 2-6
Maximum Continuous Thrust Setting, %N ₁ (PACK On; Cowl and Wing Anti-ice On)	INTERMIX 2-7

CONTENTS
SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS

PAGE

Speedcard – Legend <0006> and <Metric>	INTERMIX 2–8
Speedcard – 15400 kg <0005> or <0068> and <0006> and <Metric>	INTERMIX 2–9
Speedcard – 16000 kg <0005> or <0068> and <0006> and <Metric>	INTERMIX 2–10
Speedcard – 17000 kg <0005> or <0068> and <0006> and <Metric>	INTERMIX 2–11
Speedcard – 18000 kg <0005> or <0068> and <0006> and <Metric>	INTERMIX 2–12
Speedcard – 19000 kg <0005> or <0068> and <0006> and <Metric>	INTERMIX 2–13
Speedcard – 20000 kg <0005> or <0068> and <0006> and <Metric>	INTERMIX 2–14
Speedcard – 21000 kg <0005> or <0068> and <0006> and <Metric>	INTERMIX 2–15
Speedcard – 22000 kg <0005> or <0068> and <0006> and <Metric>	INTERMIX 2–16
Speedcard – 23000 kg <0005> or <0068> and <0006> and <Metric>	INTERMIX 2–17
Speedcard – 24000 kg <0005> or <0068> and <0006> and <Metric>	INTERMIX 2–18

MAXIMUM CLIMB THRUST SETTINGS

Maximum Climb Thrust Setting, %N ₁ (PACKs On)	INTERMIX 3–1
Maximum Climb Thrust Setting, %N ₁ (PACKs On; Cowl Anti-ice On)	INTERMIX 3–2
Maximum Climb Thrust Setting, %N ₁ (PACKs On; Cowl and Wing Anti-ice On)	INTERMIX 3–3

MAXIMUM CRUISE THRUST SETTINGS

Maximum Cruise Thrust Setting, %N ₁ (PACKs On)	INTERMIX 4–1
Maximum Cruise Thrust Setting, %N ₁ (PACKs On; Cowl Anti-ice On)	INTERMIX 4–2
Maximum Cruise Thrust Setting, %N ₁ (PACKs On; Cowl and Wing Anti-ice On)	INTERMIX 4–3

APPROACH AND LANDING DATA

Go-around Thrust Setting, %N ₁ (Zero Engine Bleed)	INTERMIX 5–1
Go-around Thrust Setting, %N ₁ (10th Stage Bleeds On – PACKs On)	INTERMIX 5–2
Go-around Thrust Setting, %N ₁ (Cowl Anti-ice On)	INTERMIX 5–3

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
-----------------	-------------

Go-around Thrust Setting, %N ₁ (Wing and Cowl Anti-ice On)	INTERMIX 5-3
Landing Speed and Actual Landing Distance (Landing Gear Down – FLAPS 45) <Metric>	INTERMIX 5-4
Actual Landing Distance Factors when Landing with Flaps Other than 45 <0006>	INTERMIX 5-4
V _{1MBE} , FLAPS 8 <0006> and <Metric>.	INTERMIX 5-5
V _{1MBE} , FLAPS 20 <Metric>	INTERMIX 5-6

REDUCED THRUST TAKE-OFF

Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), Zero Engine Bleed – Pressure Altitude, –1000 ft	INTERMIX 6-2
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), Zero Engine Bleed – Pressure Altitude 0 ft (Sea Level)	INTERMIX 6-3
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), Zero Engine Bleed – Pressure Altitude 1000 ft	INTERMIX 6-4
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), Zero Engine Bleed – Pressure Altitude 2000 ft	INTERMIX 6-5
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), Zero Engine Bleed – Pressure Altitude 3000 ft	INTERMIX 6-6
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), Zero Engine Bleed – Pressure Altitude 4000 ft	INTERMIX 6-7
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), Zero Engine Bleed – Pressure Altitude 6000 ft	INTERMIX 6-8
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), Zero Engine Bleed – Pressure Altitude 8000 ft	INTERMIX 6-9
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), Zero Engine Bleed – Pressure Altitude 10000 ft	INTERMIX 6-10
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), One Pack On Per Engine – Pressure Altitude, –1000 ft	INTERMIX 6-11
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), One PACK On Per Engine – Pressure Altitude 0 ft (Sea Level)	INTERMIX 6-12
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), One PACK On Per Engine – Pressure Altitude 1000 ft	INTERMIX 6-13

CONTENTS
SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), One PACK On Per Engine – Pressure Altitude 2000 ft	INTERMIX 6–14
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), One PACK On Per Engine – Pressure Altitude 3000 ft	INTERMIX 6–15
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), One PACK On Per Engine – Pressure Altitude 4000 ft	INTERMIX 6–16
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), One PACK On Per Engine – Pressure Altitude 6000 ft	INTERMIX 6–17
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), One PACK On Per Engine – Pressure Altitude 8000 ft	INTERMIX 6–18
Reduced Thrust Take-off Setting, %N ₁ (Two Engine Operation), One PACK On Per Engine – Pressure Altitude 10000 ft	INTERMIX 6–19

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

GENERAL

INTRODUCTION

NOTE

Refer to Chapter 7 – Supplement 10 of the Airplane Flight Manual (AFM) for the complete set of data for intermix operations. The data contained within the AFM take precedence over those contained within the QRH.

The data that follows is applicable only to CF34-3B1 equipped airplanes and contains data to cover situations wherein the airplane is fitted with intermixed engine models, i.e. a CF34-3B1 engine paired with a CF34-3A1 engine.

Effectivity:

- Airplanes **incorporating** the following Service Bulletin:
 - SB 601R-71-027 – Power Plant – General – Engine Intermix and Full Conversion from –3A1 to –3B1 Engines or from –3B1 to –3A1 Engines.

This data is applicable to either CF34-3A1 or CF34-3B1 equipped airplanes and contains data to cover situations wherein the CL 600-2B19 airplane is fitted with intermixed engine models, i.e., a C34-3B1 engine paired with a CF34-3A1 engine or vice-versa.

The pairing of two different engine models incurs performance penalties and should be regarded only as a temporary measure.

These data complement or supersede data contained in Volume 1 of the Quick Reference Handbook (QRH). These data must therefore be used in conjunction with the basic QRH.

SUPPLEMENTARY PROCEDURES

The supplementary procedures in Volume 1 of the QRH are applicable.

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



INTERMIX 1-2

Rev. 59, Mar 21/2012

LIMITATIONS

The limitations in Volume 1 of the QRH are applicable, except as modified by the following:

(1) Engine ITT Indications

ITT°C	Red	Amber	Green
Normal Take-off	900 (for first 2 min) 884 (for next 3 min)	860 to 898 (for first 2 min) 860 to 882 (for next 3 min)	0 to 860
Go-around or APR Thrust	928 (for first 2 min) 900 (for next 3 min)	860 to 926 (for first 2 min) 860 to 928 (for next 3 min)	0 to 860
MCT	— 888* 860 to 886† 888†	860 860 to 886†	0 to 860

* During MCT, amber band and red tick mark appear if ITT is greater than 888°C.

† During MCT, amber band becomes a red band if ITT remains above 860°C for more than 30 seconds or if ITT is greater than 888°C. Red tick mark stays at 888°C.

Automatic Performance Reserve

The APR system must be selected OFF for take-off and APR OFF performance must be used, if the ambient temperature is -18°C (0°F) and below, and pressure altitude is less than 1000 feet.

Engine Operating Limits

Condition	Core RPM N ₂ %	Fan RPM N ₁ %	ITT °C
Start	20	—	900
Idle	56.5 to 68.0†	—	—
Acceleration	—	—	900
Max Continuous	99.2	98.6	860
Normal Take-off	98.3	96.2	884 (5 min)* 900 (2 min out of 5 total transient)

Condition	Core RPM N ₂ %	Fan RPM N ₁ %	ITT °C
APR Thrust	99.4	98.6	900 (5 min)* 928 (2 min out of 5 total transient)
* Transient limits † Engine to engine N ₂ split at ground idle power should not be greater than 2% N ₂ . If N ₂ is 57% or less with an OAT of –20°C or greater, do not accelerate above idle.			

NORMAL PROCEDURES

Normal procedures remain unchanged, except as modified by the following:

- During intermix operations, there will nominally be a half ball throttle split at power levels greater than idle. The CF 34–3A1 engine is designated as the ‘master’ engine and its thrust lever will always lead the ‘slave’ engine (CF 34–3B1) thrust lever. When setting thrust levels above idle power, the thrust levers must be advanced such that the appropriate N1 is set on the master engine first and thereafter matched by the slave engine.
- For flight planning and cruise performance calculations, the Flight Planning and Cruise Control Manual for the CF 34–3A1 engine (CSP A–015 or CSP A–018) and Supplement 1 – Power Plant Intermix, must be used.
- The ENG TYPE MISCOMP" status message is inhibited during intermix operations.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

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CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

TAKE-OFF PERFORMANCE

INTRODUCTION

NOTE

- 1. The performance data contained herein are derived from the Airplane Flight Manual (AFM) and are to be used for reference purposes only. The data in the Performance section of Volume 1 of the QRH are applicable except as modified by INTERMIX-02 to INTERMIX-06.
- 2. It is the responsibility of the user to ascertain that the take-off and landing weights are appropriate, in accordance with AFM performance data, for the existing airport pressure altitude, ambient temperature, runway conditions and climb requirements.
- 3. Some values in the tables may be out of the airplane's operating envelope and are to be used only for interpolation.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Normal Take-off Thrust Setting

Normal Take-off Thrust Setting, %N₁ (Zero Engine Bleed)

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING ZERO ENGINE BLEED INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	80.7	81.1	81.9	82.7	83.7	84.7	85.7
-35	-31	81.6	82.0	82.8	83.6	84.6	85.6	86.6
-30	-22	82.4	82.8	83.6	84.5	85.4	86.4	87.5
-25	-13	83.2	83.6	84.5	85.3	86.3	87.3	88.4
-20	-4	84.0	84.5	85.3	86.2	87.1	88.2	89.2
-15	5	84.9	85.3	86.1	87.0	88.0	89.0	90.1
-10	14	85.7	86.1	87.0	87.8	88.8	89.9	91.0
-5	23	86.5	86.9	87.8	88.7	89.7	90.7	91.8
0	32	87.3	87.7	88.6	89.5	90.5	91.6	92.7
5	41	88.1	88.5	89.4	90.3	91.3	92.4	93.0
10	50	88.8	89.3	90.2	91.1	92.2	92.5	92.5
15	59	89.6	90.0	90.9	91.9	91.9	91.9	91.9
20	68	90.4	90.8	91.5	91.5	91.5	91.5	91.4
25	77	91.1	91.1	91.1	91.1	91.1	91.0	90.8
30	86	90.7	90.7	90.7	90.6	90.5	90.3	90.1
35	95	90.0	90.1	90.1	89.9	89.8	89.6	89.4
40	104	89.2	89.2	89.2	89.2	89.0		
45	113	88.4	88.4	88.4	88.4			
50	122	87.5	87.6	87.6				

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Normal Take-off Thrust Setting, %N₁ (10th Stage Bleeds On – PACKs On)

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING 10th STAGE BLEEDS ON – PACKS ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	–1000	0	2000	4000	6000	8000	10000
–40	–40	80.7	81.1	81.9	82.7	83.7	84.7	85.7
–35	–31	81.5	81.9	82.7	83.6	84.5	85.5	86.6
–30	–22	82.3	82.7	83.5	84.4	85.4	86.4	87.4
–25	–13	83.1	83.5	84.4	85.2	86.2	87.2	88.3
–20	–4	83.9	84.3	85.2	86.1	87.1	88.1	89.2
–15	5	84.7	85.1	86.0	86.9	87.9	89.0	90.1
–10	14	85.5	85.9	86.8	87.7	88.8	89.8	90.9
–5	23	86.3	86.8	87.6	88.6	89.6	90.7	91.8
0	32	87.1	87.6	88.5	89.4	90.5	91.5	92.3
5	41	87.9	88.4	89.3	90.2	91.3	92.0	92.1
10	50	88.7	89.2	90.1	91.1	91.6	91.7	91.6
15	59	89.5	90.0	90.9	91.2	91.2	91.1	91.1
20	68	90.3	90.6	90.8	90.8	90.7	90.6	90.5
25	77	90.4	90.4	90.4	90.3	90.2	90.0	89.8
30	86	89.9	90.0	89.9	89.7	89.6	89.3	89.0
35	95	89.3	89.3	89.3	89.1	88.8	88.6	88.3
40	104	88.4	88.3	88.3	88.3	88.0		
45	113	87.5	87.4	87.5	87.4			
50	122	86.7	86.7	86.6				

cf34–INT_nto_acu_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Normal Take-off Thrust Setting, %N₁ (Cowl Anti-ice On)

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING COWL ANTI-ICE ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	80.7	81.1	81.9	82.7	83.7	84.7	85.7
-35	-31	81.5	81.9	82.7	83.6	84.5	85.5	86.6
-30	-22	82.3	82.7	83.5	84.4	85.4	86.4	87.4
-25	-13	83.1	83.5	84.4	85.2	86.2	87.2	88.3
-20	-4	83.9	84.3	85.2	86.1	87.1	88.1	89.2
-15	5	84.7	85.1	86.0	86.9	87.9	89.0	90.1
-10	14	85.5	85.9	86.8	87.7	88.8	89.8	90.9
-5	23	86.3	86.8	87.6	88.6	89.6	90.7	91.8
0	32	87.1	87.6	88.5	89.4	90.5	91.5	92.3
5	41	87.9	88.4	89.3	90.2	91.3	92.0	92.2
10	50	88.7	89.2	90.1	91.1	91.6	91.7	91.6

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Normal Take-off Thrust Setting, %N₁ (Wing and Cowl Anti-ice On)

DISPLAYED FAN SPEED	
NORMAL TAKEOFF THRUST SETTING WING AND COWL ANTI-ICE ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	80.7	81.1	81.9	82.7	83.7	84.7	85.7
-35	-31	81.5	81.9	82.7	83.6	84.5	85.5	86.6
-30	-22	82.3	82.7	83.5	84.4	85.4	86.4	87.4
-25	-13	83.1	83.5	84.4	85.2	86.2	87.2	88.3
-20	-4	83.9	84.3	85.2	86.1	87.1	88.1	88.6
-15	5	84.7	85.1	86.0	86.9	87.6	88.1	88.6
-10	14	85.5	85.9	86.7	87.0	87.6	88.1	88.6
-5	23	86.3	86.4	86.7	87.0	87.6	88.1	88.6
0	32	86.3	86.4	86.7	87.0	87.6	88.1	88.6
5	41	86.3	86.4	86.7	87.0	87.6	88.1	88.4
10	50	86.3	86.4	86.7	87.0	87.6	87.7	87.7

cf34-INT_nto_cowl_wing_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

NOTE

In icing conditions between 0°C and +5°C, at pressure altitudes of 2000 feet and below, the take-off thrust settings with cowl and wing anti-icing on may be used when taking off with the wing and cowl anti-icing on.

Maximum Continuous Thrust Setting

Maximum Continuous Thrust Setting, %N₁ (PACK On)

DISPLAYED FAN SPEED	
MAXIMUM CONTINUOUS THRUST SETTING PACK ON INTERMIX	%N ₁

OAT		PRESSURE ALTITUDE (Feet)								
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000	36000	41000
-70	-94								89.5	90.5
-65	-85							88.6	90.6	91.6
-60	-76						88.3	89.6	91.6	92.6
-55	-67						89.3	90.6	92.7	93.7
-50	-58					88.9	90.2	91.6	93.7	94.6
-45	-49				88.2	89.9	91.1	92.6	94.7	94.0
-40	-40	82.0	84.2	86.4	89.1	90.9	92.1	93.7	94.2	93.5
-35	-31	82.8	85.0	87.2	90.0	91.8	93.0	94.7	93.6	93.0
-30	-22	83.7	85.9	88.1	90.9	92.8	94.0	94.6	93.1	
-25	-13	84.5	86.7	89.0	91.9	93.7	94.8	94.1	92.2	
-20	-4	85.3	87.6	89.9	92.8	94.7	94.6	93.3		
-15	5	86.1	88.4	90.8	93.7	94.9	93.9	92.5		
-10	14	87.0	89.3	91.7	94.6	94.2	93.2	91.5		
-5	23	87.8	90.1	92.5	94.4	93.4	92.4			
0	32	88.6	91.0	93.4	93.6	92.7	91.5			
5	41	89.4	91.8	93.2	93.0	92.0				
10	50	90.2	92.7	92.6	92.4	91.3				
15	59	91.1	92.5	92.1	91.6					
20	68	91.9	91.9	91.4	90.9					
25	77	91.6	91.3	90.7						
30	86	90.9	90.5	89.8						
35	95	90.1	89.7	88.8						
40	104	89.3	88.7							
45	113	88.4	87.8							
50	122	87.4								

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Maximum Continuous Thrust Setting, %N₁ (PACK On; Cowl Anti-ice On)

DISPLAYED FAN SPEED	
MAXIMUM CONTINUOUS THRUST SETTING PACK ON, COWL ANTI-ICE ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000
-65	-85							88.4
-60	-76						88.1	89.4
-55	-67						89.1	90.5
-50	-58					88.9	90.0	91.5
-45	-49				88.2	89.9	91.0	92.5
-40	-40	82.0	84.2	86.4	89.1	90.9	91.9	93.5
-35	-31	82.8	85.0	87.2	90.0	91.8	92.8	93.7
-30	-22	83.7	85.9	88.1	90.9	92.8	93.8	93.2
-25	-13	84.5	86.7	89.0	91.9	93.7	93.9	92.3
-20	-4	85.3	87.6	89.9	92.8	94.1	93.2	91.5
-15	5	86.1	88.4	90.8	93.7	93.6	92.4	90.4
-10	14	87.0	89.3	91.7	93.8	92.8	91.6	89.4
-5	23	87.8	90.1	92.5	93.2	92.1	90.6	
0	32	88.6	91.0	92.7	92.5	91.4	89.7	
5	41	89.4	91.8	92.3	91.9	90.6		
10	50	90.2	92.1	91.7	91.2	89.8		

cf34-INT_mct_acu_cowl_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Maximum Continuous Thrust Setting, %N₁ (PACK On; Cowl and Wing Anti-ice On)

DISPLAYED FAN SPEED	
MAXIMUM CONTINUOUS THRUST SETTING PACK ON, COWL AND WING ANTI-ICE ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000
-65	-85							87.2
-60	-76						87.3	88.2
-55	-67						88.3	89.3
-50	-58					88.5	89.4	90.3
-45	-49				88.2	89.5	90.5	91.2
-40	-40	82.0	84.2	86.4	89.1	89.9	90.6	91.2
-35	-31	82.8	85.0	87.2	89.4	89.9	90.6	90.6
-30	-22	83.7	85.9	88.1	89.4	89.9	90.6	89.6
-25	-13	84.5	86.7	88.6	89.4	89.9	90.0	88.6
-20	-4	85.3	87.6	88.6	89.4	89.9	89.2	87.7
-15	5	86.1	88.2	88.6	89.4	89.4	88.4	86.7
-10	14	87.0	88.2	88.6	89.4	88.8	87.6	85.8
-5	23	87.4	88.2	88.6	89.0	88.1	86.6	
0	32	87.4	88.2	88.6	88.4	87.4	85.5	
5	41	87.4	88.2	88.2	87.7	86.4		
10	50	87.4	88.2	87.5	86.8	85.2		

cf34-INT_mct_acu_cowl_wing_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

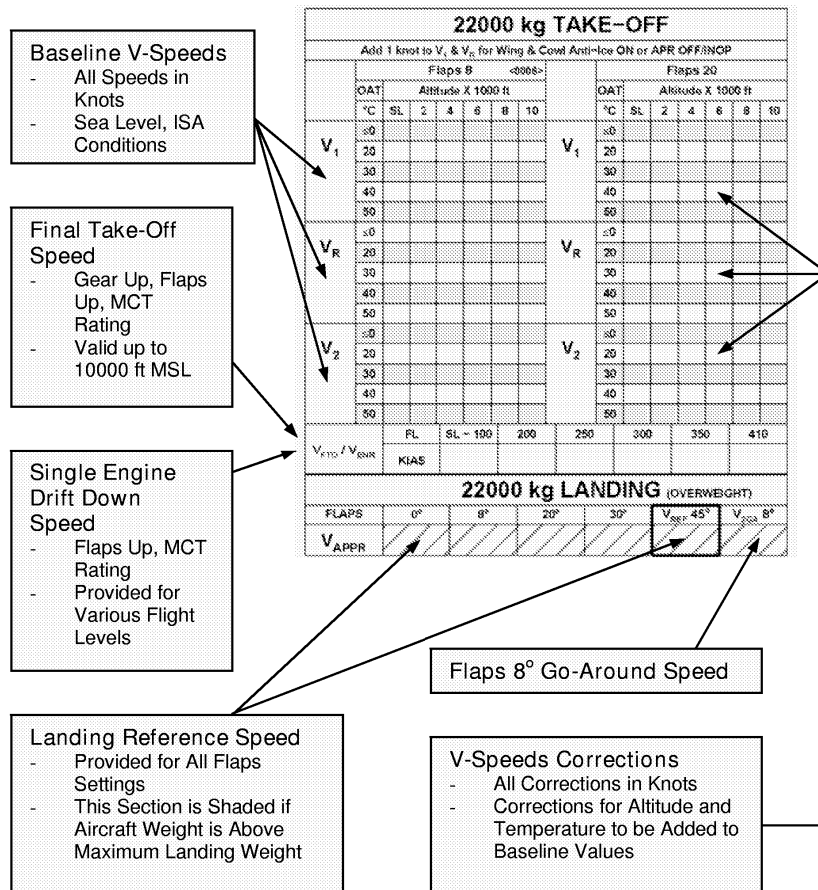


INTERMIX 2-8

Rev. 59, Mar 21/2012

Speedbook

Speedcard – Legend <0006> and <Metric>



Notes

- V_1 and V_R Data are for a Balanced Field Length
- V_1 and V_R Data are Based on 0 Runway Slope and 0 Wind
- V_1 Must Not Be Less Than V_{1MCG} Nor Greater Than V_{1MBE}
- Speed Increment with Packs On Engine Bleed is Negligible
- -- : Data is not available for interpolation

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Speedcard – 15400 kg <0005> or <0068> and <0006> and <Metric>

15400 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 108	≤0	0	2	3	5	6	8	V ₁ 98	≤0	0	2	4	5	7	8
	20	0	2	4	6	8	10		20	0	2	5	7	9	11
	30	2	4	6	8	10	12		30	2	4	7	9	11	13
	40	4	6	8	10	---	---		40	5	7	9	11	---	---
	50	6	9	---	---	---	---		50	7	9	---	---	---	---
V _R 113	≤0	0	1	2	3	4	5	V _R 105	≤0	0	1	3	4	5	6
	20	0	1	3	4	5	7		20	0	2	3	5	6	7
	30	1	2	4	5	7	8		30	1	3	4	6	7	8
	40	2	3	5	6	---	---		40	3	4	6	7	---	---
	50	3	5	---	---	---	---		50	4	6	---	---	---	---
V ₂ 127	≤0	0	0	0	0	0	0	V ₂ 117	≤0	0	0	0	0	0	0
	20	0	0	0	0	0	0		20	0	0	0	0	0	0
	30	0	0	0	0	0	0		30	0	0	0	0	0	0
	40	0	0	0	0	---	---		40	0	0	0	0	---	---
	50	0	0	---	---	---	---		50	0	0	---	---	---	---
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		152		152		159		167		175		186	
15400 kg LANDING															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		150		138		132		128		120		127			

<CRJ200_QRH_takeoff_intermix_met_Jun09_15400.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 16000 kg <0005> or <0068> and <0006> and <Metric>

16000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 -0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 111	≤0	0	2	3	5	7	8	V ₁ 101	≤0	0	2	4	5	7	8
	20	0	2	4	7	9	10		20	0	2	5	7	9	11
	30	2	4	6	8	10	12		30	2	4	7	9	11	13
	40	4	6	9	10	--	--		40	5	7	9	11	--	--
	50	7	9	--	--	--	--		50	7	9	--	--	--	--
V _R 116	≤0	0	1	2	3	4	5	V _R 108	≤0	0	1	2	4	5	6
	20	0	1	2	4	5	7		20	0	2	3	5	6	7
	30	1	2	4	5	6	8		30	1	3	4	6	7	8
	40	2	3	5	6	--	--		40	3	4	5	7	--	--
	50	3	5	--	--	--	--		50	4	6	--	--	--	--
V ₂ 129	≤0	0	0	0	0	0	0	V ₂ 119	≤0	0	0	0	0	0	0
	20	0	0	0	0	0	0		20	0	0	0	0	0	0
	30	0	0	0	0	0	0		30	0	0	0	0	0	0
	40	0	0	0	0	--	--		40	0	0	0	0	--	--
	50	0	0	--	--	--	--		50	0	0	--	--	--	--
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		154		154		162		170		179		190	
16000 kg LANDING															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		152		140		134		130		122		129			

<CRJ200 QRH takeoff intermix met Jun09 16000.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 17000 kg <0005> or <0068> and <0006> and <Metric>

17000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 117	≤0	0	1	3	4	6	7	V ₁ 107	≤0	0	1	3	5	6	7
	20	0	1	4	6	8	9		20	0	2	4	6	8	10
	30	1	3	6	8	9	11		30	1	4	6	8	10	12
	40	4	6	8	9	--	--		40	4	6	8	10	--	--
	50	6	8	--	--	--	--		50	6	8	--	--	--	--
V _R 121	≤0	-1	0	1	3	4	5	V _R 113	≤0	0	1	2	3	4	5
	20	0	1	2	4	5	6		20	0	1	3	4	5	6
	30	0	2	3	5	6	7		30	1	2	4	5	6	7
	40	2	3	5	6	--	--		40	2	4	5	6	--	--
	50	3	5	--	--	--	--		50	4	5	--	--	--	--
V ₂ 133	≤0	0	0	0	0	0	0	V ₂ 123	≤0	0	0	0	0	0	0
	20	0	0	0	0	0	0		20	0	0	0	0	0	0
	30	0	0	0	0	0	0		30	0	0	0	0	0	0
	40	0	0	0	0	--	--		40	0	0	0	0	--	--
	50	0	0	--	--	--	--		50	0	0	--	--	--	--
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		158		159		167		176		184		196	
17000 kg LANDING															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		156		144		138		134		126		133			

<CRJ200_QRH_takeoff_intermix_met_Jun09_17000.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 18000 kg <0005> or <0068> and <0006> and <Metric>

18000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 122	≤0	0	1	3	5	6	7	V ₁ 112	≤0	0	1	3	4	6	7
	20	0	2	4	6	7	9		20	0	2	4	6	8	9
	30	1	4	6	7	9	10		30	1	3	5	7	9	12
	40	4	6	7	9	--	--		40	4	6	8	9	--	--
	50	6	8	--	--	--	--		50	6	8	--	--	--	--
V _R 125	≤0	0	1	2	3	4	6	V _R 117	≤0	0	1	2	3	4	5
	20	0	1	3	4	6	7		20	1	2	3	4	5	6
	30	1	3	4	5	7	8		30	1	3	4	5	6	7
	40	3	4	5	7	--	--		40	3	4	5	6	--	--
	50	4	5	--	--	--	--		50	4	5	--	--	--	--
V ₂ 137	≤0	0	0	0	0	0	0	V ₂ 126	≤0	0	0	0	0	1	1
	20	0	0	0	0	0	0		20	0	0	0	0	1	1
	30	0	0	0	0	0	0		30	0	0	0	0	1	1
	40	0	0	0	0	--	--		40	0	0	0	0	--	--
	50	0	0	--	--	--	--		50	0	0	--	--	--	--
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		163		164		172		181		190		202	
18000 kg LANDING															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		160		148		142		138		130		137			

<CRJ200_QRH_takeoff_intermix_met_Jun09_18000.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 19000 kg <0005> or <0068> and <0006> and <Metric>

19000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 127	≤0	0	2	3	4	6	7	V ₁ 117	≤0	0	1	3	4	5	7
	20	0	2	4	6	7	9		20	0	1	3	5	7	11
	30	1	4	5	7	8	10		30	1	3	5	7	10	12
	40	4	5	7	9	---	---		40	3	5	7	10	---	---
	50	6	7	---	---	---	---		50	6	7	---	---	---	---
V _R 130	≤0	-1	1	2	3	4	5	V _R 122	≤0	0	1	2	3	3	5
	20	0	1	3	4	5	6		20	0	1	2	3	4	6
	30	1	2	4	5	6	7		30	1	2	3	4	5	7
	40	2	3	5	6	---	---		40	2	3	4	5	---	---
	50	4	5	---	---	---	---		50	3	4	---	---	---	---
V ₂ 141	≤0	0	0	0	0	0	0	V ₂ 130	≤0	0	0	0	0	0	1
	20	0	0	0	0	0	0		20	0	0	0	0	0	1
	30	0	0	0	0	0	0		30	0	0	0	0	0	1
	40	0	0	0	0	---	---		40	0	0	0	0	---	---
	50	0	0	---	---	---	---		50	0	0	---	---	---	---
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		168		169		177		186		196		208	
19000 kg LANDING															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		163		151		145		141		133		141			

<CRJ200_QRH_takeoff_intermix_met_Jun09_19000.ps>

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Speedcard – 20000 kg <0005> or <0068> and <0006> and <Metric>

20000 kg TAKE-OFF																	
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP																	
	Flaps 8 <0006>									Flaps 20							
	OAT	Altitude X 1000 ft								OAT	Altitude X 1000 ft						
	°C	SL	2	4	6	8	10	°C		SL	2	4	6	8	10		
V ₁ 132	≤0	0	1	3	4	5	6	V ₁ 121	≤0	0	2	3	4	6	9		
	20	0	2	3	5	7	8		20	0	2	4	6	8	12		
	30	1	3	5	6	8	9		30	2	4	6	7	10	13		
	40	3	5	6	8	--	--		40	4	6	8	10	--	--		
	50	5	7	--	--	--	--		50	6	8	--	--	--	--		
V _R 134	≤0	0	1	2	3	4	5	V _R 126	≤0	0	1	2	3	3	6		
	20	0	2	3	4	5	6		20	0	1	2	3	4	7		
	30	1	3	4	5	6	7		30	1	2	3	4	5	8		
	40	3	4	5	6	--	--		40	2	3	4	5	--	--		
	50	4	5	--	--	--	--		50	3	4	--	--	--	--		
V ₂ 145	≤0	0	0	0	0	0	0	V ₂ 133	≤0	0	0	0	0	0	2		
	20	0	0	0	0	0	0		20	0	0	0	0	0	2		
	30	0	0	0	0	0	0		30	0	0	0	0	0	2		
	40	0	0	0	0	--	--		40	0	0	0	0	--	--		
	50	0	0	--	--	--	--		50	0	0	--	--	--	--		
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410			
		KIAS		172		173		182		191		201		214			
20000 kg LANDING																	
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°					
V _{APPR}		167		155		149		145		137		145					

<CRJ200_QRH_takeoff_intermix_met_Jun09_20000.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 21000 kg <0005> or <0068> and <0006> and <Metric>

21000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 137	≤0	0	1	2	3	4	6	V ₁ 126	≤0	0	1	3	4	7	9
	20	0	1	3	4	6	8		20	0	1	3	5	10	11
	30	1	3	4	6	7	9		30	1	3	5	8	10	11
	40	3	4	6	7	--	--		40	3	5	8	9	--	--
	50	5	6	--	--	--	--		50	5	8	--	--	--	--
V _R 139	≤0	-1	0	1	2	3	5	V _R 130	≤0	0	1	2	2	5	6
	20	0	1	2	3	4	6		20	0	1	2	3	6	7
	30	0	2	3	4	5	7		30	1	2	3	4	6	7
	40	2	3	4	5	--	--		40	2	3	4	5	--	--
	50	3	4	--	--	--	--		50	3	4	--	--	--	--
V ₂ 148	≤0	0	0	0	0	0	0	V ₂ 136	≤0	0	0	1	1	2	3
	20	0	0	0	0	0	0		20	0	0	1	1	2	3
	30	0	0	0	0	0	0		30	0	0	1	1	2	3
	40	0	0	0	0	--	--		40	0	0	1	1	--	--
	50	0	0	--	--	--	--		50	0	0	--	--	--	--
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		176		177		186		196		206		220	
21000 kg LANDING															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		170		158		152		148		140		148			

<CRJ200_QRH_takeoff_intermix_met_Jun09_21000.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 22000 kg <0005> or <0068> and <0006> and <Metric>

22000 kg TAKE-OFF																	
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP																	
	Flaps 8 <0006>									Flaps 20							
	OAT	Altitude X 1000 ft								OAT	Altitude X 1000 ft						
	°C	SL	2	4	6	8	10	°C		SL	2	4	6	8	10		
V ₁ 141	≤0	0	1	2	3	5	6	V ₁ 130	≤0	0	1	3	4	8	10		
	20	0	1	3	4	6	8		20	0	1	3	6	10	11		
	30	1	3	4	6	7	9		30	1	3	5	8	10	—		
	40	3	4	6	7	—	—		40	3	6	7	8	—	—		
	50	5	6	—	—	—	—		50	7	7	—	—	—	—		
V _R 143	≤0	−1	1	2	3	4	5	V _R 134	≤0	0	0	1	2	5	6		
	20	0	1	2	3	5	6		20	0	1	2	3	6	7		
	30	1	2	3	4	5	7		30	0	2	3	4	6	—		
	40	2	3	4	5	—	—		40	2	2	3	4	—	—		
	50	3	4	—	—	—	—		50	3	3	—	—	—	—		
V ₂ 152	≤0	0	0	0	0	0	0	V ₂ 140	≤0	0	0	0	0	2	3		
	20	0	0	0	0	0	0		20	0	0	0	0	2	3		
	30	0	0	0	0	0	0		30	0	0	0	0	2	—		
	40	0	0	0	0	—	—		40	0	0	0	0	—	—		
	50	0	0	—	—	—	—		50	0	0	—	—	—	—		
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410			
		KIAS		180		182		191		201		211		226			
22000 kg LANDING (OVERWEIGHT)																	
FLAPS	0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°						
V _{APPR}	173		161		155		151		143		152						

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 23000 kg <0005> or <0068> and <0006> and <Metric>

23000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 145	≤0	0	1	2	3	5	7	V ₁ 134	≤0	0	1	3	6	9	12
	20	0	1	3	4	7	9		20	0	2	3	8	10	12
	30	1	3	4	6	8	9		30	1	3	6	9	10	---
	40	3	4	6	7	---	---		40	3	6	7	9	---	---
	50	5	6	---	---	---	---		50	6	7	---	---	---	---
V _R 147	≤0	0	1	2	2	4	6	V _R 137	≤0	0	1	2	5	6	9
	20	0	1	2	3	5	7		20	0	1	2	5	7	9
	30	1	2	3	4	6	7		30	1	2	3	6	7	---
	40	2	3	4	5	---	---		40	2	3	4	6	---	---
	50	3	4	---	---	---	---		50	3	4	---	---	---	---
V ₂ 155	≤0	0	0	0	0	0	1	V ₂ 143	≤0	0	0	0	2	3	4
	20	0	0	0	0	0	1		20	0	0	0	2	3	4
	30	0	0	0	0	0	1		30	0	0	0	2	3	---
	40	0	0	0	0	---	---		40	0	0	0	2	---	---
	50	0	0	---	---	---	---		50	0	0	---	---	---	---
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		184		186		195		206		216		232	
23000 kg LANDING (OVERWEIGHT)															
FLAPS		0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°			
V _{APPR}		176		164		158		154		146		155			

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Speedcard – 24000 kg <0005> or <0068> and <0006> and <Metric>

24000 kg TAKE-OFF															
Add 1 knot to V ₁ & V _R for Wing & Cowl Anti-Ice ON or APR OFF/INOP															
	Flaps 8 <0006>								Flaps 20						
	OAT	Altitude X 1000 ft							OAT	Altitude X 1000 ft					
	°C	SL	2	4	6	8	10		°C	SL	2	4	6	8	10
V ₁ 149	≤0	0	1	2	3	5	8	V ₁ 138	≤0	0	1	4	7	9	13
	20	0	1	3	4	7	9		20	0	1	5	8	9	---
	30	1	3	4	6	8	9		30	1	4	7	9	---	---
	40	3	4	6	7	---	---		40	4	5	7	---	---	---
	50	4	6	---	---	---	---		50	6	6	---	---	---	---
V _R 151	≤0	-1	0	1	3	4	6	V _R 141	≤0	0	1	3	4	6	10
	20	0	1	2	3	5	7		20	0	1	3	5	6	---
	30	0	2	3	4	6	7		30	1	2	4	6	---	---
	40	2	3	4	5	---	---		40	2	2	4	---	---	---
	50	3	4	---	---	---	---		50	3	3	---	---	---	---
V ₂ 158	≤0	0	0	0	0	1	2	V ₂ 146	≤0	0	0	1	2	3	6
	20	0	0	0	0	1	2		20	0	0	1	2	3	---
	30	0	0	0	0	1	2		30	0	0	1	2	---	---
	40	0	0	0	0	---	---		40	0	0	1	---	---	---
	50	0	0	---	---	---	---		50	0	0	---	---	---	---
V _{FTO} / V _{ENR}		FL		SL – 100		200		250		300		350		410	
		KIAS		188		190		200		210		222		237	
24000 kg LANDING (OVERWEIGHT)															
FLAPS	0°		8°		20°		30°		V _{REF} 45°		V _{2GA} 8°				
V _{APPR}	180		168		162		158		150		158				

<CRJ200_QRH_takeoff_intermix_met_Jun09_24000.ps>

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

MAXIMUM CLIMB THRUST SETTINGS

Maximum Climb Thrust Setting

Maximum Climb Thrust Setting, %N₁ (PACKs On)

DISPLAYED FAN SPEED	
MAXIMUM CLIMB THRUST SETTING PACKS ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)									
(°C)	(°F)	0	5000	10000	15000	15000+	20000	25000	30000	36000	41000
-70	-94									91.3	91.2
-65	-85								91.0	92.5	92.3
-60	-76							89.4	92.0	93.6	93.4
-55	-67							90.4	93.1	94.7	94.1
-50	-58						88.7	91.4	94.1	95.2	93.7
-45	-49					87.0	89.6	92.4	95.2	94.9	93.1
-40	-40	79.6	82.3	85.1	87.7	88.0	90.6	93.4	95.0	94.3	92.6
-35	-31	80.4	83.1	86.0	88.6	88.9	91.6	94.4	94.5	93.8	92.0
-30	-22	81.2	84.0	86.8	89.5	89.8	92.5	94.0	94.1	93.2	
-25	-13	82.0	84.8	87.7	90.4	90.7	93.5	93.6	93.5	92.4	
-20	-4	82.8	85.7	88.6	91.4	91.7	92.9	92.9	92.6		
-15	5	83.6	86.5	89.5	92.3	92.6	92.3	92.3	91.7		
-10	14	84.4	87.4	90.4	91.8	92.1	91.4	91.4	90.7		
-5	23	85.3	88.2	91.3	91.0	91.3	90.6	90.6			
0	32	86.1	89.1	90.6	90.2	90.5	89.8	89.8			
5	41	86.9	89.9	89.9	89.4	89.7	89.0				
10	50	87.7	89.2	89.1	88.7	89.0	88.3				
15	59	88.5	88.5	88.4	87.9	88.2					
20	68	87.8	87.8	87.6	87.0	87.3					
25	77	87.0	87.0	86.7							
30	86	86.0	86.0	85.7							
35	95	85.2	85.2	84.8							
40	104	84.4	84.4								
45	113	83.6	83.6								
50	122	82.9									

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Maximum Climb Thrust Setting, %N₁ (PACKs On; Cowl Anti-ice On)

DISPLAYED FAN SPEED	
MAXIMUM CLIMB THRUST SETTING PACKS ON, COWL ANTI-ICE ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)									
(°C)	(°F)	0	5000	10000	15000 ⁻	15000 ⁺	20000	25000	30000	33000	
-70	-94										88.0
-65	-85								91.0		89.6
-60	-76							89.4	92.0		91.3
-55	-67							90.4	93.1		92.9
-50	-58						88.7	91.4	94.1		93.5
-45	-49					87.0	89.6	92.4	94.3		93.5
-40	-40	79.6	82.3	85.1	87.7	88.0	90.6	93.4	93.8		93.4
-35	-31	80.4	83.1	86.0	88.6	88.9	91.6	93.4	93.4		92.8
-30	-22	81.2	84.0	86.8	89.5	89.8	92.5	93.1	92.8		92.0
-25	-13	82.0	84.8	87.7	90.4	90.7	92.5	92.4	91.9		91.0
-20	-4	82.8	85.7	88.6	91.4	91.7	91.9	91.6	90.9		89.9
-15	5	83.6	86.5	89.5	91.8	91.8	91.1	90.8	89.7		
-10	14	84.4	87.4	90.4	91.1	91.1	90.3	89.9	88.8		
-5	23	85.3	88.2	90.7	90.2	90.2	89.4	89.0			
0	32	86.1	89.1	90.0	89.5	89.5	88.6	88.1			
5	41	86.9	89.5	89.2	88.6	88.6	87.7				
10	50	87.7	88.7	88.4	87.9	87.9	86.8				

cf34-INT_mcl_acu_cowl_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Maximum Climb Thrust Setting, %N₁ (PACKs On; Cowl and Wing Anti-ice On)

DISPLAYED FAN SPEED	
MAXIMUM CLIMB THRUST SETTING PACKS ON, COWL AND WING ANTI-ICE ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)									
(°C)	(°F)	0	5000	10000	15000 ⁻	15000 ⁺	20000	25000	30000	33000	
-70	-94										84.6
-65	-85								91.0		85.3
-60	-76							89.4	91.5		86.1
-55	-67							90.4	91.5		86.8
-50	-58						88.7	90.9	91.5		87.5
-45	-49					87.0	89.6	90.9	91.5		87.5
-40	-40	79.6	82.3	85.1	87.7	88.0	90.0	90.9	91.0		87.5
-35	-31	80.4	83.1	86.0	88.6	88.9	90.0	90.9	90.1		86.9
-30	-22	81.2	84.0	86.8	89.4	89.4	90.0	90.2	89.0		86.2
-25	-13	82.0	84.8	87.7	89.4	89.4	90.0	89.3	87.8		86.1
-20	-4	82.8	85.7	88.6	89.4	89.4	89.1	88.3	86.5		84.7
-15	5	83.6	86.5	88.7	89.4	89.4	88.2	87.3	85.3		
-10	14	84.4	87.4	88.7	88.6	88.6	87.2	86.1	84.1		
-5	23	85.3	87.5	88.7	87.7	87.7	86.2	84.9			
0	32	86.1	87.5	87.9	86.8	86.8	85.2	83.6			
5	41	86.2	87.5	87.0	85.8	85.8	84.0				
10	50	86.2	86.7	86.0	84.6	84.6	82.9				

cf34-INT_mcl_acu_cowl_wing_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

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CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

MAXIMUM CRUISE THRUST SETTINGS

Maximum Cruise Thrust Setting

Maximum Cruise Thrust Setting, %N₁ (PACKs On)

DISPLAYED FAN SPEED	
MAXIMUM CRUISE THRUST SETTING PACKS ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)									
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000	36000	41000	
-70	-94								91.8	90.2	
-65	-85							90.3	92.9	91.3	
-60	-76						89.2	91.3	94.0	92.3	
-55	-67						90.3	92.4	94.7	92.9	
-50	-58					88.4	91.3	93.4	94.1	92.4	
-45	-49				86.9	89.3	92.3	94.5	93.6	91.9	
-40	-40	80.8	83.0	85.4	87.8	90.3	93.3	94.3	93.2	91.0	
-35	-31	81.7	83.8	86.2	88.7	91.3	94.3	93.9	92.3	90.1	
-30	-22	82.5	84.7	87.1	89.6	92.2	93.8	93.1	91.4		
-25	-13	83.3	85.5	88.0	90.5	93.2	93.0	92.3	90.4		
-20	-4	84.1	86.4	88.9	91.5	92.6	92.2	91.4			
-15	5	85.0	87.3	89.8	92.4	91.8	91.4	90.6			
-10	14	85.8	88.1	90.7	91.8	91.1	90.7	89.8			
-5	23	86.6	89.0	91.6	91.2	90.5	90.0				
0	32	87.4	89.8	91.0	90.6	89.8	89.2				
5	41	88.2	90.7	90.4	89.9	89.2					
10	50	89.1	90.2	89.9	89.4	88.5					
15	59	89.9	89.6	89.3	88.8						
20	68	89.3	89.1	88.8	88.2						
25	77	88.8	88.5	88.2							
30	86	88.3	88.0	87.7							
35	95	87.8	87.5	87.1							
40	104	87.2	86.9								
45	113	86.7	86.3								
50	122	86.1									

cf34-INT_mcr_acu_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Maximum Cruise Thrust Setting, %N₁ (PACKs On; Cowl Anti-ice On)

DISPLAYED FAN SPEED	
MAXIMUM CRUISE THRUST SETTING PACKS ON, COWL ANTI-ICE ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)							
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000	33000
-70	-94								90.4
-65	-85							90.3	91.6
-60	-76						89.2	91.3	92.7
-55	-67						90.3	92.4	93.5
-50	-58					88.4	91.3	93.4	93.4
-45	-49				86.9	89.3	92.3	93.6	92.8
-40	-40	80.8	83.0	85.4	87.8	90.3	93.3	93.3	92.2
-35	-31	81.7	83.8	86.2	88.7	91.3	93.5	92.6	91.5
-30	-22	82.5	84.7	87.1	89.6	92.2	92.7	91.8	90.5
-25	-13	83.3	85.5	88.0	90.5	92.4	91.9	90.8	89.4
-20	-4	84.1	86.4	88.9	91.5	91.7	91.0	89.9	88.7
-15	5	85.0	87.3	89.8	91.6	90.9	90.2	89.0	
-10	14	85.8	88.1	90.7	91.1	90.2	89.5	88.1	
-5	23	86.6	89.0	90.9	90.4	89.4	88.7		
0	32	87.4	89.8	90.3	89.7	88.7	88.0		
5	41	88.2	90.1	89.7	89.1	88.1			
10	50	89.1	89.5	89.1	88.5	87.4			

cf34-INT_mcr_acu_cowl_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Maximum Cruise Thrust Setting, %N₁ (PACKs On; Cowl and Wing Anti-ice On)

DISPLAYED FAN SPEED	
MAXIMUM CRUISE THRUST SETTING PACKS ON, COWL AND WING ANTI-ICE ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)							
(°C)	(°F)	0	5000	10000	15000	20000	25000	30000	33000
-70	-94								87.6
-65	-85							90.3	88.3
-60	-76						89.2	91.1	88.9
-55	-67						90.3	91.1	89.5
-50	-58					88.4	90.9	91.1	90.1
-45	-49				86.9	89.3	90.9	91.1	89.4
-40	-40	80.8	83.0	85.4	87.8	90.2	90.9	90.3	88.7
-35	-31	81.7	83.8	86.2	88.7	90.2	90.9	89.4	87.3
-30	-22	82.5	84.7	87.1	89.6	90.2	90.1	88.2	85.9
-25	-13	83.3	85.5	88.0	89.9	90.2	89.1	87.2	84.9
-20	-4	84.1	86.4	88.9	89.9	89.4	88.2	86.0	83.9
-15	5	85.0	87.3	89.3	89.9	88.6	87.3	85.0	
-10	14	85.8	88.1	89.3	89.2	87.8	86.4	83.9	
-5	23	86.6	88.6	89.3	88.4	87.0	85.5		
0	32	87.4	88.6	88.6	87.7	86.2	84.5		
5	41	88.0	88.6	88.0	87.0	85.4			
10	50	88.0	88.0	87.3	86.3	84.6			

cf34-INT_mcr_acu_cowl_wing_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

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CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

APPROACH AND LANDING DATA

Go-around Thrust Setting

NOTE

The following thrust setting tables are given for a go-around with either both engines operating or one engine inoperative.

Go-around Thrust Setting, %N₁ (Zero Engine Bleed)

DISPLAYED FAN SPEED	
GO-AROUND THRUST SETTING ZERO ENGINE BLEED INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	82.5	82.9	83.7	84.6	85.6	86.5	87.6
-35	-31	83.4	83.7	84.6	85.5	86.4	87.4	88.4
-30	-22	84.2	84.6	85.4	86.3	87.3	88.3	89.3
-25	-13	85.0	85.4	86.3	87.2	88.2	89.2	90.2
-20	-4	85.8	86.2	87.1	88.0	89.0	90.1	91.1
-15	5	86.7	87.1	88.0	88.9	89.9	90.9	92.0
-10	14	87.5	87.9	88.8	89.7	90.8	91.8	92.9
-5	23	88.3	88.7	89.6	90.6	91.6	92.7	93.8
0	32	89.1	89.5	90.5	91.5	92.5	93.6	94.7
5	41	89.9	90.4	91.3	92.3	93.4	94.5	95.1
10	50	90.8	91.2	92.2	93.2	94.2	94.5	94.5
15	59	91.6	92.0	93.0	94.0	94.0	94.0	94.0
20	68	92.4	92.9	93.6	93.6	93.6	93.5	93.4
25	77	93.2	93.2	93.2	93.1	93.1	93.0	92.8
30	86	92.7	92.8	92.8	92.6	92.5	92.3	92.1
35	95	92.1	92.2	92.2	92.0	91.8	91.6	91.5
40	104	91.3	91.3	91.3	91.3	91.1		
45	113	90.4	90.4	90.5	90.5			
50	122	89.6	89.7	89.7				

cf34-INT_apr_zero_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Go-around Thrust Setting, %N₁ (10th Stage Bleeds On – PACKs On)

DISPLAYED FAN SPEED	
GO-AROUND THRUST SETTING 10th STAGE BLEEDS ON – PACKS ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	–1000	0	2000	4000	6000	8000	10000
–40	–40	82.5	82.9	83.7	84.6	85.6	86.5	87.6
–35	–31	83.4	83.7	84.6	85.5	86.4	87.4	88.4
–30	–22	84.2	84.6	85.4	86.3	87.3	88.3	89.3
–25	–13	85.0	85.4	86.3	87.2	88.2	89.2	90.2
–20	–4	85.8	86.2	87.1	88.0	89.0	90.1	91.1
–15	5	86.7	87.1	88.0	88.9	89.9	90.9	92.0
–10	14	87.5	87.9	88.8	89.7	90.8	91.8	92.9
–5	23	88.3	88.7	89.6	90.6	91.6	92.7	93.8
0	32	89.1	89.5	90.5	91.5	92.5	93.6	94.0
5	41	89.9	90.4	91.3	92.3	93.4	93.7	93.8
10	50	90.8	91.2	92.2	93.1	93.4	93.4	93.4
15	59	91.6	92.0	92.8	93.1	93.0	93.0	92.9
20	68	92.4	92.5	92.7	92.6	92.5	92.4	92.2
25	77	92.3	92.3	92.2	92.1	92.0	91.8	91.6
30	86	91.9	91.9	91.8	91.6	91.4	91.1	90.8
35	95	91.2	91.2	91.1	90.9	90.6	90.3	90.1
40	104	90.3	90.2	90.2	90.1	89.8		
45	113	89.3	89.3	89.3	89.3			
50	122	88.5	88.4	88.3				

cf34–INT_apr_acu_qrh_n1.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Go-around Thrust Setting, %N₁ (Cowl Anti-ice On)

DISPLAYED FAN SPEED	
GO-AROUND THRUST SETTING COWL ANTI-ICE ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	82.5	82.9	83.7	84.6	85.6	86.5	87.6
-35	-31	83.4	83.7	84.6	85.5	86.4	87.4	88.4
-30	-22	84.2	84.6	85.4	86.3	87.3	88.3	89.3
-25	-13	85.0	85.4	86.3	87.2	88.2	89.2	90.2
-20	-4	85.8	86.2	87.1	88.0	89.0	90.1	91.1
-15	5	86.7	87.1	88.0	88.9	89.9	90.9	92.0
-10	14	87.5	87.9	88.8	89.7	90.8	91.8	92.9
-5	23	88.3	88.7	89.6	90.6	91.6	92.7	93.8
0	32	89.1	89.5	90.5	91.5	92.5	93.6	94.3
5	41	89.9	90.4	91.3	92.3	93.4	94.0	94.1
10	50	90.8	91.2	92.2	93.2	93.7	93.7	93.6

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Go-around Thrust Setting, %N₁ (Wing and Cowl Anti-ice On)

DISPLAYED FAN SPEED	
GO-AROUND THRUST SETTING WING AND COWL ANTI-ICE ON INTERMIX	%N1

OAT		PRESSURE ALTITUDE (Feet)						
(°C)	(°F)	-1000	0	2000	4000	6000	8000	10000
-40	-40	82.5	82.9	83.7	84.6	85.6	86.5	87.6
-35	-31	83.4	83.7	84.6	85.5	86.4	87.4	88.4
-30	-22	84.2	84.6	85.4	86.3	87.3	88.3	89.3
-25	-13	85.0	85.4	86.3	87.2	88.2	89.2	90.2
-20	-4	85.8	86.2	87.1	88.0	89.0	90.1	90.3
-15	5	86.7	87.1	88.0	88.9	89.9	90.1	90.3
-10	14	87.5	87.9	88.8	89.6	89.9	90.1	90.3
-5	23	88.3	88.7	89.4	89.6	89.9	90.1	90.3
0	32	88.9	89.0	89.4	89.6	89.9	90.1	90.3
5	41	88.9	89.0	89.4	89.6	89.9	90.1	90.2
10	50	88.9	89.0	89.4	89.6	89.9	89.8	89.7

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Actual Landing Distance – FLAPS 45

Landing Speed and Actual Landing Distance (Landing Gear Down – FLAPS 45) <Metric>

Weight (kg)	V _{REF} [★] (KIAS)	Pressure Altitude (Feet x 1,000)					
		0	2	4	6	8	10
		Actual Landing Distance* (meters)					
15400	120	704	733	764	799	836	878
16000	122	723	753	785	821	860	903
17000	126	754	786	819	857	898	944
18000	130	785	818	854	894	937	987
19000	133	816	850	888	930	977	1029
20000	137	846	883	923	968	1017	1072
21000	140	877	915	957	1005	1057	1118
22000	144	907	948	993	1043	1099	1170
23000	147	939	983	1031	1086	1149	1229
24000	151	975	1022	1074	1133	1208	1296

★ V_{REF} = Speed at 50' screen height (KIAS), Flaps 45°
* Actual Landing Distance = Dry Landing Field Length x 0.6
Wet Landing Field Length = Dry Landing Field Length x 1.15

Actual Landing Distance – Flaps Other than 45

Actual Landing Distance Factors when Landing with Flaps Other than 45 <0006>

Flaps Setting	Distance (No Reversers)	Distance (W/ Reversers)
0°	Dist. x 1.70 (70%)	Dist. x 1.65 (65%)
8°	Dist. x 1.55 (55%)	Dist. x 1.50 (50%)
20°	Dist. x 1.25 (25%)	Dist. x 1.22 (22%)
30°	Dist. x 1.20 (20%)	Dist. x 1.18 (18%)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Brake Cooling Requirements

V_{1MBE} , FLAPS 8 <0006> and <Metric>

V _{1MBE} , Flaps 8° (Sea Level, 0 Wind, 0 Runway Slope)								
OAT		Airplane Weight (kg x 1000)						
°C	°F	24	23	22	21	20	19	18
-40	-40	192	196	200	204	208	212	216
-30	-22	187	191	195	199	203	207	211
-20	-4	182	186	190	194	198	202	206
-10	14	178	182	186	190	194	198	202
0	32	173	177	181	185	189	193	197
10	50	169	173	177	181	185	189	193
20	68	166	170	174	178	182	186	190
30	86	163	167	171	175	179	183	187
40	104	161	165	169	173	177	181	185
50	122	159	163	167	171	175	179	183

- At weights less than 18000 kg, use V_{1MBE} for 18000 kg.
- Reduce V_{1MBE} by 3 KIAS per 1000 ft altitude above sea level.
- Wind Correction: Reduce V_{1MBE} by 1.5 KIAS/1 kt tailwind
Increase V_{1MBE} by 0.3 KIAS/1 kt headwind
- Rwy Slope Correction: Reduce V_{1MBE} by 2.0 KIAS/1% downhill.
Increase V_{1MBE} by 2.0 KIAS/1% uphill.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

V_{1MBE} , FLAPS 20 <Metric>

V_{1MBE}, Flaps 20° (Sea Level, 0 Wind, 0 Runway Slope)								
OAT		Airplane Weight (kg x 1000)						
°C	°F	24	23	22	21	20	19	18
-40	-40	195	199	203	207	211	215	219
-30	-22	190	194	198	202	206	210	214
-20	-4	185	189	193	197	201	205	209
-10	14	181	185	189	193	197	201	205
0	32	176	180	184	188	192	196	200
10	50	172	176	180	184	188	192	196
20	68	169	173	177	181	185	189	193
30	86	166	170	174	178	182	186	190
40	104	164	168	172	176	180	184	188
50	122	162	166	170	174	178	182	186

- At weights less than 18000 kg, use V_{1MBE} for 18000 kg.
- Reduce V_{1MBE} by 3 KIAS per 1000 ft altitude above sea level.
- Wind Correction: Reduce V_{1MBE} by 1.5 KIAS/1 kt tailwind
Increase V_{1MBE} by 0.3 KIAS/1 kt headwind
- Rwy Slope Correction: Reduce V_{1MBE} by 2.0 KIAS/1% downhill.
Increase V_{1MBE} by 2.0 KIAS/1% uphill.

BTMS Operative:

A minimum brake cooling time of fifteen (15) minutes must be observed between a landing stop or a rejected take-off and the subsequent take-off.

Brake temperatures, as obtained from the BTMS, must be stable or decreasing before take-off. The temperature of the warmest brake at the start of the take-off roll must not exceed the BTMS units as shown below:

V₁/V_{1MBE}	BTMS Units
≤ to 0.85	06
>0.85	05

BTMS Inoperative:

Following a rejected take-off with a brakes application speed of not more than 80 KIAS, or a landing at a weight not exceeding the maximum permissible quick turn-around landing weight, the following brake cooling times must be observed:

	INTERMIX 5-7
	Rev. 59, Mar 21/2012

V_1/V_{1MBE}	Minimum Brake Cooling Time (minutes)
<0.90	10
0.90 – 0.92	15
0.92 – 0.94	20
0.94 – 0.96	28
0.96 – 0.98	36
0.98 – 1.00	45

Following a rejected take-off with a brakes application speed greater than 80 KIAS, or a landing at a weight exceeding the maximum permissible quick turn-around landing weight, the following brake cooling times must be observed, followed by an inspection of the wheel fuse plugs:

V_1/V_{1MBE}	Minimum Brake Cooling Time (minutes)
<0.85	40
0.85 – 0.90	60
0.90 – 0.95	90
0.95 – 1.00	140

NOTE

The maintenance inspection should be carried out in accordance with Aircraft Maintenance Manual, Chapter 05-51-27, HIGH ENERGY STOP (HES)/REJECTED TAKE-OFF (RTO) – INSPECTION CHECK.

QUICK REFERENCE HANDBOOK CSP A-022-080	INTERMIX
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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

REDUCED THRUST TAKE-OFF

Reduced Thrust Take-off

To interpolate between pressure altitude, actual ambient temperature or assumed temperature, when using the reduced thrust take-off tables, the following steps must be followed:

- The pressure altitude to be used in determining the reduced %N₁ setting, for altitudes below 4000 feet, must be rounded up to the next highest altitude. For altitudes above 4000 feet, the %N₁ must be rounded down to the next lowest altitude.
- The assumed temperature to be used in determining the reduced %N₁ setting must be rounded down to the next lowest assumed temperature.
- The actual ambient temperature to be used in determining the reduced %N₁ setting must be rounded up to the next highest actual ambient temperature.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
Zero Engine Bleed – Pressure Altitude, –1000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = –1000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	25	30	31	35	40	45	50
–40	80.7	80.7	80.7	80.7	80.7	80.7	80.7
–35	81.6	81.6	81.6	81.6	81.6	81.6	81.6
–30	82.4	82.4	82.4	82.4	82.4	82.4	82.4
–25	83.2	83.2	83.2	83.2	83.2	83.2	83.2
–20	84.0	84.0	84.0	84.0	84.0	84.0	84.0
–15	84.9	84.9	84.9	84.9	84.9	84.9	84.9
–10	85.7	85.0	85.0	85.0	85.0	85.0	85.0
–5	86.5	85.3	85.0	85.0	85.0	85.0	85.0
0	87.3	86.1	85.6	85.0	85.0	85.0	85.0
5	88.1	86.9	86.4	85.5	85.0	85.0	85.0
10	88.8	87.6	87.2	86.3	85.0	85.0	85.0
15	89.6	88.4	87.9	87.0	85.5	85.0	85.0
20	90.4	89.2	88.7	87.8	86.3	85.0	85.0
25		89.9	89.5	88.5	87.0	85.5	85.0
30			90.2	89.3	87.7	86.2	85.0
31				89.6	88.0	86.5	85.1
35					88.5	87.0	85.5
40						87.7	86.2
45							86.9
50							

flex_n1_INT_zero_–1000_qrh.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
Zero Engine Bleed – Pressure Altitude 0 ft (Sea Level)

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = SEA LEVEL	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	23	25	30	35	40	45	50
–40	81.1	81.1	81.1	81.1	81.1	81.1	81.1
–35	82.0	82.0	82.0	82.0	82.0	82.0	82.0
–30	82.8	82.8	82.8	82.8	82.8	82.8	82.8
–25	83.6	83.6	83.6	83.6	83.6	83.6	83.6
–20	84.5	84.5	84.5	84.5	84.5	84.5	84.5
–15	85.3	85.0	85.0	85.0	85.0	85.0	85.0
–10	86.1	85.6	85.0	85.0	85.0	85.0	85.0
–5	86.9	86.4	85.3	85.0	85.0	85.0	85.0
0	87.7	87.2	86.1	85.0	85.0	85.0	85.0
5	88.5	88.0	86.9	85.6	85.0	85.0	85.0
10	89.3	88.8	87.6	86.4	85.0	85.0	85.0
15	90.0	89.6	88.4	87.1	85.5	85.0	85.0
20	90.8	90.4	89.2	87.9	86.3	85.0	85.0
23		90.8	89.6	88.3	86.7	85.2	85.0
25			89.9	88.6	87.0	85.5	85.0
30				89.4	87.7	86.2	85.0
35					88.5	87.0	85.5
40						87.7	86.2
45							86.9
50							

flex_n1_INT_zero_0_qrh.ps

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
Zero Engine Bleed – Pressure Altitude 1000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 1000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	21	25	30	35	40	45	48
-40	81.5	81.5	81.5	81.5	81.5	81.5	81.5
-35	82.4	82.4	82.4	82.4	82.4	82.4	82.4
-30	83.2	83.2	83.2	83.2	83.2	83.2	83.2
-25	84.0	84.0	84.0	84.0	84.0	84.0	84.0
-20	84.9	84.9	84.9	84.9	84.9	84.9	84.9
-15	85.7	85.0	85.0	85.0	85.0	85.0	85.0
-10	86.5	85.6	85.0	85.0	85.0	85.0	85.0
-5	87.3	86.4	85.3	85.0	85.0	85.0	85.0
0	88.1	87.2	86.1	85.0	85.0	85.0	85.0
5	88.9	88.0	86.8	85.5	85.0	85.0	85.0
10	89.7	88.8	87.6	86.3	85.0	85.0	85.0
15	90.5	89.6	88.4	87.1	85.6	85.0	85.0
20	91.3	90.3	89.2	87.8	86.3	85.0	85.0
21		90.5	89.3	88.0	86.5	85.0	85.0
25			89.9	88.6	87.1	85.6	85.0
30				89.3	87.8	86.3	85.4
35					88.5	87.0	86.1
40						87.7	86.8
45							87.5
48							

flex_n1_INT_zero_1000_qrh.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
Zero Engine Bleed – Pressure Altitude 2000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 2000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)							
(°C)	19	20	25	30	35	40	45	46
–40	81.9	81.9	81.9	81.9	81.9	81.9	81.9	81.9
–35	82.8	82.8	82.8	82.8	82.8	82.8	82.8	82.8
–30	83.6	83.6	83.6	83.6	83.6	83.6	83.6	83.6
–25	84.5	84.5	84.5	84.5	84.5	84.5	84.5	84.5
–20	85.3	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–15	86.1	85.9	85.0	85.0	85.0	85.0	85.0	85.0
–10	87.0	86.7	85.6	85.0	85.0	85.0	85.0	85.0
–5	87.8	87.5	86.4	85.3	85.0	85.0	85.0	85.0
0	88.6	88.3	87.2	86.1	85.0	85.0	85.0	85.0
5	89.4	89.1	88.0	86.9	85.6	85.0	85.0	85.0
10	90.2	89.9	88.8	87.6	86.4	85.0	85.0	85.0
15	90.9	90.7	89.6	88.4	87.1	85.6	85.0	85.0
19		91.3	90.2	89.0	87.7	86.2	85.0	85.0
20			90.4	89.2	87.9	86.3	85.0	85.0
25				89.9	88.6	87.1	85.6	85.3
30					89.4	87.8	86.3	86.0
35						88.5	87.0	86.7
40							87.7	87.4
45								88.1
46								

flex_n1_INT_zero_2000_qrh.ps

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
Zero Engine Bleed – Pressure Altitude 3000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 3000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	17	20	25	30	35	40	44
−40	82.3	82.3	82.3	82.3	82.3	82.3	82.3
−35	83.2	83.2	83.2	83.2	83.2	83.2	83.2
−30	84.0	84.0	84.0	84.0	84.0	84.0	84.0
−25	84.9	84.9	84.9	84.9	84.9	84.9	84.9
−20	85.7	85.0	85.0	85.0	85.0	85.0	85.0
−15	86.6	85.9	85.0	85.0	85.0	85.0	85.0
−10	87.4	86.7	85.6	85.0	85.0	85.0	85.0
−5	88.2	87.5	86.4	85.2	85.0	85.0	85.0
0	89.0	88.3	87.2	86.0	85.0	85.0	85.0
5	89.8	89.1	88.0	86.8	85.5	85.0	85.0
10	90.6	89.9	88.8	87.6	86.3	85.0	85.0
15	91.4	90.7	89.6	88.4	87.0	85.6	85.0
17		91.1	89.9	88.7	87.3	85.9	85.0
20			90.3	89.1	87.8	86.3	85.1
25				89.9	88.5	87.0	85.9
30					89.3	87.8	86.6
35						88.5	87.3
40							88.0
44							

flex_n1_INT_zero_3000_qrh.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
Zero Engine Bleed – Pressure Altitude 4000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 4000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	15	20	25	30	35	40	42
–40	82.7	82.7	82.7	82.7	82.7	82.7	82.7
–35	83.6	83.6	83.6	83.6	83.6	83.6	83.6
–30	84.5	84.5	84.5	84.5	84.5	84.5	84.5
–25	85.3	85.0	85.0	85.0	85.0	85.0	85.0
–20	86.2	85.0	85.0	85.0	85.0	85.0	85.0
–15	87.0	85.9	85.0	85.0	85.0	85.0	85.0
–10	87.8	86.7	85.6	85.0	85.0	85.0	85.0
–5	88.7	87.5	86.4	85.2	85.0	85.0	85.0
0	89.5	88.3	87.2	86.0	85.0	85.0	85.0
5	90.3	89.1	88.0	86.8	85.5	85.0	85.0
10	91.1	89.9	88.8	87.6	86.2	85.0	85.0
15		90.7	89.6	88.3	87.0	85.6	85.0
20			90.3	89.1	87.7	86.3	85.7
25				89.8	88.5	87.0	86.4
30					89.2	87.8	87.2
35						88.5	87.9
40							88.6
42							

flex_n1_INT_zero_4000_qrh.ps

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
Zero Engine Bleed – Pressure Altitude 6000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 6000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	11	15	20	25	30	35	38
-40	83.7	83.7	83.7	83.7	83.7	83.7	83.7
-35	84.6	84.6	84.6	84.6	84.6	84.6	84.6
-30	85.4	85.0	85.0	85.0	85.0	85.0	85.0
-25	86.3	85.3	85.0	85.0	85.0	85.0	85.0
-20	87.1	86.2	85.0	85.0	85.0	85.0	85.0
-15	88.0	87.0	85.9	85.0	85.0	85.0	85.0
-10	88.8	87.9	86.7	85.6	85.0	85.0	85.0
-5	89.7	88.7	87.5	86.4	85.1	85.0	85.0
0	90.5	89.5	88.3	87.2	85.9	85.0	85.0
5	91.3	90.3	89.1	88.0	86.7	85.3	85.0
10	92.2	91.1	89.9	88.7	87.5	86.0	85.2
11		91.3	90.1	88.9	87.6	86.2	85.3
15			90.7	89.5	88.2	86.8	85.9
20				90.3	89.0	87.6	86.6
25					89.7	88.3	87.4
30						89.0	88.1
35							88.8
38							

flex_n1_INT_zero_6000_qrh.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
Zero Engine Bleed – Pressure Altitude 8000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 8000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	7	10	15	20	25	30	34
–40	84.7	84.7	84.7	84.7	84.7	84.7	84.7
–35	85.6	85.0	85.0	85.0	85.0	85.0	85.0
–30	86.4	85.7	85.0	85.0	85.0	85.0	85.0
–25	87.3	86.6	85.3	85.0	85.0	85.0	85.0
–20	88.2	87.4	86.2	85.0	85.0	85.0	85.0
–15	89.0	88.3	87.0	85.8	85.0	85.0	85.0
–10	89.9	89.2	87.9	86.7	85.4	85.0	85.0
–5	90.7	90.0	88.7	87.5	86.3	85.0	85.0
0	91.6	90.8	89.5	88.3	87.1	85.7	85.0
5	92.4	91.7	90.3	89.1	87.8	86.5	85.3
7		92.0	90.7	89.4	88.2	86.8	85.7
10			91.1	89.9	88.6	87.3	86.1
15				90.7	89.4	88.0	86.9
20					90.2	88.8	87.6
25						89.5	88.3
30							89.1
34							

flex_n1_INT_zero_8000_qrh.ps

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
Zero Engine Bleed – Pressure Altitude 10000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING ZERO ENGINE BLEED PRESSURE ALTITUDE = 10000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)						
(°C)	3	5	10	15	20	25	30
–40	85.7	85.2	85.0	85.0	85.0	85.0	85.0
–35	86.6	86.1	85.0	85.0	85.0	85.0	85.0
–30	87.5	87.0	85.7	85.0	85.0	85.0	85.0
–25	88.4	87.9	86.6	85.3	85.0	85.0	85.0
–20	89.2	88.7	87.4	86.1	85.0	85.0	85.0
–15	90.1	89.6	88.3	87.0	85.7	85.0	85.0
–10	91.0	90.5	89.1	87.8	86.6	85.3	85.0
–5	91.8	91.3	90.0	88.7	87.4	86.1	85.0
0	92.7	92.2	90.8	89.5	88.2	86.9	85.5
3		92.7	91.3	90.0	88.7	87.4	86.0
5			91.6	90.3	89.0	87.7	86.3
10				91.1	89.8	88.4	87.0
15					90.6	89.2	87.8
20						90.0	88.5
25							89.3
30							

flex_n1_INT_zero_10000_qrh.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
One Pack On Per Engine – Pressure Altitude, –1000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = –1000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)							
(°C)	20	25	30	31	35	40	45	50
–40	80.7	80.7	80.7	80.7	80.7	80.7	80.7	80.7
–35	81.5	81.5	81.5	81.5	81.5	81.5	81.5	81.5
–30	82.3	82.3	82.3	82.3	82.3	82.3	82.3	82.3
–25	83.1	83.1	83.1	83.1	83.1	83.1	83.1	83.1
–20	83.9	83.9	83.9	83.9	83.9	83.9	83.9	83.9
–15	84.7	84.7	84.7	84.7	84.7	84.7	84.7	84.7
–10	85.5	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–5	86.3	85.8	85.0	85.0	85.0	85.0	85.0	85.0
0	87.1	86.6	85.4	85.0	85.0	85.0	85.0	85.0
5	87.9	87.4	86.2	85.7	85.0	85.0	85.0	85.0
10	88.7	88.1	86.9	86.4	85.6	85.0	85.0	85.0
15	89.5	88.9	87.7	87.2	86.3	85.0	85.0	85.0
20	90.3	89.7	88.4	88.0	87.1	85.5	85.0	85.0
20		89.8	88.6	88.1	87.2	85.6	85.0	85.0
25			89.2	88.7	87.8	86.2	85.0	85.0
30				89.4	88.5	86.9	85.4	85.0
31					88.8	87.2	85.7	85.0
35						87.7	86.1	85.0
40							86.8	85.3
45								86.0
50								

flex_n1_INT_acu_–1000_qrh.ps

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
One PACK On Per Engine – Pressure Altitude 0 ft (Sea Level)

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = SEA LEVEL	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	19	20	23	25	30	35	40	45	50
–40	81.1	81.1	81.1	81.1	81.1	81.1	81.1	81.1	81.1
–35	81.9	81.9	81.9	81.9	81.9	81.9	81.9	81.9	81.9
–30	82.7	82.7	82.7	82.7	82.7	82.7	82.7	82.7	82.7
–25	83.5	83.5	83.5	83.5	83.5	83.5	83.5	83.5	83.5
–20	84.3	84.3	84.3	84.3	84.3	84.3	84.3	84.3	84.3
–15	85.1	85.0	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–10	85.9	85.8	85.4	85.0	85.0	85.0	85.0	85.0	85.0
–5	86.8	86.6	86.2	85.8	85.0	85.0	85.0	85.0	85.0
0	87.6	87.4	87.0	86.5	85.4	85.0	85.0	85.0	85.0
5	88.4	88.2	87.8	87.3	86.2	85.0	85.0	85.0	85.0
10	89.2	89.0	88.6	88.1	86.9	85.6	85.0	85.0	85.0
15	90.0	89.8	89.3	88.9	87.7	86.3	85.0	85.0	85.0
19		90.4	90.0	89.5	88.3	86.9	85.3	85.0	85.0
20			90.1	89.7	88.5	87.1	85.4	85.0	85.0
23				90.1	88.9	87.5	85.9	85.0	85.0
25					89.2	87.8	86.2	85.0	85.0
30						88.6	86.9	85.4	85.0
35							87.6	86.1	85.0
40								86.8	85.3
45									86.0
50									

flex_n1_INT_acu_0_qrh.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
One PACK On Per Engine – Pressure Altitude 1000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 1000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	17	20	21	25	30	35	40	45	48
–40	81.5	81.5	81.5	81.5	81.5	81.5	81.5	81.5	81.5
–35	82.3	82.3	82.3	82.3	82.3	82.3	82.3	82.3	82.3
–30	83.1	83.1	83.1	83.1	83.1	83.1	83.1	83.1	83.1
–25	83.9	83.9	83.9	83.9	83.9	83.9	83.9	83.9	83.9
–20	84.8	84.8	84.8	84.8	84.8	84.8	84.8	84.8	84.8
–15	85.6	85.1	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–10	86.4	86.0	85.8	85.0	85.0	85.0	85.0	85.0	85.0
–5	87.2	86.8	86.6	85.7	85.0	85.0	85.0	85.0	85.0
0	88.0	87.6	87.4	86.5	85.3	85.0	85.0	85.0	85.0
5	88.8	88.4	88.2	87.3	86.1	85.0	85.0	85.0	85.0
10	89.6	89.2	89.0	88.1	86.9	85.5	85.0	85.0	85.0
15	90.4	89.9	89.8	88.9	87.6	86.3	85.0	85.0	85.0
17		90.3	90.1	89.2	87.9	86.6	85.0	85.0	85.0
20			90.6	89.6	88.4	87.0	85.5	85.0	85.0
21				89.8	88.5	87.2	85.6	85.0	85.0
25					89.1	87.7	86.2	85.0	85.0
30						88.5	86.9	85.4	85.0
35							87.6	86.1	85.2
40								86.8	85.9
45									86.6
48									

flex_n1_INT_acu_1000_qrh.ps

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
One PACK On Per Engine – Pressure Altitude 2000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 2000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	15	19	20	25	30	35	40	45	46
–40	81.9	81.9	81.9	81.9	81.9	81.9	81.9	81.9	81.9
–35	82.7	82.7	82.7	82.7	82.7	82.7	82.7	82.7	82.7
–30	83.5	83.5	83.5	83.5	83.5	83.5	83.5	83.5	83.5
–25	84.4	84.4	84.4	84.4	84.4	84.4	84.4	84.4	84.4
–20	85.2	85.0	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–15	86.0	85.4	85.2	85.0	85.0	85.0	85.0	85.0	85.0
–10	86.8	86.2	86.0	85.0	85.0	85.0	85.0	85.0	85.0
–5	87.6	87.0	86.8	85.7	85.0	85.0	85.0	85.0	85.0
0	88.5	87.9	87.6	86.5	85.3	85.0	85.0	85.0	85.0
5	89.3	88.7	88.4	87.3	86.1	85.0	85.0	85.0	85.0
10	90.1	89.5	89.2	88.1	86.9	85.6	85.0	85.0	85.0
15		90.2	90.0	88.8	87.6	86.3	85.0	85.0	85.0
19			90.6	89.5	88.2	86.9	85.3	85.0	85.0
20				89.6	88.4	87.1	85.5	85.0	85.0
25					89.1	87.8	86.2	85.0	85.0
30						88.5	86.9	85.4	85.1
35							87.6	86.1	85.8
40								86.8	86.5
45									87.2
46									

flex_n1_INT_acu_2000_qrh.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
One PACK On Per Engine – Pressure Altitude 3000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 3000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	13	15	17	20	25	30	35	40	44
–40	82.3	82.3	82.3	82.3	82.3	82.3	82.3	82.3	82.3
–35	83.1	83.1	83.1	83.1	83.1	83.1	83.1	83.1	83.1
–30	84.0	84.0	84.0	84.0	84.0	84.0	84.0	84.0	84.0
–25	84.8	84.8	84.8	84.8	84.8	84.8	84.8	84.8	84.8
–20	85.6	85.3	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–15	86.5	86.2	85.9	85.2	85.0	85.0	85.0	85.0	85.0
–10	87.3	87.0	86.7	86.0	85.0	85.0	85.0	85.0	85.0
–5	88.1	87.8	87.5	86.8	85.7	85.0	85.0	85.0	85.0
0	88.9	88.6	88.3	87.6	86.5	85.3	85.0	85.0	85.0
5	89.8	89.4	89.1	88.4	87.3	86.0	85.0	85.0	85.0
10	90.6	90.2	89.9	89.2	88.0	86.8	85.4	85.0	85.0
13		90.7	90.4	89.7	88.5	87.3	85.9	85.0	85.0
15			90.7	90.0	88.8	87.6	86.2	85.0	85.0
17				90.3	89.1	87.9	86.5	85.0	85.0
20					89.6	88.3	86.9	85.4	85.0
25						89.1	87.7	86.2	85.0
30							88.4	86.9	85.7
35								87.6	86.4
40									87.1
44									

flex_n1_INT_acu_3000_qrh.ps

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
One PACK On Per Engine – Pressure Altitude 4000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 4000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)							
(°C)	11	15	20	25	30	35	40	42
–40	82.7	82.7	82.7	82.7	82.7	82.7	82.7	82.7
–35	83.6	83.6	83.6	83.6	83.6	83.6	83.6	83.6
–30	84.4	84.4	84.4	84.4	84.4	84.4	84.4	84.4
–25	85.2	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–20	86.1	85.5	85.0	85.0	85.0	85.0	85.0	85.0
–15	86.9	86.3	85.2	85.0	85.0	85.0	85.0	85.0
–10	87.7	87.1	86.0	85.0	85.0	85.0	85.0	85.0
–5	88.6	88.0	86.8	85.6	85.0	85.0	85.0	85.0
0	89.4	88.8	87.6	86.4	85.2	85.0	85.0	85.0
5	90.2	89.6	88.4	87.2	86.0	85.0	85.0	85.0
10	91.1	90.4	89.2	88.0	86.7	85.4	85.0	85.0
11		90.6	89.4	88.2	86.9	85.5	85.0	85.0
15			90.0	88.8	87.5	86.1	85.0	85.0
20				89.5	88.2	86.9	85.4	85.0
25					89.0	87.6	86.1	85.5
30						88.3	86.8	86.2
35							87.6	86.9
40								87.6
42								

flex_n1_INT_acu_4000_qrh.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
One PACK On Per Engine – Pressure Altitude 6000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 6000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	7	10	11	15	20	25	30	35	38
–40	83.7	83.7	83.7	83.7	83.7	83.7	83.7	83.7	83.7
–35	84.5	84.5	84.5	84.5	84.5	84.5	84.5	84.5	84.5
–30	85.4	85.0	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–25	86.2	85.7	85.6	85.0	85.0	85.0	85.0	85.0	85.0
–20	87.1	86.6	86.4	85.5	85.0	85.0	85.0	85.0	85.0
–15	87.9	87.4	87.3	86.3	85.1	85.0	85.0	85.0	85.0
–10	88.8	88.3	88.1	87.1	85.9	85.0	85.0	85.0	85.0
–5	89.6	89.1	89.0	87.9	86.7	85.6	85.0	85.0	85.0
0	90.5	89.9	89.8	88.8	87.5	86.4	85.0	85.0	85.0
5	91.3	90.8	90.6	89.6	88.3	87.2	85.8	85.0	85.0
7		91.1	90.9	89.9	88.7	87.5	86.1	85.0	85.0
10			91.4	90.4	89.1	87.9	86.6	85.1	85.0
11				90.5	89.3	88.1	86.7	85.3	85.0
15					89.9	88.7	87.4	85.9	85.0
20						89.5	88.1	86.6	85.7
25							88.9	87.3	86.4
30								88.1	87.1
35									87.9
38									

flex_n1_INT_acu_6000_qrh.ps

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
One PACK On Per Engine – Pressure Altitude 8000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 8000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	3	5	7	10	15	20	25	30	34
–40	84.7	84.7	84.7	84.7	84.7	84.7	84.7	84.7	84.7
–35	85.5	85.1	85.0	85.0	85.0	85.0	85.0	85.0	85.0
–30	86.4	86.0	85.7	85.0	85.0	85.0	85.0	85.0	85.0
–25	87.2	86.9	86.5	85.8	85.0	85.0	85.0	85.0	85.0
–20	88.1	87.7	87.4	86.7	85.4	85.0	85.0	85.0	85.0
–15	89.0	88.6	88.3	87.5	86.3	85.0	85.0	85.0	85.0
–10	89.8	89.4	89.1	88.4	87.1	85.9	85.0	85.0	85.0
–5	90.7	90.3	90.0	89.2	87.9	86.7	85.4	85.0	85.0
0	91.5	91.1	90.8	90.0	88.7	87.5	86.2	85.0	85.0
3		91.6	91.3	90.5	89.2	88.0	86.7	85.2	85.0
5			91.6	90.8	89.5	88.3	87.0	85.6	85.0
7				91.2	89.9	88.6	87.3	85.9	85.0
10					90.3	89.1	87.8	86.3	85.2
15						89.9	88.5	87.1	85.9
20							89.3	87.8	86.7
25								88.6	87.4
30									88.2
34									

flex_n1_INT_acu_8000_qrh.ps

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Reduced Thrust Take-off Setting, %N₁ (Two Engine Operation),
One PACK On Per Engine – Pressure Altitude 10000 ft

DISPLAYED FAN SPEED	
INTERMIX (Two Engine Operation) REDUCED THRUST TAKEOFF SETTING 10th STAGE BLEEDS ON – PACKS ON PRESSURE ALTITUDE = 10000 FT	%N1

OAT	ASSUMED TEMPERATURE (°C)								
(°C)	-2	0	3	5	10	15	20	25	30
-40	85.7	85.3	85.0	85.0	85.0	85.0	85.0	85.0	85.0
-35	86.6	86.2	85.7	85.3	85.0	85.0	85.0	85.0	85.0
-30	87.4	87.1	86.6	86.2	85.0	85.0	85.0	85.0	85.0
-25	88.3	88.0	87.5	87.0	85.8	85.0	85.0	85.0	85.0
-20	89.2	88.9	88.4	87.9	86.6	85.4	85.0	85.0	85.0
-15	90.1	89.8	89.3	88.8	87.5	86.2	85.0	85.0	85.0
-10	90.9	90.6	90.1	89.6	88.3	87.0	85.7	85.0	85.0
-5	91.8	91.5	91.0	90.5	89.2	87.9	86.5	85.1	85.0
-2		92.0	91.5	91.0	89.6	88.3	87.0	85.6	85.0
0			91.8	91.3	90.0	88.7	87.3	85.9	85.0
3				91.8	90.5	89.2	87.8	86.4	85.0
5					90.8	89.5	88.1	86.7	85.3
10						90.3	88.9	87.5	86.0
15							89.7	88.3	86.8
20								89.0	87.5
25									88.3
30									

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
-----------------	-------------

EICAS MESSAGE LIST

EICAS Message List	EICAS MESSAGE LIST-1
General Information	EICAS MESSAGE LIST-1
Warning Messages (Red)	EICAS MESSAGE LIST-1
Caution Messages (Amber)	EICAS MESSAGE LIST-3
Advisory Messages (Green)	EICAS MESSAGE LIST-10
Status Messages (White)	EICAS MESSAGE LIST-11

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

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CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

EICAS MESSAGE LIST

General Information

- This EICAS Message List that follows provides a quick cross-reference between EICAS messages and MMEL items which may provide dispatch relief. This list in no way supersedes data outlined in the MMEL. In case of conflict, the MMEL takes precedence.
- Joint Aviation Authorities Approved CRJ MMEL Revision 20 <JAA>

NOTE

This list shall be used for guidance purposes only, and does not cover all possible combinations of system and/or components failures that may cause an EICAS message. If MMEL relief is referenced for a given EICAS message, confirm that the failure that caused the EICAS message is applicable to the referenced MMEL item(s).

- There are three possible recommendations given in the EICAS Message List as follows:
- No Dispatch: No dispatch is permitted until maintenance action has corrected the fault.
 - XX-XX-XX: This item (e.g. 22-10-01) should be consulted when the associated EICAS message is displayed.

NOTE

- An EICAS message that may have more than one possible equipment failure will have multiple MMEL/DDG items listed in this column.
- N/A: Not applicable to the MMEL. This message may be caused by a system response to a flight crew action or the result of a normal/abnormal system configuration.

After performing the applicable QRH procedure and the message persists, check dispatchability as follows:

Warning Messages (Red)

Warning Messages (Red)	
EICAS Message	Dispatch and / or MEL Ref
AFCS MSG FAIL	No Dispatch

CONTENTS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



EICAS MESSAGE LIST-2

Rev. 64, Apr 22/2016

Warning Messages (Red)

EICAS Message	Dispatch and / or MEL Ref
ANTI-ICE DUCT	No Dispatch
APU FIRE	No Dispatch
APU OVERSPEED	49-10-01
APU OVERTEMP	49-10-01
BRAKE OVHT	No Dispatch
CABIN ALT	No Dispatch
CONFIG AILERON	No Dispatch
CONFIG AP	No Dispatch
CONFIG FLAPS	No Dispatch
CONFIG RUDDER	No Dispatch
CONFIG SPLRS	No Dispatch
CONFIG STAB	No Dispatch
DIFF PRESS	No Dispatch
EMER PWR ONLY	No Dispatch
ENGINE OVERSPD <0005> and <0039>	No Dispatch
GEAR DISAGREE	32-30-01 32-50-01
L ENG FIRE	No Dispatch
L ENG OIL PRESS	No Dispatch
L JETPIPE OVHT	No Dispatch
L 10TH DUCT	No Dispatch
L 14TH DUCT	No Dispatch
MLG BAY OVHT	26-14-01
NOSE DOOR OPEN	No Dispatch
PARKING BRAKE	No Dispatch
PASSENGER DOOR	No Dispatch
R ENG FIRE	No Dispatch
R ENG OIL PRESS	No Dispatch
R JETPIPE OVHT	No Dispatch
R 10TH DUCT	No Dispatch
R 14TH DUCT	No Dispatch

Warning Messages (Red)	
EICAS Message	Dispatch and / or MEL Ref
SMOKE CARGO <0013> or <0034> or <0043> or <0053> or <0059> or <0074>	26-15-01
WING OVHT	30-12-01 30-12-03

Caution Messages (Amber)

Caution Messages (Amber)	
EICAS Message	Dispatch and / or MEL Ref
A/SKID INBD	32-44-01
A/SKID OUTBD	32-44-01
AC 1 AUTOXFER	No Dispatch
AC 2 AUTOXFER	No Dispatch
AC BUS 1	No Dispatch
AC BUS 2	No Dispatch
AC ESS BUS	No Dispatch
AC SERV BUS	24-50-01
AP PITCH TRIM	22-10-01
AP TRIM IS LWD	22-10-01
AP TRIM IS ND	22-10-01
AP TRIM IS NU	22-10-01
AP TRIM IS RWD	22-10-01
APR CMD SET	No Dispatch
APR INOP	76-13-01
APU BATT OFF	24-32-03
APU BLEED ON	N/A
APU BTL LO	26-22-01
APU FAULT	49-10-01
APU FIRE FAIL	26-12-01
APU GEN OFF	24-22-01
APU GEN OVLD	N/A
APU LCV FAIL	49-51-01
APU OIL PRESS	49-10-01

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



EICAS MESSAGE LIST-4

Rev. 65, Jan 13/2017

Caution Messages (Amber)

EICAS Message	Dispatch and / or MEL Ref
APU OIL TEMP	49-10-01
APU SOV FAIL	28-13-01
APU SOV OPEN	N/A
ARINC COOL	21-24-01
AUTO PRESS	21-31-01 21-32-01
AV BAY DOOR	52-70-02
BATTERY BUS	No Dispatch
BLEED MISCONFIG	N/A
BULK FUEL TEMP	N/A
CABIN ALT	No Dispatch
CARGO BTL LO	26-25-01
CARGO DOOR	52-70-04
CARGO OVHT	21-55-01 21-55-05
DC BUS 1	No Dispatch
DC BUS 2	No Dispatch
DC EMER BUS	No Dispatch
DC ESS BUS	No Dispatch
DC SERV BUS	No Dispatch
DISPLAY COOL	21-24-02
EFIS COMP INOP	No Dispatch
EFIS COMP MON	No Dispatch
EICAS COMP INOP	31-61-01
ELT ON	No Dispatch
EMER DEPRESS	N/A
EMER LTS OFF	No Dispatch
ENG BTL 1 LO	No Dispatch
ENG BTL 2 LO	No Dispatch
ESS TRU 1	No Dispatch
ESS TRU 2	24-31-01
FLAPS FAIL	No Dispatch
FLT SPLR DEPLOY	No Dispatch



EICAS MESSAGE LIST-5

Rev. 65, Jan 13/2017

Caution Messages (Amber)

EICAS Message	Dispatch and / or MEL Ref
FLT SPLRS	No Dispatch
FUEL CH 1/2 FAIL	No Dispatch
FUEL IMBALANCE	N/A
GEN 1 OFF	24-11-01
GEN 1 OVLD	N/A
GEN 2 OFF	24-11-01
GEN 2 OVLD	N/A
GLD NOT ARMED	N/A
GLD UNSAFE	No Dispatch
GND SPLR DEPLOY	27-65-01
HYD 1 HI TEMP	No Dispatch
HYD 1 LO PRESS	No Dispatch
HYD 2 HI TEMP	No Dispatch
HYD 2 LO PRESS	No Dispatch
HYD 3 HI TEMP	No Dispatch
HYD 3 LO PRESS	No Dispatch
HYD EDP 1A	29-11-01 29-34-01
HYD EDP 2A	29-11-01 29-34-01
HYD PUMP 1B	29-11-02 29-34-01
HYD PUMP 2B	29-11-02 29-34-01
HYD PUMP 3A	29-12-01 29-34-01
HYD PUMP 3B	29-34-01
HYD SOV 1 OPEN	N/A
HYD SOV 2 OPEN	N/A
IB BRAKE PRESS	No Dispatch
IB GND SPLRS	27-65-01
ICE <0039>	No Dispatch
ICE DET FAIL	30-81-01
IDG 1	24-11-01



EICAS MESSAGE LIST-6

Rev. 65, Jan 13/2017

Caution Messages (Amber)

EICAS Message	Dispatch and / or MEL Ref
IDG 2	24-11-01
L AOA HEAT	30-31-01 30-31-02
L COWL A/ICE	30-22-01 36-12-01
L EMER DOOR	52-21-01 52-70-03
L ENG MISCOMP	No Dispatch
L ENG SOV CLSD	N/A
L ENG SOV FAIL	No Dispatch
L ENG SOV OPEN	N/A
L FIRE FAIL	26-11-01
L FLT SPLR	No Dispatch
L FUEL FILTER	No Dispatch
L FUEL LO PRESS	73-31-03
L FUEL LO TEMP	N/A
L FUEL PUMP	28-23-01
L JET OVHT FAIL	26-11-02
L MAIN EJECTOR	No Dispatch
L PACK HI PRESS	21-51-01
L PACK HI TEMP	21-51-01
L PITOT HEAT	30-31-01 30-31-02
L REV UNLOCKED	78-30-01
L REV UNSAFE	78-30-01
L SCAV EJECTOR	No Dispatch
L SPOILERON	No Dispatch
L STATIC HEAT	30-31-01 30-31-02
L WINDOW HEAT	No Dispatch

Effectivity:

Airplanes 8102 and subsequent, or

Airplanes 7003 thru 7067, 7069 thru 7990 and 8000 thru 8101 **incorporating** the following Service Bulletin:

SB 601R-30-031, Ice and Rain Protection – Wing Anti-icing System – Installation of Low Heat Detection – Switches on the Leading Edge.

Cautions Messages (Amber)

EICAS Message

Dispatch and / or MEL Ref

L WING A/ICE

30-12-01
30-12-03
36-12-01
36-12-08

Cautions Messages (Amber)

EICAS Message

Dispatch and / or MEL Ref

L WSHLD HEAT

30-41-01

L XFER SOV

28-13-08

L XFLOW SOV

28-13-06

MACH TRIM

22-21-01

MAIN BATT OFF

24-32-01

MLG OVHT FAIL

26-14-01

NO STARTER CUTOUT

80-10-03

OB BRAKE PRESS

No Dispatch

OB GND SPLRS

27-65-01

OVBD COOL

21-24-06

OXY LO PRESS <TC> or <JAA>

35-12-01

PARK BRAKE SOV

No Dispatch

PASS OXY ON

N/A

PAX DR LATCH

52-21-01
52-70-01

PAX DR OUT HNDL

52-21-01
52-70-01

PAX DR STOW

52-21-01
52-70-01

PITOT BASE HEAT

30-31-01
30-31-02

PROX SYS CHAN

No Dispatch

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



EICAS MESSAGE LIST-8

Rev. 62, Feb 13/2015

Caution Messages (Amber)

EICAS Message	Dispatch and / or MEL Ref
R AOA HEAT	30-31-01 30-31-02
R COWL A/ICE	30-22-01 36-12-01
R EMER DOOR	52-21-01 52-70-03
R ENG MISCOMP	No Dispatch
R ENG SOV CLSD	N/A
R ENG SOV FAIL	No Dispatch
R ENG SOV OPEN	N/A
R FIRE FAIL	26-11-01
R FLT SPLR	No Dispatch
R FUEL FILTER	No Dispatch
R FUEL LO PRESS	73-31-03
R FUEL LO TEMP	N/A
R FUEL PUMP	28-23-01
R JET OVHT FAIL	26-11-02
R MAIN EJECTOR	No Dispatch
R PACK HI PRESS	21-51-01
R PACK HI TEMP	21-51-01
R PITOT HEAT	30-31-01 30-31-02
R REV UNLOCKED	78-30-01
R REV UNSAFE	78-30-01
R SCAV EJECTOR	No Dispatch
R SPOILERON	No Dispatch
R STATIC HEAT	30-31-01 30-31-02
R WINDOW HEAT	30-41-01



EICAS MESSAGE LIST-9

Rev. 64, Apr 22/2016

Effectivity:

- Airplanes 8102 and subsequent, or
- Airplanes 7003 thru 7067, 7069 thru 7990 and 8000 thru 8101 **incorporating** the following Service Bulletin:
 - SB 601R-30-031, Ice and Rain Protection – Wing Anti-icing System – Installation of Low Heat Detection – Switches on the Leading Edge.

Caution Messages (Amber)

EICAS Message	Dispatch and / or MEL Ref
R WING A/ICE	30-12-01 36-12-01 30-12-03 36-12-08

Caution Messages (Amber)

EICAS Message	Dispatch and / or MEL Ref
R WSHLD HEAT	30-41-01
R XFER SOV	28-13-08
R XFLOW SOV	28-13-06
SERVICE DOOR	52-21-01 52-70-05
SMOKE TOILET	26-16-01
SPOILERONS	No Dispatch
SPOILERONS ROLL	No Dispatch
STAB TRIM	No Dispatch
STALL FAIL	No Dispatch
STBY PITOT HEAT	30-31-01 30-31-02
STEERING INOP	32-50-01
TAT PROBE HEAT	30-31-01
WING A/ICE SNSR	30-12-03
WOW INPUT	No Dispatch
WOW OUTPUT	No Dispatch
XFLOW/APU PUMP	28-24-01
YAW DAMPER	No Dispatch

Advisory Messages (Green)

Advisory Messages (Green)	
EICAS Message	Dispatch and / or MEL Ref
APR ARM	N/A
APR TEST 1 OK	N/A
APR TEST 2 OK	N/A
APR TEST 3 OK <0019>	N/A
APU SOV CLSD	N/A
APU SQUIB 1	N/A
APU SQUIB 2	N/A
CARGO SQUIB 1	N/A
CARGO SQUIB 2	N/A
COWL A/I ON <0039>	N/A
CPLT ROLL CMD	N/A
DUCT TEST OK	N/A
FDR EVENT	N/A
FLT SPLR DEPLOY	N/A
GLD MAN ARM	N/A
GND SPLR DEPLOY	N/A
GRAV XFLOW OPEN	N/A
HYD SOV 1 CLSD	N/A
HYD SOV 2 CLSD	N/A
ICE <0039>	N/A
IGNITION A <0039>	N/A
IGNITION B <0039>	N/A
L COWL A/ICE ON <0039>	N/A
L ENG SOV CLSD	N/A
L ENG SQUIB 1	N/A
L ENG SQUIB 2	N/A
L FUEL PUMP ON	N/A
L REV ARMED	N/A
PARKING BRAKE ON	N/A
PLT ROLL CMD	N/A
R COWL A/ICE ON <0039>	N/A

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Advisory Messages (Green)	
EICAS Message	Dispatch and / or MEL Ref
R ENG SOV CLSD	N/A
R ENG SQUIB 1	N/A
R ENG SQUIB 2	N/A
R FUEL PUMP ON	N/A
R REV ARMED	N/A
SELCAL HF <0010> and <0011>	N/A
SELCAL VHF 1 <0010>	N/A
SELCAL VHF 2 <0010>	N/A
SELCAL VHF 3 <0008> or <0014> and <0010>	N/A
T/O CONFIG OK	N/A
WING A/ICE OK	N/A
WING A/ICE ON <0039>	N/A
WING/COWL A/I ON <0039>	N/A

Status Messages (White)

Status Messages (White)	
EICAS Message	Dispatch and / or MEL Ref
10TH ISOL OPEN	N/A
14TH ISOL OPEN	N/A
AC 1 AUTOXFER OFF	N/A
AC 2 AUTOXFER OFF	N/A
AC ESS ALTN	N/A
AC UTIL 1 OFF	No Dispatch
AC UTIL 2 OFF	No Dispatch
APU BATT CHGR <0039>	24-32-04
APU ECU FAIL	49-61-01
APU IN BITE	N/A
APU LCV OPEN	N/A
APU SOV OPEN	N/A
APU START	N/A
AUTO PRESS 1 FAIL	21-31-01

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



EICAS MESSAGE LIST-12

Rev. 62, Feb 13/2015

Status Messages (White)

EICAS Message	Dispatch and / or MEL Ref
AUTO PRESS 2 FAIL	21-31-01
AUTO XFLOW INHIB	28-13-06 28-24-01 28-41-01
BTMU FAIL	32-46-01
CABIN PRESS MAN	N/A
CABIN TEMP MAN	N/A
CARGO FAN FAIL	21-55-02
CARGO SOV FAIL	21-55-01 21-55-03 21-55-04
CAS MISCOMP	No Dispatch
CKPT COOL FAIL	21-24-04
CKPT TEMP MAN	N/A
CONT IGNITION	N/A
COOL EXHAUST FAIL	21-24-03
CPAM FAIL	21-33-01
DC ESS TIE CLSD	N/A
DC TIE 1 CLSD	N/A
DC TIE 2 CLSD	N/A
DCU 1 APR FAIL	76-13-01
DCU 1 AURAL INOP	31-41-01
DCU 1 INOP	31-41-01
DCU 2 APR FAIL	76-13-01
DCU 2 AURAL INOP	31-41-01
DCU 2 INOP	31-41-01
DCU 3 APR FAIL <0019>	76-13-01
DCU 3 AURAL INOP <0019>	31-41-01
DCU 3 INOP <0019>	31-41-01
DUCT MON LOOP A	N/A
DUCT MON LOOP B	N/A
EMER LTS ON	N/A
ENG TYPE MISCOMP	No Dispatch
FD 1 FAIL	22-10-02

**QUICK REFERENCE
HANDBOOK
CSP A-022-080**

EICAS MESSAGE LIST

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Status Messages (White)	
EICAS Message	Dispatch and / or MEL Ref
FD 2 FAIL	22-10-02
FDR ACCEL FAIL	31-31-01
FDR FAIL	31-31-01

Effectivity:

Airplanes 7901 and subsequent; and

Airplanes 7003 to 7067, 7069 to 7900 **incorporating** the following Service Bulletin:

SB 601R-27-116, Flight Controls – Flaps Skew Detection System (SDS) – Installation and Activation of the Skew Detection System.

Status Messages (White)	
EICAS Message	Dispatch and / or MEL Ref
FLAPS DEGRADED	27-51-03

Effectivity:

Airplanes 7003 to 7067, 7069 to 7900 **not incorporating** the following Service Bulletin:

SB 601R-27-116, Flight Controls – Flaps Skew Detection System (SDS) – Installation and Activation of the Skew Detection System.

Status Messages (White)	
EICAS Message	Dispatch and / or MEL Ref
FLAPS HALFSPEED	27-51-01

Effectivity:

Airplanes 7901 and subsequent; and

Airplanes 7003 to 7067, 7069 to 7900 **incorporating** the following Service Bulletin:

SB 601R-27-116, Flight Controls – Flaps Skew Detection System (SDS) – Installation and Activation of the Skew Detection System.

Status Messages (White)	
EICAS Message	Dispatch and / or MEL Ref
FLAPS HALFSPEED	27-51-01 27-51-03



EICAS MESSAGE LIST-14

Rev. 62, Feb 13/2015

Status Messages (White)

EICAS Message	Dispatch and / or MEL Ref
FLT SPLRS FAULT	27-62-01 27-62-02
FLUTTER DAMPER <0039>	27-15-02
FUEL CH 1 FAIL	28-41-03
FUEL CH 2 FAIL	28-41-03
GLD MAN DISARM	N/A
GPWS FAIL	34-42-01
GRAV XFLOW FAIL	No Dispatch
GS CANCEL	N/A
HGS FAIL	34-32-01
HORN MUTED	N/A
IAPS DEGRADED	22-12-02
IAPS OVERTEMP	No Dispatch
IB GND SPLR FAULT	27-62-01 27-65-01
ICE DET 1 FAIL	30-81-01
ICE DET 2 FAIL	30-81-01
IDG 1 DISC	24-11-01
IDG 2 DISC	24-11-01
IGNITION A/B	N/A
INBD COOL FAIL	21-24-05
IRS 1 DC FAIL	No Dispatch
IRS 1 IN ATT	N/A
IRS 1 ON BATT	N/A
IRS 1 OVERTEMP	No Dispatch
IRS 2 DC FAIL	No Dispatch
IRS 2 IN ATT	N/A
IRS 2 ON BATT	N/A
IRS 2 OVERTEMP	No Dispatch
L 10TH ARM OPEN	N/A
L 10TH SOV CLSD	N/A
L 14TH ARM CLSD	N/A
L 14TH SOV CLSD	N/A

**QUICK REFERENCE
HANDBOOK
CSP A-022-080**

EICAS MESSAGE LIST

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Status Messages (White)	
EICAS Message	Dispatch and / or MEL Ref
L APR ECU FAIL	76-13-01
L AUTO XFLOW ON	N/A
L ENGINE START	N/A
L PACK OFF	N/A
L XFLOW ON	N/A
MAIN BATT CHGR	24-32-02
MAN XFLOW	N/A
NO SMOKING	N/A
OB GND SPLR FAULT	27-62-01 27-65-01
OVBD COOL FAIL	21-24-06
PROX SYS FAULT	No Dispatch
R 10TH ARM OPEN	N/A
R 10TH SOV CLSD	N/A
R 14TH ARM CLSD	N/A
R 14TH SOV CLSD	N/A
R APR ECU FAIL	76-13-01
R AUTO XFLOW ON	N/A
R ENGINE START	N/A
R PACK OFF	N/A
R XFLOW ON	N/A
RAM AIR OPEN	N/A
SEAT BELTS	N/A
SPEED REFS INDEP	22-12-01
SPOILERONS FAULT	27-62-01 27-62-03
STAB CH 1 INOP	No Dispatch
STAB CH 2 INOP	No Dispatch
TERRAIN FAIL	34-42-01
TERRAIN OFF	N/A
TERRAIN NOT AVAIL	34-42-01
WINDSHEAR FAIL	34-42-01
WOW OUTPUT FAIL	No Dispatch

CONTENTS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	EICAS MESSAGE LIST-16
	Rev. 62, Feb 13/2015

Status Messages (White)	
EICAS Message	Dispatch and / or MEL Ref
YD 1 INOP	No Dispatch
YD 2 INOP	No Dispatch

QUICK REFERENCE HANDBOOK CSP A-022-080	EICAS MESSAGE LIST
---	---------------------------

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

This is revision **77** to the **A/C 7486 Quick Reference Handbook**
Vol. . 2 , CSPA-022-PRS

To bring this manual up to date, remove old pages and insert revised pages as follows:

REMOVE:		INSERT:	
Chapter Section Subject	Page	Chapter Section Subject	Page
Title Page Disclaimer	All	Title Page Disclaimer	All
00-02	00-02-3	00-02	00-02-3
		00-04	00-04-1 to 00-04-6
Abnormal Procedures	ABNORM 1-2	Abnormal Procedures	ABNORM 1-2
	ABNORM13-11		ABNORM13-11
Appendices	INDEX-8	Appendices	INDEX-8
	INDEX-13		INDEX-13

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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Model CL-600-2B19

QUICK REFERENCE HANDBOOK

Volume 2

CSP A-022-PRS

A/C 7486

This handbook contains suggested procedures only for quick reference. These procedures in no way supersede current procedures outlined in the approved Airplane Flight Manual and any revisions thereto. In case of conflict, the Airplane Flight Manual takes precedence.

CRJ SERIES TECHNICAL PUBLICATIONS

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Rev No.	Description of Revision	Incorp By	Date
REV 59	Introduces changes to the QRH to reflect Airplane Flight Manual Revision 59. • Harmonization to the other programs, Manual Changes Request (MCRs) and incorporation of all previous Temporary Revisions (TRs). • Relocation of historical data for Record of Revisions (i.e. covering REV 1 thru REV 58) and Record of Temporary Revisions (i.e. covering QRH TR RJ/1 thru TR RJ/167) to the Bombardier Aerospace online website.	BCSG	21 Mar 2012
REV 60	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 60.	BCSG	30 Nov 2012
REV 61	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 61.	BCSG	02 Apr 2013
REV 62	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 62.	BCSG	01 Oct 2014
REV 63	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 63.	BCSG	13 Feb 2015
REV 64	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 64.	BCSG	24 Mar 2015
REV 65	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 65.	BCSG	11 Jan 2016

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Rev No.	Description of Revision	Incorp By	Date
REV 66	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 66.	BCSG	22 Apr 2016
REV 67	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 67.	BCSG	13 Jan 2017
REV 67A	Not Applicable		
REV 67B	Introduces changes to Air Nostrum QRH Volume 2, to reflect Airplane Flight Manual Revision 67B.	BCSG	12 Jun 2017
REV 68	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 67.	BCSG	04 Aug 2017
REV 69	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 69.	BCSG	05 Jan 2018
REV 70	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 70.	BCSG	13 Jul 2018
REV 71	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 71.	BCSG	04 Jan 2019
REV 72	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 72.	BCSG	12 Jul 2019
REV 73	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 73.	BCSG	03 Jan 2020
REV 73A	Added tail number 07466 due to customer's recent purchase.	BCSG	26 May 2020
REV 74	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 74.	MHIRJ	03 Jul 2020

- SUPP
- LIMITS
- PERF
- MISC
- PLAN
- INTERMIX
- EICAS
- EMER
- IMM
- ABNORM
- INDEX

Rev No.	Description of Revision	Incorp By	Date
REV 74A	Removal of the MSNs 07622, 07915 and 07975 from their fleet. The MSNs that remain in the fleet are displayed on the title page as per customer's request.	MHIRJ	26 Oct 2020
REV 75	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 75.	MHIRJ	04 Jan 2021
REV 76	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 76.	MHIRJ	02 Jul 2021
REV 77	Introduces changes to the QRH Volume 2 to reflect Airplane Flight Manual Revision 77.	MHIRJ	03 Jan 2022

[SUPP](#)[LIMITS](#)[PERF](#)[MISC](#)[PLAN](#)[INTERMIX](#)[EICAS](#)[EMER](#)[IMM](#)[ABNORM](#)[INDEX](#)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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Record the date you insert each temporary revision in your manual.

Revision 59 is a major revision that incorporates all outstanding temporary revisions. A complete listing of all issued temporary revisions may be obtained from MHIRJ Customer Service.

TR No.	Subject	Incorp. By	Date

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

THIS PAGE INTENTIONALLY LEFT BLANK



Page	Date	Page	Date
Title Page / All*	Jan 03/2022	-9	Jul 02/2021
Disclaimer		-10	Jul 03/2020
		-11	Jul 03/2020
00-02 -1	Jan 13/2017	-12	Jan 04/2019
-2	Jan 04/2021	-13	Jan 04/2019
-3 *	Jan 03/2022	-14	Jan 04/2019
-4	Oct 26/2020	-15	Jan 04/2019
		-16	Jan 04/2019
00-03 -1	Jul 03/2020	-17	Jan 04/2019
-2	Mar 21/2012	-18	Jul 13/2018
00-04 -1 *	Jan 03/2022	EMER 2 -1	Jul 03/2020
-2 *	Jan 03/2022	-2	Oct 01/2014
-3 *	Jan 03/2022	-3	Jan 13/2017
-4 *	Jan 03/2022	-4	Feb 13/2015
-5 *	Jan 03/2022	-5	Jul 13/2018
-6 *	Jan 03/2022	-6	Jul 13/2018
		-7	Jul 13/2018
		-8	Jul 13/2018
		-9	Jul 13/2018
EMER TOC-1	Mar 21/2012	-10	Jul 13/2018
-2	Mar 24/2015	-11	Jun 12/2017
-3	Feb 13/2015	-12	Jun 12/2017
-4	Feb 13/2015	-13	Jun 12/2017
		-14	Jul 03/2020
IMM -1	Jul 03/2020	-15	Jul 03/2020
ACTION		-16	Jul 03/2020
ITEMS			
-2	Jan 04/2021		
-3	Jan 04/2021	EMER 3 -1	Mar 21/2012
-4	Jul 03/2020	-2	Mar 21/2012
-5	Jul 03/2020	-3	Jul 03/2020
-6	Jul 03/2020	-4	Mar 24/2015
-7	Jul 03/2020		
-8	Jul 03/2020	EMER 4 -1	Oct 01/2014
-9	Jan 04/2021	-2	Oct 01/2014
-10	Jan 04/2021	-3	Oct 01/2014
-11	Jan 04/2021	-4	Apr 02/2013
-12	Jul 03/2020		
EMER 1 -1	Jan 04/2019	EMER 5 -1	Mar 21/2012
-2	Jul 03/2020	-2	Mar 21/2012
-3	Jul 13/2018	-3	Mar 21/2012
-4	Aug 04/2017	-4	Mar 21/2012
-5	Jan 04/2019		
-6	Jan 04/2019	EMER 6 -1	Mar 21/2012
-7	Jul 02/2021	-2	Oct 01/2014
-8	Jul 02/2021	-3	Oct 01/2014

* The asterisk indicates pages changed, added or deleted.



Page	Date	Page	Date
-4	Oct 01/2014	EMER 13-1	Jul 02/2021
-5	Oct 01/2014	-2	Nov 30/2012
-6	Mar 21/2012	-3	Jan 04/2019
		-4	Jan 04/2019
EMER 7 -1	Mar 21/2012	-5	Jan 04/2019
-2	Mar 21/2012	-6	Jul 02/2021
-3	Oct 01/2014	-7	Jul 02/2021
-4	Mar 21/2012	-8	Jul 02/2021
-5	Oct 01/2014	-9	Jul 02/2021
-6	Oct 01/2014	-10	Jul 02/2021
		-11	Jul 02/2021
		-12	Jul 02/2021
EMER 8 -1	Mar 21/2012	EMER 14-1	Mar 21/2012
-2	Jul 03/2020	-2	Oct 01/2014
EMER 9 -1	Jun 12/2017	EMER 15-1	Mar 21/2012
-2	Jul 13/2018	-2	Mar 21/2012
-3	Jul 13/2018		
-4	Jul 13/2018	EMER 16-1	Mar 24/2015
-5	Jul 13/2018	-2	Mar 24/2015
-6	Aug 04/2017		
-7	Jul 13/2018	EMER 17-1	Feb 13/2015
-8	Jul 13/2018	-2	Jul 13/2018
-9	Jul 03/2020	-3	Aug 04/2017
-10	Jul 13/2018	-4	Aug 04/2017
EMER 10-1	Oct 01/2014	-5	Aug 04/2017
-2	Nov 30/2012	-6	Jul 13/2018
-3	Jul 13/2018		
-4	Nov 30/2012	-8	Jul 13/2018
-5	Nov 30/2012	-9	Jul 13/2018
-6	Oct 01/2014	-10	Jul 13/2018
-7	Jul 13/2018		
-8	Oct 01/2014	-12	Jul 13/2018
-9	Oct 01/2014		
-10	Oct 01/2014	ABNORM -1	Mar 21/2012
EMER 11-1	Mar 21/2012	TOC	
-2	Mar 21/2012	-2	Mar 21/2012
EMER 12-1	Oct 01/2014	ABNORM 1-1	Jul 13/2018
-2	Oct 01/2014	-2 *	Jan 03/2022
-3	Oct 01/2014	-3	Jul 13/2018
-4	Oct 01/2014	-4	Aug 04/2017
-5	Oct 01/2014	-5	Jul 13/2018
-6	Oct 01/2014		

* The asterisk indicates pages changed, added or deleted.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Page	Date	Page	Date
-6	Jul 13/2018	ABNORM 4-1	Mar 21/2012
-7	Jul 13/2018	-2	Mar 21/2012
-8	Jul 13/2018	-3	Mar 21/2012
-9	Jul 13/2018	-4	Mar 21/2012
-10	Jul 13/2018	-5	Mar 21/2012
-11	Jul 13/2018	-6	Mar 21/2012
-12	Jul 13/2018		
-13	Jul 13/2018	ABNORM 5-1	Mar 21/2012
-14	Jul 13/2018	-2	Mar 21/2012
-15	Jul 13/2018	-3	Mar 21/2012
-16	Jul 13/2018	-4	Mar 21/2012
-17	Jul 13/2018	-5	Mar 21/2012
-18	Jul 13/2018	-6	Mar 21/2012
-19	Jan 04/2019	-7	Mar 21/2012
-20	Jul 13/2018	-8	Mar 21/2012
-21	* Jan 03/2022	-9	Mar 21/2012
-22	* Jan 03/2022	-10	Mar 21/2012
-23	* Jan 03/2022		
-24	* Jan 03/2022	ABNORM 6-1	Jul 13/2018
-25	* Jan 03/2022	-2	Jul 13/2018
-26	* Jan 03/2022	-3	Oct 26/2020
		-4	Jul 13/2018
		-5	Oct 26/2020
ABNORM 2-1	Jan 13/2017	-6	Jul 13/2018
-2	Jan 03/2020	-7	Jul 13/2018
-3	Jan 03/2020	-8	Jul 13/2018
-4	Mar 24/2015	-9	Jul 13/2018
-5	Jan 03/2020	-10	Jul 13/2018
-6	Jan 03/2020	-11	Jul 13/2018
-7	Jan 03/2020	-12	Jul 13/2018
-8	Mar 24/2015	-13	Jan 11/2016
-9	Mar 24/2015	-14	Jan 11/2016
-10	Jul 02/2021	-15	Jan 11/2016
-11	Mar 24/2015	-16	Jan 11/2016
-12	Mar 24/2015	-17	Jan 11/2016
-13	Jan 13/2017	-18	Jan 11/2016
-14	Jan 13/2017	-19	Jul 13/2018
		-20	Jan 11/2016
ABNORM 3-1	Feb 13/2015	ABNORM 7-1	Mar 21/2012
-2	Apr 22/2016	-2	Mar 21/2012
-3	Apr 22/2016	-3	Mar 21/2012
-4	Mar 21/2012	-4	Mar 21/2012
-5	Oct 01/2014		
-6	Mar 21/2012	ABNORM 8-1	Mar 24/2015
		-2	Mar 21/2012
		-3	Feb 13/2015

* The asterisk indicates pages changed, added or deleted.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Page	Date	Page	Date
-4	Jul 13/2018	-6	Jul 02/2021
-5	Oct 01/2014	-7	Jul 02/2021
-6	Oct 01/2014	-8	Jul 02/2021
-7	Oct 01/2014	-9	Jul 02/2021
-8	Mar 24/2015	-10	Jul 02/2021
-9	Mar 24/2015	-11	Jul 02/2021
-10	Mar 24/2015	-12	Jul 02/2021
-11	Oct 01/2014	-13	Jul 02/2021
-12	Oct 01/2014	-14	Jul 02/2021
-13	Oct 01/2014	-15	Jul 02/2021
-14	Oct 01/2014	-16	Jul 02/2021
-15	Oct 01/2014	-17	Jul 02/2021
-16	Feb 13/2015	-18	Jul 02/2021
-17	Oct 01/2014	-19	Jul 02/2021
-18	Oct 01/2014	-20	Jul 02/2021
-19	Oct 01/2014	-21	Jul 02/2021
-20	Oct 01/2014	-22	Jul 02/2021
-21	Oct 01/2014	-23	Jul 02/2021
-22	Oct 01/2014	-24	Jul 02/2021
-23	Oct 01/2014	-25	Jul 02/2021
-24	Oct 01/2014	-26	Jul 02/2021
-25	Oct 01/2014	-27	Jul 02/2021
-26	Oct 01/2014	-28	Jul 02/2021
-27	Oct 01/2014		
-28	Oct 01/2014		
ABNORM 9-1	Mar 21/2012	ABNORM11-1	Apr 22/2016
-2	Apr 22/2016	-2	Nov 30/2012
-3	Oct 01/2014	-3	Nov 30/2012
-4	Aug 04/2017	-4	Oct 01/2014
-5	Oct 01/2014	-5	Oct 01/2014
-6	Jan 13/2017	-6	Oct 01/2014
-7	Mar 21/2012	-7	Mar 21/2012
-8	Oct 01/2014	-8	Jan 05/2018
-9	Oct 01/2014	-9	Oct 26/2020
-10	Aug 04/2017	-10	Oct 26/2020
-11	Aug 04/2017	-11	Jul 13/2018
-12	Aug 04/2017	-12	Jul 13/2018
-13	Mar 21/2012	-13	Jul 13/2018
-14	Jan 11/2016	-14	Aug 04/2017
-15	Aug 04/2017	-15	Aug 04/2017
-16	Mar 21/2012	-16	Aug 04/2017
		-17	Apr 22/2016
		-18	Nov 30/2012
ABNORM10-1	Jul 02/2021	ABNORM12-1	Jul 13/2018
-2	Oct 01/2014	-2	Jul 13/2018
-3	Oct 01/2014	-3	Jul 13/2018
-4	Apr 22/2016	-4	Jul 13/2018
-5	Oct 01/2014	-5	Jul 13/2018

* The asterisk indicates pages changed, added or deleted.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Page	Date	Page	Date
-6	Jul 13/2018	-3	Jul 13/2018
-7	Jul 13/2018	-4	Mar 21/2012
-8	Jul 13/2018		
-9	Jul 13/2018	STATUS -1	Jul 13/2018
-10	Jul 13/2018	-2	Jul 13/2018
-11	Jul 13/2018	-3	Jan 04/2019
-12	Jul 13/2018	-4	Jul 13/2018
-13	Jul 13/2018	-5	Jul 13/2018
-14	Jul 13/2018	-6	Jul 13/2018
-15	Jul 13/2018	-7	Jul 13/2018
-16	Jul 13/2018	-8	Jul 13/2018
ABNORM13-1	Jul 02/2021	INDEX -1	Jul 03/2020
-2	Mar 21/2012	-2	Jul 03/2020
-3	Mar 21/2012	-3	Jul 03/2020
-4	Mar 21/2012	-4	Jan 04/2021
-5	Mar 21/2012	-5	Jul 03/2020
-6	Mar 21/2012	-6	Jul 03/2020
-7	Jul 02/2021	-7	Jul 03/2020
-8	Jul 02/2021	-8 *	Jan 03/2022
-9	Jul 02/2021	-9	Jun 02/2021
-10	Jul 02/2021	-10	Jan 04/2019
-11 *	Jan 03/2022	-11	Jul 03/2020
-12	Jul 02/2021	-12	Jul 02/2021
-13	Jul 02/2021	-13 *	Jan 03/2022
-14	Jul 02/2021	-14	Jul 03/2020
		-15	Jul 03/2020
		-16	Jul 03/2020
		-17	Jan 04/2021
		-18	Jul 03/2020
ABNORM14-1	Feb 13/2015	-19	Jul 02/2021
-2	Mar 21/2012	-20	Jul 13/2018
-3	Jun 12/2017		
-4	Mar 21/2012		
ABNORM15-1	Jul 13/2018		
-2	Mar 21/2012		
-3	Mar 21/2012		
-4	Jul 13/2018		
ADVISORY-1	Jul 13/2018		
-2	Jan 13/2017		

* The asterisk indicates pages changed, added or deleted.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

CONTENTS	PAGE
-----------------	-------------

EMERGENCY PROCEDURE CONTENTS

IMMEDIATE ACTION ITEMS	
POWER PLANT	1
SMOKE/FIRE/FUMES	2
AIR-CONDITIONING AND PRESSURIZATION	3
AUTOMATIC FLIGHT CONTROL SYSTEM	4
AUXILIARY POWER UNIT	5
BLEED AIR LEAKS	6
DITCHING AND FORCED LANDING	7
EMERGENCY EVACUATION	8
ELECTRICAL	9
FLIGHT CONTROLS	10
FUEL SYSTEM	11
ICE AND RAIN PROTECTION	12
LANDING GEAR, WHEEL AND BRAKE SYSTEM	13
PASSENGER DOOR	14
AURAL/VISUAL WARNING SYSTEM	15

- SUPP
- LIMITS
- PERF
- MISC
- PLAN
- INTERMIX
- EICAS
- EMER
- IMM
- ABNORM
- INDEX

I	HYDRAULIC POWER	16
	UNRELIABLE AIRSPEED	17

- SUPP
- LIMITS
- PERF
- MISC
- PLAN
- INTERMIX
- EICAS
- EMER
- IMM
- ABNORM
- INDEX

SYMBOLOLOGY

Red warnings are followed by boxed items, for which immediate action must be taken to ensure continued flight safety. The delay involved in referring to the applicable checklist is potentially dangerous, therefore, the flight crew must be able to carry out the applicable “boxed” procedure. These “boxed” items are considered immediate action items.

A flow pattern concept is used throughout the QRH as applicable, utilizing the “decision” (rhomb) symbol ().

This decision symbol indicates a flow pattern which points to two or more possible courses of action. The procedure is completed once the (– **END** –) symbol is reached.

When several alternative actions point to a common procedure or to return/proceed to a specific step, a “return” symbol () is used.

An “arrowhead” symbol () is used to indicate that the procedure is not completed and continues on to the next page.

The end symbol (_____ **END** _____) indicates the end of the complete procedure.

Procedures contained herein assume that certain actions that follow are accomplished by the crew:

- Normal procedures have been properly accomplished.
- MASTER WARNING / MASTER CAUTION switch lights are reset.
- Obvious corrective action (if any) is taken for crew awareness items.

Immediate Action Items have been inserted at the beginning of the QRH as a learning and teaching aid for pilot training.

Diversion terms used within QRH Emergency Procedures are as follows:

- “Land immediately at the nearest suitable airport” – Land without delay at the nearest airport where a safe approach and landing is reasonably assured.
- “Land at the nearest suitable airport” – Landing airport and duration of the flight are at the discretion of the pilot-in-command (PIC). Extended flight beyond the nearest suitable airport is not recommended.
- “Suitable airport ” – Suitable airport is defined as the airport where a safe approach and landing is assured.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

CONTENTS	PAGE
----------	------

IMMEDIATE ACTION ITEMS

REJECTED TAKE-OFF

- Rejected Take-off Before Achieving V_1 IMM ACTION ITEMS-4

FIRE

- **L ENG FIRE** or **R ENG FIRE** or Severe Engine Damage (In Flight) IMM ACTION ITEMS-4
- **L ENG FIRE** or **R ENG FIRE** or Severe Engine Damage (On Ground) IMM ACTION ITEMS-5
- **L JETPIPE OVHT** or **R JETPIPE OVHT** IMM ACTION ITEMS-5
- Post-Shutdown Engine Tail Pipe Fire IMM ACTION ITEMS-5

ENGINE FAILURE

- Double Engine Failure IMM ACTION ITEMS-6
- **L REV UNLOCKED** or **R REV UNLOCKED** IMM ACTION ITEMS-6

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- Uncommanded Acceleration /
ENGINE OVERSPD IMM ACTION ITEMS-6

SMOKE/FIRE/FUMES

- SMOKE CARGO (On Ground)
<0013> or <0034> or <0043> or <0053> or
<0059> or <0074> IMM ACTION ITEMS-6

PRESSURIZATION

- CABIN**
ALT (Warning Message) or
Emergency Descent Procedure IMM ACTION ITEMS-7

AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS)

- Uncommanded Yaw Motion IMM ACTION ITEMS-7

AUXILIARY POWER UNIT

- APU FIRE** IMM ACTION ITEMS-7

DITCHING AND FORCED LANDING

- Ditching or Forced Landing
Imminent IMM ACTION ITEMS-7

EMERGENCY EVACUATION

- Passenger Evacuation IMM ACTION ITEMS-8

ELECTRICAL

- Loss of all AC Power IMM ACTION ITEMS-8

FLIGHT CONTROL

- Aileron System Jammed IMM ACTION ITEMS-8
- Elevator System Jammed IMM ACTION ITEMS-9
- Rudder System Jammed IMM ACTION ITEMS-9
- Stabilizer Trim Runaway IMM ACTION ITEMS-9
- Stall Recovery IMM ACTION ITEMS-10

LANDING GEAR

- MLG BAY OVHT** IMM ACTION ITEMS-10
- BRAKE OVHT** IMM ACTION ITEMS-10

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- During Landing – Excessive Asymmetry or Loss of Braking [IMM ACTION ITEMS-10](#)

AURAL/VISUAL WARNING

- Configuration Warning [IMM ACTION ITEMS-11](#)

UNRELIABLE AIRSPEED

- Unreliable Airspeed In-flight [IMM ACTION ITEMS-11](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



Immediate Action Items

This section contains Immediate Action Items only for quick reference and training purposes. These Immediate Action Items in no way supersede current procedures in the rest of the Quick Reference Handbook of the approved Airplane Flight Manual and any revisions thereto. In case of conflict, the Airplane Flight Manual takes precedence. To accomplish the complete Emergency procedures, refer to their appropriate sections following the Immediate Action Items.

Rejected Take-off Before Achieving V₁

Simultaneously:

- (1) Thrust leversIDLE
- (2) Wheel brakes MAXIMUM until a safe stop.
- (3) Thrust reverser(s) (operating engine[s])MAXIMUM, consistent with directional control.

----- END -----

L ENG FIRE or R ENG FIRE or Severe Engine Damage
(In Flight)

At a safe altitude, affected engine:

- (1) Thrust leverCONFIRM and IDLE
 - (2) Thrust leverCONFIRM and SHUT OFF
 - (3) ENG FIRE PUSHCONFIRM and SELECT
 - (4) FUEL, BOOST PUMPCONFIRM and SELECT OFF
- After 10 seconds L ENG FIRE or R ENG FIRE warning message persists:
- (5) Affected engine BOTTLESELECT
- After another 30 seconds L ENG FIRE or R ENG FIRE warning message still persists:
- (6) Other engine BOTTLESELECT

----- END -----



L ENG FIRE or R ENG FIRE or Severe Engine Damage
(On Ground)

NOTE

Attempt to face the airplane into the wind.

- (1) PARKING BRAKE ON
- Affected engine:**
- (2) Thrust leverSHUT OFF
- (3) ENG FIRE PUSHSELECT
- (4) FUEL, L and R BOOST PUMP SELECT OFF
- If after 10 seconds L ENG FIRE or R ENG FIRE warning message persists:**
- (5) Both engine BOTTLESELECT

----- END -----

L JETPIPE OVHT or R JETPIPE OVHT

- (1) Affected thrust lever CONFIRM and RETARD
slowly until warning message
goes out.

----- END -----

Post-Shutdown Engine Tail Pipe Fire

- (1) Affected engine DRY MOTOR until ITT is
reduced below 150°C or
starter limit.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Double Engine Failure

- (1)

IGNITION, CONT

..... ON
- (2)

Airspeed

..... Not less than 240 KIAS

----- END -----

L REV UNLOCKED or R REV UNLOCKED

- (1)

Affected thrust lever

.....CONFIRM and IDLE
- (2)

Affected THRUST REVERSER,
EMER STOW

.....SELECT

----- END -----

Uncommanded Acceleration / ENGINE OVERSPD

On the ground:

- (1)

Affected thrust lever

.....IDLE then SHUT OFF

During flight:

- (1)

Affected thrust lever

.....CONFIRM and IDLE. Check
engine response.

----- END -----

SMOKE CARGO (On Ground) <0013> or <0034> or <0043> or <0053> or
<0059> or <0074>

- (1)

PARKING BRAKE

..... ON
- (2)

EMER DEPRESS

..... ON
- (3)

CARGO switch

..... OFF
- (4)

CARGO FIREX, NORMAL and
STANDBY CARGO SMOKE
PUSH

.....SELECT
- (5)

CARGO FIREX, NORMAL and
STANDBY BOTTLE ARMED
PUSH TO DISCH

..... SELECT, to discharge

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



CABIN ALT (Warning Message) or Emergency Descent Procedure

- (1) Oxygen masks DON, SET to 100%
- (2) Crew communication ESTABLISH
- (3) PASS SIGNS (both) ON
- (4) Descent INITIATE to 10000 feet MSL
or lowest safe altitude,
whichever is higher.
- (5) Thrust levers IDLE
- (6) Flight spoilers DEPLOY

----- END -----

Uncommanded Yaw Motion

- (1) Controls Assume manual control and
counter aircraft motion using
handwheel inputs.
- (2) YAW DAMPER, DISC SELECT to disconnect both
yaw dampers.

APU FIRE

- (1) APU FIRE PUSH SELECT
- After 5 seconds APU FIRE warning message persists:
- (2) APU BOTTLE SELECT

----- END -----

Ditching or Forced Landing Imminent

- (1) L and R PACK OFF
 - (2) EMER DEPRESS ON
- Just before contact:
- (3) EMER DEPRESS SELECT OFF

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Passenger Evacuation

Pilot:

- (1) PARKING BRAKE ON
- (2) EvacuationCOMMAND
- (3) GND LIFT DUMPINGMAN DISARM
- (4) Thrust leversSHUT OFF
- (5) Evacuation INITIATE using PA system
- (6) APU, LH ENG and RH ENG FIRE
PUSHSELECT
- (7) BATTERY MASTER OFF

Copilot: On evacuation command

- (8) Air Traffic ControlNOTIFY of emergency
conditions and intent to
evacuate

NOTE

If ditching, disregard step (9).

- (9) EMER DEPRESS ON
- (10) EMER LTS (coincident with PA) ON

----- END -----

Loss of all AC Power

- (1) ADG manual deploy handlePULL

----- END -----

Aileron System Jammed

- (1) Autopilot DISENGAGE
- (2) Aileron controls (both) RELEASE pressure
- (3) ROLL DISC PULL and TURN to lock
- (4) Airplane control TRANSFER to pilot with
operative aileron.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Elevator System Jammed

- (1) Autopilot DISENGAGE

(2) Elevator controls (both) RELEASE differential pressure

(3) PITCH DISC PULL and TURN to lock

(4) Airplane control TRANSFER to pilot with operative elevator.

----- END -----

Rudder System Jammed

- (1) YAW DAMPER, DISCSELECT

(2) Rudder pedalsOVERPOWER

----- END -----

Stabilizer Trim Runaway

- (1) Control WheelASSUME manual control and OVERRIDE runaway

(2) Both STAB TRIM DISC PRESS, HOLD and RELEASE

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----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



IMM ACTION ITEMS-10

Rev. 75, Jan 04/2021

Stall Recovery

- (1) Autopilot DISENGAGE, if required
- (2) Pitch attitude Lower nose to reduce angle of attack
- (3) Thrust levers Advance to MAX POWER
- (4) Roll attitude Wings level
- (5) FLIGHT SPOILER leverRETRACT

----- END -----

MLG BAY OVHT

- (1) AirspeedNot more than 250 KIAS
- (2) LDG GEAR lever DN

----- END -----

BRAKE OVHT

During flight:

- (1) AirspeedNot more than 250 KIAS
- (2) LDG GEAR lever DN

----- END -----

During Landing – Excessive Asymmetry or Loss of Braking

- (1) Wheel brakes RELEASE momentarily
- (2) ANTI-SKID OFF
- (3) Wheel brakesRE-APPLY, as required

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

QUICK REFERENCE
HANDBOOK
CSP A-022-PRS

IMMEDIATE ACTION ITEMS



Configuration Warning

- (1) Take-offDISCONTINUE immediately

----- END -----

Unreliable Airspeed In-flight

- (1) Autopilot DISENGAGE
(2) FDsDESELECT
(3) Use standby attitude indicator / ISI for pitch reference.
Initial take-off climb or go-around is required:
(4) Pitch/N₁ 10 degrees/TOGA from SL to
15000 ft; 5 degrees/CLB
above 15000 ft
(5) Aircraft configuration At clean up altitude Gear UP
/ FLAPS 0
(6) Airplane altitude MAINTAIN lowest safe
altitude or higher



Respect stall warning/stick shaker.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

I

CONTENTS	PAGE
----------	------

POWER PLANT

REJECTED TAKE-OFF

- Rejected Take-off Before Achieving V_1 EMER 1-2

FIRE

- **L ENG FIRE** or **R ENG FIRE** or Severe Engine Damage (In Flight) EMER 1-3
- **L ENG FIRE** or **R ENG FIRE** or Severe Engine Damage (On Ground) EMER 1-3
- **L JETPIPE OVHT** or **R JETPIPE OVHT** EMER 1-4
- Post-Shutdown Engine Tail Pipe Fire EMER 1-4

ENGINE FAILURE

- Double Engine Failure EMER 1-5
- **L REV UNLOCKED** or **R REV UNLOCKED** EMER 1-14
- **L ENG OIL PRESS** or **R ENG OIL PRESS** or Low Engine Oil Pressure Indication EMER 1-16
- Uncommanded Acceleration / **ENGINE OVERSPD** EMER 1-16

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Rejected Take-off Before Achieving V₁

- Simultaneously:
- (1) Thrust leversIDLE

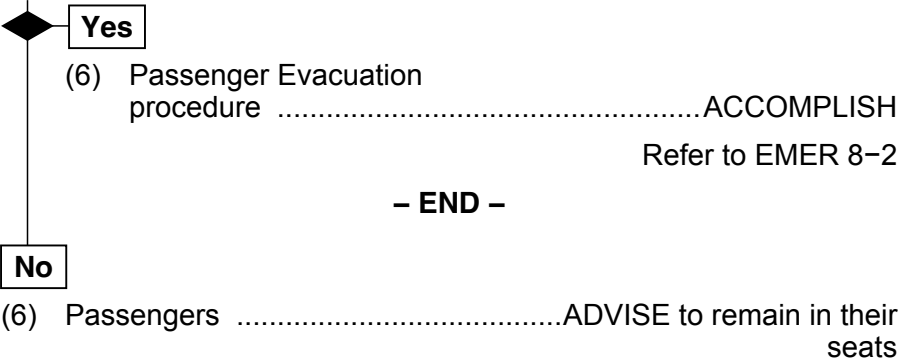
(2) Wheel brakes MAXIMUM until a safe stop.

(3) Thrust reverser(s) (operating engine[s])MAXIMUM, consistent with directional control.

After the airplane has been safely brought to a stop:

- (4) PARKING BRAKE ON
- (5) Applicable Immediate Actions or procedures Accomplish

Evacuation required:



NOTE

After a rejected take-off, a minimum brake cooling time of 15 minutes must be observed between a rejected take-off and the subsequent take-off. If a brake overheat warning is displayed on EICAS, an inspection of the wheel fuse plugs is required before the next take-off.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

L ENG FIRE or R ENG FIRE or Severe Engine Damage
(In Flight)

At a safe altitude, affected engine:

- (1) Thrust leverCONFIRM and IDLE

(2) Thrust leverCONFIRM and SHUT OFF

(3) ENG FIRE PUSHCONFIRM and SELECT

(4) FUEL, BOOST PUMPCONFIRM and SELECT OFF

After 10 seconds L ENG FIRE or R ENG FIRE warning message persists:

(5) Affected engine BOTTLESELECT

After another 30 seconds L ENG FIRE or R ENG FIRE warning message still persists:

(6) Other engine BOTTLESELECT

- (7) Single Engine procedures, In-flight

Engine ShutdownACCOMPLISH

Refer to ABNORM 1-4

----- END -----

L ENG FIRE or R ENG FIRE or Severe Engine Damage
(On Ground)

NOTE

Attempt to face the airplane into the wind.

- (1) PARKING BRAKE ON

Affected engine:

(2) Thrust leverSHUT OFF

(3) ENG FIRE PUSHSELECT

(4) FUEL, L and R BOOST PUMP SELECT OFF

If after 10 seconds L ENG FIRE or R ENG FIRE warning message persists:

(5) Both engine BOTTLESELECT

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(6) Passenger Evacuation procedureAS REQUIRED

Refer to EMER 8-2

----- END -----

L JETPIPE OVHT or R JETPIPE OVHT

(1) Affected thrust leverCONFIRM and RETARD
slowly until warning message
goes out.

L JETPIPE OVHT or R JETPIPE OVHT warning message
persists:

Yes

(2) Single Engine Procedures,
In-flight Engine ShutdownACCOMPLISH

Refer to ABNORM 1-4

- END -

No

(2) Engine indicationsMONITOR

----- END -----

Post-Shutdown Engine Tail Pipe Fire

(1) Affected engine DRY MOTOR until ITT is
reduced below 150°C or
starter limit.

Tail pipe fire indications persist:

Yes

(2) Affected ENG FIRE PUSHCONFIRM and SELECT

(3) Passenger Evacuation
procedureACCOMPLISH

Refer to EMER 8-2

- END -

No

(2) Normal operationsRESUME

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Double Engine Failure

- (1) IGNITION, CONT ON
- (2) Airspeed Not less than 240 KIAS

- (3) Turn towards nearest suitable airport.
- (4) Engine instrumentsMONITOR

WARNING

Increase airspeed as required to maintain at least 4% N₂ indicated. If the N₂ is allowed to drop to 0%, the engine may thermally seize and may not rotate even at higher airspeeds or with starter engagement.

If engines continue to run-down:

- (5) Thrust levers (both)SHUT OFF
- (6) ADG manual deploy handlePULL

When ADG power is established:

- (7) STAB TRIM, CH 2SELECT
- (8) Oxygen masks (if required)DON
- (9) Crew communicationsESTABLISH
- (10) PASS SIGNS (both) ON
- (11) APU (if available, at 30000 feet and below)START
- (12) APU GEN (if APU available)CHECK ON

- If relight of either engine is not considered feasible, proceed to step (13).
 - If windmilling relight is considered feasible, proceed to step (16).
 - If APU bleed air relight is considered feasible, proceed to step (43).
- (13) RELIGHT OF EITHER ENGINE IS NOT CONSIDERED FEASIBLE

(14) Target glide airspeedESTABLISH

Airplane Weight	Target Airspeed
20000 kg (44090 lb) or below	190 KIAS
22000 kg (48500 lb)	200 KIAS
24000 kg (52910 lb)	210 KIAS

(15) Proceed to step (72).

(16) **WINDMILLING RELIGHT PROCEDURE**

(17) DescentInitiate to 21000 feet or below

NOTE

1. Between 21000 feet and 15000 feet, a minimum of 12% N₂ is necessary for a windmill relight.
2. At 15000 feet and below, a minimum of 9% N₂ is necessary for a windmill relight.
3. Disconnecting an IDG can gain approximately 2.5% N₂ on the associated engine. It will take 40 to 60 seconds after disconnecting an IDG to realize the 2.5% N₂ increase on the associated engine.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



If the IDGs are disconnected, normal electrical power cannot be re-established even after a successful engine start.

NOTE

1.
- Acceleration to V_{MO} is recommended to attain the necessary N_2 for a windmill relight.
2.
- The altitude loss when accelerating from 240 KIAS to V_{MO} could be more than 5000 feet.
3.
- A push-over to as steep as 15 degrees nose down may be required.

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- (18)
- FUEL, L and R BOOST PUMP CONFIRM ON
- (19)
- ANTI-ICE, WING and COWL ALL OFF
- (20)
- IGNITION, CONT CONFIRM ON

When ITT is 90°C or less and N_2 is at least 12% (between 21000 feet and 15000 feet) or N_2 is at least 9% (15000 feet and below):

- (21)
- Thrust levers (both) IDLE
- (22)
- Engine indications MONITOR

- If at least one engine relights within 25 seconds (stable IDLE within 2 minutes), proceed to step (23).
- If neither engine relight within 25 seconds, proceed to step (39).

At least one engine relights within 25 seconds (stable IDLE within 2 minutes):

- (23)
- Thrust lever(s) As required
- (24)
- 14TH STAGE ISOL AS REQUIRED
- (25)
- ANTI-ICE, WING and COWL AS REQUIRED

(26) IGNITION, CONT SELECT OFF

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Effectivity:

- Airplanes 7305 and subsequent:

(27) No airspeed limitation with the ADG deployed.

Operative engine:

(28) L and/or R 10TH STAGE bleed(s)CHECK OPEN

(29) Applicable PACK(s)CHECK ON

Re-establish normal power:

(30) Affected GEN(s)CHECK ON

(31) ADG manual deploy handleSTOW

(32) ADG CONTROL, PWR TXFR
OVERRIDESELECT

NOTE

If cabin altitude exceeds 14500 feet, the **AUTO PRESS** caution message may be posted and automatic pressurization control will not be functional.

AUTO PRESS caution message is not displayed:

(33) Proceed to step (38).

AUTO PRESS caution message is displayed:

Attempt to regain automatic pressurization control as follows:

(34) Cabin Altitude Check below 14500 feet

(35) PRESS CONTSelect MAN, then AUTO

AUTO PRESS caution message goes out:

(36) Proceed to step (38).

AUTO PRESS caution message persists:

(37) Manual Pressurization Control
ProcedureACCOMPLISH
Refer to ABNORM 2-11

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(38) Single Engine ProceduresACCOMPLISH, if required
Refer to ABNORM 1-1

– END –

Neither engine relight within 25 seconds:

(39) Thrust levers (both)SHUT OFF

Another windmill relight attempt is still possible:

(40) Airspeed MAINTAIN V_{MO}

(41) Wait 30 seconds, then proceed to step (16) and repeat
WINDMILLING RELIGHT PROCEDURE.

Another windmill relight attempt is not possible:

(42) Proceed to step (43) or to step (72).

(43) **APU BLEED AIR RELIGHT PROCEDURE**

(44) Airspeed Not less than 240 KIAS

(45) FUEL, L and R BOOST PUMP CONFIRM ON

(46) ANTI-ICE, WING and COWL ALL OFF

(47) L and R 10TH STAGE bleedSELECT CLOSED

(48) APU LCV SELECT OPEN

Attempt to start one engine at a time (13000 feet and below):

(49) IGNITION, CONT CONFIRM ON

(50) L or R ENG STARTSELECT

When N_2 is at least 28% and ITT is 90°C or less:

(51) Applicable thrust leverIDLE

(52) Engine indicationsMONITOR

- **If engine relights within 25 seconds (stable IDLE within 2 minutes), proceed to step (53).**
- **If engine does not relight within 25 seconds, proceed to step (67).**

Engine relights within 25 seconds (stable IDLE within 2 minutes):

(53) Thrust lever As required

(54) IGNITION, CONT SELECT OFF

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



EMER 1-10

Rev. 74, Jul 03/2020

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Effectivity:

- Airplanes 7305 and subsequent:

(55) No airspeed limitation with the ADG deployed.

Operative engine:

(56) Applicable 10TH STAGE Bleed SELECT OPEN

(57) Applicable PACKCHECK ON

Re-establish normal power:

(58) Affected GENCHECK ON

(59) ADG manual deploy handleSTOW

(60) ADG CONTROL, PWR TXFR
OVERRIDESELECT

NOTE

If cabin altitude exceeds 14500 feet, the **AUTO PRESS** caution message may be posted and automatic pressurization control will not be functional.

AUTO PRESS caution message is not displayed:

(61) Proceed to step (66).

AUTO PRESS caution message is displayed:

Attempt to regain automatic pressurization control as follows:

(62) Cabin Altitude Check below 14500 feet

(63) PRESS CONTSelect MAN, then AUTO

AUTO PRESS caution message goes out:

(64) Proceed to step (66).

AUTO PRESS caution message persists:

(65) Manual Pressurization Control
ProcedureACCOMPLISH

Refer to ABNORM 2-11

(66) Single Engine ProceduresACCOMPLISH, if required

Refer to ABNORM 1-1

– END –



QUICK REFERENCE
HANDBOOK
CSP A-022-PRS

POWER PLANT

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Engine does not relight within 25 seconds:

- (67) Affected thrust leverSHUT OFF
- (68) Affected ENG STOPSELECT
- (69) Attempt relight on other engine.

Neither engine restarted:

- (70) Target glide airspeedESTABLISH

Airplane Weight	Target Airspeed
20000 kg (44090 lb) or below	190 KIAS
22000 kg (48500 lb)	200 KIAS
24000 kg (52910 lb)	210 KIAS

- (71) Proceed to step (72).
- (72) **ALL ENGINE OUT PROCEDURE**
- (73) Determine if APU GEN is available:
 - If APU GEN is available, proceed to step (74).
 - If APU GEN is not available, proceed to step (77).

APU GEN is available:

- (74) ADG manual deploy handleSTOW
- (75) ADG CONTROL, PWR TXFR
OVERRIDESELECT
- (76) Proceed to step (77).

APU GEN is not available:

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

(77) Determine if a suitable landing site is within gliding range.

NOTE

1. Target glide speed provides a range of approximately 2.5 NM for every 1000 feet of altitude lost.
2. Descent rate will be approximately 1200 – 1600 fpm.
3. Recommended approach is a descending 360-degree turn started at 5000 feet above the landing area.
4. With both engines inoperative, the gliding distance and time from 10000 feet above the landing area to the initiation of the recommended approach maneuver at 5000 feet above the landing area, will be approximately:

Airplane Weight	Target Airspeed	Gliding Distance and Time
20000 kg (44090 lb)	190 KIAS	12.5 NM – 4 Minutes
24000 kg (52910 lb)	210 KIAS	12.5 NM – 3.5 Minutes

5. If a turn is required before 5000 feet above the landing area, the descent rates for the following bank angles will be approximately:

Bank Angle	20 Degrees	25 Degrees	30 Degrees
Descent Rate (fpm)	1300–1700	1375–1800	1450–1900

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(78) If time allows, affected aircraft systemsReview

NOTE

The following significant systems are not available when all engines are out:

- Both PACKs,
- Yaw damper 2 and autopilot,
- Ground spoilers, flight spoilers, flaps, anti-skid system and nosewheel steering,
- Stabilizer trim channel 1, Mach trim, aileron and rudder trims,
- Hydraulic EDP 1A and EDP 2A, hydraulic pumps 1B, 2B and 3A,
- PFD 2, MFD 2, VHF COM 2, RTU 2,
- FD 2, VHF NAV 2, ADF 2, ATC 2, DME 1 and 2,
- Radio altimeter 2, <0045>
- FMS 1 and 2, <0024>
- Copilot's instrument lights, NAV lights, left and right landing and taxi lights,
- ADC 2 (probe heaters inoperative) and ice detector 2,
- Windshield wipers, both windshield heaters and right window heater.

- If suitable landing site is in range, proceed to step (79).
- If no suitable site is in range, proceed to step (91).

Suitable landing site is in range:

At 10000 feet MSL or lowest safe altitude:

- (79) EMER DEPRESS ON
- (80) RAM-AIR OPEN
- (81) GRND PROX, FLAP <0040> OVRD
- (82) Approach start pointESTABLISH at 5000 feet
AGL over landing area.
- (83) 20 to 30-degree angle of bank
turn START over landing area.
- (84) Target airspeed MAINTAIN

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- (85) Downwind leg ESTABLISH at 2500 feet
AGL abeam landing area and
180 degrees from final
approach heading.
- (86) Final approach ESTABLISH at 1500 feet
AGL.
- (87) LDG GEAR lever DN when runway is assured.
- (88) Landing FLAPS (if available) 20 Recommended.
- (89) Final approach speed $V_{REF (FLAPS 45)} + \Delta V_{REF}$ (refer
to the table that follows):

FLAPS Position	ΔV_{REF} (kts)
0	30
20	12

- (90) Actual landing distance Increase

FLAPS Position	APU GEN available	EMER PWR only
0	1.70 (70%)	3.30 (230%)
20	1.25 (25%)	N/A

No suitable landing site is in range:

- (91) Ditching or Forced Landing
Procedure Accomplish

L REV UNLOCKED or R REV UNLOCKED

In flight:

- (1) Affected thrust lever CONFIRM and IDLE
- (2) Affected THRUST REVERSER,
EMER STOW SELECT

NOTE

The affected LH or RH EMER STOW switch/light
and the corresponding **L REV UNLOCKED** or **R
REV UNLOCKED** caution message will remain
on.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Affected thrust reverser stows (stow indicated by improved airplane handling and normal operation of affected thrust lever):

Yes

- (3) Affected THRUST REVERSER OFF
- (4) Thrust leversAS REQUIRED
- (5) Land at the nearest suitable airport.



Use remaining thrust reverser carefully upon landing.

- END -

No

- (3) AirspeedNot more than 200 KIAS
- (4) Affected THRUST REVERSER OFF
- (5) APU (if available, at 30000 feet and below)START
- (6) APU GEN ON
- (7) Single Engine Procedures, In-flight Engine ShutdownACCOMPLISH

Refer to ABNORM 1-4

- END -

On the ground:



If the L REV UNLOCKED or R REV UNLOCKED caution message is posted during ground operations, ensure that the affected thrust lever is at forward IDLE. Do not cycle the thrust reverser(s). Maintenance inspection is required.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

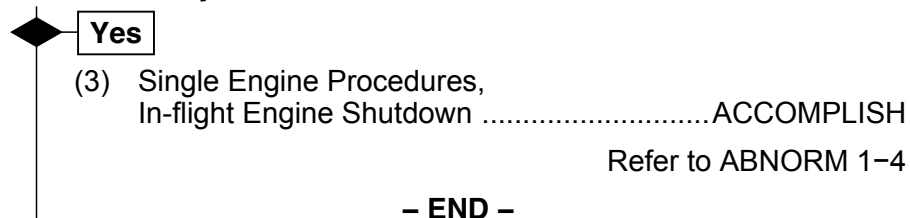


L ENG OIL PRESS or R ENG OIL PRESS or Low Engine Oil Pressure Indication

- (1) Affected engine oil pressure CHECK
(2) Affected thrust lever CONFIRM and RETARD to IDLE

Any 2 of the following 3 indications are displayed:

- L ENG OIL PRESS or R ENG OIL PRESS warning message;
- Affected engine oil pressure is below limits;
- Affected engine oil temperature is increasing or decreasing abnormally.



- No
- Only 1 of the above indications is displayed:
- (3) Affected thrust lever ADJUST as required
(4) Engine indications MONITOR

----- END -----

Uncommanded Acceleration / ENGINE OVERSPD

On the ground:

- (1) Affected thrust lever IDLE then SHUT OFF

– END –

During flight:

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



- (1) Affected thrust leverCONFIRM and IDLE
Check engine response.

Engine responds:

Yes

NOTE

To prevent engine overspeed when selecting
ENG SPEED switches OFF, N_2 indications
should be referenced instead of N_1 .

- (2) Affected engineADJUST thrust
below 79% N_2
(3) Affected ENG SPEED OFF
(4) Affected thrust lever ADVANCE
(5) Affected engine indicationsMONITOR



Do not change power settings abruptly as
engine response is degraded.

NOTE

1. To minimize asymmetric thrust, both
ENG SPEED switches should be
selected to OFF prior to approach and
landing.
2. Exercise caution, to prevent engine
overspeed during thrust reverser use,
when operating with ENG SPEED
switches OFF.
3. The APR system is inoperative when
the engine speed switches are
selected off.

– END –

No

- (2) Single Engine Procedures,
In-flight Engine ShutdownACCOMPLISH
Refer to ABNORM 1-4

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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CONTENTS

PAGE

SMOKE/FIRE/FUMES

- General Notes, Cautions and WarningsEMER 2-2
- Smoke/Fire/Fumes ProcedureEMER 2-2
- Smoke or Fumes Removal ProcedureEMER 2-13
- SMOKE TOILETEMER 2-14
- I • SMOKE CARGO (In Flight) <0013>
or <0034> or <0043> or <0053> or <0059> or <0074>EMER 2-15
- I • SMOKE CARGO (On Ground)
<0013> or <0034> or <0043> or <0053> or <0059>
or <0074>EMER 2-16

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

General Notes, Cautions and Warnings

WARNING

Time is critical during smoke/fire/fumes emergencies. The flight crew should consider an immediate landing anytime the situation cannot be controlled.

CAUTION

Passenger masks should not be deployed when performing smoke or fire procedures.

Smoke/Fire/Fumes Procedure

- (1)

Oxygen masks / smoke goggles (if required)ON, 100%
- (2)

Crew and cabin communicationESTABLISH
- (3)

Land at the nearest suitable airport.

CAUTION

Continuing to destination is not recommended. If the decision to continue is made the flight crew must acknowledge that the threat has been positively identified, confirmed to be extinguished and the smoke/fumes have dissipated.

- (4)

CARGO OFF

WARNING

Anytime smoke or fumes becomes the greatest threat accomplish the smoke or fumes removal procedure. Refer to EMER 2-13

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Time and conditions permit and source of smoke/fire/fumes can be quickly identified and eliminated:

Yes

(5) Appropriate procedureACCOMPLISH

- Air-conditioning (Smoke coming from gaspers or vents) See

“SOURCE IS IDENTIFIED AS AIR-CONDITIONING SYSTEM”

Refer to EMER 2-4

- Electrical (Electrical Fire Smell, Potential Loss of Associated Systems) See

“SOURCE IS IDENTIFIED AS ELECTRICAL SYSTEM”

Refer to EMER 2-5

- Galley See

“SOURCE IS IDENTIFIED AS GALLEY FIRE/SMOKE”

Refer to EMER 2-10

- Cabin See

“SOURCE IS IDENTIFIED AS CABIN FIRE/SMOKE”

Refer to EMER 2-12

– END –

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

(5) Land immediately at the nearest suitable airport.

CAUTION

Dependent upon the severity of the situation, the flight crew should expedite the landing. The crew should also consider an overweight landing, tailwind landing, ditching or a forced off-airport landing.

----- END -----

SOURCE IS IDENTIFIED AS AIR-CONDITIONING SYSTEM:

- (1) L and R 10TH STAGE BLEEDS SELECT open
- (2) APU LCV SELECT closed
- (3) 10TH STAGE ISOL SELECT closed
- (4) L PACK OFF

Smoke/fumes conditions persists:

Yes

- (5) L PACK ON
- (6) R PACK OFF

Yes

- (7) R PACK ON
- (8) Land immediately at the nearest suitable airport.

CAUTION

Dependent upon the severity of the situation, the flight crew should expedite the landing. The crew should also consider an overweight landing, tailwind landing, ditching or a forced off-airport landing.

- END -

No

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

NOTE

Airplane altitude not above 25000 feet during single pack operations.

- (5) Smoke or Fumes Removal
ProcedureACCOMPLISH, if required
Refer to EMER 2-13

----- END -----

SOURCE IS IDENTIFIED AS ELECTRICAL SYSTEM:

- (1) AC and DC Electrical loads MONITOR
Source of electrical smoke/fire/fumes is positively identified:

Yes

WARNING

The series of steps that follow provide guidance on how to isolate each electrical bus. Select only the steps required to isolate the positively identified bus. The steps below are not to be used for trial and error in an attempt to identify the source bus.

- (2) Affected electrical bus ISOLATE

AC BUS 1:

- (a) AC ESS XFER ALTN
(b) Left AUTO XFER OFF
(c) GEN 1 OFF/RESET

AC BUS 2:

- (a) Right AUTO XFER OFF
(b) GEN 2 OFF/RESET

AC ESS BUS:

- (a) AC ESS FEED
circuit breaker (1C2) OPEN
(b) AC ESS FEED
circuit breaker (2C2) OPEN



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

AC SERV BUS:

- (a) AC SERV BUS
FEED
circuit breaker (2R2) OPEN

AC UTLY 1:

- (a) AC UTLY BUS
FEED
circuit breaker (1E2) OPEN

AC UTLY 2:

- (a) AC UTLY BUS
FEED
circuit breaker (2E2) OPEN

DC BUS 1:

- (a) DC BUS 1 FEED
circuit breaker (1E12)..... OPEN

DC BUS 2:

- (a) DC BUS 2 FEED
circuit breaker (2E12)..... OPEN

DC ESS BUS:

- (a) FEED 1 DC ESS
circuit breaker (2M6)..... OPEN
- (b) FEED 2 DC ESS
circuit breaker (2M8)..... OPEN

DC SERV BUS:

- (a) DC SERV BUS
FEED
circuit breaker (2T5)..... OPEN
- (b) DC SERVICE..... OFF

DC BATT BUS:

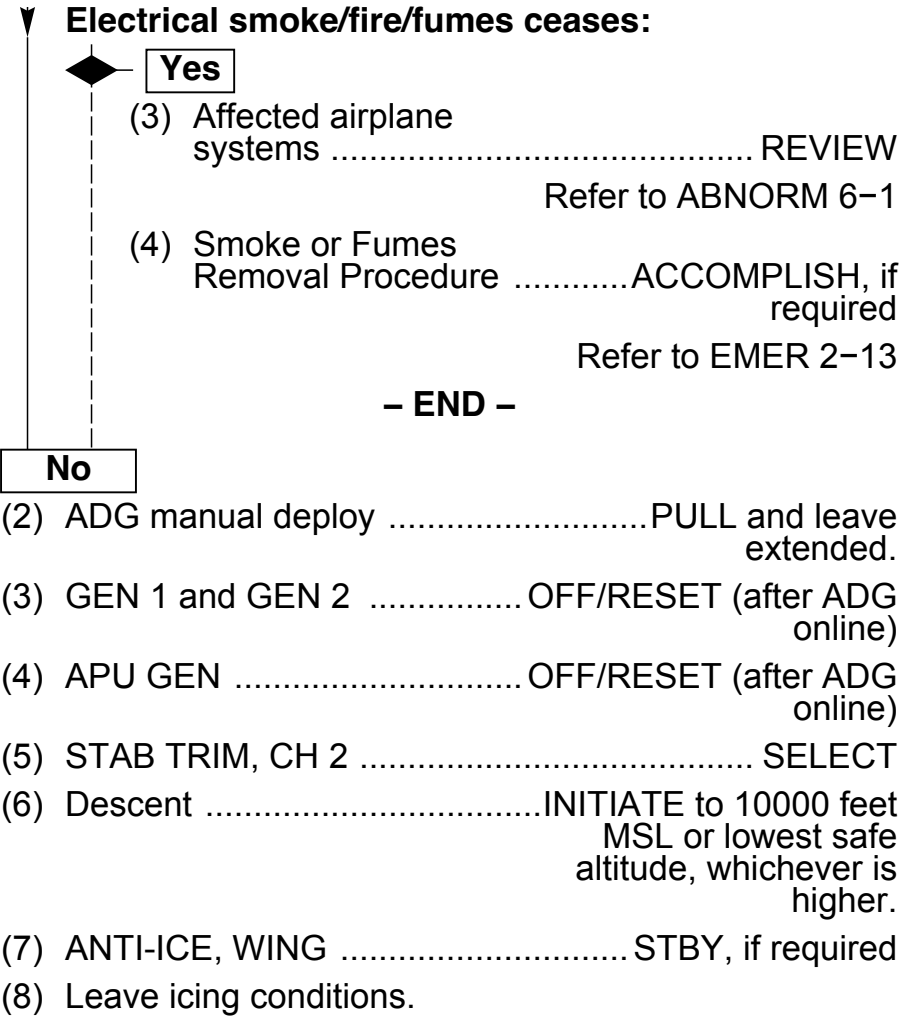
- (a) FEED 1 BATT BUS
circuit breaker (2M5)..... OPEN
- (b) FEED 2 BATT BUS
circuit breaker (2M7)..... OPEN

DC UTLY 1:

- (a) UTLY BUS 1 FEED
circuit breaker (1E13)..... OPEN

DC UTLY 2:

- (a) TIE AND UTLY
circuit breaker (2E13)..... OPEN



NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is -40°C (-40°F) or below.

(9) Land immediately at the nearest suitable airport.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Dependent upon the severity of the situation, the flight crew should expedite the landing. The crew should also consider an overweight landing, tailwind landing, ditching or a forced off-airport landing.

Smoke/fire/fumes persists:

Yes

(10)GEN 1, GEN 2, APU
GEN ON

(11)ADG manual deploy
handleSTOW

(12)ADG CONTROL, PWR
TXFR OVERRIDE SELECT

(13)CAS messages REVIEW

(14)Affected systems RESTORE

- END -

No

Smoke/fire/fumes ceases:

At 10000 feet MSL or lowest safe altitude:

(10)RAM-AIR OPEN

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

NOTE

- 1. 250 KIAS airspeed recommended during cruise to provide sufficient airflow to passengers within cabin.
- 2. The following significant systems are not available when on emergency power only:
 - Both PACKs,
 - Yaw damper 2 and autopilot,
 - Flaps, anti-skid system and nosewheel steering,
 - Stabilizer trim channel 1, Mach trim, aileron and rudder trims,
 - Hydraulic pumps 1B, 2B and 3A,
 - PFD 2, MFD 2, VHF COM 2, RTU 2,
 - FD 2, VHF NAV 2, ADF 2, ATC 2, DME 1 and 2,
 - Radio altimeter 2, <0045>
 - FMS 1 and 2, <0024>
 - Copilot's instrument lights, NAV lights, left and right landing and taxi lights,
 - ADC 2 (probe heaters inoperative) and ice detector 2,
 - Windshield wipers, both windshield heaters and right window heater.

Prior to landing:

(11)ANTI-SKID OFF
(12)N/W STRG OFF

Effectivity: <0040>
(13)GRND PROX, FLAP OVRD

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



EMER 2-10

Rev. 70, Jul 13/2018

(14)Final approach speed $V_{REF (FLAPS 45)} + \Delta V_{REF}$

Flaps Setting (Degrees)	$\Delta V_{REF}(kt)$
0	30
8	18
20	12
30	8

(15)Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Final Approach Speed $\Delta V_{REF}(kt)$	Runway Surface	
		Dry	Wet or Contaminated
With Two Thrust Reversers	0	2.60 (160%)	3.30 (230%)
	8	2.40 (140%)	3.00 (200%)
	20	2.10 (110%)	2.65 (165%)
	30	2.00 (100%)	2.55 (155%)
	45	1.80 (80%)	2.20 (120%)
Without Thrust Reversers	0	3.30 (230%)	3.30 (230%)
	8	3.00 (200%)	3.00 (200%)
	20	2.65 (165%)	2.65 (165%)
	30	2.55 (155%)	2.55 (155%)
	45	2.20 (120%)	2.20 (120%)

----- END -----

SOURCE IS IDENTIFIED AS GALLEY FIRE/SMOKE:

(1) Designated crew member ADVISE

▼

QUICK REFERENCE HANDBOOK CSP A-022-PRS	SMOKE/FIRE/FUMES
--	------------------

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



- (a) To isolate and extinguish source of smoke or fire, and to secure the area.
- (b) Galley control panel circuit breakers (all) OPEN
- (2) Galley, galley heating system and water system DE-ENERGIZE
 - (a) AC UTLY BUS FEED circuit breakers (1E2) OPEN
 - (b) AC UTLY BUS FEED circuit breakers (2E2) OPEN
 - (c) GALLEY / LAVATORY FAN circuit breaker (2C5) OPEN
 - (d) GALLEY HTRS circuit breakers (2B11) OPEN
 - (e) GALLEY HTRS circuit breakers (2G9) OPEN
 - (f) WATER SYSTEM, LINES HTRS, GALLEY / TANKS HTRS circuit breakers (2D13) OPEN
 - (g) WATER SYSTEM, LINES HTRS, GALLEY / TANKS HTRS circuit breakers (2D15) OPEN
 - (h) GALLEY DOME circuit breaker (2U3) OPEN

Smoke/fire/fumes condition persists:

Yes

- (3) Land immediately at the nearest suitable airport.



Dependent upon the severity of the situation, the flight crew should expedite the landing. The crew should also consider an overweight landing, tailwind landing, ditching or a forced off-airport landing.

– END –

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



EMER 2-12

Rev. 67B, Jun 12/2017

No

- (3) Smoke or Fumes Removal ProcedureACCOMPLISH, if required

Refer to EMER 2-13

----- END -----

SOURCE IS IDENTIFIED AS CABIN FIRE/SMOKE:

- (1) EMER LTS ON
- (2) Designated crew member ADVISE
- (a) To isolate and extinguish source of smoke or fire, and to secure the area.
- (3) PASS SIGNS OFF
- (4) Cabin lighting systemDE-ENERGIZE
- (a) LIGHTING CEILING
circuit breaker (2S4) OPEN
- (b) LIGHTING SIDEWALL
circuit breaker (2S5) OPEN
- (c) Left CABIN
READING LTS
circuit breakers (1L5,
1L6, 1L7, and 1L8) OPEN
- (d) Right CABIN
READING LTS
circuit breakers (2L5,
2L6, 2L7 and 2L8) OPEN
- (5) Affected boarding lights /
door actuator system DE-ENERGIZE, if required
- (a) BOARD
circuit breaker (2T4) OPEN
- (b) DOOR SYST
circuit breaker (1L2) OPEN

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

QUICK REFERENCE
HANDBOOK
CSP A-022-PRS

SMOKE/FIRE/FUMES

**Smoke/fire/fumes condition persists:****Yes**

(6) Land immediately at the nearest suitable airport.



Dependent upon the severity of the situation, the flight crew should expedite the landing. The crew should also consider an overweight landing, tailwind landing, ditching or a forced off-airport landing.

– END –**No**

(6) Smoke or Fumes Removal
ProcedureACCOMPLISH, if
required
Refer to EMER 2-13

----- END -----**Smoke or Fumes Removal Procedure**

- (1) CARGO OFF
(2) DescentINITIATE to 10000 feet
MSL or lowest safe
altitude, whichever is
higher.
(3) PRESS CONT MAN
(4) MAN ALT UP
(5) MAN RATEINCR as required to
depressurize the cabin.

To evacuate smoke/fumes at a faster rate:

- (6) EMER DEPRESS ON
(7) Smoke/Fire/Fumes
ProcedureCONTINUE, if required
Refer to EMER 2-2

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



SMOKE TOILET

- (1) Designated crew memberADVISE to isolate and extinguish source of smoke or fire, and to secure the area.
- (2) Toilet/lavatory area and water systemDE-ENERGIZE
 - (a) GALLEY / LAVATORY FAN circuit breaker (2C5) OPEN
 - (b) WATER SYSTEM, LINE HTRS, LAV / TANKS HTRS circuit breakers (2D14 and 2D15) OPEN
 - (c) PWR SENS / TOILET circuit breaker (2S2) OPEN
 - (d) LIGHTS – TOILET circuit breaker (2U2) OPEN

Smoke/fire/fumes condition persists:

Yes

- (3) Land immediately at the nearest suitable airport.

CAUTION

Dependent upon the severity of the situation, the flight crew should expedite the landing. The crew should also consider an overweight landing, tailwind landing, ditching or a forced off-airport landing.

– END –

No

- (3) Land at the nearest suitable airport.

– – – – END – – – –

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

I

SMOKE CARGO (In Flight) <0013> or <0034> or <0043> or <0053> or <0059> or <0074>

- (1) CARGO FIREX, NORMAL
and STANDBY CARGO
SMOKE PUSH SELECT

(2) CARGO FIREX, NORMAL
and STANDBY BOTTLE
ARMED PUSH TO DISCHSELECT, to discharge

(3) Squib / bottle dischargeCHECK

(a) FIREX MONITOR,
CARGO BOTTLETEST 1 and check that
CARGO SQUIB 1 advisory
message does not come
on.

(b) FIREX MONITOR,
CARGO BOTTLETEST 2 and check that
CARGO SQUIB 2 advisory
message does not come
on.

(4) LDG ELEV SET to 8000 feet or
landing field elevation,
whichever is higher.

(5) DescentINITIATE to 10000 feet
MSL or minimum
enroute altitude,
whichever is higher.

At 10000 feet MSL or lowest safe altitude, whichever
is higher:

(6) LDG ELEV SET to landing field
elevation, if not already
done.

(7) Land immediately at the nearest suitable airport.
-
- Dependent upon the severity of the situation, the flight crew should expedite the landing. The crew should also consider an overweight landing, tailwind landing, ditching or a forced off-airport landing.
- END -----
- | |
|----------|
| SUPP |
| LIMITS |
| PERF |
| MISC |
| PLAN |
| INTERMIX |
| EICAS |
| EMER |
| IMM |
| ABNORM |
| INDEX |
- QUICK REFERENCE
HANDBOOK
CSP A-022-PRS

SMOKE/FIRE/FUMES

SMOKE CARGO (On Ground) <0013> or <0034> or <0043> or <0053> or <0059> or <0074>

- (1) PARKING BRAKE ON

(2) EMER DEPRESS ON

(3) CARGO OFF

(4) CARGO FIREX, NORMAL
and STANDBY CARGO
SMOKE PUSH SELECT

(5) CARGO FIREX, NORMAL
and STANDBY BOTTLE
ARMED PUSH TO DISCHSELECT, to discharge

(6) Squib / bottle dischargeCHECK

(7) Passenger Evacuation
procedureAS REQUIRED

Refer to EMER 8-2

----- END -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
----------	------

AIR-CONDITIONING AND PRESSURIZATION

- **CABIN ALT** or Emergency Descent
Procedure [EMER 3-2](#)
- **DIFF PRESS** [EMER 3-4](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CABIN ALT (Warning Message) or Emergency Descent Procedure

- (1) Oxygen masks DON, SET to 100%

(2) Crew communication ESTABLISH

(3) PASS SIGNS (both) ON

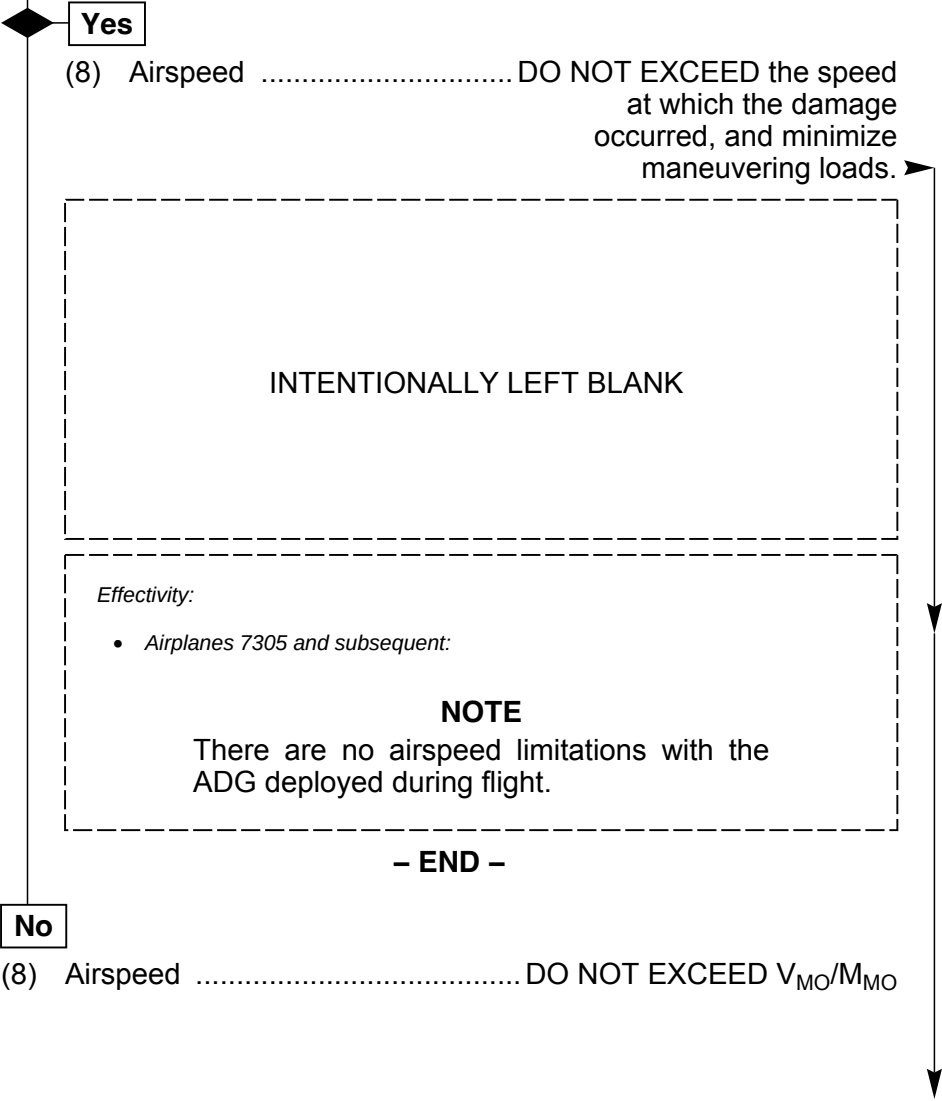
(4) Descent INITIATE to 10000 feet MSL
or lowest safe altitude,
whichever is higher.

(5) Thrust levers IDLE

(6) Flight spoilers DEPLOY

- (7) PASS OXY ON

Structural damage is suspected:



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

At a safe altitude:

(9) Oxygen and masks As required



Closing the doors on the crew mask stowage compartments and pressing RESET will stop the flow of oxygen to the masks.

NOTE

If supplemental crew oxygen is still required, setting masks to normal (N) will reduce consumption.

Door/hatch message(s) displayed/unsafe condition:

Yes

(10) Do not attempt to repressurize the airplane.

(11) Applicable door procedure ACCOMPLISH when at a safe cabin altitude.

Refer to EMER 14-2

Refer to ABNORM 5-1

– END –

No

(10) PRESS CONTMAN

(11) MAN ALT DN

(12) MAN RATE MAXIMUM INCR

Control of cabin pressurization is regained:

Yes

(13) Manual Pressurization Control ProcedureCONTINUE

Refer to ABNORM 2-11

– END –

No

(13) Unpressurized Flight Procedure (PACKs on)ACCOMPLISH

Refer to ABNORM 2-13

----- END -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

DIFF PRESS

- (1) PRESS CONTMAN
- (2) MAN ALTUP
- (3) MAN RATEINCR as required

Control of cabin pressurization is regained:

Yes

(4) Manual Pressurization Control ProcedureCONTINUE

Refer to ABNORM 2-11

- END -

- No

(4) Descent INITIATE to 10000 feet MSL or lowest safe altitude, whichever is higher.

(5) L or R PACK OFF

(6) Airplane altitude Not above 25000 feet

DIFF PRESS warning message persists:

Yes

(7) L and R PACK OFF

(8) Descent INITIATE to lowest possible altitude.

(9) Unpressurized Flight Procedure (PACKs off)ACCOMPLISH

Refer to ABNORM 2-12

- END -

- No

(7) PressurizationMONITOR

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
----------	------

AUTOMATIC FLIGHT CONTROL SYSTEM

I	• General Notes, Cautions and Warnings.....	EMER 4-2
	• AFCS MSG FAIL or Autopilot Failure	EMER 4-2
I	• Uncommanded Yaw Motion	EMER 4-2

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

General Notes, Cautions and Warnings

NOTE

After airplane generator transfers, flight control computer 2 (FCC 2) may momentarily fail and the following must be manually re-engaged:

- Yaw damper 2 (YD 2), and
- Autopilot (if it was coupled before the power interruption).

AFCS MSG FAIL or Autopilot Failure

- (1) Autopilot DISENGAGE using AP/SP DISC switch on control wheel or AP DISC switch on flight control panel.

NOTE

1. Do not use the autopilot when an **AFCS MSG FAIL** warning message is displayed.
2. RVSM operations are affected. Review the requirements. <0030>

----- END -----

Uncommanded Yaw Motion

- (1) Controls Assume manual control and counter aircraft motion using handwheel inputs.
- (2) YAW DAMPER, DISC SELECT to disconnect both yaw dampers.

NOTE

Disregard **YAW DAMPER** caution message.

Uncommanded motion persists:

- (3) AirspeedNot more than 250 KIAS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(4) Land at nearest suitable airport.

NOTE

Select the runway available with minimum cross-wind and turbulence.

Airplane handling is acceptable with flaps 45:

Yes

(5) No further action required.

– END –

No

- (5) GRND PROX, FLAP <0040> OVRD
- (6) Landing FLAPS20
- (7) Approach speedNot less than
V_{REF (FLAPS 45)} + 12 KIAS.
- (8) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.22 (22%)	1.25 (25%)
Without Thrust Reverser(s)	1.25 (25%)	1.25 (25%)

----- END -----

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
----------	------

AUXILIARY POWER UNIT

- **APU FIRE** [EMER 5-2](#)
- **APU OVERSPEED** [EMER 5-2](#)
- **APU OVERTEMP** [EMER 5-2](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

APU FIRE

- (1) APU FIRE PUSHSELECT
- After 5 seconds APU FIRE warning message persists:
- (2) APU BOTTLESELECT



Do not restart the APU.

- (3) L and R PACKsTRANSFER to engine bleeds
- (4) APU, START/STOP SELECT OFF
- (5) APU, PWR FUEL SELECT OFF
- (6) Land immediately at the nearest suitable airport.

----- END -----

APU OVERSPEED

- (1) L and R PACKsTRANSFER to engine bleeds
- (2) APU, START/STOP SELECT OFF
- (3) APU, PWR FUEL SELECT OFF



Do not restart the APU.

----- END -----

APU OVERTEMP

On the ground:



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

▼ (3) APU, PWR FUEL SELECT OFF



Do not restart the APU.

– END –

No

In flight:

(1) L and R PACKsTRANSFER to engine bleeds

APU generator is required:

◆ Yes

- (2) APU indications MONITOR
- (3) Land at the nearest suitable airport.

NOTE

Upon landing, if the APU overtemperature condition persists, the APU will shut down automatically after 60 seconds.

After landing:

- (4) APU GEN OFF
- (5) APU, START/STOP SELECT OFF
- (6) APU, PWR FUEL SELECT OFF

– END –

No

- (2) APU GEN OFF
- (3) APU, START/STOP SELECT OFF
- (4) APU, PWR FUEL SELECT OFF

----- END -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

THIS PAGE INTENTIONALLY LEFT BLANK

CONTENTS	PAGE
----------	------

BLEED AIR LEAKS

- L 10TH DUCT EMER 6-2
- R 10TH DUCT EMER 6-3
- L 14TH DUCT or R 14TH DUCT EMER 6-4

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

L 10TH DUCT

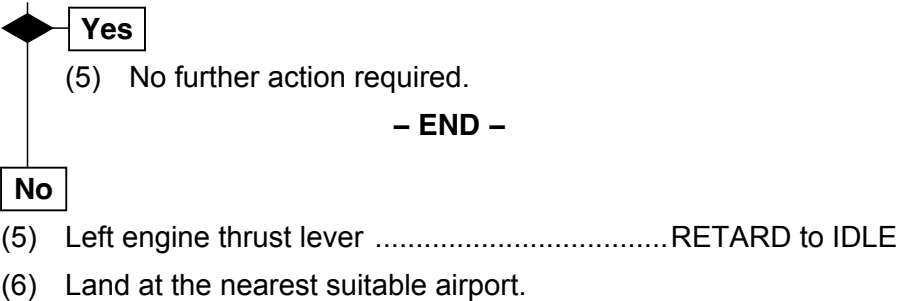
Isolate affected bleed duct:

- (1) L 10TH STAGECLOSED
- (2) APU LCVSELECT CLOSED
- (3) 10TH STAGE ISOLSELECT CLOSED
- (4) L PACK OFF

NOTE

Airplane maximum altitude during single-pack operations is 25000 feet.

After 40 seconds and L 10TH DUCT warning message goes out:



Approach and landing:

- (7) Go-around thrust reference SET N₁ bug accordingly
Refer to Go-around Thrust Settings – PERF-04.
- (8) GRND PROX, FLAP <0040> OVRD
- (9) Approach and landing FLAPS20
- (10) Final approach speed V_{REF} (FLAPS 45) + 12 KIAS
- (11) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	Runway Surface	
	Dry	Wet or Contaminated
With Thrust Reversers	1.20 (20%)	1.25 (25%)
Without Thrust Reverser(s)	1.25 (25%)	1.25 (25%)

NOTE

If a go-around is required, accomplish a Single Engine Go-around Procedure.

----- END -----

R 10TH DUCT

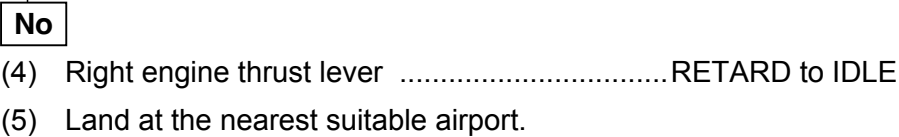
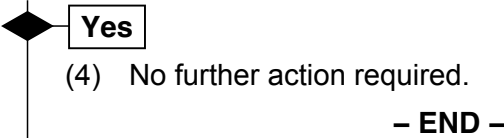
Isolate affected bleed duct:

- (1) R 10TH STAGECLOSED
- (2) 10TH STAGE ISOLSELECT CLOSED
- (3) R PACK OFF

NOTE

Airplane maximum altitude during single-pack operations is 25000 feet.

After 40 seconds and R 10TH DUCT warning message goes out:



Approach and landing:

- (6) Go-around thrust reference SET N₁ bug accordingly
Refer to Go-around Thrust Settings - PERF-04.

- (7) GRND PROX, FLAP <0040> OVRD
- (8) Approach and landing FLAPS20
- (9) Final approach speed V_{REF} (FLAPS 45) + 12 KIAS
- (10) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Thrust Reversers	1.20 (20%)	1.25 (25%)
Without Thrust Reverser(s)	1.25 (25%)	1.25 (25%)

NOTE

If a go-around is required, accomplish a Single Engine Go-around Procedure.

----- END -----

L 14TH DUCT or R 14TH DUCT

Isolate affected bleed duct:

- (1) L or R 14TH STAGE BLEEDCLOSED

NOTE

The thrust reverser is inoperative on the affected engine.

- (2) 14TH STAGE ISOL OPEN
- (3) Affected side ANTI-ICE, COWL OFF

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

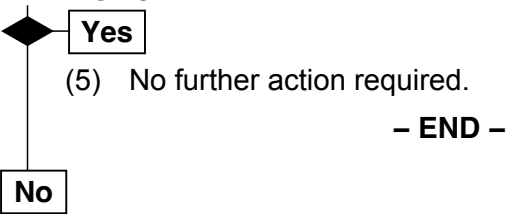
INDEX

(4) Leave icing conditions.

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is -40°C (-40°F) or below.

After 40 seconds and L 14TH DUCT or R 14TH DUCT warning message goes out:



Approach and landing:

- (7) Go-around thrust reference SET N₁ bug accordingly
Refer to Go-around Thrust
Setting - PERF-04.
- I

(8) GRND PROX, FLAP <0040> OVRD

(9) Approach and landing FLAPS20

(10) Final approach speed V_{REF} (FLAPS 45) + 12 KIAS

(11) Actual landing distance INCREASE by a factor of
1.25 (25%)

(12) Do not use the thrust reverser on the affected engine.

NOTE

If a go-around is required, accomplish a Single Engine Go-around Procedure.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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CONTENTS	PAGE
----------	------

DITCHING AND FORCED LANDING

- Recommended Configuration [EMER 7-2](#)
- Ditching or Forced Landing Imminent [EMER 7-2](#)
- Planned Ditching [EMER 7-3](#)
- Forced Landing [EMER 7-5](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Recommended Configuration

The recommended configuration for safe water landing or safe forced landing is given in the table that follows:

Parameters	Safe Water Landing	Safe Forced Landing
Approach speed	V _{REF}	V _{REF}
Descent rate (if thrust available)	200-300 fpm	200-300 fpm
Landing gear	Retracted	As required
FLAPS	45	45
If flaps 45 is not possible, use the most extended flap setting available to determine the final approach speed: • Flaps 0: V_{REF} + 30 Kts. • Flaps 8: V_{REF} + 18 Kts. • Flaps 20: V_{REF} + 12 Kts. • Flaps 30: V_{REF} + 8 Kts.		

----- END -----

Ditching or Forced Landing Imminent

- (1) L and R PACKs OFF

(2) EMER DEPRESS ON

Just before contact:

(3) EMER DEPRESS SELECT OFF

- (4) Thrust leversSHUT OFF

(5) APU, LH ENG and RH ENG FIRE
PUSHSELECT

(6) Water (or terrain) CONTACT with minimum
forward speed, but not less
than stick shaker speed, and
at minimum sink rate.

When the airplane has stopped:

- (7) APU BOTTLESELECT

(8) Both engine BOTTLESELECT

(9) Doors and overwing exits OPEN

(10) Passenger Evacuation procedureACCOMPLISH

Refer to EMER 8-2

----- END -----

Planned Ditching

- NOTE
- This procedure is intended for use where sufficient time is available.
- A. Preliminary
- (1) Descent PLAN
- (2) Crew and flight attendants ALERT and BRIEF
- (3) Air Traffic Control NOTIFY
- (4) RTU SELECT transponder code 7700
- (5) ELT <Type Spec> or <0092> ON
- (6) PASS SIGNS (both) ON
- (7) Loose equipment SECURE and STOW
- (8) Flight compartment door UNLOCKED
- (9) GND PROX WARN circuit breaker (1A15) OPEN
- (10) AUDIO WARNING (all) DISABLE
- (11) Survival equipment CHECK
- (12) Life vest, harness and belts ON and TIGHTENED

- NOTE
1. Life vests should be donned but not inflated until outside of the airplane.
2. Light plugs should be removed only if ditching at night.
- (13) Shoulder harness TIGHT and LOCKED
- (14) LDG GEAR lever UP
- (15) FLAPS 45

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

B. Approach

NOTE

If possible, ditch in the vicinity of rescue vessels, near coastlines or islands.

At approximately 2000 feet AGL:

- (1) Sea conditions and wind direction DETERMINE
- (2) Ditching heading ESTABLISH
 - Wind speed <15 kt: contact parallel to swells.
 - Wind speed between 15 and 45 kt: compromise between wind and swell.
 - Wind speed >45 kt: land into the wind.
- (3) Descent rate / approach speed ESTABLISH
- (4) L and R PACK OFF
- (5) PRESS CONT MAN
- (6) MAN ALT UP
- (7) MAN RATE MAXIMUM INCR

When the airplane is completely depressurized:

- (8) L and R 10th STAGE CLOSED
- (9) MAN ALT DN
- (10) FUEL, L and R BOOST PUMP SELECT OFF
- (11) APU (if not required) SHUT DOWN

At approximately 500 feet AGL:

- (12) Crew and flight attendants ALERT that ditching is imminent
- (13) Radio TRANSMIT final position
- (14) Brace for impact ORDER over PA system
- (15) EMER LTS ON
- (16) LANDING LTS, LEFT, NOSE and RIGHT ON

C. Prior to Water Contact

Just before contact:

- (1) Thrust levers SHUT OFF
- (2) APU, LH ENG and RH ENG FIRE PUSH SELECT

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(3) Water CONTACT with minimum forward speed, but not less than stick shaker speed, and at minimum sink rate.

D. After Water Contact

When the airplane has stopped:

- (1) APU BOTTLESELECT
- (2) Both engine BOTTLESELECT
- (3) Doors and overwing exits OPEN
- (4) Passenger Evacuation procedureACCOMPLISH

Refer to EMER 8-2

----- END -----

Forced Landing

NOTE

This procedure is intended for use where sufficient time is available.

A. Preliminary

- (1) Descent PLAN
- (2) Crew and flight attendants ALERT and BRIEF
- (3) Air Traffic Control NOTIFY
- (4) RTU SELECT transponder code 7700
- (5) ELT <Type Spec> or <0092> ON
- (6) PASS SIGNS (both) ON
- (7) Loose equipment SECURE and STOW
- (8) Flight compartment doorUNLOCKED
- (9) GND PROX WARN circuit breaker (1A15) OPEN
- (10) AUDIO WARNING (all) DISABLE
- (11) Survival equipment CHECK
- (12) Harness and belts ON and TIGHTENED
- (13) Shoulder harness TIGHT and LOCKED
- (14) LDG GEAR leverAS REQUIRED
- (15) FLAPS45

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

B. Approach

At approximately 2000 feet AGL:

- (1)

Landing area conditions and wind direction

DETERMINE
- (2)

Descent rate / approach speed

ESTABLISH
- (3)

L and R PACKs

OFF
- (4)

PRESS CONT

MAN
- (5)

MAN ALT

UP
- (6)

MAN RATE

MAXIMUM INCR

When the airplane is completely depressurized:

- (7)

L and R 10TH STAGE

CLOSED
- (8)

FUEL, L and R BOOST PUMP

SELECT OFF
- (9)

APU (if not required)

SHUT DOWN

At approximately 500 feet AGL:

- (10)

Crew and flight attendants

ALERT that forced landing is imminent
- (11)

Radio

TRANSMIT final position
- (12)

Brace for impact

ORDER over PA system
- (13)

EMER LTS

ON
- (14)

LANDING LTS, LEFT, NOSE and RIGHT

ON

C. Prior to Contact

Just before contact:

- (1)

Thrust levers

SHUT OFF
- (2)

APU, LH ENG and RH ENG FIRE PUSH

SELECT
- (3)

Terrain

CONTACT with minimum forward speed, but not less than stick shaker speed, and at minimum sink rate.

D. After Contact

When the airplane has stopped:

- (1)

APU BOTTLE

SELECT
- (2)

Both engine BOTTLE

SELECT
- (3)

Doors and overwing exits

OPEN
- (4)

Passenger Evacuation procedure

ACCOMPLISH

Refer to EMER 8-2

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
----------	------

EMERGENCY EVACUATION

- Passenger Evacuation [EMER 8-2](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Passenger Evacuation

This procedure contains the steps required to prepare for passenger evacuation and is initiated when a condition potentially endangering life or physical well-being of passengers and crew exists.

- Pilot:
- (1) PARKING BRAKE ON

(2) EvacuationCOMMAND

(3) GND LIFT DUMPINGMAN DISARM

(4) Thrust leversSHUT OFF

(5) Evacuation INITIATE using PA system

(6) APU, LH ENG and RH ENG FIRE
PUSHSELECT

(7) BATTERY MASTER OFF

- Copilot: On evacuation command
- (8) Air Traffic ControlNOTIFY of emergency
conditions and intent to
evacuate

NOTE

If ditching, disregard step (9).

(9) EMER DEPRESS ON

(10) EMER LTS (coincident with PA) ON

I

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
----------	------

ELECTRICAL

- **EMER PWR ONLY** [EMER 9-2](#)
- **Loss of All AC Power** [EMER 9-6](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

EMER PWR ONLY

- (1) STAB TRIM, CH 2SELECT
- (2) Engine instruments VERIFY N₁, N₂, ITT

Engine instruments indicate that engines are not operating:

Yes

(3) Double Engine Failure
procedureACCOMPLISH

Refer to EMER 1-5

- END -

No

(3) Cabin altitudeMONITOR

(4) DescentAS REQUIRED

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Effectivity:

Airplanes 7305 and subsequent:

(5) No airspeed limitation with the ADG deployed.

(6) GEN 1 and GEN 2 OFF/RESET then ON

(7) APU (if available, 30000 feet and
below)START

(8) APU GEN (if APU available) ON

Any generator comes on line:

Yes

Re-establish normal power:

(9) ADG manual deploy handle STOW, if ADG
deployed manually

(10) ADG CONTROL, PWR
TXFR OVERRIDESELECT

(11) CAS messages REVIEW

(12) Affected systems RESTORE

- END -

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

Continue flight on emergency power only:

- (9) Descent CONTINUE to 10000 feet
MSL or lowest safe altitude.
- (10) ANTI-ICE, WING STBY, if required
- (11) Leave icing conditions.

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is -40°C (-40°F) or below.

- (12) Land at the nearest suitable airport.

At 10000 feet MSL or lowest safe altitude:

- (13) RAM-AIR OPEN

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

NOTE

1. 250 KIAS airspeed recommended during cruise to provide sufficient airflow to passengers within cabin.
2. The following significant systems are not available when on emergency power only:
 - Both PACKs,
 - Yaw damper 2 and autopilot,
 - Flaps, anti-skid system and nosewheel steering,
 - Stabilizer trim channel 1, Mach trim, aileron and rudder trims,
 - Hydraulic pumps 1B, 2B and 3A,
 - PFD 2, MFD 2, VHF COM 2, RTU 2,
 - FD 2, VHF NAV 2, ADF 2, ATC 2, DME 1 and 2,
 - Radio altimeter 2, <0045>
 - FMS 1 and 2, <0024>
 - Copilot’s instrument lights, NAV lights, left and right landing and taxi lights,
 - ADC 2 (probe heaters inoperative) and ice detector 2,
 - Windshield wipers, both windshield heaters and right window heater.
3. If the ADG is not operating properly and the airplane is on battery power only, then all electrical power may be lost after 30 minutes.
4. Hydraulic pump 3B will be inoperative when on battery power only.

Prior to landing:

- (14) ADG manual deploy handlePULL
- (15) ANTI-SKID OFF
- (16) N/W STRG OFF
- (17) GRND PROX, FLAP <0040> OVRD

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

(18) Final approach speed $V_{REF} \text{ (FLAPS 45)} + \Delta V_{REF}$

Flaps Setting (Degrees)	$\Delta V_{REF} \text{ (kt)}$
0	30
8	18
20	12
30	8

(19) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Final Approach Speed $\Delta V_{REF} \text{ (kt)}$	Runway Surface	
		Dry	Wet or Contaminated
With Two Thrust Reversers	0	2.60 (160%)	3.30 (230%)
	8	2.40 (140%)	3.00 (200%)
	20	2.10 (110%)	2.65 (165%)
	30	2.00 (100%)	2.55 (155%)
	45	1.80 (80%)	2.20 (120%)
Without Thrust Reverser(s)	0	3.30 (230%)	3.30 (230%)
	8	3.00 (200%)	3.00 (200%)
	20	2.65 (165%)	2.65 (165%)
	30	2.55 (155%)	2.55 (155%)
	45	2.20 (120%)	2.20 (120%)

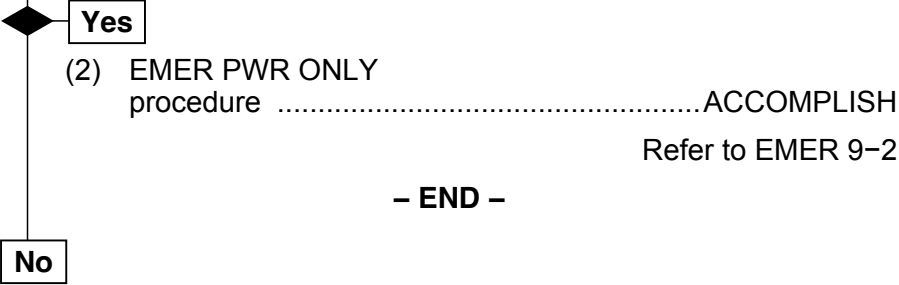
----- END -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Loss of All AC Power

(1) ADG manual deploy handlePULL

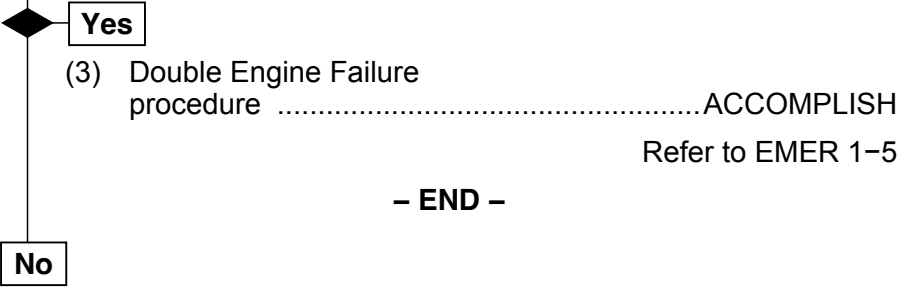
On emergency power:



Battery power only persists:

(2) Engine instrumentsVERIFY N₁, N₂, ITT

Engine instruments indicate that engines are not operating:



(3) Cabin altitudeMONITOR

(4) DescentAS REQUIRED

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Effectivity:

- Airplanes 7305 and subsequent:

(5) No airspeed limitation with the ADG deployed.

(6) GEN 1 and GEN 2OFF/RESET then ON

(7) APU (if available, 30000 feet and below)START

(8) APU GEN (if APU available)ON

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Any generator comes on-line:

Yes

Re-establish normal power:

- (9) ADG manual deploy handleSTOW
- (10) ADG CONTROL, PWR
TXFR OVERRIDESELECT
- (11) CAS messages REVIEW
- (12) Affected systems RESTORE

NOTE

IRU 2 may have to be re-aligned in ATT mode: <0025>

- Maintain aircraft in straight and level flight for 30 seconds,
- Select FMS INDEX.
- Select IRU Control page,
- Enter current heading, IRS should align within 34 seconds.

– END –

No

Continue flight on battery power only:

NOTE

Electrical power may be lost after 30 minutes.

- (9) DescentCONTINUE to 10000 feet
MSL or lowest safe altitude.
- (10) ANTI-ICE, WING STBY, if required
- (11) Leave icing conditions.

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is –40°C (–40°F) or below.

- (12) Land at the nearest suitable airport.

At 10000 feet MSL or lowest safe altitude:

- (13) RAM-AIR OPEN

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

NOTE

1. 250 KIAS airspeed recommended during cruise to provide sufficient airflow to passengers within cabin.
2. The following significant systems are not available when on battery power only:
 - Both PACKs,
 - Yaw damper 2 and autopilot,
 - Flaps, anti-skid system and nosewheel steering,
 - Both stabilizer trim channels, Mach trim, aileron and rudder trims,
 - Normal landing gear extension,
 - Hydraulic pumps 1B, 2B 3A and 3B,
 - PFD 2, MFD 2, VHF COM 2, RTU 2,
 - FD 2, VHF NAV 2, ADF 2, ATC 2, DME 1 and 2,
 - Radio altimeter 2, <0045>
 - FMS 1 and 2, <0024>
 - Copilot’s instrument lights, NAV lights, landing and taxi lights,
 - ADC 2 (probe heaters inoperative) and ice detectors,
 - Windshield wipers, both windshield heaters and both window heaters.

Prior to landing:

- (14) ANTI-SKID OFF
- (15) N/W STRG OFF
- (16) GRND PROX, FLAP <0040> OVRD
- (17) Final approach speed $V_{REF (FLAPS 45)} + \Delta V_{REF}$

Flaps Setting (Degrees)	ΔV_{REF} (kt)
0	30
8	18
20	12
30	8

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

(18) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Flap Setting (Degrees)	Runway Surface	
		Dry	Wet or Contaminated
With Two Thrust Reversers	0	2.60 (160%)	3.30 (230%)
	8	2.40 (140%)	3.00 (200%)
	20	2.10 (110%)	2.65 (165%)
	30	2.00 (100%)	2.55 (155%)
	45	1.80 (80%)	2.20 (120%)
Without Thrust Reverser(s)	0	3.30 (230%)	3.30 (230%)
	8	3.00 (200%)	3.00 (200%)
	20	2.65 (165%)	2.65 (165%)
	30	2.55 (155%)	2.55 (155%)
	45	2.20 (120%)	2.20 (120%)

(19) Landing Gear Manual Extension
ProcedureACCOMPLISH

Refer to ABNORM 13-7

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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CONTENTS	PAGE
----------	------

FLIGHT CONTROLS

- Aileron System Jammed [EMER 10-2](#)
- Elevator System Jammed [EMER 10-4](#)
- Rudder System Jammed [EMER 10-5](#)
- Stabilizer Trim Runaway [EMER 10-7](#)
- Stall Recovery [EMER 10-9](#)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Aileron System Jammed

- (1) Autopilot DISENGAGE

(2) Aileron controls (both) RELEASE pressure

(3) ROLL DISC PULL and TURN to lock

(4) Airplane control TRANSFER to pilot with
operative aileron

NOTE

Roll controllability is reduced. The roll disconnect will result in half feel during airplane handling.

- (5) PLT ROLL or CPLT ROLL SELECT operative side

NOTE

1. If the PLT ROLL or CPLT ROLL switch is not selected within 20 seconds of pulling the ROLL DISC handle the **SPOILERONS ROLL** caution message will come on.

2. Use aileron trim to compensate if the aileron jam has occurred away from neutral.

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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(6) Land at the nearest suitable airport.

NOTE

Select the longest runway available with minimum turbulence and cross-wind.

Prior to landing:

- (7) GRND PROX, FLAP <0040> OVRD
- (8) Landing FLAPS20
- (9) Approach speedNot less than
V_{REF} (FLAPS 45) + 12 KIAS
- (10) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



EMER 10-4

Rev. 60, Nov 30/2012

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.20 (20%)	1.25 (25%)
Without Thrust Reverser(s)	1.25 (25%)	1.25 (25%)

----- END -----

Elevator System Jammed

- (1) Autopilot DISENGAGE

(2) Elevator controls (both) RELEASE differential pressure

(3) PITCH DISC PULL and TURN to lock

(4) Airplane control TRANSFER to pilot with operative elevator

NOTE

1. Pitch controllability is reduced. The pitch disconnect will result in half feel during airplane handling.

2. Stick pusher is inoperative, if the right side is jammed.

- (5) Land at the nearest suitable airport.

NOTE

Select the longest runway available with minimum turbulence and cross-wind.

Prior to landing:

- (6) GRND PROX, FLAP <0040> OVRD

(7) Landing FLAPS20

(8) Approach speedNot less than
V_{REF} (FLAPS 45) + 12 KIAS

QUICK REFERENCE
HANDBOOK
CSP A-022-PRS

FLIGHT CONTROLS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

I

	EMER 10-5
	Rev. 60, Nov 30/2012

(9) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.20 (20%)	1.25 (25%)
Without Thrust Reverser(s)	1.25 (25%)	1.25 (25%)

----- END -----

Rudder System Jammed

(1) YAW DAMPER, DISCSELECT
(2) Rudder pedalsOVERPOWER

NOTE

- 1. Rudder may be available at higher pedal forces.
- 2. Limited travel may be available through the use of the rudder trim.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

QUICK REFERENCE HANDBOOK CSP A-022-PRS	FLIGHT CONTROLS
---	------------------------

(3) Land at the nearest suitable airport. Select the longest runway available with minimum cross-wind and turbulence.

NOTE

- 1. If the rudder is jammed out of the neutral position, use aileron and differential thrust to maintain straight flight until touchdown.
- 2. After touchdown, use nosewheel steering and, if necessary, differential braking to maintain directional control, as the airplane will turn in the direction of the jammed rudder.

----- **END** -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Stabilizer Trim Runaway

- (1) Control wheel

ASSUME manual control and
OVERRIDE runaway
- (2) Both STAB TRIM DISC

PRESS, HOLD and
RELEASE

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- (5) Airspeed

ADJUST to minimize
out-of-trim condition

NOTE

1. If a pull force is required, increase
airspeed. If a push force is required,
decrease airspeed.
2. Do not exceed V_{MO} / M_{MO} or minimum
maneuvering speed as appropriate.
3. Flight spoilers will reduce airspeed and
provide nose-up pitch.
4. Reducing thrust will reduce airspeed and
provide nose-up pitch.
5. Autopilot, stabilizer trim and Mach trim are
not available.

- (6) PASS SIGN(both)

ON
- (7) Land at the nearest suitable airport.

Prior to landing:

- (8) GRND PROX, FLAP <0040>

OVRD

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(9) Recommended landing FLAPS20 consistent with acceptable forces

NOTE

Control column force gradient is significantly reduced with flap deployment from FLAP 0. The combined effects of reduced approach airspeed, flap configuration and changes to the pitch force gradient with flap selection, make FLAP 20 the recommended landing configuration for both a nose-up and nose-down out-of-trim condition.



If control forces are excessive, some relief may be gained by adjusting airspeed or flap position. Increased flap settings generally reduce pull forces for a given airspeed. Decreased airspeed generally reduces push forces for a given flap configuration.

- (10) Approach speedNot less than
 $V_{REF} \text{ (FLAPS 45)} + \Delta V_{REF}$
- (11) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Flaps Setting (Degrees)	ΔV_{REF} (kt)	Runway Surface	
			Dry	Wet or Contaminated
With Two Thrust Reversers	0	30	1.65 (65%)	1.70 (70%)
	8	18	1.50 (50%)	1.55 (55%)
	20	12	1.20 (20%)	1.25 (25%)
	30	8	1.15 (15%)	1.20 (20%)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	Flaps Setting (Degrees)	ΔV_{REF} (kt)	Runway Surface	
			Dry	Wet or Contaminated
Without Thrust Reverser(s)	0	30	1.70 (70%)	1.70 (70%)
	8	18	1.55 (55%)	1.55 (55%)
	20	12	1.25 (25%)	1.25 (25%)
	30	8	1.20 (20%)	1.20 (20%)

WARNING

In the event of a go-around, increasing thrust will increase nose-down forces. Increase thrust slowly to avoid excessive pitch forces.

----- END -----

Stall Recovery

- (1) Autopilot Disengage, if required

(2) Pitch attitudeLOWER nose to reduce angle of attack

(3) Thrust levers Advance to MAX POWER

(4) Roll attitude Wings level

(5) FLIGHT SPOILER leverRETRACT

After airspeed increases and stall warning goes out:

- (6) Pitch attitude Adjust to minimize altitude loss

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(7) Thrust levers and aircraft configurationADJUST as required.

NOTE

- 1. It is essential that the AOA be immediately reduced, even if this means a loss of altitude.
- 2. Avoid abrupt or aggressive pitch control inputs during recovery. Inappropriate recovery inputs can result in a secondary stall.
- 3. Height loss resulting from high AOA recovery, especially at cruise altitude and/or low initial thrust conditions, can be significant.

----- **END** -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
----------	------

FUEL

- Low Fuel [EMER 11-2](#)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Low Fuel

Quantity in either main tank is less than 204 kg (450 lb) or the total fuel quantity is less than 408 kg (900 lb):

- (1) Airplane attitude Not more than 10 degrees nose up
- (2) Fuel balance CHECK
- (3) Land at the nearest suitable airport.

NOTE

- 1. Do not attempt go-around.
- 2. The minimum fuel quantity for go-around is 204 kg (450 lb) per wing (with the airplane level) and assuming a maximum airplane climb attitude of 10 degrees nose up.

----- **END** -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
----------	------

ICE AND RAIN PROTECTION

- WING OVHT [EMER 12-2](#)
- ANTI-ICE DUCT [EMER 12-3](#)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

WING OVHT

(1) ANTI-ICE, WING OFF

After 40 seconds, WING OVHT warning message persists:

Yes

(2) Leave icing conditions

– END –

No

(2) Leave Icing conditions.

Within 40 seconds, WING OVHT warning message goes out:

Yes

(2) ANTI-ICE, WING STBY

– END –

No

(2) Leave Icing conditions.

WING OVHT warning message re-occurs:

Yes

(3) Affected side DETERMINE

(4) L or R 14TH STAGE Affected side CLOSED

NOTE

The thrust reverser is inoperative on the affected engine.

(5) 14TH STAGE ISOL OPEN

– END –

No

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

After 40 seconds, WING OVHT warning message persists:

- Yes

(6) ANTI-ICE, WING OFF

(7) Leave icing conditions.

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is -40°C (-40°F) or below.

– END –

- No

(6) No further action required.

----- END -----

ANTI-ICE DUCT

- (1) ANTI-ICE, WING OFF
- (2) L and R 14TH STAGECLOSED
- (3) ANTI-ICE, LH and RH COWL OFF
- (4) Leave icing conditions.

WARNING

Even small accumulations of ice on the wing leading edge can change the stall speed or stall characteristics or the warning margins provided by the stall protection system.

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is -40°C (-40°F) or below.

After leaving icing conditions, ice accumulation is observed on the heated portion of the wing leading edge:

Yes

(5) Airspeed INCREASE to V_{MO}/M_{MO} to disperse ice, if possible.

Ice accumulation on wing leading edge persists:

Yes

(6) L and R 14TH STAGESELECT open

(7) ANTI-ICE, WING STBY

When ANTI-ICE DUCT warning message comes on:

(8) ANTI-ICE, WING OFF

(9) L and R 14TH STAGECLOSED

Repeat steps (6) to (9), as required (maximum 5 times).

Ice accumulation on the wing leading edge still persists:

Yes

(10) Maneuvering speedNot less than 200 KIAS

Prior to landing:

(11) GRND PROX, FLAP <0040> OVRD

(12) Landing FLAPS20

(13) Approach speedNot less than $V_{REF (FLAPS 45)} + 30$ KIAS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

NOTE

The landing distance factors that follow are based upon an ice accumulation and FLAPS 20 landing.

(14) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
Without Thrust Reverser(s)	1.65 (65%)	1.65 (65%)

NOTE

Thrust reversers are inoperative when L and R 14 STAGE SOVs are CLOSED.

– END –

No

(5) No further action required.

----- END -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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CONTENTS	PAGE
----------	------

LANDING GEAR, WHEEL AND BRAKE SYSTEM

BRAKES

- **BRAKE OVHT** EMER 13-2
- **PARKING BRAKE** EMER 13-11
- During Landing – Excessive Asymmetry or Loss of Braking EMER 13-12

LANDING GEAR

- **MLG BAY OVHT** EMER 13-2
- **GEAR DISAGREE** EMER 13-3
- LDG GEAR Lever Jammed in the UP Position EMER 13-7
- Landing Gear Up / Unsafe Landing Procedure EMER 13-9

NOSE LANDING GEAR

- **NOSE DOOR OPEN** EMER 13-12

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

MLG BAY OVHT

- (1) AirspeedNot more than 250 KIAS

(2) LDG GEAR lever DN

- (3) BTMS indicatorsMONITOR

MLG BAY OVHT warning message goes out and all BTMS indicators are displaying 06 or less:

Yes

(4) BTMS OVHT WARN RESETSELECT

(5) AirspeedNot more than 200 KIAS

(6) LDG GEAR leverUP

- END -

No

(4) Land at the nearest suitable airport.

----- END -----

BRAKE OVHT

During flight:

- (1) AirspeedNot more than 250 KIAS

(2) LDG GEAR lever DN

- (3) BTMS indicatorsMONITOR

All BTMS indicators are decreasing:

Yes

(4) BTMS indicators MONITOR until 10 or less

(5) BTMS OVHT WARN RESETSELECT

(6) BTMS indicators MONITOR until 06 or less

(7) AirspeedNot more than 200 KIAS

(8) LDG GEAR leverUP

- END -

No

(4) Land at the nearest suitable airport.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

On the ground:

- (1) Brakes MINIMIZE

NOTE

Use thrust reversers or shut down one engine to minimize brake usage.

- (2) Do not take-off.



Suspect fire. There is a potential that the wheel fuse plug may release. Keep personnel clear of the wheel bay area until the brake temperatures have cooled down.

----- END -----

GEAR DISAGREE

- If Gear Disagree (after selecting LDG GEAR Lever UP), proceed to step (1).
- If Gear Disagree (after Selecting LDG GEAR Lever DN), proceed to step (11).

Gear Disagree (after selecting LDG GEAR lever UP)

- (1) AirspeedNot more than 200 KIAS
- (2) HYDRAULIC 2 and 3B pump ON

GEAR DISAGREE warning message goes out:

- (3) No further action required.

– END –

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

GEAR DISAGREE warning message persists:



1. Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.
2. Nosewheel steering may not be available upon landing, even if the **STEERING INOP** caution message is not displayed.

NOTE

The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 lbs. The flight crew may choose to reposition the seat prior to manual release handle deployment. In safe flight condition, the flight crew may also choose to stand up to pull the manual release handle.

- (4) LANDING GEAR MANUAL
RELEASE handlePULL to full extension

NOTE

The **NOSE DOOR OPEN** warning message is displayed when the LANDING GEAR MANUAL RELEASE handle is pulled.

- (5) LDG GEAR lever DN

GEAR DISAGREE warning message goes out:

- (6) Proceed to step (8).

GEAR DISAGREE warning message still persists:

- (7) LANDING GEAR MANUAL
RELEASE handle STOW and PULL to full
extension

NOTE

Step (7) may be repeated as many times as desired.

GEAR DISAGREE warning message goes out:

- (8) LANDING GEAR MANUAL
RELEASE handleSTOW

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

GEAR DISAGREE warning message remains out:

(9) Land at the nearest suitable airport.

– END –

GEAR DISAGREE warning message still persists:

(10) Proceed to step (22).

Gear Disagree (after selecting LDG GEAR lever DN)

(11) AirspeedNot more than 200 KIAS

(12) HYDRAULIC 2 and 3B pump ON

GEAR DISAGREE warning message goes out:

(13) No further action required.

– END –

GEAR DISAGREE warning message persists:

(14) LDG GEAR leverUP

Wait 30 seconds, then:

(15) LDG GEAR lever DN

NOTE

- Steps (14) and (15) may be repeated as many times as desired.
- Quickly cycling the landing gear lever UP then DN may also provide a down and locked indication. For the quick cycling, do not wait for the landing gear UP indications before reselecting the landing gear lever down.

GEAR DISAGREE warning message goes out:

(16) Land at the nearest suitable airport.

– END –

GEAR DISAGREE warning message still persists:

(17) LDG GEAR leverUP

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Wait 30 seconds, then:



1.

Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.
2.

Nosewheel steering may not be available upon landing, even if the **STEERING INOP** caution message is not displayed.

NOTE

The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 lbs. The flight crew may choose to reposition the seat prior to manual release handle deployment. In safe flight condition, the flight crew may also choose to stand up to pull the manual release handle.

(18) LANDING GEAR MANUAL
RELEASE handlePULL to full extension

NOTE

The **NOSE DOOR OPEN** warning message is displayed when the LANDING GEAR MANUAL RELEASE handle is pulled.

(19) LDG GEAR lever DN

GEAR DISAGREE warning message goes out:

(20) LANDING GEAR MANUAL
RELEASE handleSTOW



In some cases, stowing the LANDING GEAR MANUAL RELEASE handle may cause the main or nose gear to retract. If this occurs, the LANDING GEAR MANUAL RELEASE handle must again be pulled to full extension and left in this position until the landing gear locking pins have been installed.

(21) Land at the nearest suitable airport.

– END –

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

GEAR DISAGREE warning message still persists:

(22) HYDRAULIC 3A and 3B pump OFF

When hydraulic system 3 pressure is less than 200 psi:

(23) LANDING GEAR MANUAL
RELEASE handle STOW and PULL to full
extension

(24) LANDING GEAR MANUAL
RELEASE handle STOW

(25) LDG GEAR lever DN

(26) HYDRAULIC 3A and 3B pump ON

GEAR DISAGREE warning message goes out:

(27) Land at the nearest suitable airport.

– END –

GEAR DISAGREE warning message still persists:

(28) Landing Gear Up / Unsafe
Landing procedure ACCOMPLISH

Refer to EMER 13-9

----- END -----

LDG GEAR Lever Jammed in the UP Position

(1) Airspeed Not more than 250 KIAS

(2) HYDRAULIC 2 and 3B pump ON



Failure to pull the LANDING GEAR MANUAL
RELEASE handle to its full extension may
prevent successful landing gear extension.

NOTE

The force required to operate the LANDING
GEAR MANUAL RELEASE handle is greater
than 40 lbs. The flight crew may choose to
reposition the seat prior to manual release
handle deployment. In safe flight condition, the
flight crew may also choose to stand up to pull
the manual release handle.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- SUPP

LIMITS

- PERF

- MISC

- ## PLAN

- INTERMIX**

EICAS

- EMER

- IMM

ABNORM

- ## INDEX

Step (7) may be repeated as many times as desired.

- If landing gear is down-and-locked, proceed to step (9).

- Landing gear is down-and-locked:**

- (9) No further action required.

Any landing gear still fails to down-and-lock:

- (10) Landing Gear Up / Unsafe
Landing procedureACCOMPLISH
- Refer to EMER 13-9

Landing Gear Up / Unsafe Landing Procedure

A. Preliminary

- (1) Descent PLAN

NOTE

Reduce fuel to the minimum, if possible, while retaining sufficient fuel for a controlled, powered approach.

B. Preparation

NOTE

1. **If one main landing gear is up or unsafe**, hold applicable wing up for as long as possible. Maintain directional control with rudder and nosewheel steering (if considered safe). When wing touches the ground, apply asymmetrical braking for directional control.
2. **If nose landing gear is up or unsafe**, trim stabilizer nose-up after touchdown. Gently lower the nose before elevator effectiveness is lost.
3. **If all wheels are up or unsafe**, perform a nose high attitude touchdown but do not reduce touchdown speed below stick shaker speed.
4. **If both main landing gear cannot be locked down**, consideration should be given to landing with all wheels up.

- (1) Crew and flight attendants ALERT and BRIEF

NOTE

The briefing should include the type of emergency, time available, airplane attitude after landing and exits available for use.

- (2) Air Traffic Control NOTIFY



EMER 13-10

Rev. 76, Jul 02/2021

- (3) PASS SIGNS (both) ON
- (4) Loose equipmentSECURE
- (5) GND PROX WARN circuit breaker
(1A15) OPEN
- (6) AUDIO WARNING (all) DISABLE

NOTE

Radio altitude callouts are not available.

- (7) Flight compartment doorLATCH OPEN
- (8) Shoulder harness TIGHT and LOCKED
- (9) Plan to land with FLAPS 45.

C. Approach

- (1) L and R PACK OFF
- (2) PRESS CONTMAN
- (3) MAN ALTUP

When the airplane is completely depressurized:

- (4) L and R 10TH STAGECLOSED
- (5) EMER LTS ON

At approximately 500 feet AGL:

- (6) Brace for impact ORDER over PA system

BEFORE TOUCHDOWN

- (1) APU FIRE PUSHSELECT
- (2) Airplane attitude MAINTAIN nose high attitude

NOTE

Ground/landing field contact should be accomplished using minimum forward speed, but not less than stick shaker speed, and at a minimum sink rate.

D. After Landing

Landing gear has collapsed or failed to extend:



QUICK REFERENCE
HANDBOOK
CSP A-022-PRS

LANDING GEAR, WHEEL AND
BRAKE SYSTEM

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



- (3) Passenger Evacuation
procedure AS REQUIRED

Refer to EMER 8-2

– END –

No

- (1) HYDRAULIC 2 and 3B pump
(both) ON
- (2) Landing gear locking pins INSTALL

– – – – END – – – –

PARKING BRAKE

- (1) PARKING BRAKE handle CHECK released

PARKING BRAKE warning message persists:

Yes

Prior to landing:

NOTE

The following landing distance factors are
based upon both anti-skid systems
inoperative.

- (2) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45
actual landing distances for either Dry
surfaces in PERF-04 or to Wet or
Contaminated surfaces in PERF-06.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.75 (75%)	1.95 (95%)
Without Thrust Reverser(s)	1.95 (95%)	1.95 (95%)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

– END –

No

(2) No further action required.

----- END -----

During Landing – Excessive Asymmetry or Loss of Braking

- (1) Wheel brakes RELEASE momentarily
- (2) ANTI-SKID OFF
- (3) Wheel brakes RE-APPLY, as required



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

----- END -----

NOSE DOOR OPEN

- (1) Airspeed Not more than 250 KIAS
- (2) LDG GEAR lever DN
- (3) Land at the nearest suitable airport.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
----------	------

PASSENGER DOOR

- **PASSENGER DOOR** [EMER 14-2](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

PASSENGER DOOR

WARNING

Door failure may be indicated by a loud noise, pressurization leak or rumble emanating from the door area. If any of these indications are present, do not approach the door.

- (1) PASS SIGNS (both) ON

(2) DescentINITIATE to 10000 feet or lowest safe altitude, as soon as possible.

NOTE

1. Prepare to land at the nearest suitable airport.

2. Accomplish CABIN ALT or Emergency Descent Procedure, if required. Refer to EMER 3-2.

- (3) MAN ALTCHECK mid-position / centre

(4) PRESS CONTMAN

(5) MAN ALTUP

(6) MAN RATE MAXIMUM INCR

(7) Land at the nearest suitable airport.

NOTE

After landing, ensure the airplane is completely depressurized before opening any airplane doors.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
----------	------

AURAL/VISUAL WARNING SYSTEM

- Configuration Warning [EMER 15-2](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Configuration Warning

(1) Take-offDISCONTINUE IMMEDIATELY

NOTE

Check the position/status of the following:

- Ailerons trim,
- Autopilot,
- Flaps,
- Rudder trim,
- Spoilers,
- Stabilizer,
- Parking brake.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
----------	------

HYDRAULIC POWER

- This Section Not Applicable [EMER 16-2](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

THIS SECTION NOT APPLICABLE

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
----------	------

UNRELIABLE AIRSPEED

- Unreliable Airspeed In-flight [EMER 17-2](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Unreliable Airspeed In-flight

- (1) Autopilot DISENGAGE

(2) FDs DESELECT

(3) Use standby attitude indicator / ISI for pitch reference.

Initial take-off climb or go-around is required:

(4) Pitch/N₁ 10 degrees/TOGA from SL to 15000 ft; 5 degrees/CLB above 15000 ft

(5) Aircraft configuration At clean up altitude Gear UP / FLAPS 0

(6) Airplane altitude MAINTAIN lowest safe altitude or higher



Respect stall warning/stick shaker.

- (7) Set and monitor pitch and roll using PFD. <0025>

At desired altitude:

- (8) Pitch/N₁ Set as per Table A – Level Flight – FLAPS 0

(9) STALL PTCT, PUSHER (left or right) OFF



1. Do not use any FD/autopilot SPEED mode.

2. Do not use flight spoilers (flight spoiler retraction may be asymmetric).

- (10) Flight Director PTCH/ROLL or HDG modes

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(11) AutopilotENGAGE (if available)

NOTE

1.

Autopilot may not be available.
2.

No crew action is required for the following messages:

• Caution messages: **EFIS COMP MON** with amber IAS flag, **EFIS COMP INOP** with red IAS flag, **STALL FAIL**, **MACH TRIM**.
3.

Airspeed Trend Vector, Ground Speed indications and FMS time and fuel prediction information may not be reliable.
4.

ILS/LOC/VOR course deviation indicator information are reliable.
5.

Disengage MACH TRIM if erratic stabilizer motion is observed.
6.

Select the affected AUDIO WARNING switch to DISABLE if overspeed warning sounds.
7.

Altitude, Mode C and TCAS may be in error by up to 600 feet.

(12) Leave current environmental conditions if they are considered to be a reason for unreliable airspeed. Refer to “PITCH / THRUST TARGETS” Tables B or C as required for climb or descent.

(13) ANTI-ICE, LH and RH COWLAS REQUIRED

(14) ANTI-ICE, WINGAS REQUIRED

Evaluate indicated airspeed sources as following:



Airspeed should be considered reliable only if the following conditions exist:

1.

Airspeed indication is consistent with pitch, thrust and airspeed range from “PITCH / THRUST TARGETS” Table A, B, C or D.
2.

Airspeed indication changes and speed trend vector is consistent with pitch and thrust adjustments.

(15) AIR DATA source selectorSELECT to NORM

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(16) Both PFDs and standby airspeed indicator / ISICOMPARE airspeed to expected speed range from Table A.

- (17) Determine which of the following conditions apply:
- If no indicated airspeed is considered reliable, proceed to step (18).
 - If one or more airspeed source is considered reliable, proceed to step (23).

(18) **NO RELIABLE AIRSPEED SOURCE**

(19) Airspeed Indications DISREGARD

(20) Pitch/N₁Set using Table A, B, C, D, or E as appropriate for phase of flight.

(21) Land at the nearest suitable airport.

(22) When ready to start descent, proceed to step (29).

(23) **ONE OR MORE AIRSPEED SOURCE IS CONSIDERED RELIABLE**

(24) AIR DATA source selectorCONFIRM NORM

(25) Reliable airspeed source MONITOR WITH CAUTION

(26) Pitch/N₁ Set using Table A, B, C, D or E as appropriate for phase of flight and continue to monitor reliable airspeed source and airspeed range.



1. Disregard airspeed indication if at any time it becomes unreliable.
2. AIR DATA source selector must remain in NORM position.

(27) Land at the nearest suitable airport.

(28) When ready to start descent, proceed to step (29).

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

(29) Descent INITIATE when ready



To avoid significant flap overspeed and flap damage:

1. Transition from FLAPS 0 to FLAPS 30 must be accomplished in level flight.
2. Select FLAPS 45 at the start of final descent on the glideslope/glidepath.

NOTE

Plan to intercept a long final to allow time to re-configure for approach/landing and set pitch/N₁.

(30) Unreliable aircraft systems REVIEW

- Enhanced ground proximity warning, <0040>
- Windshear detection,
- Landing gear warning horn,
- Flap overspeed,
- **NOSE DOOR OPEN** warning message
- Automatic function of passenger signs.

(31) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAPS 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.20 (20%)	1.25 (25%)
Without Thrust Reverser(s)	1.25 (25%)	1.25 (25%)

Flight Technique:

Set thrust and pitch to target, then adjust pitch to maintain required altitude or path:

- Allow sufficient time to stabilize altitude or path between pitch adjustments.
- If still descending below desired altitude or path, INCREASE pitch using 0.5-degree increments.
- If still climbing above desired altitude or path, DECREASE pitch using 0.5-degree increments.
- If displayed, VSI can be used to monitor required altitude or path.

NOTE

1. Initial pitch target is for reference only. See Flight Technique to adjust pitch as required up to a maximum of +/- 1.5 degrees.
2. If pitch is increased, REDUCE expected speed range by ΔKIAS/ΔMach.
3. If pitch is decreased, INCREASE expected speed range by ΔKIAS/ΔMachs.

Table A – LEVEL FLIGHT – FLAPS 0						
ALT (ft)	EXPECTED SPD RANGE KIAS/Mach	EXPECTED SPD CHANGE FOR 0.5 DEGREE CHANGE IN PITCH ΔKIAS/ΔMach	Weight lb Pitch (degree) / Initial Thrust (%NT)			
			35000	40000	45000	50000
39000	220 – 230/ .71 – .75	5/0.01	1.0 / 89.3	1.5 / 91.8	-- / --	-- / --
37000	225 – 250/ .70 – .78	10/0.03	1.0 / 88.0	1.0 / 90.0	1.5 / 92.4	2.0 / 95.9
35000	235 – 265/ .70 – .78	5/0.01	0.5 / 87.5	1.0 / 89.0	1.5 / 90.9	1.5 / 93.2
30000	255 – 285	10	0.0 / 85.8	0.5 / 86.8	1.0 / 87.9	1.0 / 89.3
25000	265 – 285	10	0.0 / 82.4	0.5 / 83.1	1.0 / 83.9	1.0 / 84.9
20000	265 – 285	10	0.0 / 79.6	0.5 / 80.3	1.0 / 81.0	1.0 / 81.8
15000*	255 – 285	10	0.0 / 76.3	0.5 / 76.9	1.0 / 77.5	1.0 / 78.3
15000**	200 – 225	5	1.5 / 67.1	2.0 / 68.4	3.0 / 69.7	3.5 / 71.1
10000	200 – 225	5	1.5 / 63.4	2.0 / 64.5	3.0 / 65.8	3.5 / 67.1
5000	205 – 225	5	1.5 / 59.8	2.0 / 60.9	3.0 / 62.0	3.5 / 63.2
0	205 – 225	5	1.5 / 56.5	2.0 / 57.6	3.0 / 58.6	3.5 / 57.9
If any indicated speed is outside the expected indicated airspeed range, it should be considered UNRELIABLE.						

* Pitch/N; settings for cruise phase

** Pitch/N; settings for transition to approach phase

Interpolation is acceptable

INDEX

ABNORM

IMM

EMER

EICAS

INTERMIX

PLAN

MISC

PERF

LIMITS

SUPP

Flight Technique:

- 1. Use FD in PITCH mode only.
- 2. Use AP if available, otherwise handfly airplane to the desired altitude

CAUTION

Do not use any AP/FD SPEED mode.

Table B – CLIMB – FLAPS 0			
ALT (ft)	Pitch	N1%	EXPECTED SPD RANGE KIAS/Mach
SL	10.0	TOGA	217 – 284
5000	8.0	TOGA	232 – 278
10000	6.0	CLB	236 – 279
15000	4.0	CLB	251 – 284
20000	2.5	CLB	258 – 286
25000	2.0	CLB	251 – 282
30000	1.0	CLB	254 – 283 / .67 – .75
35000	0.5	CLB	249 – 277 / .73 – .81

CAUTION

Do not attempt to climb above 30,000ft for aircraft weights above 38,000lbs.

Flight Technique:

1. Use FD in PITCH mode only.

2. Use AP if available, otherwise handfly airplane to the desired altitude

CAUTION

Do not use any AP/FD SPEED mode.

Table C - DESCENT - FLAPS 0 - IDLE			
ALT (ft)	Pitch	N1%	EXPECTED SPD RANGE KIAS/Mach
40000	- 1.5	IDLE	219 - 235 / .73 - .78
35000	- 2.5	IDLE	228 - 272 / .68 - .80
30000	- 3.0	IDLE	235 - 297 / .66 - .81
25000	- 3.5	IDLE	227 - 294
20000	- 2.0	IDLE	213 - 275
15000	- 1.5	IDLE	206 - 260
10000	- 1.5	IDLE	201 - 255
5000	- 1.5	IDLE	198 - 252
0	- 1.0	IDLE	193 - 243

NOTE

- If anti-ice is required, adjust thrust as required to maintain a minimum of 75% N₂.
- For an N1 increase of 5%, add 0.5 degree to the pitch from Table C.
 - For an N1 increase of 10%, add 1.0 degree to the pitch from Table C.
 - For an N1 increase of 15%, add 1.5 degree to the pitch from Table C.

Flight Technique:

Allow sufficient time to stabilize altitude or path between pitch adjustments or configuration change.
If displayed, VSI can be used to monitor required altitude or path.

For LEVEL FLIGHT:

Set N1 and pitch to target, then adjust pitch to maintain required altitude:

- If still descending below desired altitude or path, INCREASE pitch using 0.5-degree increments.
- If still climbing above desired altitude or path, DECREASE pitch using 0.5-degree increments.

CAUTION

1. Flaps transition from 0 to 30 must be accomplished in level flight.
2. FLAPS 45 must be selected at the start of final descent.

Table D – APPROACH – LEVEL FLIGHT							
ALT (ft)	EXPECTED SPD RANGE KIAS/Mach	EXPECTED SPD CHANGE FOR 0.5 DEGREE CHANGE IN PITCH ΔKIAS	Weight lb Pitch (degree) / Initial Thrust (%N1)				
			35000	40000	45000	50000	53000
15000 to 0			Level Flight - FLAPS 8 - Gear UP				
	170 - 205	5	1.5 / 57.4	2.0 / 58.8	3.0 / 60.3	4.0 / 62.0	4.5 / 63.1
			Level Flight - FLAPS 20 - Gear UP				
	155 - 190	5	-1.0 / 62.5	0.0 / 64.5	1.5 / 66.7	2.0 / 67.0	2.5 / 68.0
			Level Flight - FLAPS 30 - Gear DN				
	135 - 175	5	-1.5 / 70.2	0.5 / 72.1	1.0 / 72.7	2.0 / 74.0	3.0 / 75.2
			Level Flight - FLAPS 45 - Gear DN				
	130 - 175	5	-4.0 / 81.0	-3.0 / 82.0	-1.5 / 83.0	-0.5 / 84.0	0.5 / 85.2

NOTE

1. Initial pitch target is for reference only. See Flight Technique to adjust pitch as required up to a maximum of +/- 1.5 degree.
2. If pitch is increased, REDUCE expected speed range byΔKIAS.
3. If pitch is decreased, INCREASE expected speed range byΔKIAS.

Interpolation is acceptable



Flight Technique:

Allow sufficient time to stabilize altitude or path between pitch adjustments or configuration change.
Set N1 and pitch to target, then adjust pitch to maintain required glidepath:
Do not change N1.
• Adjust pitch as necessary to maintain required glideslope guidance/glidepath.
•

CAUTION

FLAPS 45 must be selected at the start of final descent.

Table E – APPROACH – FLAPS 45 – GEAR DN – 3 degree G/S						
ALT (ft)	EXPECTED SPD RANGE KIAS	Weight lb Pitch (degree) / Initial Thrust (%N1)				
		35000	40000	45000	50000	
FLAPS 45 – GEAR DN – 3 degree G/S						
15000	V _{REF} ±10	- 2.5 / 67.8	- 2.5 / 71.7	- 2.5 / 74.6	- 2.5 / 77.5	- 2.5 / 79.4
10000		- 2.5 / 64.1	- 2.5 / 67.7	- 2.5 / 70.7	- 2.5 / 73.6	- 2.5 / 75.4
5000		- 2.5 / 60.2	- 2.5 / 64.0	- 2.5 / 66.9	- 2.5 / 69.7	- 2.5 / 71.4
0		- 2.5 / 56.7	- 2.5 / 60.3	- 2.5 / 63.2	- 2.5 / 66.0	- 2.5 / 67.8

NOTE

Initial pitch target is for reference only. See Flight Technique to adjust pitch as required.
Interpolation is acceptable

CONTENTS	PAGE
-----------------	-------------

ABNORMAL PROCEDURE CONTENTS

POWER PLANT	1
AIR-CONDITIONING, BLEED AND PRESSURIZATION	2
AUTOMATIC FLIGHT CONTROL SYSTEM	3
AUXILIARY POWER UNIT	4
DOORS	5
ELECTRICAL	6
FIRE PROTECTION	7
FLIGHT CONTROLS	8
FUEL	9
HYDRAULIC POWER	10
ICE AND RAIN PROTECTION	11
INSTRUMENTS SYSTEM	12
LANDING GEAR, WHEEL AND BRAKE SYSTEM	13
MISCELLANEOUS SYSTEMS	14
AURAL/VISUAL WARNING SYSTEM	15

- SUPP
- LIMITS
- PERF
- MISC
- PLAN
- INTERMIX
- EICAS
- EMER
- IMM
- ABNORM
- INDEX

ABNORMAL PROCEDURE FORMAT

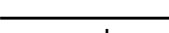
Amber cautions are items for which future action may be required to ensure airplane safety

A flow pattern concept is used throughout the Quick Reference Handbook (QRH) as applicable, utilizing the “decision” (rhomb) symbol ().

This decision symbol indicates a flow pattern which points to two or more possible courses of action. The procedure is completed once the (– **END** –) symbol is reached.

When several alternative actions point to a common procedure or to return / proceed to a specific step, a “return” symbol () is used.

An “arrowhead” symbol () is used to indicated that the procedure is not completed and continues on to the next page.

The “end” symbol ( **END** ) indicates the end of the complete procedure.

Procedures contained herein assume that certain actions listed below are accomplished by the crew:

- Normal procedures have been properly accomplished.
- MASTER WARNING / MASTER CAUTION switch lights are reset.
- Obvious corrective action (if any) is taken for crew awareness items.

Diversion terms used within the QRH abnormal procedures are as follows:

- “Land at the nearest suitable airport” – Landing airport and duration of the flight are at the discretion of the pilot-in-command (PIC). Extended flight beyond the nearest suitable airport in not recommended.
- “Land immediately at the nearest suitable airport” – Land without delay at the nearest airport where a safe approach and landing is reasonably assured.
- “Suitable airport” – Suitable airport is defined as the airport where a safe approach and landing is assured.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



CONTENTS	PAGE
----------	------

POWER PLANT

IN-FLIGHT ENGINE FAILURES

- Engine Failure During Approach ABNORM 1-3
- Engine Failure in Climb During ALTS
CAP ABNORM 1-4

SINGLE ENGINE PROCEDURES

- In-flight Engine Shutdown ABNORM 1-4
- Starter-assisted Cross Bleed Relight ABNORM 1-6
- Windmilling Relight ABNORM 1-10
- Single Engine Approach and Landing ABNORM 1-12

THRUST REVERSER FAILURES

- L REV UNSAFE or R REV UNSAFE ABNORM 1-13

POWER PLANT FAILURES

- APR INOP ABNORM 1-14
- APR CMD SET ABNORM 1-14
- Uncommanded Deceleration,
Abnormal Engine
Oscillation/Fluctuation ABNORM 1-15
- L ENG MISCOMP or R ENG
MISCOMP ABNORM 1-16
- Thrust Lever Jammed or Inoperative ABNORM 1-16
- Engine Oil Level Indication System
Failure ABNORM 1-18
- Fluctuating Engine Oil Pressure ABNORM 1-18
- Engine Starter Fails to Cutout / NO
STRTR CUTOUT ABNORM 1-19
- Interrupted Start ABNORM 1-20
- No Lightoff ABNORM 1-20
- Engine Hot Start ABNORM 1-21
- N₂ Stagnation ABNORM 1-21
- Low Idling ABNORM 1-21
- N₁ Fan Vibration ABNORM 1-21

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- N₂ Core Vibration [ABNORM 1-24](#)
- High Oil Temperature [ABNORM 1-24](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



Engine Failure During Approach

- (1) Autopilot (if engaged) DISENGAGE
- (2) Operating engine INCREASE thrust as required
- (3) Flight spoilers (if extended)RETRACT
- (4) Approach and landing FLAPS20
- (5) Airspeed INCREASE to $V_{REF(FLAPS\ 45)} + 12\ KIAS$
- (6) Airplane Retrim and continue approach, or go-around at pilot's discretion

NOTE

The autopilot may be re-engaged, if above 400 feet AGL.

If continuing the approach:

- (7) L and R PACKs TRANSFER to APU bleed
- (8) Go-around thrust referenceSET
- (9) GRND PROX, FLAP switch <0040> OVRD
- (10) Final approach speedMAINTAIN $V_{REF(FLAPS\ 45)} + 12\ KIAS$
- (11) Actual landing distance INCREASE



If required, use remaining thrust reverser carefully upon landing.

NOTE

- 1. If Ground Lift Dumping fails to deploy upon touchdown, select FLIGHT SPOILER lever to MAX extend.
- 2. Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	Runway Surface	
	Dry	Wet or Contaminated
With One Thrust Reverser	1.25 (25%)	1.25 (25%)
Without Thrust Reverser(s)	1.25 (25%)	1.25 (25%)

After landing:

(12) Affected thrust leverSHUT OFF

----- END -----

Engine Failure in Climb During ALTS CAP

- (1) Autopilot (if engaged) DISENGAGE
- (2) Pitch attitude Adjust to maintain the required single engine operating airspeed.

NOTE

Vertical FD commands may be used and autopilot re-engaged after FCC mode changes to ALTS at desired altitude.

----- END -----

In-flight Engine Shutdown

- Accomplish an engine shutdown only when flight conditions permit:
- (1) Affected thrust lever CONFIRM and IDLE
- (2) Affected thrust lever CONFIRM and SHUT OFF
- (3) 14TH STAGE ISOL OPEN
- (4) Affected HYDRAULIC B pump ON
- (a) Left engine shut down HYDRAULIC 1 ON
- (b) Right engine shut down HYDRAULIC 2 ON
- (5) Affected FUEL, BOOST PUMPCONFIRM and SELECT OFF

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

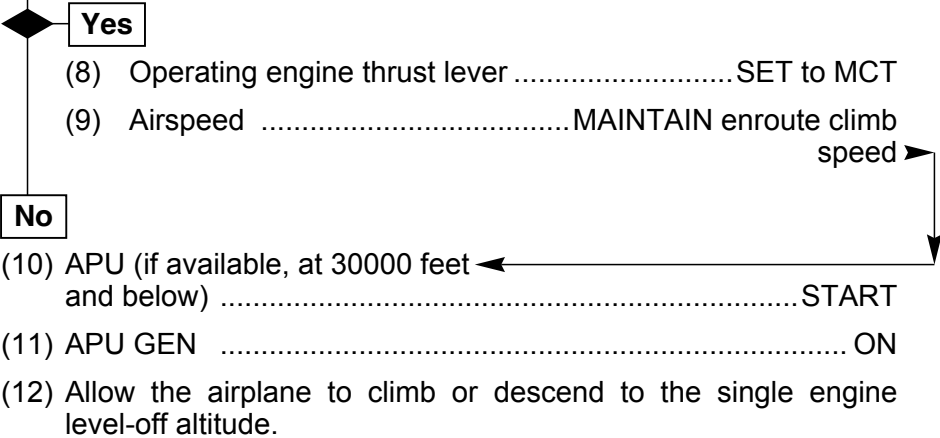
INDEX

- (6) ANTI-ICE, LH or RH COWLAffected side OFF
- (7) Fuel system CHECK
- (a) Cross-flowAUTO
- (b) Quantity / Balance CHECK

NOTE

1. Leave icing conditions to prevent ice accumulation on the engine cowl with the inoperative anti-icing system.
2. Powered cross-flow may not be able to correct fuel imbalance during single engine operations.
3. Crew may have to perform the Gravity Cross-feed procedure when required and time permits. Refer to ABNORM 9-10.

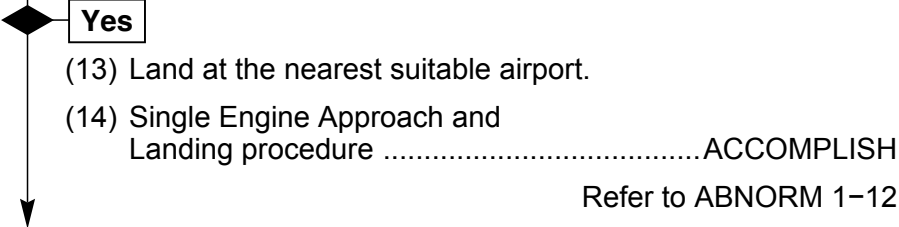
Enroute terrain clearance is a consideration:



NOTE

Do not attempt to relight an engine that is suspected to be damaged (engine fire, rotor burst, reverser deployed, etc...).

Engine damage is suspected / intentional shutdown:



- END -

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- No
- (13) DescentINITIATE
- (14) Engine Relight procedureACCOMPLISH as required
 - Starter-assisted Cross Bleed Relight procedure. Refer to ABNORM 1-6
 - Windmilling Relight procedure. Refer to ABNORM 1-10

Relight engine using starter-assisted start whenever possible.

----- END -----

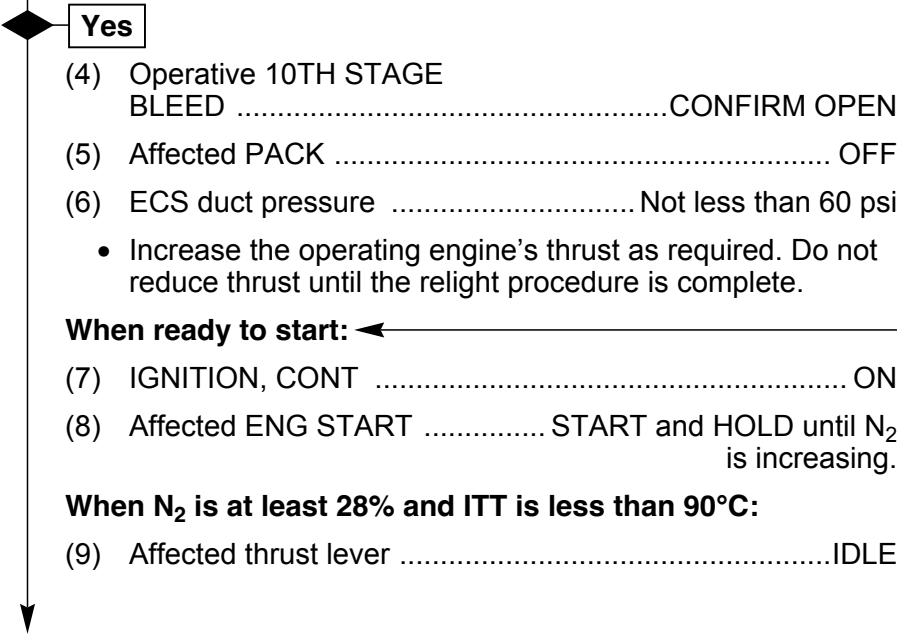
Starter-assisted Cross Bleed Relight

- (1) Relight envelope REVIEW

Starter-assisted Cross Bleed Relight Envelope	
Altitude	Airspeed
21000 feet to 15000 feet	180 KIAS to V _{MO}
15000 feet to SL	V _{REF} to V _{MO}

- (2) Affected GEN OFF
- (3) FUEL, L and R BOOST PUMP CONFIRM ON

PACKs on engine bleeds:



SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

- (8) ECS duct pressure Not less than 60 psi
- Increase the operating engine’s thrust as required. Do not reduce thrust until the relight procedure is complete.

When ready to start: ←

- (9) IGNITION, CONT ON
- (10) Affected ENG START START and hold until N₂ is increasing.

When N₂ is at least 28% and ITT is less than 90°C:

- (11) Affected thrust lever IDLE
- (12) Engine indications MONITOR

Engine relights (within 25 seconds) and stabilizes at IDLE:

- Yes

(13) Thrust levers AS REQUIRED

(14) 14TH STAGE ISOL SELECT closed

(15) L and R 10TH STAGE BLEED CLOSED

(16) APU LCV OPEN

(17) 10TH STAGE ISOL OPEN

(18) Affected PACK SELECT on

(19) Affected GEN ON

(20) IGNITION, CONT SELECT off

(21) ANTI-ICE, LH and RH COWL AS REQUIRED

(22) ANTI-ICE, WING AS REQUIRED

(23) Affected HYDRAULIC pump AUTO

– END –

No

- (13) Affected thrust lever CONFIRM and SHUT OFF
- (14) Affected ENG STOP STOP
- (15) Wait 30 seconds, then repeat relight procedure if desired. →

Engine relight is still not successful:

- (16) IGNITION, CONT SELECT off
- (17) Affected FUEL, BOOST PUMP CONFIRM and SELECT OFF
- (18) L and R 10TH STAGE BLEED CLOSED
- (19) APU LCV OPEN
- (20) 10TH STAGE ISOL OPEN

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(21) L and R PACK As required

NOTE

- 1. Use only one air-conditioning pack during single engine operations, when the operating engine is the only 10th stage bleed source.
- 2. Airplane altitude not above 25000 feet during single pack operations.

(22) Land at the nearest suitable airport.

(23) Single Engine Approach and Landing procedureACCOMPLISH

I Refer to ABNORM 1-12

----- **END** -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



Windmilling Relight

(1) Relight envelope REVIEW

Windmill Relight Envelope	
Altitude	Airspeed
21000 feet to 15000 feet	300 KIAS to V _{MO}
15000 feet to SL	

NOTE

1. Windmill airstart efficiency is enhanced by attaining the highest practical airspeed and N₂ within the relight envelope.
2. Between 21000 feet and 15000 feet, a minimum of 12% N₂ is necessary for a windmill relight.
3. At 15000 feet and below, a minimum of 9% N₂ is necessary for a windmill relight.
4. If the minimum N₂ cannot be achieved, an approximate 1.5% to 2.5% increase in N₂ may be obtained by disconnecting the corresponding IDG using the IDG DISC. Once a disconnect has been performed that IDG cannot be re-connected in flight.

(2) Affected GEN OFF

(3) Affected PACK OFF

(4) FUEL, L and R BOOST PUMP CONFIRM ON

When ready to start: ←

(5) IGNITION, CONT ON

(6) Airspeed INCREASE to 300 KIAS
or greater

NOTE

Maintain airspeed throughout light-off until engine start is complete (stable idle). Monitor engine parameters carefully.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

When relight envelope is established:

(7) Affected thrust leverIDLE

NOTE

N₂ acceleration should be positive and uninterrupted.

Engine relights within 25 seconds (stable IDLE within 2 minutes):

Yes

(8) Thrust leversAS REQUIRED

INTENTIONALLY LEFT BLANK

Effectivity:

- Airplanes 7305 and subsequent:

(9) No airspeed limitation with the ADG deployed.

(10) 14TH STAGE ISOL SELECT closed

(11) Affected GEN ON

(12) Affected PACKSELECT on

(13) IGNITION, CONT SELECT OFF

(14) ANTI-ICE, LH and RH COWL.....AS REQUIRED

(15) ANTI-ICE, WINGAS REQUIRED

(16) Affected HYDRAULIC B pumpAUTO

- END -

No

(8) Affected thrust leverCONFIRM and SHUT OFF

(9) Wait 30 seconds, then repeat relight procedure if desired.

Engine relight is still not successful:

- (10) IGNITION, CONT SELECT OFF
- (11) Affected FUEL, BOOST PUMPCONFIRM and SELECT OFF
- (12) Land at the nearest suitable airport.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(13) Single Engine Approach and Landing procedureACCOMPLISH

Refer to ABNORM 1-12

----- END -----

Single Engine Approach and Landing

(1) APU (if available, at 30000 feet and below)START

NOTE

1. Use only one air-conditioning pack during single engine operations, when the operating engine is the only 10th stage bleed source.
2. Airplane altitude not above 25000 feet during single pack operations.

(2) APU GEN ON

Below 15000 feet:

- (3) L and R PACK TRANSFER to APU bleed
- (4) Go-around thrust reference SET N₁ bug accordingly.
Refer to Go-around Thrust Settings – PERF-04.
- (5) GRND PROX, FLAP <0040> OVRD

Approach and landing:

- (6) Approach and landing FLAPS20
- (7) Final approach speedNot less than
V_{REF} (FLAPS 45) + 12 KIAS
- (8) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	Runway Surface	
	Dry	Wet or Contaminated
With One Thrust Reverser	1.25 (25%)	1.25 (25%)
Without Thrust Reversers	1.25 (25%)	1.25 (25%)



If required, use remaining thrust reverser carefully upon landing.

NOTE

If Ground Lift Dumping fails to deploy upon touchdown, select FLIGHT SPOILER lever to MAX deployment.

----- END -----

L REV UNSAFE or R REV UNSAFE

Thrust reverser is unsafe to arm to flight. The thrust reverser deploy switches have failed in a commanded deploy state but thrust reversers are not unlocked and are not armed.

- (1) Affected THRUST REVERSER,
ARMED OFF



If required, use remaining thrust reverser carefully upon landing.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

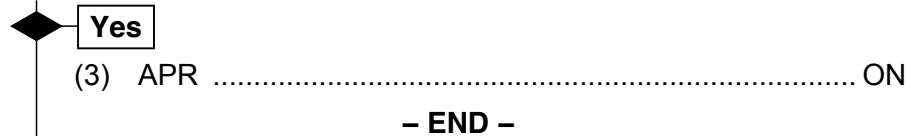
INDEX



APR INOP

- (1) APR OFF
- (2) ENG SPEED (both)CHECK ON

Engine indications are normal:



- Decision diamond: No
- (3) Thrust levers RETARD to below 79% N₁
 - (4) ENG SPEED (both) OFF
 - (5) Use APR-off performance (APR is not available).
 - (6) Thrust levers ADVANCE as required

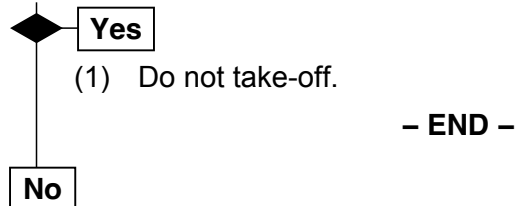
NOTE

Due to the possibility of N₁ overspeed, monitor the N₁ if more than idle reverse thrust is used after landing. Do not exceed 90% N₁.

– – – – – END – – – – –

APR CMD SET

On the ground:



In flight:

- (1) Thrust levers ADJUST to within normal range
- (2) APR OFF

– – – – – END – – – – –

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Uncommanded Deceleration, Abnormal Engine Oscillation / Fluctuation

NOTE

To prevent engine overspeed when selecting ENG SPEED switches OFF, N₂ indications should be referenced instead of N₁.

- (1) Affected engine ADJUST thrust below 79% N₂ or IDLE (whichever comes first)
- (2) Affected engine, ENG SPEED OFF
- (3) Affected thrust lever ADVANCE
- (4) Affected engine indicationsMONITOR



Do not change power settings abruptly as engine response is degraded.

Engine responds:

Yes

NOTE

1. To minimize asymmetric thrust, both ENG SPEED switches should be selected to OFF prior to approach and landing.
2. Exercise caution, to prevent engine overspeed during thrust reverser use, when operating with ENG SPEED switches OFF.
3. The APR system is inoperative when the ENG SPEED switches are selected OFF.

– END –

No

- (5) Single Engine Procedures, In-flight Engine ShutdownACCOMPLISH
- Refer to ABNORM 1-4

----- END -----

L ENG MISCOMP or R ENG MISCOMP

- (1) EICAS displays (both)SELECT to PRI
(engines) page
- (2) Affected engine’s parameters CROSS-CHECK to find
disagreement
- (3) Affected engine MAINTAIN within limits using
reliable PRI page and its
associated DCU.

----- END -----

Thrust Lever Jammed or Inoperative

At a safe altitude, before commencing the approach:

- (1) Affected engine, ENG FIRE PUSH.....CONFIRM and SELECT
- (2) 14TH STAGE ISOL OPEN
- (3) Affected HYDRAULIC B pump ON
- (a) Left engine shut down HYDRAULIC 1 switch ON
- (b) Right engine shut down HYDRAULIC 2 switch ON
- (4) APU (if available, at 30000 feet
and below)START

NOTE

1. Use only one air-conditioning pack during
single engine operations, when the
operating engine is the only 10th-stage
bleed source.
2. Airplane altitude maximum 25000 feet
during single pack operation.

- (5) APU GEN ON
- (6) Fuel system CHECK
- (a) Cross-flowAUTO
- (b) Quantity / Balance CHECK
- (7) L and R PACK TRANSFER to APU bleed
- (8) Go-around thrust reference SET N₁ bug accordingly.
Refer to Go-around Thrust
Settings – PERF-04.
- (9) GRND PROX, FLAP <0040> OVRD

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Approach and landing:

- (10) Approach and landing FLAPS20
- (11) Final approach speedNot less than
V_{REF (FLAPS 45)} +12 KIAS
- (12) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With One Thrust Reverser	1.45 (45%)	1.50 (50%)
Without Thrust Reversers	1.50 (50%)	1.50 (50%)

Upon touchdown:

- (13) FLIGHT SPOILER lever SELECT MAX deployment



If required, use remaining thrust reverser carefully upon landing.

----- **END** -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Engine Oil Level Indication System Failure

LH FAIL or RH FAIL light(s) on during engine oil level check.

NOTE

The MASTER CAUTION lights will not come on for this condition.

- (1) ENGINE OIL LEVEL, STOP/STARTSELECT off
- (2) Affected engine oil systemSERVICE immediately

----- END -----

Fluctuating Engine Oil Pressure

Affected engine oil pressure readout is above 25 psi and fluctuating in excess of ±5 psi:

Yes

- (1) Affected thrust leverCONFIRM and IDLE
- (2) Descent (if required)INITIATE to 29000 feet or below or until oil pressure stabilizes.

When affected engine oil pressure stabilizes:

- (3) Affected thrust leverAS REQUIRED

NOTE

- 1. If necessary, keep descending to a lower altitude to maintain a stable oil pressure, as cruise power setting is resumed.
- 2. If oil pressure fluctuations in excess of ±5 psi resume, select affected thrust lever to IDLE and descend to a lower altitude.

- END -

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

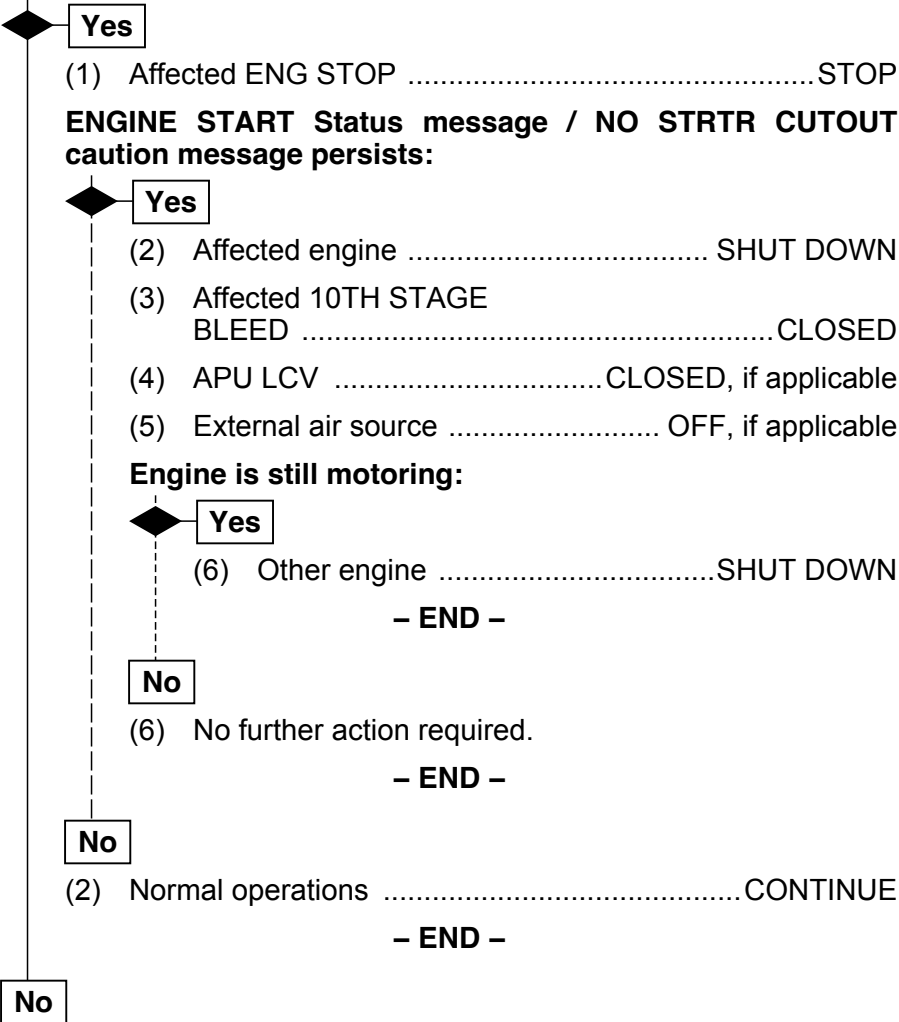
- (1) **L ENG OIL PRESS** or **R ENG OIL PRESS** or Low Engine Oil Pressure Indication procedureACCOMPLISH

Refer to EMER 1-16

----- END -----

Engine Starter Fails to Cutout / NO STRTR CUTOUT

On the ground:



In flight:

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



- (1) Affected ENG STOPSTOP
- (2) IGNITION, CONT OFF
- (3) Affected ENG START L circuit
breaker (1M5) OPEN
- (4) Affected ENG START R circuit
breaker (1M4) OPEN

ENGINE START status message / NO STRTR CUTOUT caution
message persists:

Yes

- (5) Affected 10 ST B/AIR
SOV L circuit breaker
(1F11) OPEN
- (6) Affected 10 ST B/AIR
SOV R circuit breaker
(2F11) OPEN
- (7) Land at the nearest suitable airport.

- END -

No

- (5) Normal operationsCONTINUE

----- END -----

Interrupted Start

- (1) Affected thrust leverSHUT OFF
 - (2) IGNITION OFF
 - (3) Dry motor Until ITT <120°C or starter
time limit, whichever comes
first.
- END -----

No Lightoff

- (1) Affected thrust leverSHUT OFF
 - (2) IGNITION OFF
 - (3) Dry motor until starter time limit.
- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Engine Hot Start

- (1) Affected thrust leverSHUT OFF
- (2) IGNITION SELECT all off
- (3) Dry motor Until ITT < 120°C or starter
time limit, whichever comes
first

----- END -----

N₂ Stagnation

- (1) Affected thrust leverSHUT OFF
- (2) IGNITION OFF
- (3) Dry motor Until ITT <120°C or starter
time limit, whichever comes
first

----- END -----

Low Idling

- (1) Affected engineDo not advance thrust lever
above 70% N₂ until N₂ has
stabilized within normal limits

----- END -----

N₁ Fan Vibration



It is not recommended that an engine be shut down unless there is another indication of a severe engine abnormality including high oil temperature, low oil pressure, uncontrolled excessive ITT and/or continuous abnormal airframe vibration.

- If icing conditions or ice accumulation is suspected, proceed to step (1).
- If in non-icing conditions, proceed to step (9).

- (1) In icing conditions or ice accumulation is suspected:
- (2) N₁ thrust target SELECT "CLB" on FMS
MENU - THRUST LIMIT
page

NOTE

If both engines are affected the following steps need to be done for one engine at a time.



(3) Affected thrust leverCONFIRM and IDLE, then immediately set thrust to CLB

(4) Wait 30 seconds with affected thrust lever at CLB position.

NOTE

1. In colder conditions several attempts/minutes may be required to shed ice.
2. If more than one attempt is required to reduce fan vibration to within normal operating range, icing conditions should be considered severe.
3. Ice shedding process may be accompanied by the following:
 - an airframe vibration and change in engine noise;
 - sharp metallic noise as ice impacts inside of the nacelle;
 - vibrations may increase;
 - vibrations may momentarily indicate full scale.

- If after 30 seconds, vibration can be controlled or reduced to within normal operating range, proceed to step (12).
- If after 30 seconds, vibration cannot be controlled or reduced to within normal operating range, proceed to step (5).

(5) Repeat steps (3) and (4) above as many times as required to shed the ice.

(6) Leave icing conditions.

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is -40°C (-40°F) or below.

(7) Engine indicationsMONITOR

– END –

(8) If vibrations still cannot be controlled or reduced to within normal operating range, proceed to step (14).

(9) Non-icing conditions:

(10) Affected thrust leverCONFIRM and IDLE, then immediately adjust thrust to the required setting.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- If vibration can be controlled or reduced to within normal operating range, proceed to step (12).
 - If vibration cannot be controlled or reduced to within normal operating range, proceed to step (11).
- (11) Airplane altitude DESCEND by 4000 feet or to lowest safe altitude, whichever is higher.
- If after altitude reduction vibration can be controlled or reduced to within normal operating range, proceed to step (12).
 - If after altitude reduction vibration cannot be controlled or reduced to within normal operating range, proceed to step (14).
- (12) Vibration can be controlled or reduced to within normal operating range:
- (13) Engine indicationsMONITOR
- END –
- (14) Vibration cannot be controlled or reduced to within normal operating range:
- (15) Affected engine ADJUST thrust to maintain vibration levels within normal range.
- If vibration can be maintained within normal operating range, proceed to step (16).
 - If vibration still cannot be maintained within normal operating range, proceed to step (18).
- (16) Vibration can be maintained within normal operating range:
- (17) Engine indicationsMONITOR
- END –
- (18) Vibration still cannot be maintained within normal operating range:
- (19) Single Engine Procedures,
In-flight Engine ShutdownACCOMPLISH
- Refer to ABNORM 1-4
- – – – END – – – –

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



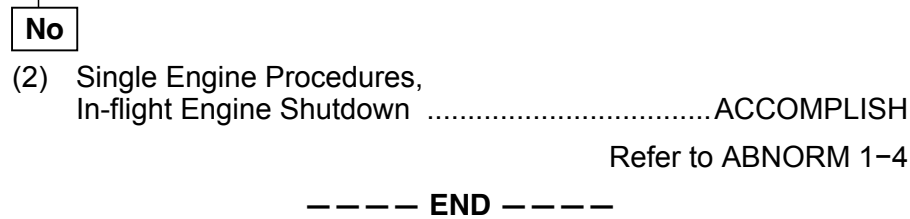
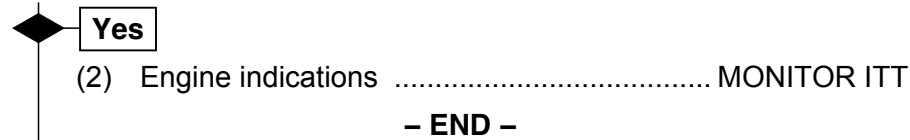
N₂ Core Vibration



It is not recommended that an engine be shut down unless there is another indication of a severe engine abnormality, i.e., high oil temperature, high oil pressure, or abnormal engine vibration is felt through the airframe.

- (1) Affected engine ADJUST thrust to maintain vibration levels within normal range.

Vibration can be controlled or reduced to within normal operating range:



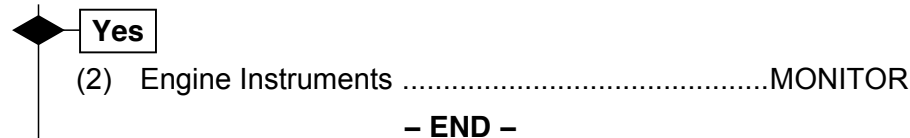
High Oil Temperature

Readout of 150°C to 162°C (amber range)

Readout of 163°C and above (red range)

- (1) Affected thrust lever ADJUST to maintain oil temperature within normal range

Oil temperature can be reduced below red range:



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

I

	ABNORM 1-25
	Rev. 77, Jan 03/2022

No

(2) Single Engine Procedures,
In-flight Engine ShutdownACCOMPLISH

Refer to ABNORM 1-4

----- **END** -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

QUICK REFERENCE HANDBOOK CSP A-022-PRS	POWER PLANT
---	--------------------

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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CONTENTS	PAGE
----------	------

AIR-CONDITIONING SYSTEM, BLEED AND PRESSURIZATION

AIR-CONDITIONING

- L PACK HI PRESS or R PACK HI PRESS (Bleed source is APU) ABNORM 2-2
- L PACK HI PRESS or R PACK HI PRESS (Bleed source is engine) ABNORM 2-3
- L PACK HI TEMP or R PACK HI TEMP (Bleed source is APU) ABNORM 2-5
- L PACK HI TEMP or R PACK HI TEMP (Bleed source is engine) ABNORM 2-6

AVIONICS COOLING

- ARINC COOL ABNORM 2-7
- DISPLAY COOL ABNORM 2-8
- OVBD COOL ABNORM 2-9
- OVBD COOL FAIL Status Message (On the Ground only) ABNORM 2-9

BLEED SYSTEM

- BLEED MISCONFIG <0039> ABNORM 2-10

PRESSURIZATION

- AUTO PRESS ABNORM 2-10
- CABIN ALT ABNORM 2-11
- Manual Pressurization Control Procedure ABNORM 2-11
- Unpressurized Flight Procedure (PACKs Off) ABNORM 2-12
- Unpressurized Flight Procedure (PACKs On) ABNORM 2-13
- CARGO OVHT <0013> or <0034> or <0074> ABNORM 2-13

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

L PACK HI PRESS or R PACK HI PRESS (Bleed source is APU)

NOTE

Airplane altitude not above 25000 feet during single pack operations.

- (1) Affected PACK PRESS out
- After 10 seconds:
- (2) Affected PACKSELECT on
- L PACK HI PRESS or R PACK HI PRESS caution message persists:

Yes



If message persists do not attempt to transfer pack source to engine on the affected side. Attempting transfer may cause loss of remaining pack.

- (3) Affected PACK PRESS out
- Operative PACK transfer from APU to engine bleed is required:

Yes

- At a safe altitude, below 15000 ft ASL and when flight conditions permit:
- (1) Thrust levers SET to 60% N₁ to reduce pressure at the bleed air manifold.

NOTE

The following steps need to be done one after the other without delay to avoid remaining pack going off line inadvertently between steps.

- (2) 10TH STAGE ISOL SELECT closed
- (3) APU LCV SELECT closed
- (4) Non-affected 10TH STAGE BLEEDSELECT open

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

When bleed source transfer from APU to engine is confirmed:

(5) Thrust levers ADVANCE as required

NOTE

When the 10th stage bleed valves are selected open, there is a significant time delay before the valve position indicators (on the ECS synoptic page) actually indicates open.

– END –

No

(3) No further action required.

----- END -----

L PACK HI PRESS or R PACK HI PRESS (Bleed source is engine)

NOTE

1.

Airplane altitude not above 25000 feet during single pack operations.
2.

The isolation valve, pack valve and the 10th stage bleed valve will close automatically during an overpressure condition (when the engine is supplying bleed air).

- (1)

Affected PACK PRESS out
- (2)

Affected 10TH STAGE BLEEDSELECT CLOSED

After 10 seconds:

- (3)

Affected 10TH STAGE BLEEDSELECT open

NOTE

When the 10th stage bleed valves are selected open, there is a significant time delay before the valve position indicators (on the ECS synoptic page) actually indicates open.

- (4)

Affected PACKSELECT on

L PACK HI PRESS or R PACK HI PRESS caution message persists:

Yes

- (5) Affected PACK PRESS out
- (6) Affected 10TH STAGE BLEEDSELECT CLOSED

Operative PACK transfer from engine to APU bleed is required:

Yes

NOTE

1. Bleed air extraction from the APU is not permitted above 15000 feet.
2. The following steps need to be done one after the other without delay to avoid remaining pack going off line inadvertently between steps.

- (1) Non-affected 10TH STAGE BLEEDSELECT CLOSED
- (2) APU LCV SELECT OPEN
- (3) 10TH STAGE ISOL SELECT OPEN

NOTE

The 10th stage engine bleeds must be closed for landing when the cowl and/or wing anti-ice system are in use. Refer to LIMITATIONS – Systems Limitations.

– END –

No

- (5) No further action required.

----- END -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

L PACK HI TEMP or R PACK HI TEMP (Bleed source is APU)

NOTE

Airplane altitude not above 25000 feet during single pack operations.

(1) Affected PACK PRESS out

After 15 seconds and the L PACK HI TEMP or R PACK HI TEMP caution message goes out:

Yes

(2) Affected PACKSELECT on



If message persists do not attempt to transfer pack source to engine on the affected side. Attempting transfer may cause loss of remaining pack.

– END –

No

After 15 seconds and the L PACK HI TEMP or R PACK HI TEMP caution message persists and operative PACK transfer from APU to engine bleed is required:

At a safe altitude, below 15000 ft ASL and when flight conditions permit:

(1) Thrust levers SET to 60% N₁ to reduce pressure at the bleed air manifold.

NOTE

The following steps need to be done one after the other without delay to avoid remaining pack going off line inadvertently between steps.

- (2) 10TH STAGE ISOL SELECT closed
- (3) APU LCV SELECT closed
- (4) Non-affected 10TH STAGE BLEEDSELECT open

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

When bleed source transfer from APU to engine is confirmed:

- (5) ADVANCE as requiredThrust
levers

NOTE

When the 10th stage bleed valves are selected open, there is a significant time delay before the valve position indicators (on the ECS synoptic page) actually indicates open.

----- END -----

L **PACK HI TEMP** or R **PACK HI TEMP** (Bleed source is engine)

NOTE

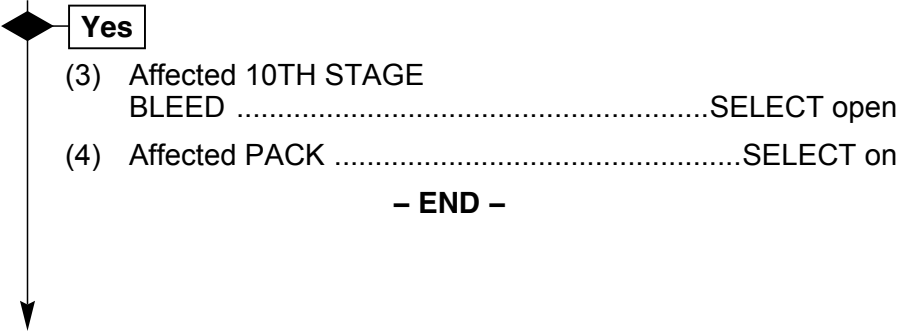
- 1. Airplane altitude not above 25000 feet during single pack operations.
- 2. The isolation valve, pack valve and the 10th stage bleed valve will close automatically during an overtemperature condition (when the engine is supplying bleed air).

- (1) Affected PACK PRESS out
(2) Affected 10TH STAGE BLEEDSELECT CLOSED

NOTE

When the 10th stage bleed valves are selected open, there is a significant time delay before the valve position indicators (on the ECS synoptic page) actually indicates open.

After 15 seconds and the L **PACK HI TEMP** or R **PACK HI TEMP** caution message goes out:



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

(3) No further action required

NOTE

Airplane altitude not above 25000 feet during single pack operations.

Operative PACK transfer from engine to APU bleed is required:

NOTE

1.

Bleed air extraction from the APU is not permitted above 15000 feet.
2.

The following steps need to be done one after the other without delay to avoid remaining pack going off line inadvertently between steps.

- (1)

Non-affected 10TH STAGE BLEEDSELECT CLOSED
- (2)

APU LCV SELECT OPEN
- (3)

10TH STAGE ISOL SELECT OPEN

NOTE

The 10th stage engine bleeds must be closed for landing when the cowl and/or wing anti-ice system are in use. Refer to LIMITATIONS – Systems Limitations.

----- END -----

ARINC COOL

On the ground:

Yes

- (1)

ARINC FAN GND ALTN

- END -

No

In flight:

- (1)

ARINC FAN FLT ALTN

----- END -----

DISPLAY COOL

On the ground:

Yes

(1) DSPLY FAN GND ALTN

After 60 seconds DISPLAY COOL caution message persists:

Yes

(2) DSPLY FANSTDBY

- END -

No

(2) No further action required.

- END -

No

In flight:

(1) DSPLY FAN FLT ALTN

After 60 seconds DISPLAY COOL caution message persists:

Yes

(2) DSPLY FAN STDBY

(3) CKPT temperature COLD

After another 60 seconds DISPLAY COOL caution message still persists:

Yes

NOTE

All displays will keep functioning for at least 30 minutes after fan failure.

(4) Applicable PFD/EICAS
reversionAS REQUIRED

(5) Land at the nearest suitable airport.

- END -

No

(2) No further action required.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

OVBD COOL

NOTE

The airplane will not pressurize to normal levels when the overboard cooling SOV is failed open.

On the ground:

Yes

(1) Do not take-off.

- END -

No

In flight:

(1) Cabin pressureCHECK

Cabin pressure is normal:

Yes

(2) No further action required.

- END -

No

(2) Unpressurized Flight Procedure (PACKs ON)ACCOMPLISH

Refer to ABNORM 2-13

----- END -----

OVBD COOL FAIL Status Message (On the Ground only)

(1) Passenger door and service doorENSURE correctly closed and fully locked.

The following door messages should be out:

PASSENGER DOOR

warning message,

PAX DR LATCH

caution message,

PAX DR STOW

caution message,

PAX DR OUT HNDL

caution message,

SERVICE DOOR

caution message.

----- END -----

QUICK REFERENCE HANDBOOK CSP A-022-PRS

AIR-CONDITIONING, BLEED AND PRESSURIZATION

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

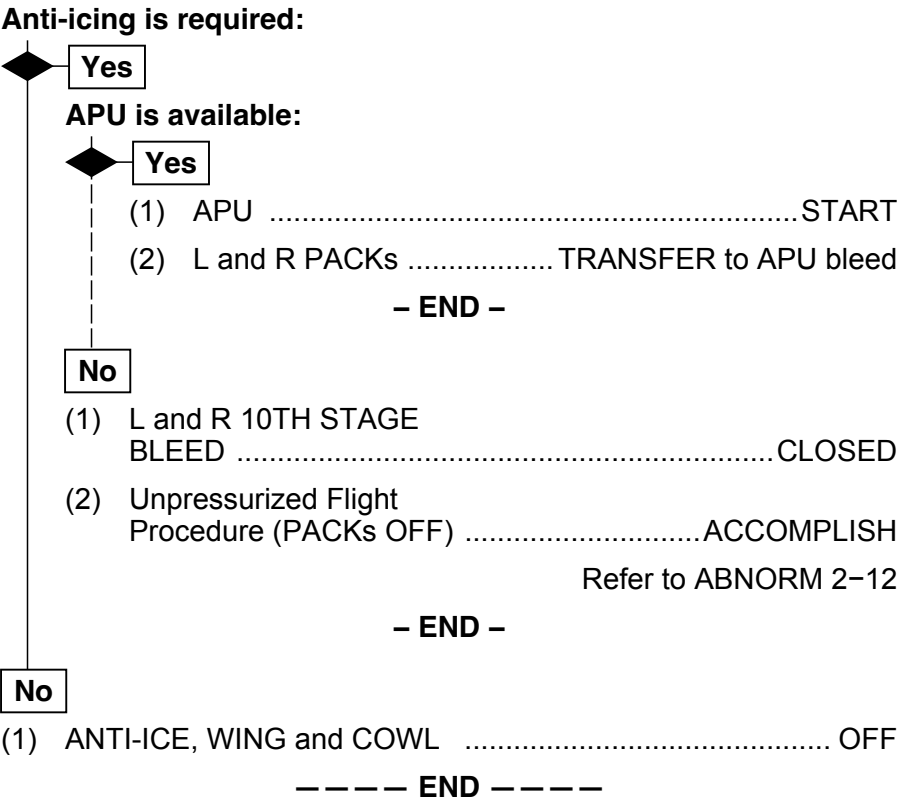
EMER

IMM

ABNORM

INDEX

BLEED MISCONFIG <0039>



AUTO PRESS

- NOTE
- I

1.

If pressurization problems occur that are accompanied with false take-off configuration warning, then a WOW OUTPUT fault has occurred. Refer to ABNORM 13-13.

2.

Associated with PRESS CONT FAULT light on. Do not attempt to reselect auto mode.
- (1) Manual Pressurization Control ProcedureACCOMPLISH

Refer to ABNORM 2-11

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CABIN ALT

- (1)

Thrust

INCREASE
- (2)

L and R 10TH STAGE BLEED

CHECK open
- (3)

L and R PACK

CHECK ON
- (4)

EMER DEPRESS

CHECK OFF
- (5)

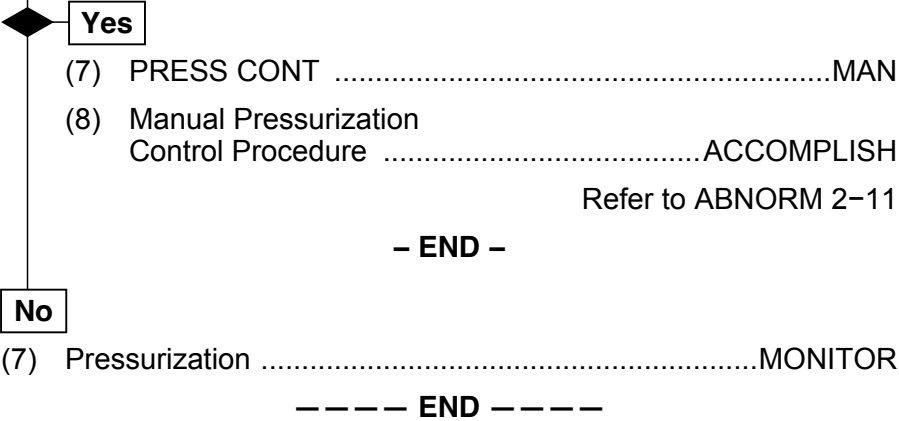
MAN ALT

SET to mid position / centre
- (6)

PRESS CONT

MAN then reset

CABIN ALT caution message persists:



Manual Pressurization Control Procedure

- (1)

MAN ALT

Mid position / centre
- (2)

PRESS CONT

MAN

To increase cabin altitude:

- (3)

MAN ALT

UP
- (4)

MAN RATE

AS REQUIRED

To decrease cabin altitude:

- (3)

MAN ALT

DN
- (4)

MAN RATE

AS REQUIRED

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



To maintain cabin altitude:

- (5) MAN ALT SET to mid position / centre
when reaching target cabin
altitude (refer to the table that
follows)

Cruise Flight Level	180	200	220	240	260	280	290	310	330	350	370	390	410
Target Cabin Altitude	1700	2000	2400	2800	3200	3700	4100	4600	5300	6000	6700	7400	8000

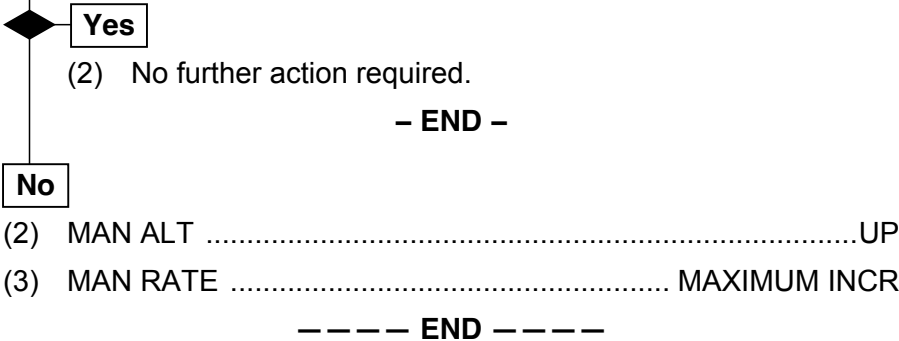
Before landing:

- (1) Cabin altitude ADJUST to landing field
elevation

NOTE

Do not set cabin altitude below landing field
elevation.

Differential pressure is zero upon landing:



Unpressurized Flight Procedure (PACKs off)

- (1) Airplane altitudeNot above 10000 feet MSL or
lowest safe altitude,
whichever is higher.
- (2) Airspeed Not less than 250 KIAS

NOTE

250 KIAS airspeed recommended during cruise
to provide sufficient airflow to passengers within
cabin.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- (3)

L and R PACK

OFF
- (4)

AIR-CONDITIONING, CARGO

OFF
- (5)

DSPLY FAN

STDBY
- (6)

RAM-AIR

OPEN
- (7)

EMER DEPRESS

ON
- END

- Unpressurized Flight Procedure (PACKs on)
- (1)

Airplane altitude

Not above 10000 feet MSL or
lowest safe altitude,
whichever is higher.
- (2)

EMER DEPRESS

ON
- END

- CARGO OVHT <0013> or <0034> or <0074>
- (1)

AIR-CONDITIONING, CARGO

FAN
- CARGO OVHT caution message persists:
- Yes

(2)

AIR-CONDITIONING,
CARGO

OFF

Live cargo are loaded in the cargo bay:

Yes

(3)

Land at the nearest suitable airport.

END

No

(2)

No further action required.

END

- SUPP
- LIMITS
- PERF
- MISC
- PLAN
- INTERMIX
- EICAS
- EMER
- IMM
- ABNORM
- INDEX
- QUICK REFERENCE
HANDBOOK
CSP A-022-PRS

AIR-CONDITIONING, BLEED
AND PRESSURIZATION

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
----------	------

AUTOMATIC FLIGHT CONTROL SYSTEM

- **AP TRIM IS LWD** or **AP TRIM IS RWD**
or **AP TRIM IS NU** or **AP TRIM IS ND** [ABNORM 3-2](#)
- **AP PITCH TRIM** [ABNORM 3-3](#)
- **YAW DAMPER** [ABNORM 3-3](#)
- Flight Director Guidance Failure [ABNORM 3-5](#)
- Flight Control Computer Failure [ABNORM 3-5](#)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Effectivity:

Airplanes that comply with Transport Canada Airworthiness Directive CF-2002-35R2, Aileron Stiffness, Part C – Termination of Control Checks, or paragraph (i) of FAA Airworthiness Directive 2005-18-06:

AP TRIM IS LWD or AP TRIM IS RWD or AP TRIM IS NU or AP TRIM IS ND

NOTE

Anticipate out-of-trim condition when disengaging autopilot.

(1) Autopilot DISENGAGE

(2) Airplane RETRIM

(3) Autopilot ENGAGE

(4) Autopilot operation MONITOR

AP TRIM IS LWD or AP TRIM IS RWD or AP TRIM IS NU or AP TRIM IS ND caution message persists:

Yes

NOTE

1. Anticipate out-of-trim condition when disengaging autopilot.

2. RVSM operations may be affected. Review the requirements. <0030>

3. Autopilot disengaged warnings are indicated by a “Cavalry Charge” aural and a flashing red AP message on the primary flight display.

(5) Autopilot DISENGAGE

– END –

No

(5) No further action required

----- END -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

QUICK REFERENCE HANDBOOK CSP A-022-PRS

AUTOMATIC FLIGHT CONTROL SYSTEM

AP PITCH TRIM

NOTE

Anticipate elevator out-of-trim condition when disengaging autopilot.

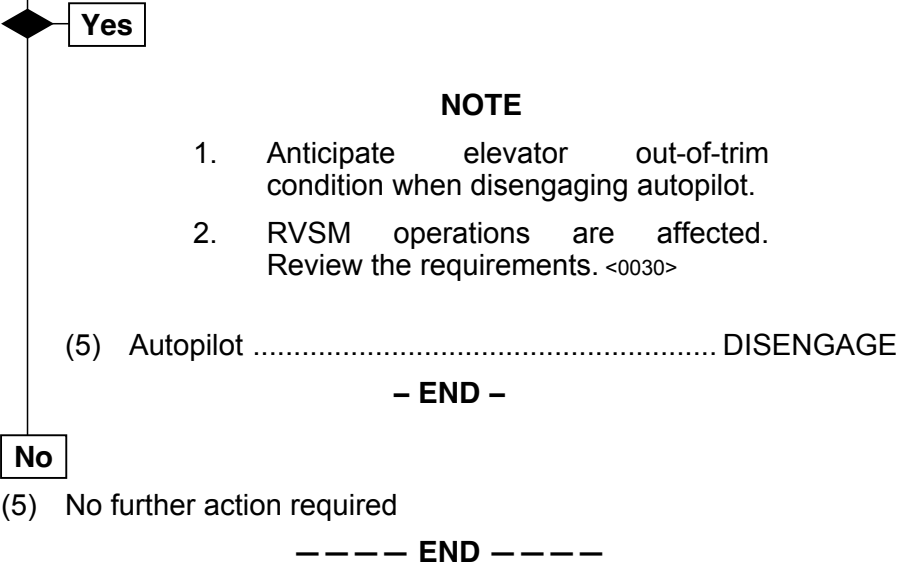
(1) Autopilot DISENGAGE

(2) Airplane RETRIM

(3) Autopilot ENGAGE

(4) Autopilot operation MONITOR

AP PITCH TRIM caution message persists:



YAW DAMPER

(1) YAW DAMPER, YD 1 and YD 2 ENGAGE

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

YAW DAMPER caution message persists:

Yes

(2) YAW DAMPER, DISCSELECT

NOTE

1.

Select the runway available with minimum cross-wind.
2.

Category II or IIIa operations may be affected. Review the requirements.

Airplane handling is acceptable with FLAPS at 45:

Yes

(3) No further action required.

– END –

No

(3) GRND PROX, FLAP <0040> OVRD

(4) Landing FLAPS20

(5) Approach speedNot less than
 $V_{REF} \text{ (FLAPS 45)} + 12 \text{ KIAS}$

(6) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.22 (22%)	1.25 (25%)
Without Thrust Reverser(s)	1.25 (25%)	1.25 (25%)

– END –

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

(2) AutopilotAS REQUIRED

----- END -----

Flight Director Guidance Failure

FD annunciator flag displayed on PFD and autopilot/FD coupled to affected side:

Yes

(1) AP/FD XFRTRANSFER

- END -

No

Sensor failure indicated (red line across affected mode indication):

(1) Affected FD modeDESELECT

----- END -----

Flight Control Computer Failure

(1) Autopilot XFRTRANSFER to operable side

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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CONTENTS	PAGE
----------	------

AUXILIARY POWER UNIT

- APU LCV FAIL ABNORM 4-2
- APU BLEED ON ABNORM 4-2
- APU FAULT ABNORM 4-3
- APU OIL PRESS ABNORM 4-3
- APU OIL TEMP ABNORM 4-4
- APU Door Failure ABNORM 4-5
- APU SOV FAIL ABNORM 4-6
- APU SOV OPEN ABNORM 4-6

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

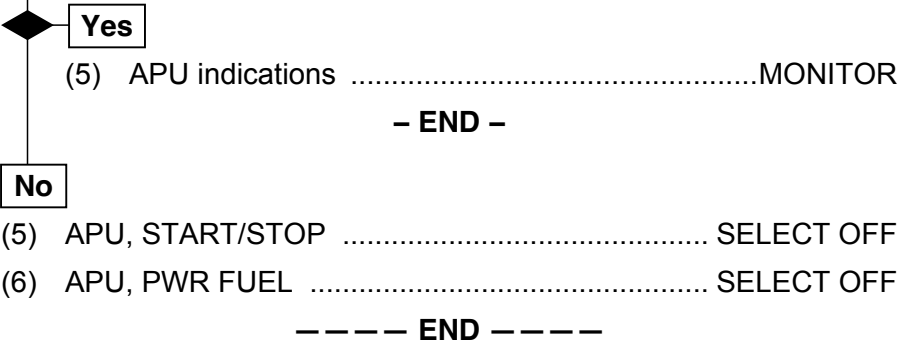
APU LCV FAIL

- (1) APU LCV SELECT closed
- (2) L PACK OFF
- (3) L 10TH STAGE BLEEDCLOSED
- (4) 10TH STAGE ISOL SELECT closed

NOTE

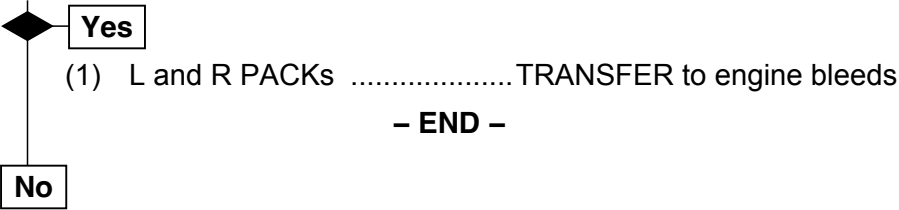
Airplane altitude not above 25000 feet during
single pack operations.

APU generator is required:



APU BLEED ON

APU BLEED ON caution message occurs during normal ECS
operation:



APU BLEED ON caution message occurs during a
“Starter-Assisted APU Bleed Relight”:

- (1) No action required.
- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

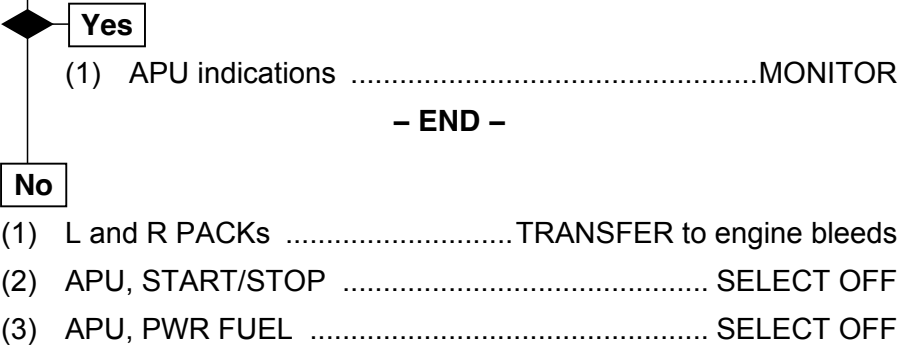
IMM

ABNORM

INDEX

APU FAULT

APU generator is required:

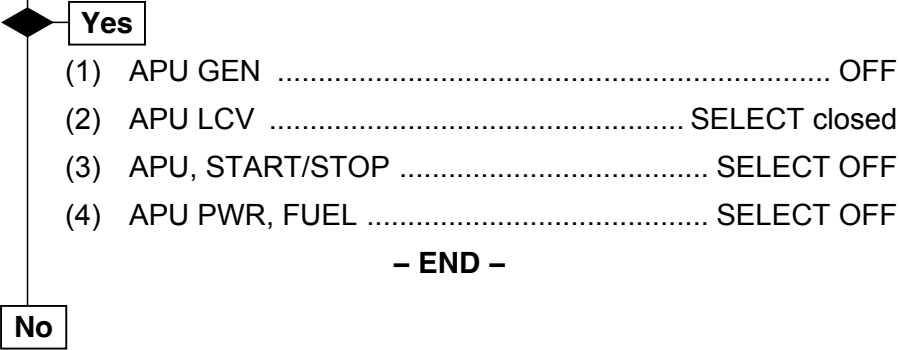


Do not restart APU.

----- END -----

APU OIL PRESS

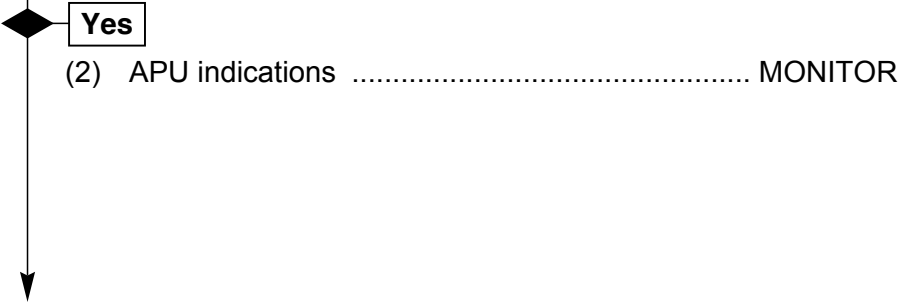
On the ground:



In flight:

- (1) L and R PACKsTRANSFER to engine bleeds

The APU generator is required:



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

▼

(3) Land at the nearest suitable airport.

NOTE

Upon landing, the APU will shutdown after 60 seconds.

– END –

No

(2) APU GEN OFF

(3) APU, START/STOP SELECT OFF

(4) APU, PWR FUEL SELECT OFF

– – – – END – – – –

APU OIL TEMP

On the ground:

◆

Yes

(1) APU GEN OFF

(2) APU LCV SELECT closed

(3) APU, START/STOP SELECT OFF

(4) APU, PWR FUEL SELECT OFF

– END –

No

In flight:

(1) L and R PACKs TRANSFER to engine bleeds

The APU generator is required:

◆

Yes

(2) APU indications MONITOR

(3) Land at the nearest suitable airport.

NOTE

Upon landing, the APU will shutdown after 60 seconds.

– END –

▼

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

(2) APU GEN OFF

(3) APU, START/STOP SELECT OFF

(4) APU, PWR FUEL SELECT OFF

----- END -----

APU Door Failure

APU message area indicates DOOR INHIB/CLSD:

(1) APU, START/STOP SELECT OFF

(2) APU, PWR FUEL SELECT OFF

- END -

APU message area indicates DOOR INHIB/OPEN:

APU is operating:

Yes

(1) No further action required.

- END -

No

(1) AirspeedNot more than 300 KIAS

----- END -----

APU message area indicates DOOR INHIB/MID or DOOR INHIB ----- or DOOR -----:

APU is operating:

Yes

(1) L and R PACKsTRANSFER to engine bleeds

(2) APU LCV SELECT closed

APU generator is required:

Yes

(3) APU indicationsMONITOR

- END -

No

(3) APU GEN OFF

(4) APU, START/STOP SELECT OFF

(5) APU, PWR FUEL SELECT OFF

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(6) AirspeedNot more than 300 KIAS

— END —

No

(1) Do not attempt to start the APU.

(2) AirspeedNot more than 300 KIAS

----- END -----

APU SOV FAIL

(1) APU, START/STOP SELECT OFF

(2) APU, PWR FUEL SELECT OFF

----- END -----

APU SOV OPEN

(1) Land at the nearest suitable airport.



The negative G valve will open and fuel may be supplied through the right engine feed line to the APU fuel control unit.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
----------	------

DOORS

- PAX DOOR OUT HNDL(Phase IV) ABNORM 5-2
- PAX DOOR OUT HNDL(Phase III) ABNORM 5-3
- PAX DOOR LATCH ABNORM 5-5
- PAX DOOR STOW(Phase III) ABNORM 5-6

MISCELLANEOUS DOORS

- CARGO DOOR ABNORM 5-7
- AV BAY DOOR ABNORM 5-7
- SERVICE DOOR ABNORM 5-7
- L EMER DOOR ABNORM 5-7
- R EMER DOOR ABNORM 5-7
- Crew Escape Hatch Unsafe ABNORM 5-7

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Effectivity:

Airplanes 7390, 7396, 7409, 7412, 7416, 7423, 7432, 7444, 7452, 7455, 7457 and subsequent:

PAX DR OUT HNDL

(1) SEAT BLTS ON

(2) Cabin pressure CHECK normal

(3) Passenger door CHECK visually

NOTE

Check that green witness marks (registers) on all latches (rotary and sliding) are correctly aligned.

Cabin pressure is normal and green witness marks are aligned:

Yes

(4) No further action required.

- END -

No

(4) MAN ALT Mid position / centre

(5) PRESS CONT MAN

(6) MAN ALT AS REQUIRED

NOTE

1. Maintain cabin altitude at 8000 feet or landing field elevation, whichever is higher, to minimize the pressure differential across the affected door.

2. Set the MAN ALT switch to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

(7) MAN RATE AS REQUIRED

commensurate with crew and passenger comfort.

When reaching 8000 feet cabin altitude or landing field elevation, whichever is higher:

(8) MAN ALT SET to mid position / centre

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

(9) Descent INITIATE to 10000 feet MSL
or lowest safe altitude,
whichever is higher.

(10) Land at the nearest suitable airport.

At 10000 feet MSL or lowest safe altitude, whichever is higher:

(11) PRESS CONTAUTO

(12) LDG ELEV SET to landing field
elevation

NOTE

After landing ensure that the airplane is
completely depressurized prior to opening any
airplane doors.

----- END -----

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

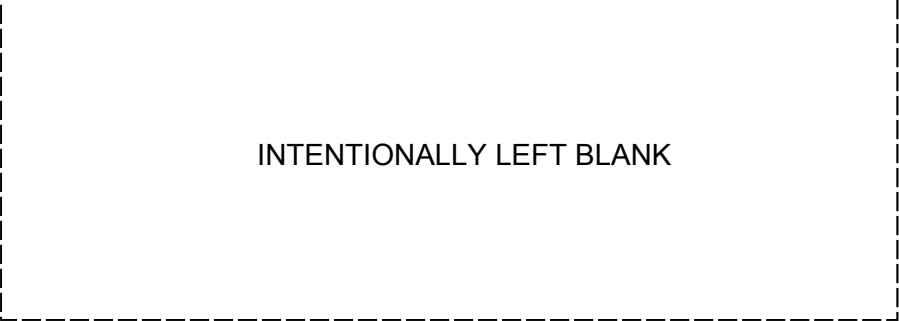
EICAS

EMER

IMM

ABNORM

INDEX



PAX DR LATCH

- (1) SEAT BLTS ON
- (2) Cabin pressure CHECK normal

Cabin pressure is normal:

Yes

- (3) No further action required.

– END –

No

- (3) MAN ALT Mid position / centre
- (4) PRESS CONTMAN
- (5) MAN ALTAS REQUIRED

NOTE

1. Maintain cabin altitude at 8000 feet or landing field elevation, whichever is higher, to minimize the pressure differential across the affected door.
2. Set the MAN ALT switch to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

- (6) MAN RATEAS REQUIRED
commensurate with crew and passenger comfort.

When reaching 8000 feet cabin altitude or landing field elevation, whichever is higher:

- (7) MAN ALT SET to mid position / centre
- (8) Descent INITIATE to 10000 feet MSL
or lowest safe altitude,
whichever is higher

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

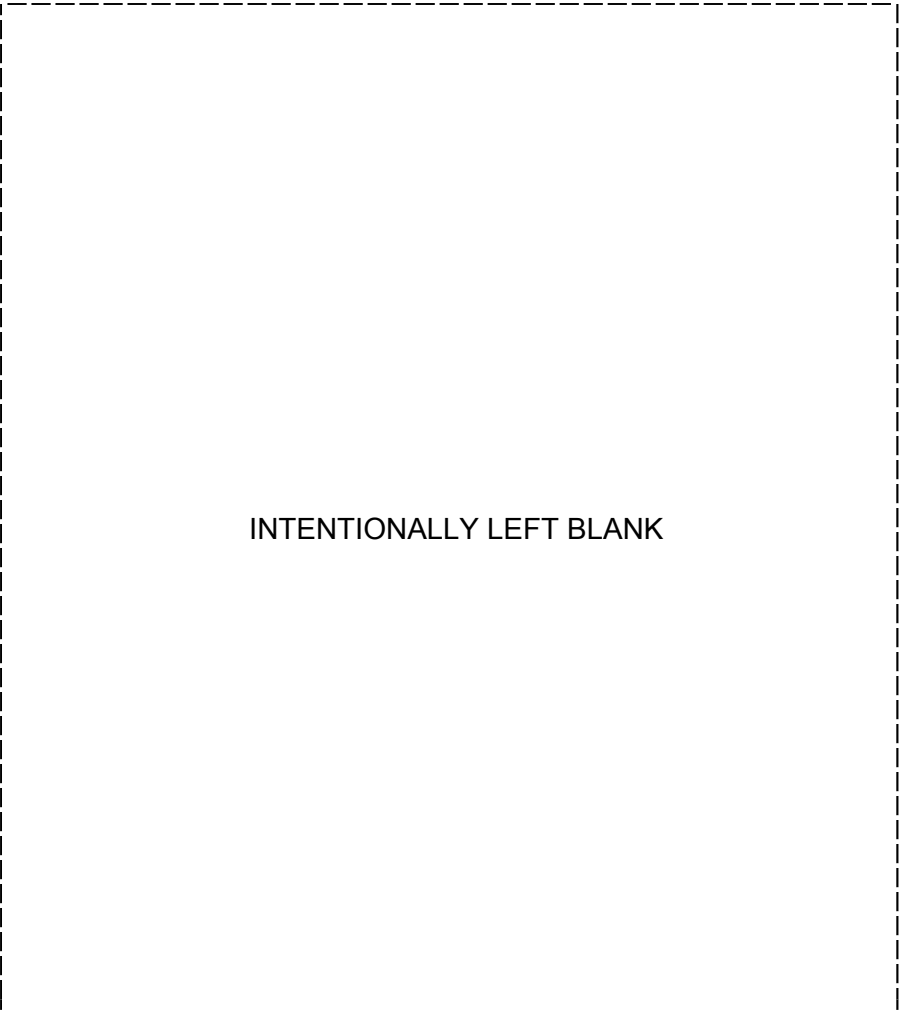
INDEX

- (9) Land at the nearest suitable airport.
- At 10000 feet MSL or lowest safe altitude, whichever is higher:
- (10) PRESS CONTAUTO
- (11) LDG ELEV SET to landing field elevation

NOTE

After landing ensure that the airplane is completely depressurized prior to opening any airplane doors.

----- END -----



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

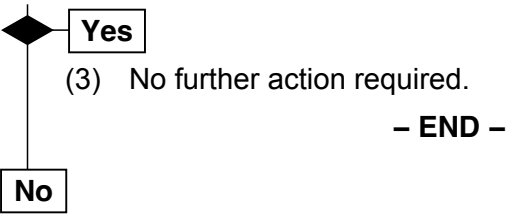
INDEX

AV BAY DOOR or CARGO DOOR or L EMER DOOR or R EMER DOOR or SERVICE DOOR or Crew Escape Hatch
Unsafe

- (1) SEAT BLTS ON

(2) Cabin pressure CHECK normal

Cabin pressure is normal:



- (3) MAN ALT Mid position / centre
- (4) PRESS CONTMAN
- (5) MAN ALTAS REQUIRED

NOTE

1. Maintain cabin altitude at 8000 feet or landing field elevation, whichever is higher, to minimize the pressure differential across the affected door.
2. Set the MAN ALT switch to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

- (6) MAN RATEAS REQUIRED
- commensurate with crew and passenger comfort.

When reaching 8000 feet cabin altitude or landing field elevation, whichever is higher:

- (7) MAN ALT SET to mid position / centre

NOTE

Full cabin depressurization is not recommended.

- (8) Descent INITIATE to 10000 feet MSL
- or lowest safe altitude,
- whichever is higher.
- (9) Land at the nearest suitable airport.

At 10000 feet MSL or lowest safe altitude, whichever is higher:

- (10) PRESS CONTAUTO

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(11) LDG ELEV SET to landing field elevation

NOTE

After landing ensure that the airplane is completely depressurized prior to opening any airplane doors.

----- **END** -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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CONTENTS	PAGE
----------	------

ELECTRICAL

- AC 1 AUTOXFER ABNORM 6-2
- AC 2 AUTOXFER ABNORM 6-2
- AC BUS 1 ABNORM 6-3
- AC BUS 2 ABNORM 6-4
- AC ESS BUS ABNORM 6-5
- AC SERV BUS ABNORM 6-6
- APU BATT CHGR ABNORM 6-6
- APU BATT OFF ABNORM 6-7
- APU GEN OFF ABNORM 6-7
- APU GEN OVLD ABNORM 6-8
- ADG Inadvertent Deployment ABNORM 6-8
- BATTERY BUS ABNORM 6-9
- DC BUS 1 ABNORM 6-10
- DC BUS 2 ABNORM 6-11
- DC EMER BUS ABNORM 6-13
- DC ESS BUS ABNORM 6-13
- DC SERV BUS ABNORM 6-14
- ESS TRU 1 ABNORM 6-14
- ESS TRU 2 ABNORM 6-15
- GEN 1 OFF ABNORM 6-15
- GEN 2 OFF ABNORM 6-15
- GEN 1 OVLD ABNORM 6-16
- GEN 2 OVLD ABNORM 6-16
- IDG 1 ABNORM 6-17
- IDG 2 ABNORM 6-17
- MAIN BATT CHGR ABNORM 6-18
- MAIN BATT OFF ABNORM 6-18
- TIE CLOSED Light(s) On ABNORM 6-20

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

AC 1 AUTOXFER

AC BUS 1 caution message is also displayed:

Yes

(1) APU (if available, at 30000 feet and below)START

(2) GEN 1 OFF/RESET

(3) APU GEN ON

(4) AC BUS 1 inoperative, affected airplane systemsREVIEW, as follows:

AC BUS 1

• Hydraulic Pump 2B and 3B

• Flap Power Drive Unit 1

• EGPWS <0040>

• Hydraulic System Fan

• Left Windshield Heater

• TAT Probe Heater

• Right AOA Heater

• Left Landing Light

• Right Pitot Heater

• Right Display Cooling Fan

• Avionics Exhaust and Cockpit Fans

• Engine Vibration Monitor

• Flight Data Recorder

• TRU 1

• Cargo Heater <0013> or <0034> or <0074>

• Left Taxi Lights

– END –

No

(1) No action required.

– END –

AC 2 AUTOXFER

AC BUS 2 caution message is also displayed:

Yes

(1) APU (if available, at 30000 feet and below)START

(2) GEN 2 OFF/RESET

(3) APU GEN ON

QUICK REFERENCE HANDBOOK CSP A-022-PRS

ELECTRICAL

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

▼ (4) AC BUS 2 inoperative, affected airplane systemsREVIEW, as follows:

AC BUS 2

- Hydraulic Pumps 1B and 3A
 - Horizontal Stabilizer Trim Actuator (HSTA), Channel 1
 - Right Windshield Heater
 - Flap Power Drive Unit 2
 - Ice Detector 2
 - TRU 2
 - Right Landing Light
- Galley Heaters
 - Copilot's Instruments Lights
 - Right Window Heater
 - Right ARINC Display Fan
 - ESS TRU 2
 - Galley, Lavatory and Cabin Fans
 - Right Taxi Light

– END –

No

(1) No action required.

----- END -----

AC BUS 1

(1) GEN 1 OFF/RESET then ON

AC BUS 1 caution message persists:

Yes

- (2) APU (if available, 30000 feet and below) START

(3) GEN 1 OFF/RESET

(4) APU GEN ON

AC BUS 1 caution message still persists:

Yes

- (5) AC BUS 1 inoperative, affected airplane systemsREVIEW, as follows:

AC BUS 1

- Hydraulic Pump 2B and 3B
- Right Pitot Heater



AC BUS 1

- Avionics Exhaust and Cockpit Fans
- Flap Power Drive Unit 1
- EGPWS <0040>
- Hydraulic System Fan
- Left Windshield Heater
- TAT Probe Heater
- Right AOA Heater
- Left Landing Light
- Right Display Cooling Fan
- Engine Vibration Monitor
- Flight Data Recorder
- TRU 1
- Cargo Heater <0013> or <0034> or <0074>
- Left Taxi Lights

– END –

No

(2) No further action required.

----- END -----

AC BUS 2

(1) GEN 2 OFF/RESET then ON

AC BUS 2 caution message persists:

Yes

- (2) APU (if available, 30000 feet and below) START
- (3) GEN 2 OFF/RESET
- (4) APU GEN ON

AC BUS 2 caution message still persists:

Yes

- (5) HYDRAULIC 3B pump ON
- (6) AC BUS 2 inoperative, affected airplane systems REVIEW, as follows:

AC BUS 2

- Hydraulic Pump 1B and 3A
- Galley Heaters

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

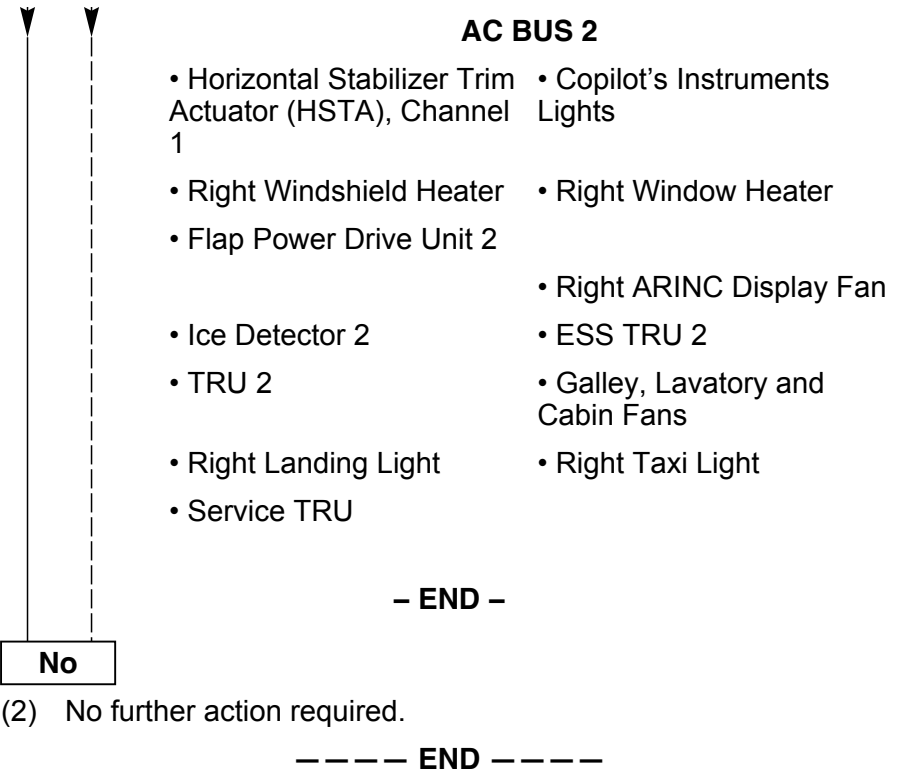
EICAS

EMER

IMM

ABNORM

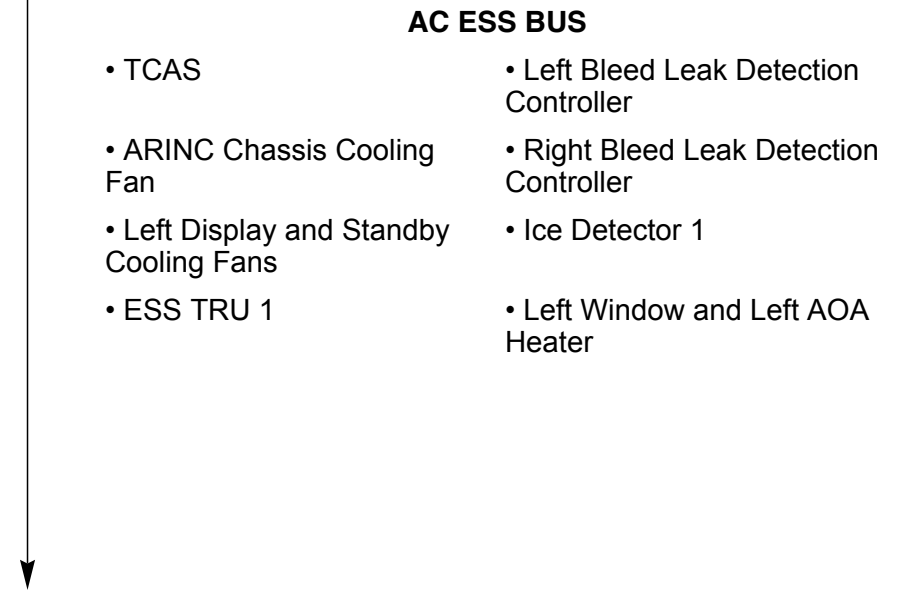
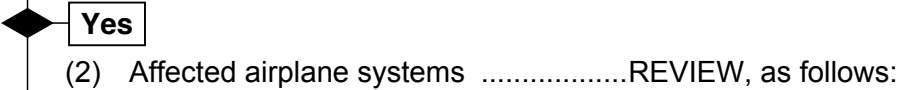
INDEX



AC ESS BUS

(1) AC ESS XFER ALTN

AC ESS BUS caution message persists:



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



AC ESS BUS

- Horizontal Stabilizer Trim Actuator (HSTA), Channel 2
- Left and Standby Pitot Heaters
- Circuit Breaker Panel Lights
- Pilot's and Centre Instrument Lights

– END –

No

(2) No further action required.

– END –

AC SERV BUS

(1) Affected airplane systemsREVIEW, as follows:

AC SERV BUS

- Service TRU
- Toilet Motor / Pump
- AC Serv Bus Feed
- Lighting, Side Wall
- 115V / 400 AC Outlet for Cabin Servicing
- Lighting, Ceiling
- Logo Lights <0020>

– END –

Effectivity:

- Airplanes **not incorporating** the following Service Bulletin:
 - SB 601R-31-022, Indicating and Recording – Engine Indication and Crew Alert System (EICAS) – EICAS 2000 Avionics System Upgrade (CSUs, DCPs, DCUs, EFDs, IOC, FMC, and MDC).

APU BATT CHGR

(1) Assume loss of APU battery charger. No action required.

– END –

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



APU BATT OFF

(1) BATTERY MASTER ON

APU BATT OFF caution message persists:

Yes

(2) DC ELECTRICAL pageMONITOR

(3) Affected airplane systemsREVIEW, as follows:

NOTE

1. APU start is not possible.
2. The APU shutdown sequence is altered. Allow 60 seconds for the APU ECU to log all events. The APU door will remain open.

APU BATT DIRECT BUS

- APU Battery Control
- Emergency Refuel
- External DC Power
- Service Bus Feed
- Oil Bypass Indicator
- Refuel/Defuel Panel
- ESS Power Control
- Engine Oil Replenishing System Power

– END –

No

(2) No further action required.

– END –

APU GEN OFF

(1) APU GENOFF/RESET then ON

(2) AC ELECTRICAL pageMONITOR

APU GEN OFF caution message persists:

Yes

(3) APU GEN OFF/RESET

– END –

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

(3) No further action required.

----- END -----

APU GEN OVLD

(1) AC ELECTRICAL pageMONITOR

(2) AC loads REDUCE as necessary

----- END -----

ADG Inadvertent Deployment

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Effectivity:

- Airplanes 7305 and subsequent:

(1) No airspeed limitation with the ADG deployed

To re-establish normal operations:

(2) ADG manual release handleSTOW

(3) ADG PWR TXFR OVERRIDESELECT

Effectivity:

- Airplanes **incorporating** the following Service Bulletin:
 - SB A601R-29-031, Hydraulic Power – Hydraulic System No. 3 – Deactivation and Removal of the Accumulator Assembly.

(4) HYDRAULIC 3B pump switch ON

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----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

BATTERY BUS

(1) Affected airplane systemsREVIEW, as follows:

BATTERY BUS

- | | |
|--|---|
| • Engine Ignition B | • EICAS Control Panel and Right Lamp Driver Unit |
| • L Engine Start and R Engine Start | • EICAS / RTU Dimming |
| • Fuel System Control | • Standby Horizon Lights |
| • Left Fuel Pump Power | • Integrated Standby Instrument Lights |
| • Left Fuel Pump Control | • APU Fuel Pump |
| • Fuel Cross-flow and Transfer | • APU Transfer |
| • Left Engine Oil Pressure Indication | • APU Control |
| • Engine Oil Level | • IDG 1 and 2 Disconnect |
| • Fire Detection (A, B and Test) | • Main Battery Contactor |
| • MLG Bay Overheat Detection | • Generator Control Unit (1, 2 and 3) |
| • Cargo Bay Fire Extinguishing | • Feed 1 (Batt and DC ESS) |
| • CPAM | • Ess AC Transfer Control |
| • Ram Air SOV | • Feed 2 (Batt and DC ESS) |
| • Crew Oxygen Quantity Indicators | • Stick Pusher, Left Channel |
| • Passenger Oxygen (Automatic and Manual Deployment) | • Spoiler Electronic Control Unit, 1B and 2A |
| • IAPS, Left AFCS and Fan | • Hydraulic System 3 Indication |
| • Copilot Audio Control | • Cowl Anti-icing and Right Standby Wing Anti-icing |
| • VHF 1 | • Dome Lights |
| • Emergency Tuning Unit | • Copilot Observer Map Lights |
| • PA | • Pilot Map Light |

(2) Land at the nearest suitable airport.

----- **END** -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

DC BUS 1

(1) TIE 1CLOSED

DC BUS 1 caution message persists:

Yes

(2) Affected airplane systemsREVIEW, as follows:

DC BUS 1

- Standby Cargo Bay Fire Extinguishing
 - Toilet Smoke Detector
 - Overboard Shutoff Valve
 - Left PACK and Left 10th Stage Bleed Air SOV
 - Left 14th Stage SOV and 14th Stage Isolation Valve
 - Cargo Bay Ventilation System Controller
 - Cockpit Auto and Manual Temperature Control
 - Radio Altimeter, DME 1, Weather Radar
 - IAPS Left FMS/MDC <0024>
 - GPS 1 <0027> or <0047>
 - Flight Data Recorder Control
- DC 1 Power Sensing
 - TRU 1 Power Sensing
 - Flaps Control Channel 1 and Pitch Feel Channel 1
 - Spoiler Electronic Control System Power 1
 - Hydraulic 2 Indicator, Hydraulic Pump Controllers 2B and 3B
 - Hydraulic System Fan Controller
 - Brake Temperature Monitor
 - Left Normal Wing Anti-ice Overheat Protection
 - Right Static and ADS Controller Heaters
 - Left Windshield Heater Controller and Pilot Wiper
 - Cockpit Floor, Rear Anti-collision and Wing Inspection Lights
 - Pax Door System
 - Pass Signs

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



DC BUS 1

- Data Loader
 - Utility Bus 1 Feed
- Left Lamp Driver Unit
 - DC Bus 1 Feed

NOTE

The left normal wing anti-icing overheat protection is not available. If wing anti-icing is required, use the standby mode.

– END –

No

(2) No further action required.

----- END -----

DC BUS 2

(1) TIE 2CLOSED

DC BUS 2 caution message persists:

Yes

(2) Affected airplane systemsREVIEW, as follows:

DC BUS 2

- Right Engine Fuel Pump
 - Right PACK and Right 10th Stage Bleed Air SOV
 - Right 14th Stage Bleed Air SOV
 - IAPS Right Fan
 - IAPS Right FMS <0024>
 - Copilot PFD, MFD, EFIS Control
 - VHF 2, Radio Tuning Unit 2 and HF <0011> or <0060>
- Spoiler Electronic Control System Power 2
 - Horizontal Stabilizer Trim Control Unit (HSTCU) Channel 1
 - Hydraulic 1 Indicator, Hydraulic Pumps 1B and 3A Controllers
 - Brake Pressure Indicators
 - Right Normal Wing Anti-icing Overheat Protection
 - Bus 2 Feed Tie and Utility
 - DC 2 Power Sensing
 - TRU 2 Power Sensing

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



DC BUS 2

- NAV 2, ADF 2, DME 2, and ATC 2
- Radio Altimeter 2 <0045>
- GPS 2 <0027>
- Avionic Cooling Ckpt SOV
- AHRS FAN 2
- Avionic Cooling Cont 2
- Avionic Cooling I/B SOV
- Rudder and Aileron Trims
- DC BUS 2 FEED
- Right Window Heater Controller and Copilot Wiper
- Right Windshield Heater Controller
- Cabin Auto and Manual Temperature Control
- Galley Heater Controller
- Copilot Map Lights
- Wing Anti-collision Lights
- Flaps Control Channel 2 and Pitch Feel Channel 2

NOTE

The right normal wing anti-icing overheat protection is not available. If wing anti-icing is required, use the standby mode.

– END –

No

(2) No further action required.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



DC EMER BUS

(1) Affected airplane systemsREVIEW, as follows:

- Engine and APU fire extinguishing systems are inoperative,
- Firex bottles are inoperative,
- Engine and APU fuel feed shut-off valves will not close, and
- Engine hydraulic shut-off valves will not close.

DC EMER BUS

- Left and Right Engine Fire Extinguisher 1 and 2
- Left and Right Engine Fuel SOV
- Left and Right Engine Hydraulic SOV
- Power Sensing
- APU Fire Extinguisher
- APU Fuel SOV
- APU Battery Direct Feed

(2) Land at the nearest suitable airport.

----- END -----

DC ESS BUS

(1) Affected airplane systemsREVIEW, as follows:

DC ESSENTIAL BUS

- Thrust Reversers (Both)
- Right Oil Pressure Indicator
- Fuel System Control
- Fuel Gravity Cross-flow
- Pressurization Controllers (1 and 2) and Control Panel
- 10th Stage Isolation Valve
- Cargo Bay Air-conditioning Shut-off Valves
- Pilot PFD, MFD, EFIS Control
- EFIS, Dimming Panel 1
- 28 VDC Essential Bus Sensing
- Bleed Air Leak Test
- Spoiler Electronic Control System 1 and 2 (PWR 3)
- Stall Protection, Right Channel
- Horizontal Stabilizer Trim Control Unit (HSTCU) Channel 2
- Left Standby Wing Anti-ice
- Left Window Heater Controller

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



DC ESSENTIAL BUS

- Avionic Cooling Controller (1)
- Left Static and ADS Controller Heaters
- AHRS Fan 1
- Cockpit Voice Recorder
- NAV 1, ADF 1, and ATC 1
- Instrument Flood Lights
- Radio Tuning Unit 1
- Emergency Lights

(2) Land at the nearest suitable airport.

----- END -----

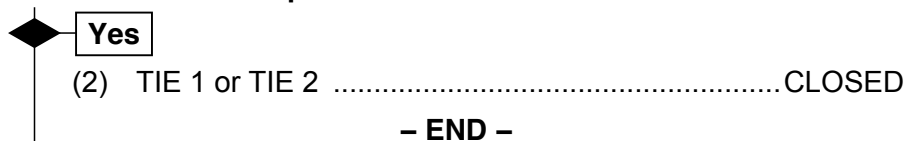
DC SERV BUS

(1) Affected aircraft systemsREVIEW, as follows:

DC SERVICE BUS

- Beacon Lights <0021>
- Waste System
- CKPT Dome Lights
- Water Control
- Chart Holder
- Service Bus Feed
- General Cabin Lighting
- PWR Sens Serv Bus
- Toilet and Galley Dome Lights
- PWR Sens Serv TRU

DC SERV BUS is required:



No

(2) No further action required.

----- END -----

ESS TRU 1

(1) DC ELECTRICAL pageMONITOR DC loads

NOTE

During generator switching, power interruptions will occur on the DC essential bus.

----- END -----



ESS TRU 2

(1) ESS TIE switchCLOSED

NOTE

Service TRU is tied to the essential and battery busses.

----- END -----

GEN 1 OFF

(1) GEN 1 OFF/RESET then ON

GEN 1 OFF caution message persists:

Yes

(2) GEN 1 OFF/RESET

NOTE

1. With GEN 1 shut down, HYD PUMP 2B is not available.
2. During single generator operations, AC utility busses are load shed. Both battery chargers are not available.

(3) APU (if available, at 30000 feet and below)START

(4) APU GEN ON

(5) HYDRAULIC 2 pump OFF

- END -

No

(2) Electrical loadsMONITOR

----- END -----

GEN 2 OFF

(1) GEN 2 OFF/RESET then ON

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



GEN 2 OFF caution message persists:

Yes

(2) GEN 2 OFF/RESET

NOTE

1. With GEN 2 shut down, HYD PUMP 1B is not available.
2. During single generator operations, AC utility busses are load-shed. Both battery chargers are not available.

(3) APU (if available, at 30000 feet and below)START

(4) APU GEN ON

(5) HYDRAULIC 1 pump OFF

— END —

No

(2) Electrical loadsMONITOR

— — — — END — — — —

GEN 1 OVLD

(1) AC ELECTRICAL page MONITOR GEN 1 LOAD

(2) AC loads REDUCE as necessary

— — — — END — — — —

GEN 2 OVLD

(1) AC ELECTRICAL page MONITOR GEN 2 LOAD

(2) AC loads REDUCE as necessary

— — — — END — — — —

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

IDG 1

(1) GEN 1 OFF/RESET

NOTE

- 1. With GEN 1 shut down, HYD PUMP 2B is not available.
- 2. During single generator operations, AC utility busses are load-shed. Both battery chargers are not available.

(2) IDG 1 DISC CONFIRM and DISC
(for not more than 3 seconds)

NOTE

- 1. If **IDG 1 DISC** status message is not displayed within 5 seconds, monitor oil pressure and oil temperature on the affected engine.
- 2. The IDG cannot be re-connected during flight.

(3) APU (if available, at 30000 feet and below)START

(4) APU GEN ON

(5) HYDRAULIC 2 pump OFF

----- **END** -----

IDG 2

(1) GEN 2 OFF/RESET

NOTE

- 1. With GEN 2 shut down, HYD PUMP 1B is not available.
- 2. During single generator operations, AC utility busses are load-shed. Both battery chargers are not available.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



ABNORM 6-18

Rev. 65, Jan 11/2016

(2) IDG 2 DISC CONFIRM and DISC
(for not more than 3 seconds)

NOTE

- 1. If **IDG 2 DISC** status message is not displayed within 5 seconds, monitor oil pressure and oil temperature on the affected engine.
- 2. The IDG cannot be re-connected during flight.

(3) APU (if available, at 30000 feet and below) START
(4) APU GEN ON
(5) HYDRAULIC 1 pump OFF

----- END -----

Effectivity:

- Airplanes **not incorporating** the following Service Bulletin:
 - SB 601R-31-022, *Indicating and Recording – Engine Indication and Crew Alert System (EICAS) – EICAS 2000 Avionics System Upgrade (CSUs, DCPs, DCUs, EFDs, IOCs, FMC, and MDC).*

MAIN BATT CHGR

(1) Assume loss of main battery charger. No action required.

----- END -----

MAIN BATT OFF

(1) BATTERY MASTER ON

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

MAIN BATT OFF caution message persists:

Yes

- (2) DC ELECTRICAL pageMONITOR main battery voltage and assume loss of main battery if <18 volts.
- (3) Affected airplane systemsREVIEW, as follows:

MAIN BATTERY DIRECT BUS

- Engine APR System

• Main Battery Contactor
- Standby Power Controller

NOTE

1. Engine APR system is inoperative.
2. Standby power within the IRS is not available. <0025> or <0049>
3. During generator switching, **ENG MISCOMP** warning messages will be displayed.
4. The APU shutdown sequence is altered. Allow 60 seconds for the APU ECU to log all events. The APU door will remain open.

– END –

No

- (2) No further action required.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

TIE CLOSED Light(s) On

(1) Associated DC TIECLOSED

NOTE

- 1. DC TIE 1 will close automatically with failure of TRU 1 or failure of TRU 2 and SERV TRU.
- 2. DC TIE 2 will close automatically with failure of TRU 2 or failure of TRU 1 and SERV TRU.
- 3. With a DC TIE 1 or DC TIE 2 closed, the associated utility bus is not powered.
- 4. DC ESS TIE can only be closed manually.

----- **END** -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
----------	------

FIRE PROTECTION

- APU FIRE FAIL ABNORM 7-2
- L FIRE FAIL or R FIRE FAIL ABNORM 7-2
- L JET OVHT FAIL or R JET OVHT FAIL ABNORM 7-3

FIREX BOTTLES

- APU BTL LO ABNORM 7-3
- ENG BTL 1 LO or ENG BTL 2 LO ABNORM 7-4
- CARGO BTL LO <0013> or <0034> or <0043> or <0053> or <0059> or <0074> ABNORM 7-4

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

APU FIRE FAIL

(1) FIRE DETECTION TESTWARN

APU FIRE warning message comes on:

Yes

(2) No further action required.

NOTE

APU fire protection is still available.

- END -

No

(2) APU fire protection is not available.

APU generator is required:

Yes

(3) APU operationMONITOR

- END -

No

(3) APU, START/STOP SELECT OFF

(4) APU, PWR FUEL SELECT OFF

----- END -----

L FIRE FAIL or R FIRE FAIL

(1) FIRE DETECTION TESTWARN

L ENG FIRE or R ENG FIRE warning message comes on:

Yes

(2) No further action required.

NOTE

Engine fire protection is still available.

- END -

No

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(2) Affected engineMONITOR

NOTE

Affected engine fire protection is not available.

----- END -----

L JET OVHT FAIL or R JET OVHT FAIL

(1) FIRE DETECTION TESTWARN

L JETPIPE OVHT or R JETPIPE OVHT warning message comes on:

Yes

(2) No further action required.

NOTE

Overheat protection is still available.

- END -

No

(2) Affected engineMONITOR

NOTE

Affected engine overheat protection is not available.

----- END -----

APU BTL LO

(1) FIRE DETECTION TESTWARN

APU BOTTLE PUSH TO DISCH light comes on:

Yes

(2) No further action required.

NOTE

APU bottle is charged.

- END -

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

(2) APU BottleASSUME discharged

APU is required:

Yes

(3) APU operationMONITOR

- END -

No

(3) APU, START/STOP SELECT OFF

(4) APU, PWR FUEL SELECT OFF

----- END -----

ENG BTL 1 LO or ENG BTL 2 LO

(1) Affected engine BottleASSUME discharged

----- END -----

CARGO BTL LO <0013> or <0034> or <0043> or <0053> or <0059> or <0074>

(1) CARGO BOTTLEASSUME discharged

(2) AIR CONDITIONING, CARGO OFF

(3) Land at the nearest suitable airport.

NOTE

Advise ground crew of the possible presence of Halon vapours and smoke in the cargo compartment.

----- END -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
----------	------

FLIGHT CONTROLS

STALL PROTECTION

- **STALL FAIL** [ABNORM 8-2](#)

PRIMARY FLIGHT CONTROL

- **Aileron PCU Runaway** [ABNORM 8-3](#)

TRIM CONTROL

- **MACH TRIM** [ABNORM 8-5](#)
- **STAB TRIM** [ABNORM 8-5](#)
- **Horizontal Stabilizer Trim Malfunction** [ABNORM 8-8](#)
- **Aileron or Rudder Trim Runaway** [ABNORM 8-8](#)
- **Stab Trim Channel Failure** [ABNORM 8-9](#)

FLAPS

- **Flap Twist (In Flight)** [ABNORM 8-9](#)
- **FLAPS FAIL** [ABNORM 8-13](#)

SPOILERS SYSTEM

- **FLT SPLR DEPLOY** <JAA> [ABNORM 8-17](#)
- **FLT SPLRS** [ABNORM 8-18](#)
- **L FLT SPLRS or R FLT SPLRS** [ABNORM 8-19](#)
- **SPOILERONS ROLL** [ABNORM 8-20](#)
- **L SPOILERON or R SPOILERON** [ABNORM 8-21](#)
- **SPOILERONS** [ABNORM 8-21](#)
- **FLIGHT SPOILER Lever Jammed (Spoilers Deployed)** [ABNORM 8-22](#)
- **GND SPLR DEPLOY** [ABNORM 8-23](#)
- **IB GND SPLRS** [ABNORM 8-24](#)
- **OB GND SPLRS** [ABNORM 8-25](#)
- **GLD UNSAFE** [ABNORM 8-26](#)
- **GLD NOT ARMED** [ABNORM 8-27](#)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

STALL FAIL

STALL FAIL caution message displayed:

- Yes

(1) STALL PTCT PUSHER
(left or right) OFF

(2) Approach speedNot less than
V_{REF} (FLAPS 45) + 10 KIAS

(3) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.22 (22%)	1.25 (25%)
Without Thrust Reverser(s)	1.25 (25%)	1.25 (25%)



The low speed awareness cue may represent preset/default settings and should not be relied upon for proximity to stall shaker. Increase all reference speeds by 10 KIAS.

NOTE

Windshear guidance is inoperative on both sides, **WINDSHEAR FAIL** status message comes on.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

With nuisance shaker operation and/or warbler aural warning:

- (4)

STALL PROT L CH circuit
breaker (1Q2) OPEN
- (5)

STALL PROT R CH circuit
breaker (4C7) OPEN

– END –

No

- (1) No further action required.

----- END -----

Aileron PCU Runaway

The following indications of an aileron PCU runaway may be present singularly or in combination:

- Abnormal or uncommanded change in roll attitude and/or higher than normal control forces required to maintain level flight.
- PLT ROLL or CPLT ROLL light on the glareshield and/or the PLT ROLL or the **PLT ROLL CMD** or **CPLT ROLL CMD** advisory message may be displayed.
- AP TRIM IS LWD** or **AP TRIM IS RWD** caution message may be displayed.

- (1) Aileron controls OVERPOWER to maintain control of the operating aileron and both spoilerons. Additional control forces will be experienced with control wheel movement.

NOTE

1.

The PLT ROLL or CPLT ROLL light on the glareshield or the **PLT ROLL CMD** or **CPLT ROLL CMD** advisory message indicates the operative aileron.
2.

If the PLT ROLL or CPLT ROLL light is not on, identify the failed side by verifying aileron response to handwheel movements in the FLIGHT CONTROLS synoptic page.
3.

The pilot with the operative aileron will take control of the airplane.

- (2) Airplane control Transfer to pilot with operative aileron

- (3) Autopilot Disengage
- (4) ROLL DISC PULL and TURN to lock

NOTE

1. Following disconnect, the inoperative side handwheel will go hardover in the direction of the failure.
2. Roll controllability is reduced. The roll disconnect will result in half feel during airplane handling.

- (5) PLT ROLL or CPLT ROLL SELECT operative side

NOTE

If the PLT ROLL or CPLT ROLL switch is not selected within 20 seconds of pulling the ROLL DISC handle, the **SPOILERONS ROLL** caution message will come on.

- (6) Land at the nearest suitable airport.

Prior to landing:

NOTE

Controllability is reduced. Select the longest runway available with minimum cross-wind and turbulence.

- (7) GRND PROX, FLAP <0040> OVRD
- (8) Landing FLAPS20
- (9) Approach speedNot less than

$V_{REF (FLAPS 45)} + 12 \text{ KIAS}$
- (10) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.22 (22%)	1.25 (25%)
Without Thrust Reverser(s)	1.25 (25%)	1.25 (25%)

I
----- END -----

MACH TRIM

With at least one STAB TRIM channel engaged:

(1) MACH TRIM ENGAGE

NOTE

1.

With a Mach trim failure, do not exceed 250 KIAS (0.7 M) unless the autopilot is engaged and operating normally.
2.

Two Air Data Computers are required for MACH TRIM operation.

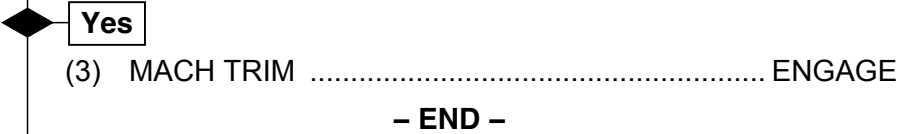
----- END -----

STAB TRIM

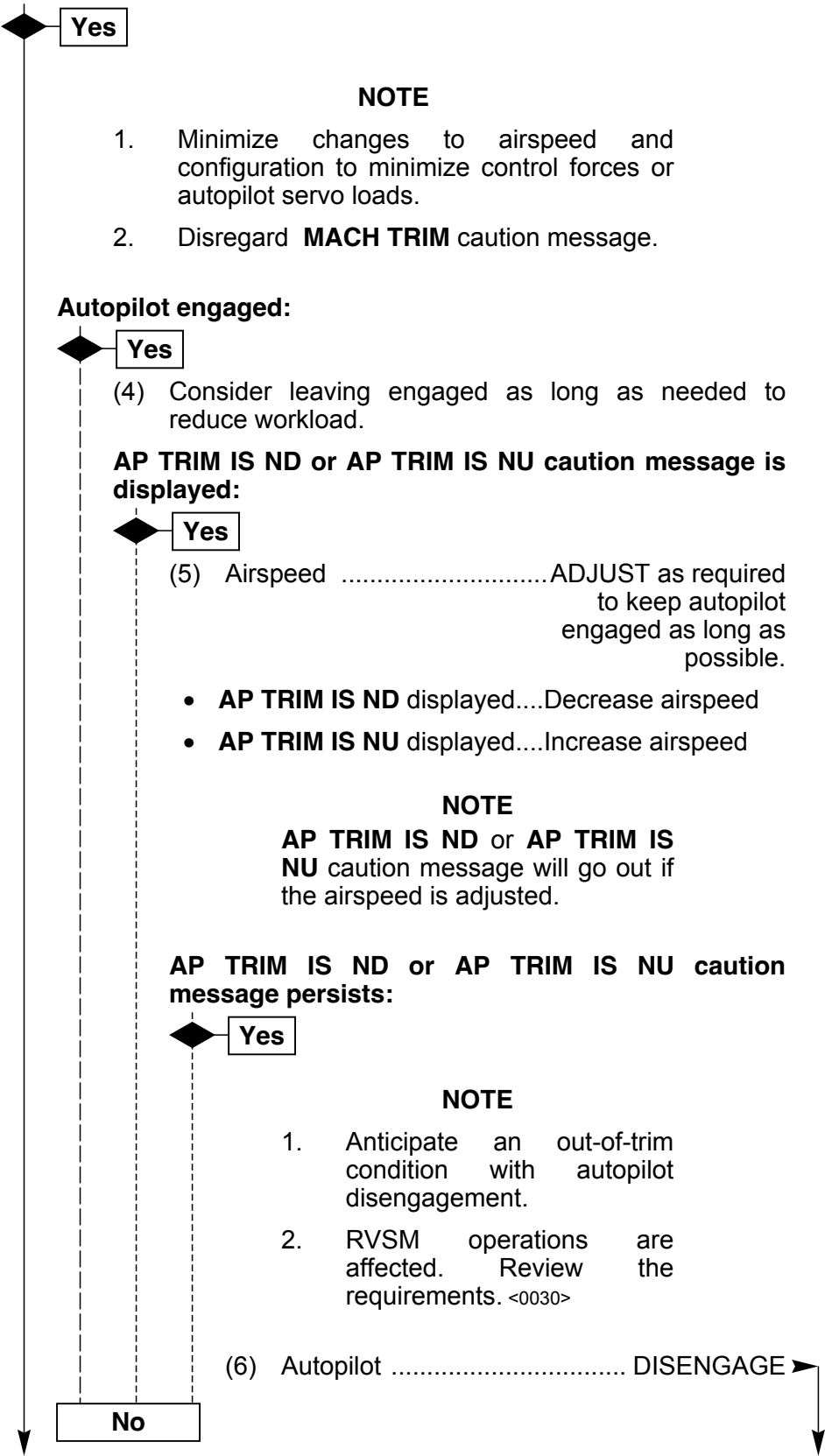
(1) STAB TRIM DISC PRESS, HOLD and RELEASE

(2) STAB TRIM, CH 1 and CH 2 ENGAGE

STAB TRIM caution message goes out:



STAB TRIM caution message persists:



Prior to landing: ←

Failure occurred at FLAPS 0, 8, 20 or 30:

Yes

- (5) GRND PROX, FLAP
<0040> OVRD
- (6) Landing FLAPS20
- (7) Approach speedNot less than
V_{REF} (FLAPS 45) + 12 KIAS
- (8) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.22 (22%)	1.25 (25%)
Without Thrust Reversers	1.25 (25%)	1.25 (25%)

No

Failure occurred at FLAPS 45:

- (5) Landing FLAPSLeave at 45
- END -

No

- (4) MACH TRIM ENGAGE
- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Horizontal Stabilizer Trim Malfunction

Horizontal stabilizer cannot be trimmed to the desired position and STAB CH 1 INOP status message is not displayed:

- (1) STAB TRIM DISC PRESS, HOLD and
RELEASE
- (2) STAB TRIM, CH 2 ENGAGE

NOTE

Do not engage STAB TRIM, CH 1.

- (3) MACH TRIM ENGAGE
- END -----

Aileron or Rudder Trim Runaway

NOTE

Some system failures may result in an inaccurate trim position indication. Flight crew should compare the trim position indication with the actual aircraft behavior to assess if the trim indication is accurate.

- (1) Aileron (or rudder) OVERPOWER
- (2) Affected trim NEUTRAL

Trim runaway persists:

- Yes

(3) Applicable trim circuit
breaker OPEN

Rudder trim circuit breaker (2F6)

Aileron trim circuit breaker (2F7)

NOTE

Use of autopilot may alleviate aileron trim problem.

– END –

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



No

- (3) Operate the trim switch with caution for the remainder of the flight.

----- END -----

Stab Trim Channel Failure

- (1) Affected STAB TRIM, CH 1 or
CH 2 ENGAGE

STAB CH 1 or STAB CH 2 INOP status message persists:

Yes

- (2) Assume one stabilizer trim channel inoperative. Pitch trim automatically transfers to the other channel.

- END -

No

- (2) No further action required.

----- END -----

Flap Twist (In Flight)

Abnormal aileron control wheel angles during flap operation with the autopilot on or the airplane rolls without pilot input with the autopilot off. This can occur with or without a **FLAPS FAIL** caution message.

Flaps are being extended:

Yes

- (1) FLAPS lever SELECT to previously
selected position (e.g.
FLAPS selected from 8 to 20;
reselect 8)

No

Flaps are being retracted:

- (1) FLAPS lever MAINTAIN in current
selected position (e.g.
FLAPS selected from 20 to 8;
leave selector at 8)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Do not attempt to operate the flaps any further.

(2) Autopilot DISENGAGE

NOTE

Anticipate out-of-trim condition and hold the aileron control wheel in its current position when disengaging autopilot.

Flaps failed at 0:

Yes

(3) HYDRAULIC pumps 1, 2 and 3B ON

NOTE

No reduction in cruise speed is required unless flap overspeed warning comes on.

No

(3) Maximum enroute airspeed V_{FE} for next greater flap setting from failed position.

Flaps failed at 45:

Yes

(4) Airspeed V_{FE} and V_{REF} for FLAPS 45

NOTE

1. If an abnormal aileron control wheel angle to the left develops, do not land with a cross-wind from the left of greater than 20 knots.
2. If an abnormal aileron control wheel angle to the right develops, do not land with a cross-wind from the right of greater than 20 knots.

- END -

No

Prior to landing:

(4) GRND PROX, FLAP <0040> OVRD

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(5) Approach speed $V_{REF (FLAPS 45)} + \Delta V_{REF}$ for next smaller flap setting from failed position.

NOTE

- 1. If failure occurs with the flaps at a detented position, use V_{FE} , V_{REF} and landing distance factors relative to that flap position.
- 2. If flaps fail at greater than 5 degrees, land at the nearest suitable airport.
- 3. If an abnormal aileron control wheel angle to the left develops, do not land with a cross-wind from the left of greater than 20 knots.
- 4. If an abnormal aileron control wheel angle to the right develops, do not land with a cross-wind from the right of greater than 20 knots.

(6) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	FLAPS Position (Degrees)	Final Approach Speed $\Delta V_{REF}(kt)$	Runway Surface	
			Dry	Wet or Contaminated
With Two Thrust Reversers	0	30	1.65 (65%)	1.70 (70%)
	8	18	1.50 (50%)	1.55 (55%)
	20	12	1.22 (22%)	1.25 (25%)
	30	8	1.18 (18%)	1.20 (20%)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



	FLAPS Position (Degrees)	Final Approach Speed $\Delta V_{REF}(kt)$	Runway Surface	
			Dry	Wet or Contaminated
Without Thrust Reverser(s)	0	24	1.70 (70%)	1.70 (70%)
	8	18	1.55 (55%)	1.55 (55%)
	20	12	1.25 (25%)	1.25 (25%)
	30	8	1.20 (20%)	1.20 (20%)

NOTE

1. Accomplish a normal descent. Reduce vertical speed at touchdown to 400 fpm or less.
2. Improper landing technique during a flaps failure can generate nose gear loads sufficient to cause structural damage.
3. To prevent damage:
 - after main landing gear touchdown, gently lower the nose gear to the runway.
 - apply brakes only after nose wheel touchdown.

After landing:

- (7) Do not attempt to retract the flaps. Report to maintenance.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

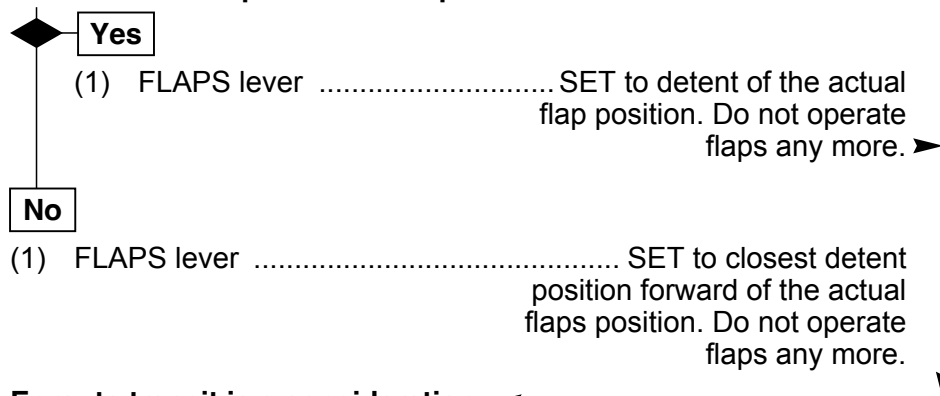


FLAPS FAIL

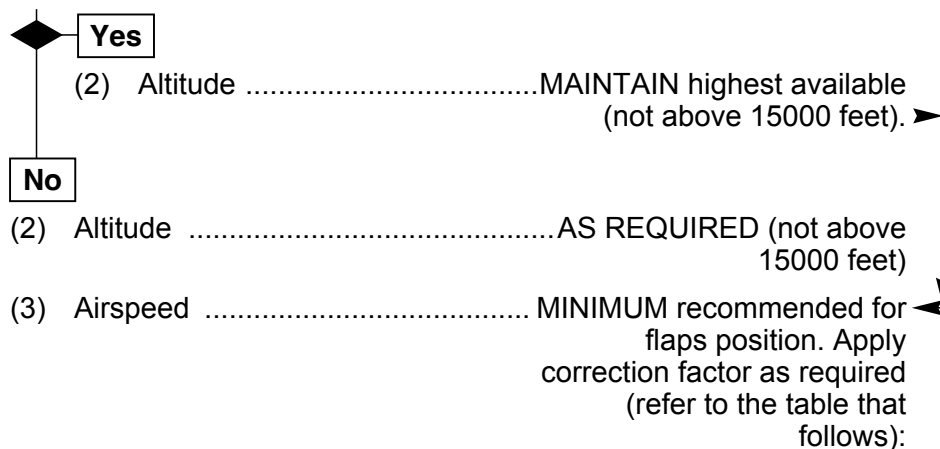
WARNING

1. If the flaps fail at greater than 30 degrees with the landing gear retracted, the landing gear warning horn may sound continuously. To silence ALL audio warnings, set the AUDIO WARNING switches to DISABLE.
2. Range is significantly reduced with flaps failures at positions above 0 degrees. Check Fuel Flow (FF) indications on EICAS. Consider diversion to a closer suitable airport.
3. Determine the fuel or range requirements using the FMS for normal operations without altitude or speed restriction and apply the fuel burn factor or range factor.

EICAS shows flaps at a detent position:



Enroute transit is a consideration:



FLAPS Position (Degrees)	Maximum Speed (kts)	Minimum Recommended Speed (kts)	Fuel Burn Factor (Fixed Distance)	Range Factor (Fixed Fuel Quantity)
1 – 7	230	V _{REF} (FLAPS 45) + 30	1.50	0.65
8	230	V _{REF} (FLAPS 45) + 18	1.50	0.65
9 – 19	230	V _{REF} (FLAPS 45) + 18	1.80	0.55
20	230	V _{REF} (FLAPS 45) + 12	1.80	0.55
21 – 29	185	V _{REF} (FLAPS 45) + 12	2.25	0.45
30	185	V _{REF} (FLAPS 45) + 8	2.25	0.45
31 – 44	170	V _{REF} (FLAPS 45) + 8	3.20	0.30
45	170	V _{REF} (FLAPS 45)	3.20	0.30
NOTE Factors are based on landing gear up configuration.				

NOTE
If the flaps are confirmed retracted (0 degrees),
reduction of cruise airspeed and altitude is not
required.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Flaps failure occurred at greater than 0:

Yes



If possible, avoid prolonged flight in icing. If icing is encountered for an extended period of time, aerodynamic buffet may be experienced.

NOTE

If flaps failure at 45 and aerodynamic buffet is encountered after flying in icing conditions, to reduce buffet, increase the minimum approach speed to $V_{REF (FLAPS 45)} + 5 \text{ KIAS}$.

(4) Land at the nearest suitable airport.

No

Prior to landing:

(5) AUDIO WARNING (all) NORMAL

(6) HYDRAULIC pumps 1, 2 and 3B ON

NOTE

Higher than normal flight control feel force loads may exist.

EICAS shows flaps position is at 45:

Yes

(7) No further action required.

– END –

No

(7) GRND PROX, FLAP <0040> OVRD

(8) Final approach speed $V_{REF (FLAPS 45)} + \Delta V_{REF}$
(refer to the table that follows)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(9) Actual landing distanceINCREASE (refer to the table that follows)

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	FLAPS Position (Degrees)	Final Approach Speed $\Delta V_{REF}(kt)$	Runway Surface	
			Dry	Wet or Contaminated
With Two Thrust Reversers	0 – 7	30	1.65 (65%)	1.70 (70%)
	8 – 19	18	1.50 (50%)	1.55 (55%)
	20 – 29	12	1.22 (22%)	1.25 (25%)
	30 – 44	8	1.18 (18%)	1.20 (20%)
Without Thrust Reverser(s)	0 – 7	30	1.70 (70%)	1.70 (70%)
	8 – 19	18	1.55 (55%)	1.55 (55%)
	20 – 29	12	1.25 (25%)	1.25 (25%)
	30 – 44	8	1.20 (20%)	1.20 (20%)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

NOTE

- 1. When the runway is wet or contaminated and additional accuracy of the actual landing distance is required, refer to the Quick Reference Handbook, Volume 1, Wet and Contaminated Runway Data, for the specific flaps setting table.
- 2. Accomplish a normal descent. Reduce vertical speed at touchdown to 400 fpm or less.
- 3. Improper landing technique during a flaps failure can generate nose gear loads sufficient to cause structural damage.
- 4. To prevent damage:
 - after main gear touchdown, gently lower the nose gear to the runway.
 - apply brakes only after nose wheel touchdown.

EXAMPLE:

Failure occurs at FLAPS 28:

Maximum enroute airspeedV_{FE} for FLAPS 30
Approach and landing speedV_{REF} (FLAPS 45)
+ 12 KIAS
Actual landing distance factor 1.25 (25%)
Actual landing distance factor with thrust
reversers 1.22 (22%)

----- **END** -----

FLT SPLR DEPLOY <JAA>

At less than 800 feet but more than 300 feet:

(1) FLIGHT SPOILER lever DO NOT RELEASE

At 300 feet:

(1) FLIGHT SPOILER leverRETRACT

----- **END** -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



FLT SPLRS

Flight spoilers are deployed:

Yes
(1) FLIGHT SPOILER lever RETRACT. Do not deploy flight spoilers manually.
- END -

No
(1) Airplane altitude Not above 32000 feet
(1) Airplane altitude <0071> NO RESTRICTION

NOTE

Landing distance factors for failed flight spoilers are based upon serviceable ground spoilers.

(2) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.15 (15%)	1.20 (20%)
Without Thrust Reverser(s)	1.20 (20%)	1.20 (20%)

----- END -----



L FLT SPLR or R FLT SPLR

- (1) Flight spoilers MONITOR position
to identify malfunctioning surface.

One flight spoiler is deployed:

Yes

- (2) FLIGHT SPOILER leverSET to match
position of deployed flight spoiler.

- (3) Approach speedNot less than
 $V_{REF} (FLAPS 45) + 17 \text{ KIAS}$

NOTE

Landing distance factors for failed flight
spoilers are based upon serviceable ground
spoilers.

- (4) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45
actual landing distances for either Dry
surfaces in PERF-04 or to Wet or
Contaminated surfaces in PERF-06. If
dispatched with Anti-skid inoperative, apply
to dry or wet actual landing distances found
in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.30 (30%)	1.35 (35%)
Without Thrust Reverser(s)	1.35 (35%)	1.35 (35%)

– END –



No

No flight spoilers are deployed:

- (2) FLIGHT SPOILER lever CHECK set to RETRACT.
Do not deploy flight
spoilers manually.

NOTE

Landing distance factors for failed flight spoilers
are based upon serviceable ground spoilers.

- (3) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual
landing distances for either Dry surfaces in
PERF-04 or to Wet or Contaminated surfaces in
PERF-06. If dispatched with Anti-skid
inoperative, apply to dry or wet actual landing
distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.10 (10%)	1.15 (15%)
Without Thrust Reverser(s)	1.15 (15%)	1.15 (15%)

----- END -----

SPOILERONS ROLL

On the side with the operative roll control:

- (1) PLT ROLL or CPLT ROLL SELECT operative side

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

L SPOILERON or R SPOILERON

Prior to landing:

- (1) GRND PROX, FLAP <0040> OVRD
- (2) Landing FLAPS20
- (3) Approach speedNot less than
V_{REF} (FLAPS 45) + 12 KIAS
- (4) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.25 (25%)	1.30 (30%)
Without Thrust Reverser(s)	1.30 (30%)	1.30 (30%)

NOTE

Select the runway available with minimum cross-wind.

----- END -----

SPOILERONS

Prior to landing:

- (1) GRND PROX, FLAP <0040> OVRD
- (2) Landing FLAPS20
- (3) Approach speedNot less than
V_{REF} (FLAPS 45)+ 12 KIAS

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



(4) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.25 (25%)	1.30 (30%)
Without Thrust Reverser(s)	1.30 (30%)	1.30 (30%)

NOTE

1. Select the runway available with minimum cross-wind.
2. During cross-wind landings in gusty conditions, anticipate increased pilot workload.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

FLIGHT SPOILER Lever Jammed (Spoilers Deployed)

- (1) SECU 1 circuit breakers (1N8) OPEN
- (2) SECU 2 circuit breakers (2N8) OPEN
- (3) SECU 1A circuit breakers (4A2) OPEN

(4) SECU 2B circuit breakers (4A3) OPEN

NOTE

- 1. Opening the above SECU circuit breakers will retract the flight spoilers and disable the inboard ground spoilers.
- 2. Landing distance factors are based upon inoperative flight spoilers and inoperative inboard ground spoilers.

(5) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.30 (30%)	1.35 (35%)
Without Thrust Reverser(s)	1.35 (35%)	1.35 (35%)

----- END -----

GND SPLR DEPLOY

(1) GND LIFT DUMPINGMAN DISARM

Prior to landing:

(2) GRND PROX, FLAP <0040> OVRD

(3) Landing FLAPS20

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



(4) Approach speedNot less than
 $V_{REF} (FLAPS 45) + 12 \text{ KIAS}$

NOTE

Landing distance factors for failed ground
spoilers are based upon serviceable flight
spoilers.

(5) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual
landing distances for either Dry surfaces in
PERF-04 or to Wet or Contaminated surfaces in
PERF-06. If dispatched with Anti-skid
inoperative, apply to dry or wet actual landing
distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.35 (35%)	1.40 (40%)
Without Thrust Reverser(s)	1.40 (40%)	1.40 (40%)

After touchdown:

(6) GND LIFT DUMPINGMAN ARM

----- END -----

IB GND SPLRS

(1) GND LIFT DUMPINGMAN DISARM

NOTE

Landing distance factors for failed ground
spoilers are based upon serviceable flight
spoilers.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(2) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.13 (13%)	1.15 (15%)
Without Thrust Reverser(s)	1.15 (15%)	1.15 (15%)

After touchdown:

(3) GND LIFT DUMPINGMAN ARM

----- **END** -----

OB GND SPLRS

(1) GND LIFT DUMPINGMAN DISARM

NOTE

Landing distance factors for failed ground spoilers are based upon serviceable flight spoilers.

(2) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.13 (13%)	1.15 (15%)
Without Thrust Reverser(s)	1.15 (15%)	1.15 (15%)

After touchdown:

(3) GND LIFT DUMPINGMAN ARM

----- END -----

GLD UNSAFE

(1) GND LIFT DUMPINGMAN DISARM

GLD UNSAFE caution message persists:

Yes

Prior to landing:

- (2) GRND PROX, FLAP <0040> OVRD
- (3) Landing FLAPS20
- (4) Approach speedNot less than
V_{REF} (FLAPS 45) + 12 KIAS

NOTE

Landing distance factors for failed ground spoilers are based upon serviceable flight spoilers.

(5) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



ABNORM 8-27

Rev. 62, Oct 01/2014

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.28 (28%)	1.30 (30%)
Without Thrust Reverser(s)	1.30 (30%)	1.30 (30%)

– END –

No

After touchdown:

(2) GND LIFT DUMPINGMAN ARM

----- END -----

GLD NOT ARMED

(1) GND LIFT DUMPINGMAN ARM

GLD NOT ARMED caution message persists:

Yes

Prior to landing:

NOTE

Landing distance factors for failed ground spoilers are based upon serviceable flight spoilers.

(2) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

QUICK REFERENCE
HANDBOOK
CSP A-022-PRS

FLIGHT CONTROLS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

I



ABNORM 8-28

Rev. 62, Oct 01/2014

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.40 (40%)	1.60 (60%)
Without Thrust Reverser(s)	1.60 (60%)	1.60 (60%)

After touchdown:

- (3) FLIGHT SPOILER lever SELECT MAX deployment
- END -

No

- (2) No further action required.
----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
----------	------

FUEL

- **FUEL CH 1 FAIL or FUEL CH 2 FAIL**
 <0039> ABNORM 9-2
- **L MAIN EJECTOR and L SCAV EJECTOR or R MAIN EJECTOR and R SCAV EJECTOR** ABNORM 9-2
- **FUEL IMBALANCE** ABNORM 9-2
- **L ENG SOV OPEN or R ENG SOV OPEN** ABNORM 9-4
- **L ENG SOV CLSD or R ENG SOV CLSD** ABNORM 9-4
- **L ENG SOV FAIL or R ENG SOV FAIL**..... ABNORM 9-4
- **L FUEL LO PRESS or R FUEL LO PRESS** ABNORM 9-5
- **L MAIN EJECTOR or R MAIN EJECTOR** ABNORM 9-5
- **L SCAV EJECTOR or R SCAV EJECTOR** ABNORM 9-6
- **L FUEL PUMP or R FUEL PUMP** ABNORM 9-6
- **L XFER SOV or R XFER SOV** ABNORM 9-6
- **L XFLOW SOV or R XFLOW SOV** ABNORM 9-7
- **BULK FUEL TEMP** ABNORM 9-8
- **L FUEL LO TEMP or R FUEL LO TEMP** ABNORM 9-8
- **L FUEL FILTER or R FUEL FILTER** ABNORM 9-8
- **XFLOW/APU PUMP** ABNORM 9-9
- Gravity Cross-feed Procedure ABNORM 9-10
- Abnormal Increase of Centre Tank Quantity or Suspected Leak into Centre Tank ABNORM 9-11
- Fuel Leak Procedure ABNORM 9-13

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

I

FUEL CH 1/2 FAIL <0039>

(1) Engine thrustADJUST as required to maintain equal fuel flow to the engines.

(2) Fuel balanceMAINTAIN using the FUEL USED indication on the FMS PERF MENU – FUEL MGMT page 2/3.

(3) Land at the nearest suitable airport.

(4) Gravity Cross-feed ProcedureACCOMPLISH as required
Refer to ABNORM 9–10

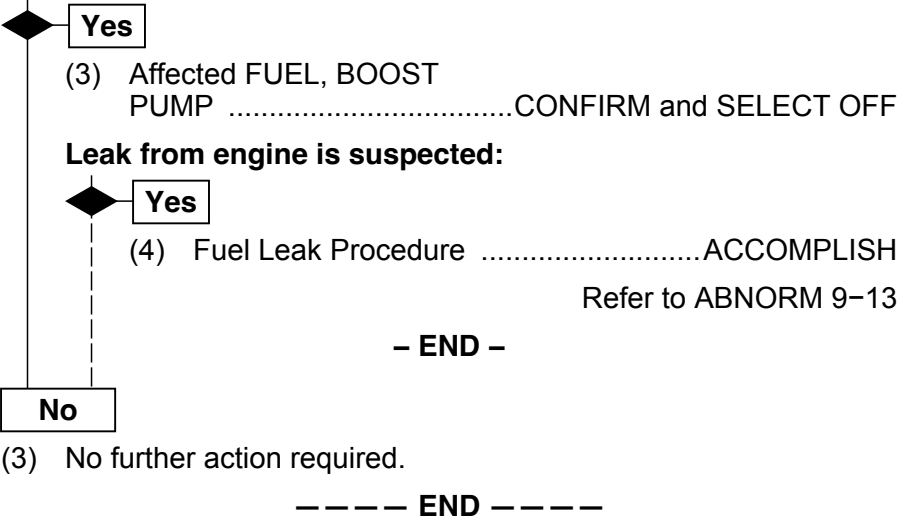
----- END -----

L MAIN EJECTOR and L SCAV EJECTOR or R MAIN EJECTOR and R SCAV EJECTOR

- (1) Left and right fuel boost pumpsCONFIRM operating

(2) Fuel tank quantitiesMONITOR

Total fuel quantity is depleting abnormally:

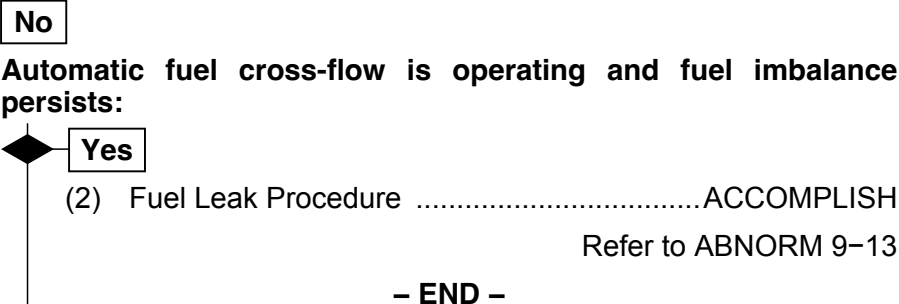
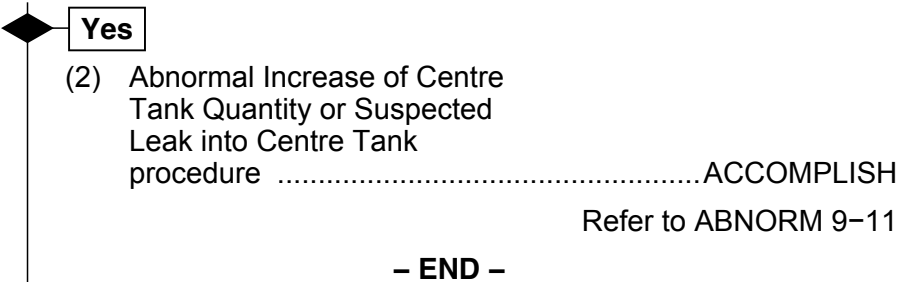


FUEL IMBALANCE

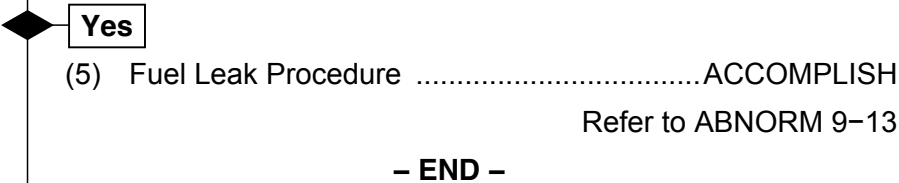
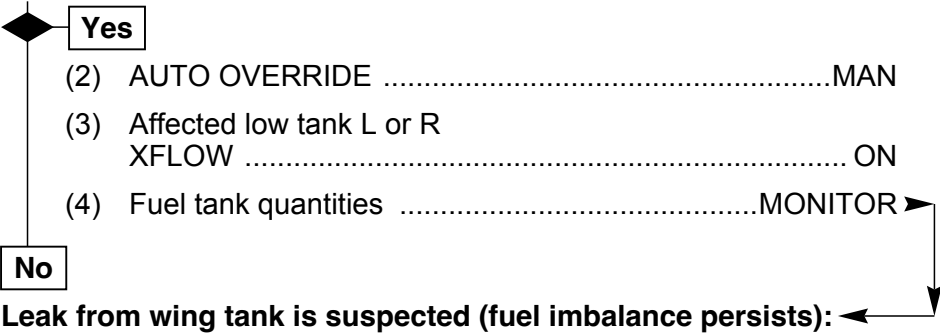
- (1) Automatic fuel cross-flowCONFIRM operating

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Leak into centre tank is suspected:



Automatic fuel cross-flow is inoperative:



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

L ENG SOV OPEN or R ENG SOV OPEN

- (1) Affected FUEL SOV circuit breakerCLOSED
- L ENG circuit breaker (1S2)

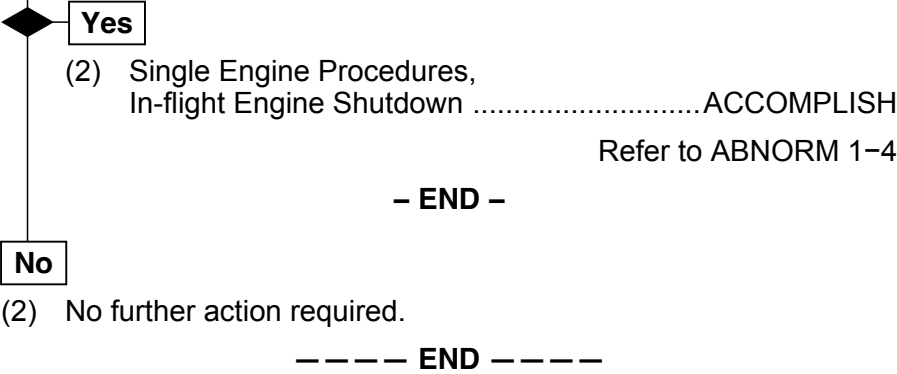
R ENG circuit breaker (1S1)

----- END -----

L ENG SOV CLSD or R ENG SOV CLSD

- (1) Engine instrumentsCONFIRM engine shutdown using N₁, FUEL FLOW, ITT and N₂.

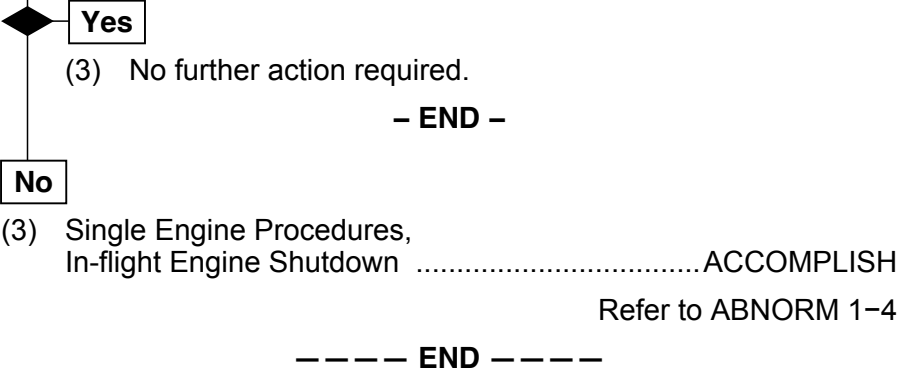
Engine failure is confirmed:



L ENG SOV FAIL or R ENG SOV FAIL

- (1) Normal engine operationCONFIRM
- (2) Affected engine instrumentsMONITOR

Engine parameters can be maintained within normal limits:



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

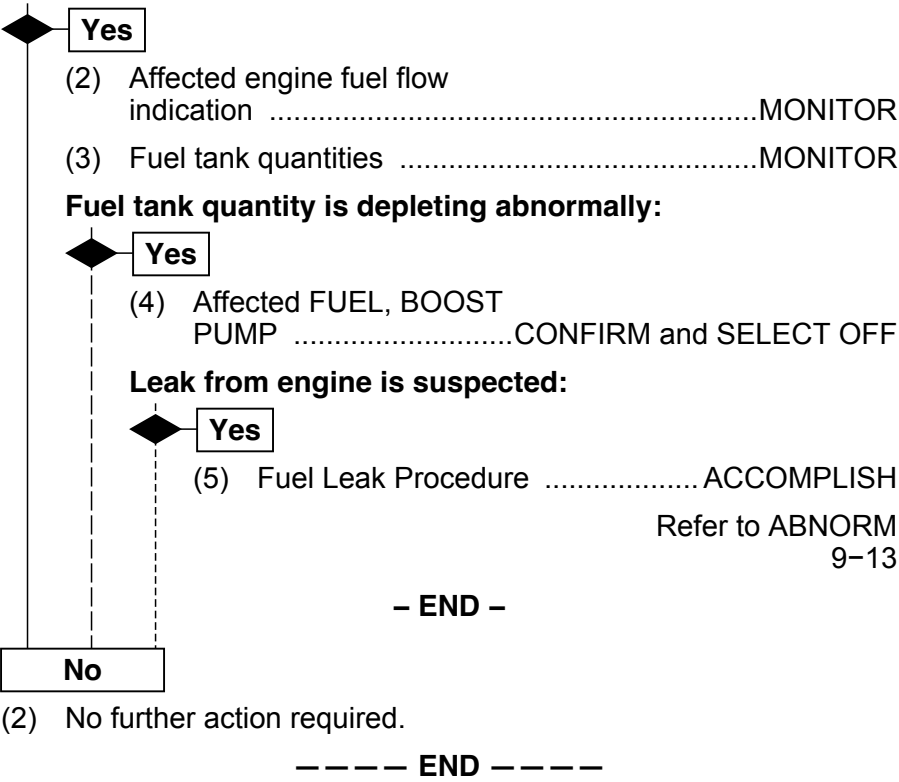
ABNORM

INDEX

L FUEL LO PRESS or R FUEL LO PRESS

(1) Affected engine instrumentsMONITOR

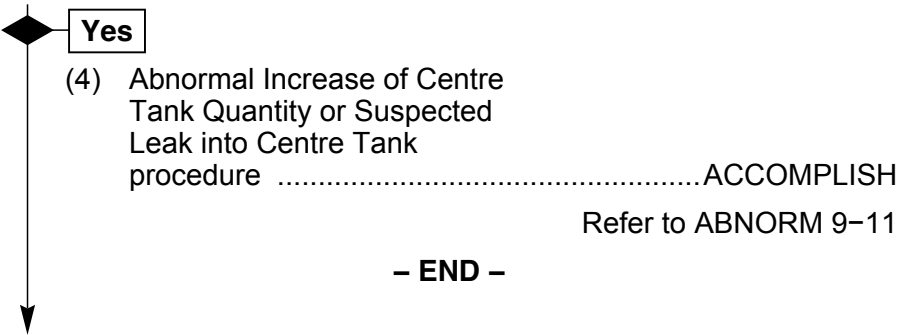
Accompanied by the same side MAIN EJECTOR caution message and/or SCAV EJECTOR caution message:



L MAIN EJECTOR or R MAIN EJECTOR

- (1) Left and right fuel boost pumpsCONFIRM operating
- (2) Affected engine instrumentsMONITOR
- (3) Fuel tank quantities MONITOR and BALANCE, if required

Centre tank quantity increasing or leak into centre tank suspected:



SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

No

(4) No further action required.

----- END -----

L SCAV EJECTOR or R SCAV EJECTOR

(1) Fuel quantity and balanceMONITOR

NOTE

Collector tank quantity may decrease at high thrust settings. This will be indicated by the fuel quantity indications being displayed amber. If this occurs, reduce pitch attitude and/or thrust on the affected engine to avoid engine flameout.

----- END -----

L FUEL PUMP or R FUEL PUMP

(1) Affected FUEL, BOOST PUMPCONFIRM and SELECT OFF, then ON

L FUEL PUMP or R FUEL PUMP caution message persists:

Yes

(2) Engine instrumentsMONITOR

- END -

No

(2) No further action required.

----- END -----

L XFER SOV or R XFER SOV

(1) Left, right and centre fuel tank quantities COMPARE

(2) Left and right wing fuel tank quantitiesBALANCE, if required

(3) Centre tank quantityMONITOR

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

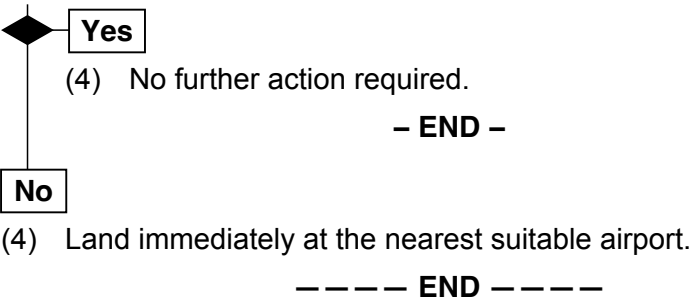
EMER

IMM

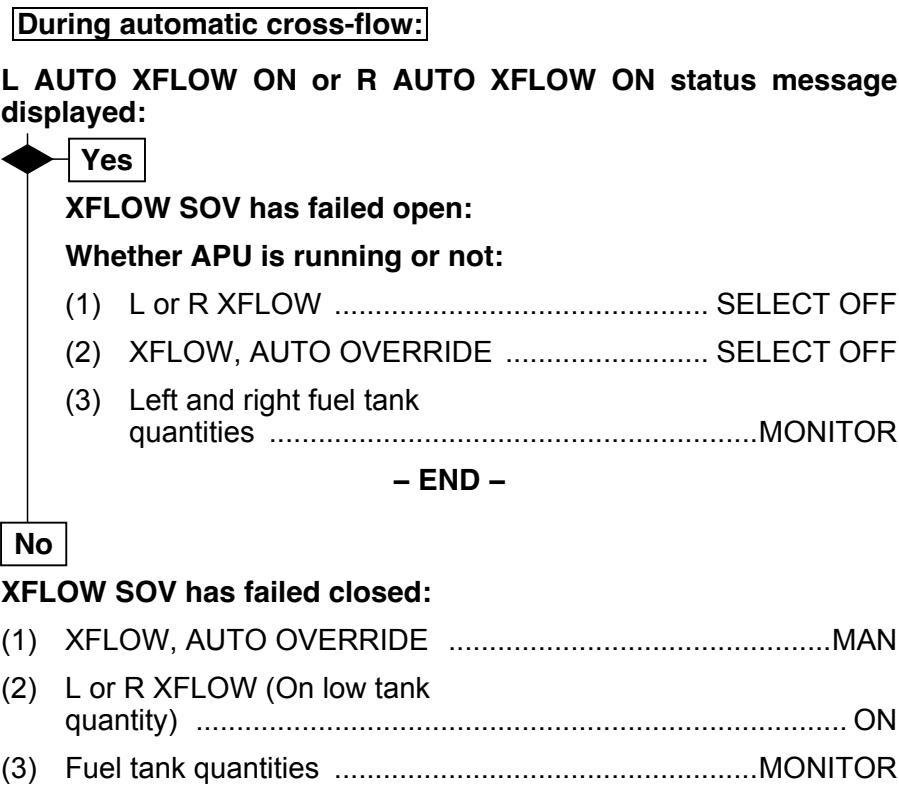
ABNORM

INDEX

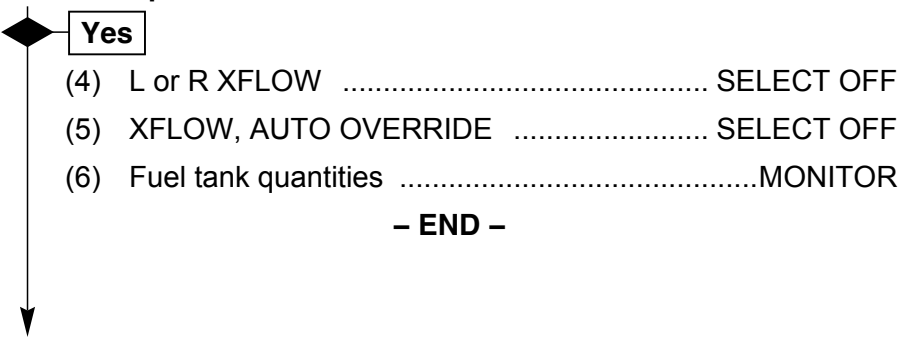
Centre tank quantity depletes with fuel burn:



L XFLOW SOV or R XFLOW SOV



Fuel tank quantities are balanced:



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

(4) Gravity Cross-feed ProcedureACCOMPLISH

Refer to ABNORM 9-10

----- END -----

During manual cross-flow:

L XFLOW ON or R XFLOW ON status message displayed:

Yes

XFLOW SOV has failed open:

(1) Left and right fuel tank quantitiesMONITOR

- END -

No

XFLOW SOV has failed closed:

(1) Gravity Cross-feed ProcedureACCOMPLISH

Refer to ABNORM 9-10

----- END -----

BULK FUEL TEMP

- (1) Descend or deviate to warmer air mass and/or increase
airspeed.
- (2) Engine instrumentsMONITOR

----- END -----

L FUEL LO TEMP or R FUEL LO TEMP

- (1) Affected engine instrumentsMONITOR

----- END -----

L FUEL FILTER or R FUEL FILTER

- (1) Affected engine instrumentsMONITOR

NOTE

Fuel filter may be in bypass mode.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

XFLOW/APU PUMP

(1) Fuel tank quantitiesMONITOR

APU is running, with the right-hand engine operating:

Yes
(2) APU indicationsMONITOR

NOTE
APU operation will be sustained through
fuel feed from right-hand engine.

Fuel imbalance persists:

Yes
(3) Gravity Cross-feed
ProcedureACCOMPLISH
Refer to ABNORM 9-10

- END -

No
(3) Fuel tank quantitiesMONITOR
- END -

No
APU is running, with the right-hand engine not operating:

Yes
APU is required:
Yes
(2) APU indicationsMONITOR

NOTE
APU power may fluctuate or APU may
flameout due to fuel starvation.

- END -

No
(2) APU, START/STOP SELECT OFF
(3) APU, PWR FUEL SELECT OFF
- END -

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

(2) Fuel tank quantitiesMONITOR

----- END -----

Gravity Cross-feed Procedure

- (1) L and R XFLOW SELECT OFF
- (2) AUTO OVERRIDE SELECT OFF
- (3) GRAVITY XFLOW OPEN
- (4) Steady heading sideslip
maneuverACCOMPLISH
 - Establish a bank angle of 10 degrees down on the low quantity side. Use rudder pedal/trim to maintain a constant heading/course.

NOTE

During the maneuver, fuel will transfer at a rate of up to 45.4 kg (100 lb) per minute. Fuel tank quantity gauging accuracy will be degraded. Accurate fuel indications will occur after 30 seconds of stabilized and coordinated flight.

(5) Fuel tank quantitiesMONITOR

Main tank quantities are balanced:

- Yes

(6) Coordinated flightESTABLISH

(7) GRAVITY XFLOWSELECT CLOSED

- END -

No

Fuel imbalance persists and cannot be controlled within limits:

- Yes

On the low quantity side:

(6) Affected thrust leverCONFIRM and IDLE

(7) Affected thrust leverCONFIRM and SHUT OFF

(8) Affected engine, ENG FIRE
PUSHCONFIRM and SELECT

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- I

▼

No
- (9)

Single Engine Procedures,
In-flight Engine Shutdown

ACCOMPLISH

Refer to ABNORM 1-4

- END -
- (6)

Fuel tank quantities

MONITOR

----- END -----

Abnormal Increase of Centre Tank Quantity or Suspected Leak into Centre Tank

- NOTE

It is acceptable to have a temporary limited fuel variance of up to 300 lb (140 kg) observed in the centre tank during airplane pitch, roll and acceleration maneuvers.
- (1)

Fuel tank quantities

MONITOR

On the low quantity side:

(2)

Affected thrust lever

CONFIRM and IDLE

(3)

Land immediately at the nearest suitable airport.

(4)

GRAVITY XFLOW

OPEN

(5)

Steady heading sideslip
maneuver

ACCOMPLISH

•

Establish a bank angle of 10 degrees down on the low quantity side. Use rudder pedal/trim to maintain a constant heading/course.

- NOTE

During the maneuver, fuel will transfer at a rate of up to 45.4 kg (100 lb) per minute. Fuel tank quantity gauging accuracy will be degraded. Accurate fuel indications will occur after 30 seconds of stabilized and coordinated flight.
- (6)

Fuel tank quantities

MONITOR

Main tank quantities are balanced:

Yes

(7) Coordinated flightESTABLISH

(8) GRAVITY XFLOWSELECT CLOSED

- END -

No

Centre tank quantity continues to increase (more than 454 kg [1000 lb]) or abnormally full:

Yes

(7) Affected FUEL, BOOST PUMPCONFIRM and SELECT OFF

NOTE

Do not action **L MAIN EJECTOR** or **R MAIN EJECTOR** caution message (if displayed).

- END -

No

Low quantity (affected side) wing tank depletes to 900 kg (2000 lb) and centre tank quantity is increasing or abnormally full:

Yes

On the low quantity side:

(7) Affected thrust leverCONFIRM and IDLE

(8) Affected thrust leverCONFIRM and SHUT OFF

(9) Affected engine, ENG FIRE PUSHCONFIRM and SELECT

(10) Single Engine Procedures, In-flight Engine ShutdownACCOMPLISH

Refer to ABNORM 1-4

- END -

No

(7) Fuel tank quantities MONITOR
----- END -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

Fuel Leak Procedure

NOTE

1. Diversion may be required.

2. If visibility permits, a visual check from the cabin may enable identification of the leak source.

Leak is confirmed or suspected:

(1) Land immediately at the nearest suitable airport.

• Do not delay landing while attempting to determine location of the leak.

• Expedite landing if tank quantity gauge indications amber.

NOTE

1. The minimum fuel quantity for go-around is 204 kg (450 lb) per wing (with the airplane level) and assuming a maximum airplane climb attitude of 10 degrees nose up.

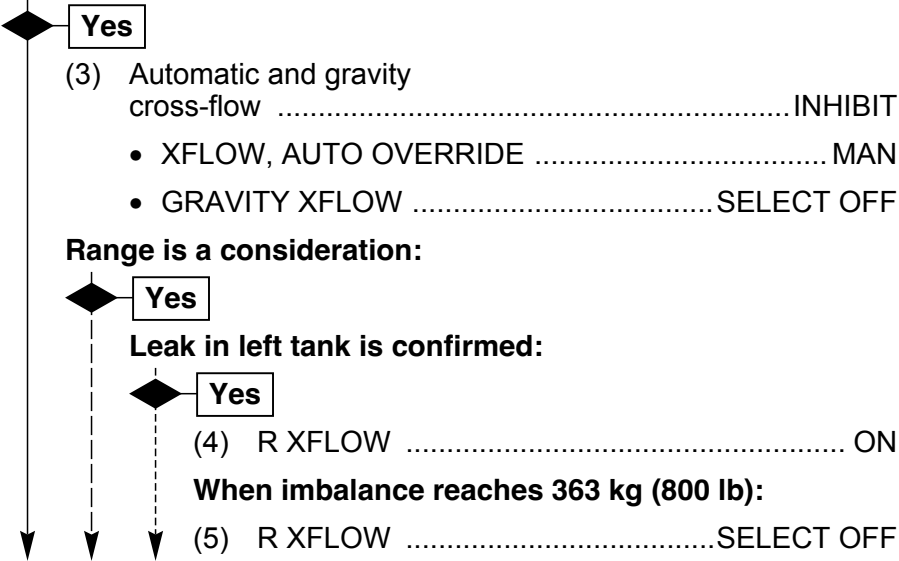
2. Do not action **FUEL IMBALANCE**, **MAIN EJECTOR**, **SCAV EJECTOR**, or **FUEL LO PRESS** caution messages.

(2) Autopilot operation

..... MONITOR, anticipate out-of-trim situation when disengaging the autopilot.

Leak from wing tank is suspected:

Wing tank isolation is required:





(6) Proceed to “Landing with a fuel imbalance”

No

Leak in right tank is confirmed:

(4) L XFLOWON

When imbalance reaches 363 kg (800 lb):

(5) L XFLOWSELECT OFF

(6) Proceed to “Landing with a fuel imbalance”

– END –

No

Landing with a fuel imbalance:

Yes

(4) Select the runway available with minimum turbulence and cross-wind.

(5) GRND PROX, FLAP
<0040> OVRD

(6) Approach and landing
FLAPS20

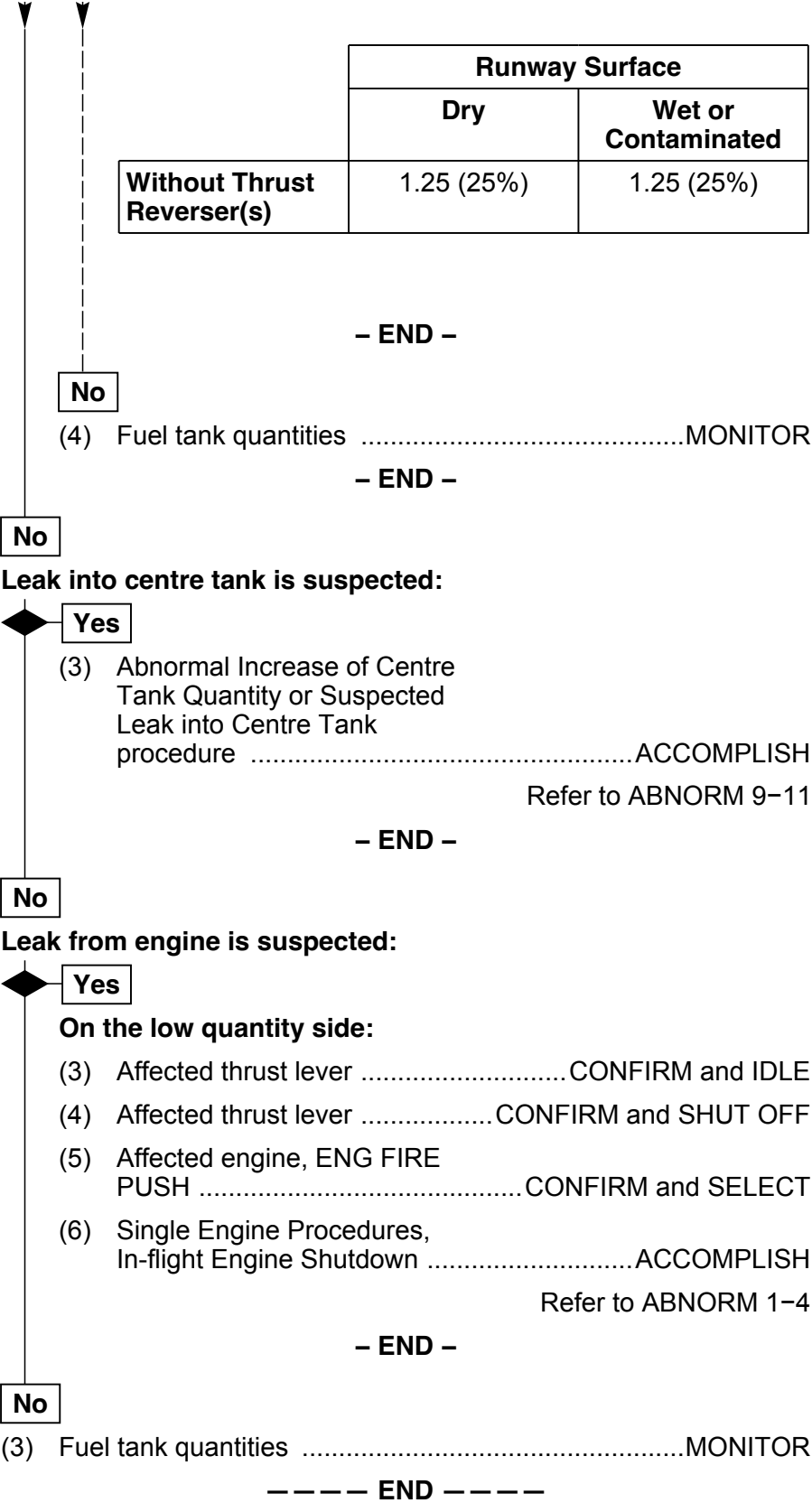
(7) Final approach speedNot less than
 $V_{REF} (FLAPS 45) + 12 KIAS$

(8) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.20 (20%)	1.25 (25%)



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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CONTENTS	PAGE
----------	------

HYDRAULIC POWER

I SINGLE SYSTEM FAILURES

- **HYD 1 LO PRESS** [ABNORM 10-2](#)
- **HYD 2 LO PRESS** [ABNORM 10-3](#)
- **HYD 3 LO PRESS** [ABNORM 10-5](#)
- **HYD 1 HI TEMP** [ABNORM 10-9](#)
- **HYD 2 HI TEMP** [ABNORM 10-12](#)
- **HYD 3 HI TEMP** [ABNORM 10-14](#)
- **HYD EDP 1A or HYD EDP 2A** [ABNORM 10-15](#)
- **HYD PUMP 1B** [ABNORM 10-16](#)
- **HYD PUMP 2B** [ABNORM 10-16](#)
- **HYD PUMP 3A** [ABNORM 10-17](#)
- **HYD PUMP 3B** [ABNORM 10-17](#)
- **HYD SOV 1 OPEN or HYD SOV 2 OPEN** [ABNORM 10-17](#)

COMBINED FAILURES

- **HYD 1 LO PRESS and HYD 2 LO PRESS** [ABNORM 10-17](#)
- **HYD 1 LO PRESS and HYD 3 LO PRESS** [ABNORM 10-20](#)
- **HYD 2 LO PRESS and HYD 3 LO PRESS** [ABNORM 10-24](#)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

HYD 1 LO PRESS

- NOTE

If during the accomplishment of a hydraulic system low pressure procedure, a second system also fails, disregard both single system failures and proceed directly to the applicable double system failure procedure.
- (1)

HYDRAULIC 1 pump

ON
- (2)

Hydraulic pressure and fluid quantity

MONITOR

If system 1 pressure is less than 1800 psi:

- Yes

(3)

HYDRAULIC 1 pump

OFF

- NOTE

The Engine Driven Pump (EDP) 1A is operating without sufficient hydraulic fluid. Log the length of time that the message remains on display. Monitor hydraulic system 1 fluid temperature readout.
- (4)

HYDRAULIC and FLIGHT CONTROLS pages

REVIEW affected systems
- | Hydraulic Synoptic | |
|---|-------------|
| Component | System 1 |
| L and R Ground Spoilers (outboard only) | Inoperative |
- (5)

Land at the nearest suitable airport.
- (6)

Actual landing distance

INCREASE

- NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.13 (13%)	1.15 (15%)
Without Thrust Reverser(s)	1.15 (15%)	1.15 (15%)

NOTE

Maximize the use of reverse thrust.

– END –

No

(3) No further action required.

----- END -----

HYD 2 LO PRESS

NOTE

If during the accomplishment of a hydraulic system low pressure procedure, a second system also fails, disregard both single system failures and proceed directly to the applicable double system failure procedure.

- (1) HYDRAULIC 2 pump ON
- (2) Hydraulic pressure and fluid quantityMONITOR

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

If system 2 pressure is less than 1800 psi:

Yes

(3) HYDRAULIC 2 pump OFF

NOTE

The Engine Driven Pump (EDP) 2A is operating without sufficient hydraulic fluid. Log the length of time that the message remains on display. Monitor hydraulic system 2 fluid temperature readout.

(4) HYDRAULIC and FLIGHT CONTROLS pages REVIEW affected systems

Hydraulic Synoptic	
Component	System 2
L and R Ground Spoilers (inboard only)	Inoperative
Outboard Brakes (when system 2 accumulator pressure is depleted)	Inoperative
Main Landing Gear	Downlock assist actuator is inoperative

(5) Land at the nearest suitable airport.

(6) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.55 (55%)	1.65 (65%)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	Runway Surface	
	Dry	Wet or Contaminated
Without Thrust Reversers	1.65 (65%)	1.65 (65%)

NOTE

Apply brakes carefully and maximize the use of reverse thrust.



1.

Anticipate the loss of outboard brakes during landing when system 2 brake accumulator depressurizes.
2.

A steady brake application is recommended upon landing. Do not cycle the brakes.

– END –

No

(3) No further action required.

----- END -----

HYD 3 LO PRESS

NOTE

If during the accomplishment of a hydraulic system low pressure procedure, a second system also fails, disregard both single system failures and proceed directly to the applicable double system failure procedure.

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

WARNING

Failure to complete the entire procedure that follows, could result in an uncommanded nosewheel deflection after touchdown with the possibility of a runway excursion.

- (1) HYDRAULIC 3B pump ON
- (2) N/W STRG OFF
- (3) Hydraulic pressure and fluid quantityMONITOR

- If system pressure is equal to or more than 1800 psi, proceed to step (4).

• If system pressure is less than 1800 psi, proceed to step (5).

System pressure is equal to or more than 1800 psi:

NOTE

1. Select the longest runway available with minimum cross-wind and turbulence.
2. Use differential braking, rudder and engine thrust as required to assist in directional control.
3. In high cross-wind conditions, rudder effectiveness may be limited after landing with maximum reverse thrust selected.

- (4) No further action required.

System pressure is less than 1800 psi:

- (5) HYDRAULIC 3A and 3B pump OFF

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(6) HYDRAULIC and FLIGHT CONTROLS pages REVIEW affected systems

Hydraulic Synoptic	
Component	System 3
Inboard Brakes (when system 3 accumulator pressure is depleted)	Inoperative
Landing Gear (extension and retraction)	Inoperative
Nose door will remain open	—
N/W Steering	Inoperative (May result in nose wheel shimmy)
Parking Brake	Inoperative

(7) Land at the nearest suitable airport.

Prior to landing:

(8) LDG GEAR lever DN



Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE

The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 lbs. The flight crew may choose to reposition the seat prior to manual release handle deployment. In safe flight condition, the flight crew may also choose to stand up to pull the manual release handle.

(9) LANDING GEAR MANUAL RELEASE handlePULL to full extension

NOTE

The **NOSE DOOR OPEN** warning message is displayed when the LANDING GEAR MANUAL RELEASE handle is pulled.

- If any landing gear fails to down-and-lock, proceed to step (10).

- If landing gear is down-and-locked, proceed to step (11).

Any landing gear fails to down-and-lock:

(10) LANDING GEAR MANUAL
RELEASE handle STOW and PULL to full extension

NOTE

Step (10) may be repeated as many times as desired.

- If landing gear is down-and-locked, proceed to step (11).
- If any landing gear still fails to down-and-lock, proceed to step (13).

Landing gear is down-and-locked:

(11) LANDING GEAR MANUAL
RELEASE handleSTOW

(12) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.50 (50%)	1.60 (60%)
Without Thrust Reverser(s)	1.60 (60%)	1.60 (60%)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CAUTION

1. Anticipate the loss of inboard brakes during landing when system 3 brake accumulator depressurizes.
2. A steady brake application is recommended upon landing. Do not cycle the brakes.

NOTE

1. Select the longest runway available with minimum cross-wind and turbulence.
2. Use differential braking, rudder and engine thrust as required to assist in directional control.
3. In high cross-wind conditions, rudder effectiveness may be limited after landing with maximum reverse thrust selected.
4. Apply brakes carefully and maximize the use of reverse thrust.

Any landing gear still fails to down-and-lock:

(13) Landing Gear Up / Unsafe
Landing procedureACCOMPLISH
Refer to EMER 13-9

HYD 1 HI TEMP

NOTE

Disregard **HYD 1 HI TEMP** caution if one other hydraulic system has already failed.

(1) HYDRAULIC 1 pump OFF
(2) System 1 temperatureMONITOR

Temperature is less than 135°C (275°F):

Yes

(3) Land at the nearest suitable airport.

- END -

**QUICK REFERENCE
HANDBOOK
CSP A-022-PRS**

HYDRAULIC POWER

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



No

- (3) Left engine thrust lever CONFIRM and IDLE
- (4) Left engine thrust lever CONFIRM and SHUT OFF
- (5) 14TH STAGE ISOL OPEN
- (6) L FUEL BOOST PUMP CONFIRM and OFF
- (7) ANTI-ICE, LH COWL OFF
- (8) APU (if available, 30000 feet and below) START
- (9) APU GEN ON
- (10) Fuel system CHECK
 - Cross-flow AUTO
 - Quantity / Balance CHECK

NOTE

1. Leave icing conditions to prevent ice accumulation on the engine cowl with the inoperative anti-icing system.
2. Powered cross-flow may not be able to correct fuel imbalance during single engine operations.
3. Do not attempt to relight an engine that is suspected to be damaged (engine fire, rotor burst, reverser deployed, etc...).

- (11) Land at the nearest suitable airport.
- (12) HYDRAULIC and FLIGHT CONTROLS pages REVIEW affected systems

Hydraulic Synoptic	
Component	System 1
L and R Ground Spoilers (outboard only)	Inoperative

Below 15000 feet:

- (13) L and R PACKs TRANSFER to APU Bleed
- (14) Go-around thrust reference SET N₁ bug accordingly.
Refer to Go-around Thrust Settings – PERF-04.

Prior to landing:

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- (15) GRND PROX, FLAP <0040> OVRD
- (16) Landing FLAPS20
- (17) Approach speedNot less than
V_{REF} (FLAPS 45) + 12 KIAS
- (18) Actual landing distance (without
thrust reversers) INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With One Thrust Reverser	1.30 (30%)	1.30 (30%)
Without Thrust Reverser	1.30 (30%)	1.30 (30%)

NOTE

The actual landing distance factors shown above must be used instead of the factors stated in the Single Engine Procedures.



If required, use remaining thrust reverser carefully upon landing.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



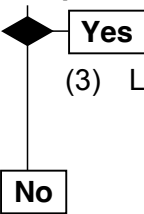
HYD 2 HI TEMP

NOTE

Disregard **HYD 2 HI TEMP** caution if one other hydraulic system has already failed.

- (1) HYDRAULIC 2 pump OFF
- (2) System 2 temperatureMONITOR

Temperature is less than 135°C (275°F):



- (3) Land at the nearest suitable airport.

– END –

- (3) Right engine thrust leverCONFIRM and IDLE
- (4) Right engine thrust leverCONFIRM and SHUT OFF
- (5) 14TH STAGE ISOL OPEN
- (6) R FUEL BOOST PUMPCONFIRM and OFF
- (7) ANTI-ICE, RH COWL OFF
- (8) APU (if available, 30000 feet and below)START
- (9) APU GEN ON
- (10) Fuel systemCHECK
 - Cross-flowAUTO
 - Quantity / Balance CHECK

NOTE

1. Leave icing conditions to prevent ice accumulation on the engine cowl with the inoperative anti-icing system.
2. Powered cross-flow may not be able to correct fuel imbalance during single engine operations.
3. Do not attempt to relight an engine that is suspected to be damaged (engine fire, rotor burst, reverser deployed, etc...).

- (11) Land at the nearest suitable airport.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(12) HYDRAULIC and FLIGHT
CONTROLS pages REVIEW affected systems

Hydraulic Synoptic	
Component	System 2
L and R Ground Spoilers (inboard only)	Inoperative
Outboard Brakes (when system 2 accumulator pressure is depleted)	Inoperative
Main Landing Gear	Downlock assist actuator is inoperative

Below 15000 feet:

- (13) L and R PACKsTRANSFER to APU Bleed
- (14) Go-around thrust reference SET N₁ bug accordingly.
Refer to Go-around Thrust
Settings – PERF-04.

Prior to landing:

- (15) GRND PROX, FLAP <0040> OVRD
- (16) Landing FLAPS20
- (17) Approach speedNot less than
V_{REF} (FLAPS 45) + 12 KIAS
- (18) Actual landing distance (without
thrust reversers) INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With One Thrust Reverser	2.00 (100%)	2.00 (100%)

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

	Runway Surface	
	Dry	Wet or Contaminated
Without Thrust Reverser	2.00 (100%)	2.00 (100%)

NOTE

The actual landing distance factors shown above must be used instead of the factors stated in the Single Engine Procedures.



- 1. Anticipate the loss of outboard brakes during landing when the system 2 brake accumulator depressurizes.
- 2. A steady brake application is recommended upon landing. Do not cycle the brakes.

----- END -----

HYD 3 HI TEMP

NOTE

If the HYDRAULIC 3A and 3B pumps were running at the time the **HYD 3 HI TEMP** caution message appeared, attempt to isolate the malfunctioning pump by switching each pump to OFF, in turn.

- (1) HYDRAULIC 3B pump ON
- (2) HYDRAULIC 3A pump OFF
- (3) System 3 temperatureMONITOR

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



Temperature can be controlled:

Yes
(4) Affected HYDRAULIC pump LEAVE OFF
- END -

No
(4) HYDRAULIC 3A and 3B pump OFF
(5) System 3 temperature MONITOR

NOTE

No action is required for the **HYD 3 LO PRESS** caution message, continue HYD 3 HI TEMP procedure until landing.

Temperature is less than 107°C (225°F):

Yes
Prior to landing:
(6) HYDRAULIC 3A and 3B pump ON
- END -

No
(6) Land at the nearest suitable airport.
Prior to landing:
(7) HYDRAULIC 3A and 3B pump ON
----- END -----

HYD EDP 1A or HYD EDP 2A

(1) Associated HYDRAULIC (1 or 2) pump ON
(2) Hydraulic pressure and fluid quantity MONITOR

Both system pressure and quantity are normal:

Yes
(3) No further action required.
- END -



No

- (3) Associated HYDRAULIC (1 or 2)
pump OFF

NOTE

The applicable EDP may be operating without sufficient hydraulic fluid. Log the length of time that the message remains on display. Monitor the applicable hydraulic system fluid temperature readout.

----- END -----

HYD PUMP 1B

- (1) HYDRAULIC 1 pump ON
(2) Hydraulic pressure and fluid
quantityMONITOR

HYD PUMP 1B caution message persists:

Yes

- (3) HYDRAULIC 1 pump OFF

- END -

No

- (3) No further action required.

----- END -----

HYD PUMP 2B

- (1) HYDRAULIC 2 pump ON
(2) Hydraulic pressure and fluid
quantityMONITOR

HYD PUMP 2B caution message persists:

Yes

- (3) HYDRAULIC 2 pump OFF

- END -

No

- (3) No further action required.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



HYD PUMP 3A

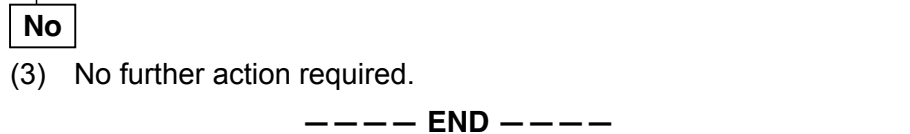
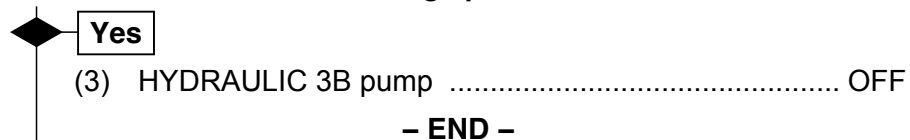
- (1) HYDRAULIC 3B pump ON
- (2) HYDRAULIC 3A pump OFF
- (3) Hydraulic pressure and fluid quantityMONITOR

----- END -----

HYD PUMP 3B

- (1) HYDRAULIC 3B pump ON
- (2) Hydraulic pressure and fluid quantityMONITOR

HYD PUMP 3B caution message persists:



HYD SOV 1 OPEN or HYD SOV 2 OPEN

NOTE

Hydraulic shut-off valve failed after engine shutdown due to fire.

- (1) Land at the nearest suitable airport.

----- END -----

HYD 1 LO PRESS and HYD 2 LO PRESS

- (1) HYDRAULIC pump (all) ON
- (2) Systems 1 and 2 pressure and fluid quantity CHECK

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

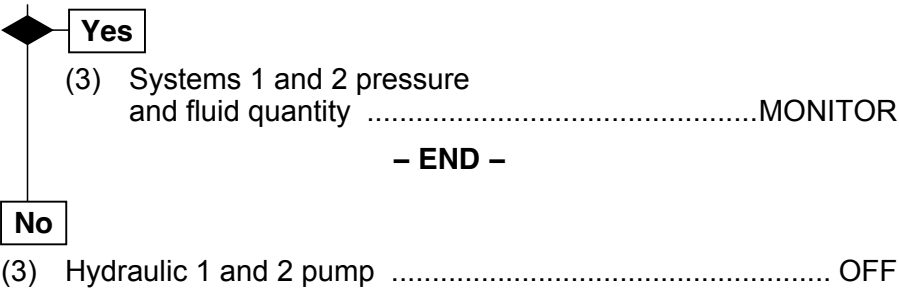
EMER

IMM

ABNORM

INDEX

Systems 1 and 2 pressure and fluid quantity normal:



NOTE

The Engine Driven Pumps (EDPs) 1A and 2A are operating without sufficient hydraulic fluid. Log the length of time that the messages remain on display. Monitor hydraulic systems 1 and 2 fluid temperature readouts.

- (4) HYDRAULIC and FLIGHT CONTROLS pages REVIEW affected systems

Hydraulic Synoptic	
Component	System 1 and 2
L and R Flight Spoilers	Inoperative
L and R Ground Spoilers	Inoperative
Landing Gear	Downlock assist actuator is inoperative
Outboard Brakes	Inoperative once system 2 accumulator pressure is depleted.

- (5) Land at the nearest suitable airport.

Prior to landing:

- (6) GRND PROX, FLAP <0040> OVRD
- (7) Landing FLAPS20
- (8) Approach speedNot less than $V_{REF} \text{ (FLAPS 45)} + 12 \text{ KIAS}$
- (9) Actual landing distance INCREASE

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.90 (90%)	2.40 (140%)
Without Thrust Reverser(s)	2.40 (140%)	2.40 (140%)

NOTE

Apply brakes carefully and maximize the use of reverse thrust.



- 1. Anticipate the loss of outboard brakes during landing when system 2 brake accumulator depressurizes.
- 2. A steady brake application is recommended upon landing. Do not cycle the brakes.

----- END -----

HYD 1 LO PRESS and HYD 3 LO PRESS

WARNING

Failure to complete the entire procedure that follows, could result in an uncommanded nosewheel deflection after touchdown with the possibility of a runway excursion.

- (1) HYDRAULIC pump (all) ON
- (2) N/W STRG OFF
- (3) Systems 1 and 3 pressure and fluid quantity CHECK
 - If systems 1 and 3 pressure and fluid quantity are normal, proceed to step (4).
 - If systems 1 and 3 pressure and fluid quantity are not normal, proceed to step (6).
- Systems 1 and 3 pressure and fluid quantity are normal:**
- (4) Systems 1 and 3 pressure and fluid quantityMONITOR

Prior to landing:

NOTE

1. Select the longest runway available with minimum cross-wind and turbulence.
2. Use differential braking, rudder and engine thrust as required to assist in directional control.
3. In high cross-wind conditions, rudder effectiveness may be limited after landing with maximum reverse thrust selected.

- (5) No further action required.

Systems 1 and 3 pressure and fluid quantity are not normal:

- (6) Hydraulic 1, 3A and 3B pump OFF

NOTE

The Engine Driven Pump (EDP) 1A is operating without sufficient hydraulic fluid. Log the length of time that the messages remain on display. Monitor the hydraulic system 1 fluid temperature readout.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(7) HYDRAULIC and FLIGHT CONTROLS pages REVIEW affected systems

Hydraulic Synoptic	
Component	System 1 and 3
L Aileron	Inoperative
L Spoileron	Inoperative
L and R Ground Spoilers (outboard only)	Inoperative
Landing Gear	Normal extension and retraction inoperative.
N/W Steering	Inoperative (May result in nose wheel shimmy).
Inboard Brakes	Inoperative once system 3 accumulator pressure is depleted.
Parking Brake	Inoperative

(8) Land at the nearest suitable airport.

Prior to landing:

(9) GRND PROX, FLAP <0040> OVRD

(10) Landing FLAPS20

(11) LDG GEAR lever DN



Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE

The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 lbs. The flight crew may choose to reposition the seat prior to manual release handle deployment. In safe flight condition, the flight crew may also choose to stand up to pull the manual release handle.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(12) LANDING GEAR MANUAL
RELEASE handlePULL to full extension

NOTE

The **NOSE DOOR OPEN** warning message is displayed when the LANDING GEAR MANUAL RELEASE handle is pulled.

- If any landing gear fails to down-and-lock, proceed to step (13).
- If landing gear is down-and-locked, proceed to step (14).

Any landing gear fails to down-and-lock:

(13) LANDING GEAR MANUAL
RELEASE handle STOW and PULL to full extension

NOTE

Step (13) may be repeated as many times as desired.

- If landing gear is down-and-locked, proceed to step (14).
- If any landing gear still fails to down-and-lock, proceed to step (17).

Landing gear is down-and-locked:

(14) LANDING GEAR MANUAL
RELEASE handleSTOW

(15) Approach speedNot less than
 $V_{REF} (FLAPS 45) + 12 \text{ KIAS}$

(16) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.75 (75%)	2.00 (100%)
Without Thrust Reverser(s)	2.00 (100%)	2.00 (100%)

NOTE

- Flight path control is limited with hydraulic systems 1 and 3 failed.
- Select the longest runway available with minimum cross-wind and turbulence.
- Rudder control is adequate for normal flight and should be used in coordination with aileron, if necessary, during turns.
- Apply brakes carefully and maximize the use of reverse thrust.



- Anticipate the loss of inboard brakes upon landing, when the system 3 brake accumulator depressurizes.
- A steady brake application is recommended upon landing. Do not cycle the brakes.

Any landing gear still fails to down-and-lock:

(17) Landing Gear Up / Unsafe
Landing procedureACCOMPLISH
Refer to EMER 13-9

HYD 2 LO PRESS and HYD 3 LO PRESS

WARNING

Failure to complete the entire procedure that follows could result in an uncommanded nosewheel deflection after touchdown with the possibility of a runway excursion.

- (1) HYDRAULIC pump (all) ON
- (2) N/W STRG OFF
- (3) Systems 2 and 3 pressure and fluid quantity CHECK
- If systems 2 and 3 pressure and fluid quantity are normal, proceed to step (4).
 - If systems 2 and 3 pressure and fluid quantity are not normal, proceed to step (6).
- Systems 2 and 3 pressure and fluid quantity are normal:**
- (4) Systems 2 and 3 pressure and fluid quantityMONITOR

Prior to landing:

NOTE

1.

Select the longest runway available with minimum cross-wind and turbulence.
2.

Use differential braking, rudder and engine thrust as required to assist in directional control.
3.

In high cross-wind conditions, rudder effectiveness may be limited after landing with maximum reverse thrust selected.

- (5) No further action required.

Systems 2 and 3 pressure and fluid quantity are not normal:

- (6) Hydraulic 2, 3A and 3B pump OFF

NOTE

The Engine Driven Pump (EDP) 2A is operating without sufficient hydraulic fluid. Log the length of time that the messages remain on display. Monitor hydraulic system 2 fluid temperature readout.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(7) HYDRAULIC and FLIGHT CONTROLS pages REVIEW affected systems

Hydraulic Synoptic	
Component	System 2 and 3
R Aileron	Inoperative
R Spoileron	Inoperative
L and R Ground Spoilers (inboard only)	Inoperative
Landing Gear	Normal extension and retraction inoperative. (MLG may not lock down during manual extension).
N/W Steering	Inoperative (May result in nose wheel shimmy).
Inboard and Outboard Brakes	With accumulator pressure only, use a slow, steady application of brakes. DO NOT cycle the brakes.
Parking Brake	Inoperative once system 3 accumulator pressure is depleted.

(8) Land at the nearest suitable airport.

Prior to landing:

(9) GRND PROX, FLAP <0040> OVRD

(10) ANTI-SKID OFF

(11) Landing FLAPS20

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(12) LDG GEAR lever DN



Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE

The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 lbs. The flight crew may choose to reposition the seat prior to manual release handle deployment. In safe flight condition, the flight crew may also choose to stand up to pull the manual release handle.

(13) LANDING GEAR MANUAL RELEASE handlePULL to full extension

NOTE

1.

Landing gear extension relies upon gear free fall. A sideslip, for each side, may be required in order to achieve MLG down-lock.
2.

The **NOSE DOOR OPEN** warning message is displayed when the LANDING GEAR MANUAL RELEASE handle is pulled.

• If any landing gear fails to down-and-lock, proceed to step (14).

• If landing gear is down-and-locked, proceed to step (15).

Any landing gear fails to down-and-lock:

(14) LANDING GEAR MANUAL RELEASE handle STOW and PULL to full extension

NOTE

Step (14) may be repeated as many times as desired.

• If landing gear is down-and-locked, proceed to step (15).

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- If any landing gear still fails to down-and-lock, proceed to step (21).

Landing gear is down-and-locked:

- (15) LANDING GEAR MANUAL
RELEASE handleSTOW
- (16) Approach speedNot less than
 $V_{REF} (FLAPS 45) + 12 \text{ KIAS}$
- (17) Actual landing distanceINCREASE by 3.50 (250%).

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

Upon landing:

- (18) Maximum reverse thrust (both engines)APPLY
- (19) BrakesAPPLY light to moderate braking until safe taxi speed is maintained.
- (20) Brake pressureMONITOR

NOTE

1. Approximately six (6) full brake applications are possible, using brake accumulator pressure.
2. A steady brake application is recommended upon landing. Do not cycle the brakes. Avoid hard braking.
3. **IB BRAKE PRESS** and **OB BRAKE PRESS** caution messages come on at less than 1030 psi.

Any landing gear still fails to down-and-lock:

- (21) Landing Gear Up / Unsafe
Landing procedureACCOMPLISH
- Refer to EMER 13-9

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
----------	------

ICE AND RAIN PROTECTION

ICE DISPERSAL

- Ice Dispersal Procedure ABNORM 11-2

COWL/WING ANTI-ICE

- L COWL A/ICE or R COWL A/ICE ABNORM 11-3
- L WING A/ICE or R WING A/ICE ABNORM 11-4
- WING A/ SNSR ABNORM 11-6
- Wing Anti-ice Piccolo Duct Damage
Suspected Procedure ABNORM 11-6
- SUPP GND WING ANTI-ICE, FAIL LO
Light or FAIL HI Light ABNORM 11-8

ICE DETECTOR

- ICE ABNORM 11-8
- ICE DET FAIL (Caution Message) or
ICE DET 1 FAIL or ICE DET 2 FAIL
(Status Message) ABNORM 11-8

PROBE HEATERS

- L AOA HEAT or R AOA HEAT ABNORM 11-9
- L AOA HEAT and R AOA HEAT ABNORM 11-9
- L PITOT HEAT ABNORM 11-11
- R PITOT HEAT ABNORM 11-12
- PITOT BASE HEAT ABNORM 11-13
- L STATIC HEAT ABNORM 11-14
- R STATIC HEAT ABNORM 11-14
- STBY PITOT HEAT ABNORM 11-15
- TAT PROBE HEAT ABNORM 11-15

WINDSHIELD/WINDOW HEAT FAILURE

- L WINDOW HEAT or R WINDOW
HEAT ABNORM 11-16
- L WSHLD HEAT or R WSHLD HEAT ABNORM 11-16
- Arcing, Delaminated, Shattered, or
Cracked Window or Windshield ABNORM 11-17

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Ice Dispersal Procedure

After leaving icing conditions and if ice accumulation is observed on the heated portion of the wing leading edge:

- (1) AirspeedINCREASE to V_{MO}/M_{MO} , if possible, to disperse ice

Not able to remove ice from the wing leading edge and in order to assure adequate stall margin:

Yes

- (2) Maneuvering airspeed Not less than 200 KIAS

WARNING

Even small accumulations of ice on the wing leading edge can change the stall speed, stall characteristics or the warning margins provided by the stall protection system.

Prior to landing:

- (3) GRND PROX, FLAP <0040> OVRD
- (4) Landing FLAPS20
- (5) Approach speedNot less than $V_{REF} \text{ (FLAPS 45)} + 30 \text{ KIAS}$
- (6) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.50 (50%)	1.65 (65%)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	Runway Surface	
	Dry	Wet or Contaminated
Without Thrust Reverser(s)	1.65 (65%)	1.65 (65%)

Just prior to touchdown:

- (7) Thrust leversRETARD to IDLE and do not prolong the flare.

NOTE

A slight pitch-up tendency may occur upon selection of reverse thrust. This can be readily corrected by the application of nose-down elevator and/or brakes.

– END –

No

- (2) No further action required.

----- END -----

L COWL A/ICE or R COWL A/ICE

Cowl anti-ice selected ON:

Yes

(May indicate a low pressure condition.)

- (1) Affected 14TH STAGE BLEEDCHECK OPEN
- (2) Affected engine thrust Not less than 78% N₂
- (3) IGNITION, CONT ON
- (4) Leave icing conditions.

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

I

▼

(5) Ice Dispersal ProceduresACCOMPLISH, if required

Refer to ABNORM 11-2

- END -

No

(May indicate a high pressure condition.)

- (1) Affected engine thrust REDUCE N₂ to < 85%
- (2) Affected 14TH STAGE BLEEDCLOSED
- (3) Leave icing conditions.

NOTE

1. Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is -40°C (-40°F) or below.
2. If wing anti-ice is required, open the 14th stage isolation valve.
3. Cowl anti-ice and thrust reverser will not be available on the affected engine.

----- END -----

L WING A/ICE or R WING A/ICE

- (1) Affected 14TH STAGE BLEEDCHECK OPEN
- (2) Affected engine thrust INCREASE

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

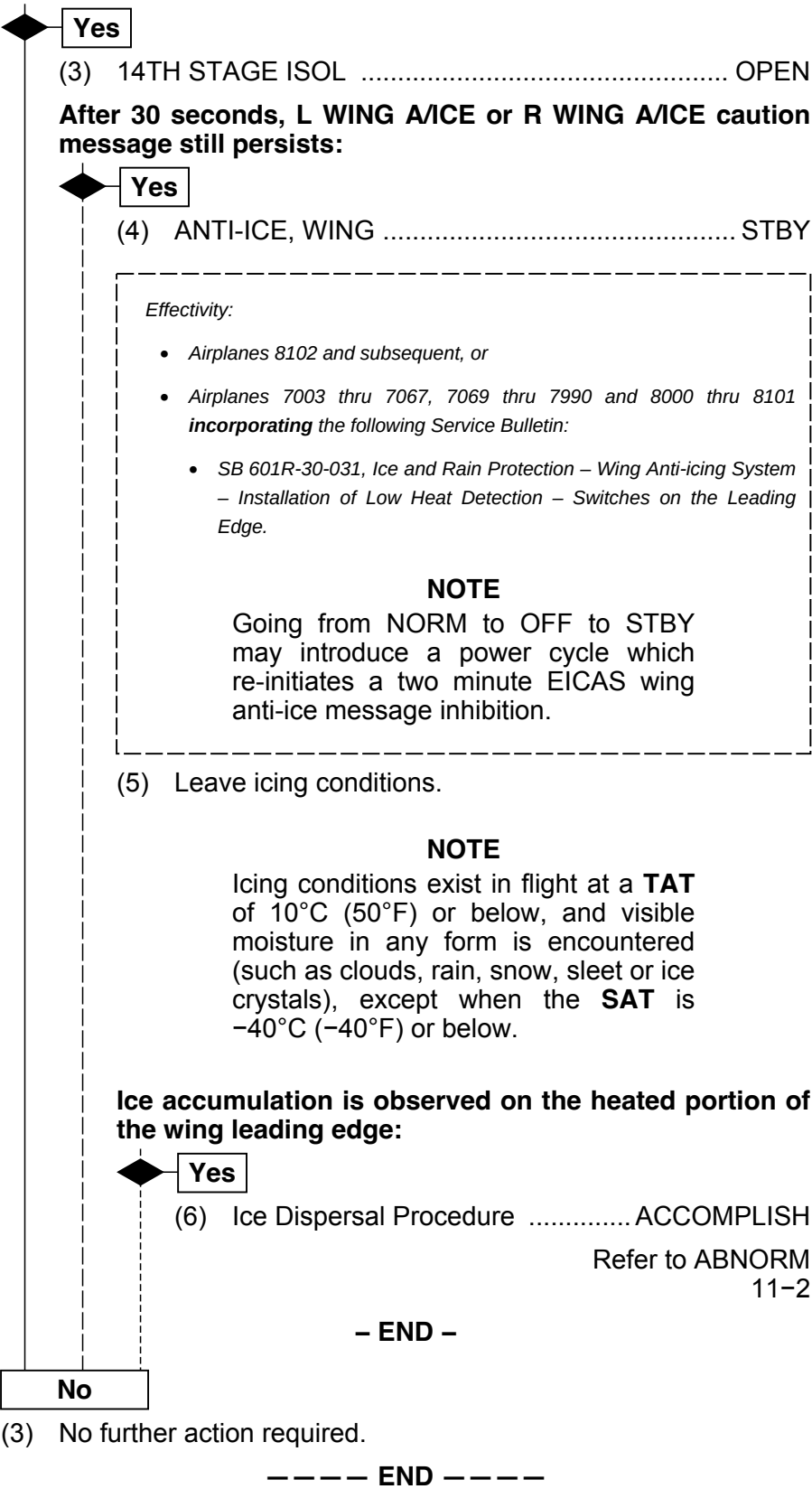
EMER

IMM

ABNORM

INDEX

L WING A/ICE or R WING A/ICE caution message persists:



WING A/ICE SNSR

(1) ANTI-ICE, WING STBY

ICE caution message and light come on:

Yes

(2) 14TH STAGE ISOL OPEN

ICE caution message and light persists:

Yes

(3) Leave icing conditions.

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is -40°C (-40°F) or below.

Ice accumulation is observed on the heated portion of the wing leading edge:

Yes

(4) Ice Dispersal Procedure ACCOMPLISH

Refer to ABNORM
11-2

- END -

No

(2) No further action required.

----- END -----

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

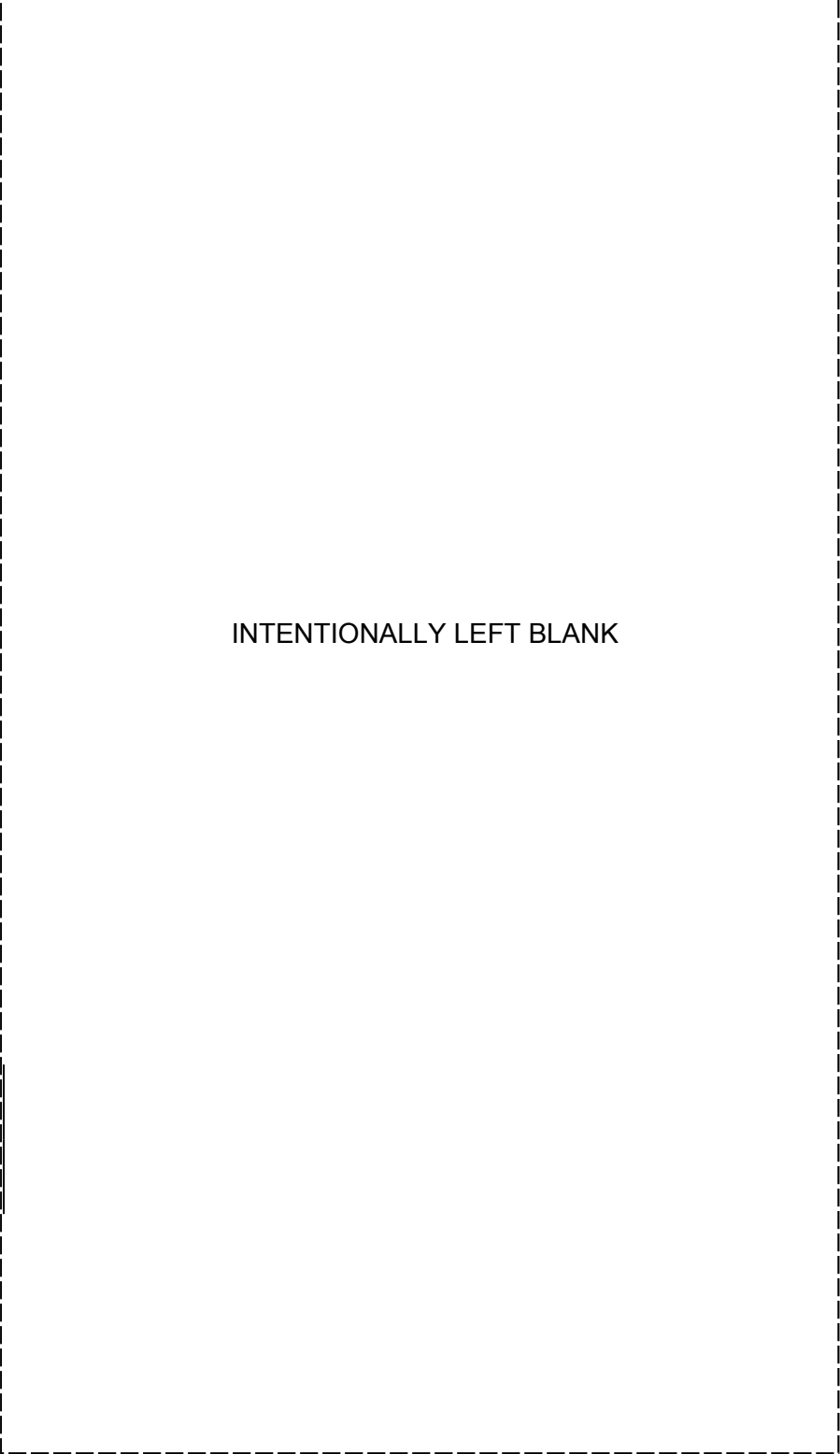
EICAS

EMER

IMM

ABNORM

INDEX



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



ICE

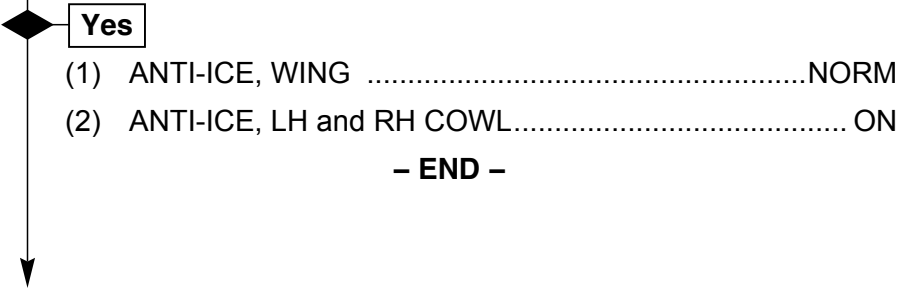
- (1) ANTI-ICE, WINGNORM
- (2) ANTI-ICE, LH and RH COWL ON
- END -----

ICE DETECT FAIL (Caution Message) or ICE DET 1 FAIL or ICE DET 2 FAIL (Status Message)

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is -40°C (-40°F) or below.

Icing conditions are present or are anticipated:



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

(1) No action required.

----- END -----

L AOA HEAT or R AOA HEAT

(1) ANTI-ICE, LH or RH PROBES Affected side OFF then ON

L AOA HEAT or R AOA HEAT caution message persists:

Yes

Effectivity:

- Airplanes 7003 thru 7890:

(2) After flight in icing conditions, do not rely on stall protection and windshear guidance systems.

– END –

No

(2) No action required.

----- END -----

L AOA HEAT and R AOA HEAT

(1) ANTI-ICE, LH and RH PROBES OFF then ON

L AOA HEAT and R AOA HEAT caution messages persist:

Yes

(2) Do not rely on stall protection and windshear recovery guidance systems.

(3) STALL PTCT PUSHER
(either left or right) OFF

(4) Avoid icing conditions.

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is -40°C (-40°F) or below.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Prior to landing:

- (5) Approach speedNot less than
V_{REF} (FLAPS 45) + 10 KIAS
- (6) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.22 (22%)	1.25 (25%)
Without Thrust Reverser(s)	1.25 (25%)	1.25 (25%)

– END –

No

- (2) No further action required.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



L PITOT HEAT

L STATIC HEAT, PITOT BASE HEAT and L AOA HEAT caution messages are also displayed:

Yes

(1) No further action required.

NOTE

The heater controller #1 is reporting a fault.
The heaters are still powered, but the
EICAS will continue to display the caution
message.

– END –

No

(1) ANTI-ICE, LH PROBES OFF then ON.

L PITOT HEAT caution message persists:

Yes

(2) AIR DATA source selector SELECT operative side

NOTE

MACH TRIM caution message comes on.

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– END –

No

(2) No further action required.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



R PITOT HEAT

R STATIC HEAT, TAT PROBE HEAT and R AOA HEAT caution messages are also displayed:

Yes

(1) No further action required.

NOTE

The heater controller #2 is reporting a fault. The heaters are still powered, but the EICAS will continue to display the caution message.

– END –

No

(1) ANTI-ICE, RH PROBES OFF then ON.

R PITOT HEAT caution message persists:

Yes

(2) AIR DATA source selector SELECT operative side

NOTE

MACH TRIM caution message comes on.

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– END –

No

(2) No further action required.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

PITOT BASE HEAT

L STATIC HEAT, L PITOT HEAT and L AOA HEAT caution messages are also displayed:

Yes

(1) No further action required.

NOTE

The heater controller #1 is reporting a fault. The heaters are still powered, but the EICAS will continue to display the caution message.

- END -

No

(1) ANTI-ICE, LH PROBES OFF then ON

PITOT BASE HEAT caution message persists:

Yes

(2) EFIS and standby flight instruments CROSS-CHECK to determine inoperative side.

(3) AIR DATA source selector SELECT operative side

NOTE

MACH TRIM caution message comes on.

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- END -

No

(2) No further action required.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

L STATIC HEAT

PITOT BASE HEAT, L PITOT HEAT and L AOA HEAT caution messages are also displayed:

Yes

(1) No further action required.

NOTE

The heater controller #1 is reporting a fault. The heaters are still powered, but the EICAS will continue to display the caution message.

- END -

No

(1) ANTI-ICE, LH PROBES OFF then ON.

L STATIC HEAT caution message persists:

Yes

(2) Do not rely on the standby altitude and airspeed indications.

(3) Do not rely on the cabin pressure acquisition module (CPAM).

- END -

No

(2) No further action required.

----- END -----

R STATIC HEAT

TAT PROBE HEAT, R PITOT HEAT and R AOA HEAT caution messages are also displayed:

Yes

(1) No further action required.

NOTE

The heater controller #2 is reporting a fault. The heaters are still powered, but the EICAS will continue to display the caution message.

- END -

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

No

(1) ANTI-ICE, RH PROBES OFF then ON.

R STATIC HEAT caution message persists:

Yes

- (2) Do not rely on the standby altitude and airspeed indications.
- (3) Do not rely on the cabin pressure acquisition module (CPAM).

– END –

No

(2) No further action required.

----- END -----

STBY PITOT HEAT

(1) ANTI-ICE, LH PROBES OFF then ON

STBY PITOT HEAT caution message persists:

Yes

- (2) Do not rely on the standby airspeed indication

– END –

No

(2) No further action required.

----- END -----

TAT PROBE HEAT

R PITOT HEAT, R STATIC HEAT and R AOA HEAT caution messages are also displayed:

Yes

- (1) No further action required.

NOTE

The heater controller #2 is reporting a fault. The heaters are still powered, but the EICAS will continue to display the caution message.

– END –

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



No

(1) ANTI-ICE, RH PROBES OFF then ON

TAT PROBE HEAT caution message persists:

Yes

(2) Do not rely on the air temperature indications.

– END –

No

(2) No further action required.

– – – – – END – – – – –

L WINDOW HEAT or R WINDOW HEAT

(1) ANTI-ICE, LH or RH WSHLDAffected side OFF/RESET
then LOW or HI, as required.

L WINDOW HEAT or R WINDOW HEAT caution message persists:

Yes

(2) Leave icing conditions.

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is –40°C (–40°F) or below.

– END –

No

(2) No further action required.

– – – – – END – – – – –

L WSHLD HEAT or R WSHLD HEAT

(1) ANTI-ICE, LH or RH WSHLDAffected side OFF/RESET
then LOW or HI, as required.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



L WSHLD HEAT or R WSHLD HEAT caution message persists:

Yes

- (2) Leave icing conditions.

NOTE

Icing conditions exist in flight at a **TAT** of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the **SAT** is -40°C (-40°F) or below.

To remove ice accumulation from affected windshield:

- (3) AirspeedINCREASE to V_{MO}/M_{MO} , if possible, to disperse ice.

— END —

No

- (2) No further action required.

— — — — — END — — — — —

Arcing, Delaminated, Shattered, or Cracked Window or Windshield

- (1) ANTI-ICE, LH or RH WSHLDAffected side OFF
(2) PRESS CONTMAN
(3) MAN RATE selector INCR MAX
(4) MAN ALT UP (position), to achieve 6.2 psid or less
(5) Crew and passenger oxygen ON, if required
(6) Descent INITIATE, if required

Windshield core ply or inboard ply shattered:

Yes

- (7) AirspeedREDUCE to 205 KIAS when operating below 8000 feet

No

When below 8000 feet:

- (7) Cabin altitude SET to destination airport elevation

— — — — — END — — — — —

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

CONTENTS	PAGE
----------	------

INSTRUMENTS SYSTEM

- EFIS COMP MON <0025>.....ABNORM 12-2
- EFIS COMP INOP ABNORM 12-5
- Primary Flight Display FailureABNORM 12-5
- Any Two Displays Indicate DISPLAY
TEMP Flag ABNORM 12-5
- Display Control Panel FailureABNORM 12-6
- ADC 1 FailureABNORM 12-6
- ADC 2 FailureABNORM 12-8
- Radio Altimeter Failure <0045>ABNORM 12-9
- Inertial Reference System Failure<0025>..... ABNORM 12-11
- Radio Tuning Unit FailureABNORM 12-14
- Air Traffic Control Transponder Failure
<0030>ABNORM 12-14
- Altitude Alerting System Failure <0030>.....ABNORM 12-15
- Position Information Unreliable <0040>.....ABNORM 12-15
- Uncommanded True Heading
Indication <0025> ABNORM 12-15

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

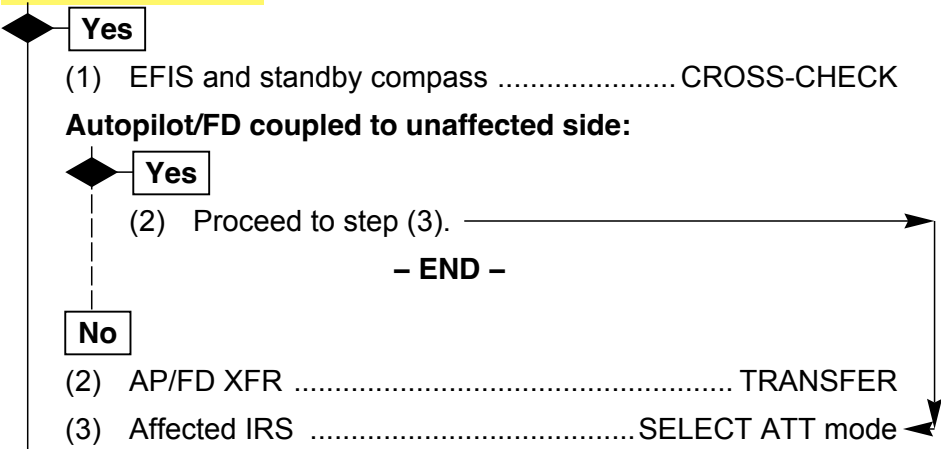
INDEX

EFIS COMP MON <0025>

NOTE

RVSM operations may be affected. Review the requirements. <0030>

EFIS COMP MON caution message on, and PFD displays HDG:

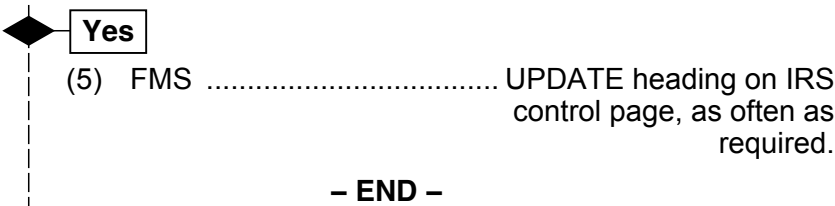


NOTE

1. With the IRS in ATT mode, a heading drift rate of up to 15 degrees per hour can occur.
2. With the IRS in ATT mode, windshear guidance is inoperative on the affected side.
3. The attitude indicator will be removed from PFD for 30 seconds after the IRS is selected to ATT mode.

- (4) FMSENTER present heading on IRS control page.

IRS operation in ATT mode is acceptable (AP remains operational):



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



No

- (5) ATTD HDG source selector SELECT to reliable side
- (6) Inertial Reference System
Failure procedure REVIEW

Refer to ABNORM 12-11

– END –

No

EFIS COMP MON caution message on, and PFD displays ROL or PIT:

Yes

- (1) EFIS and standby flight
instruments CROSS-CHECK
- (2) ATTD HDG source selector SELECT to reliable side
- (3) Inertial Reference System
Failure procedure REVIEW

Refer to ABNORM 12-11

– END –

No

EFIS COMP MON caution message on, and PFD displays ALT and/or IAS:

Yes

- (1) EFIS and standby flight
instruments CROSS-CHECK

NOTE

Use **Unreliable Airspeed In-flight procedure:**
Refer to EMER 17-2

- If pitch attitude, thrust setting or external noise not consistent with indicated airspeed,
- If no reliable airspeed side can be determined.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

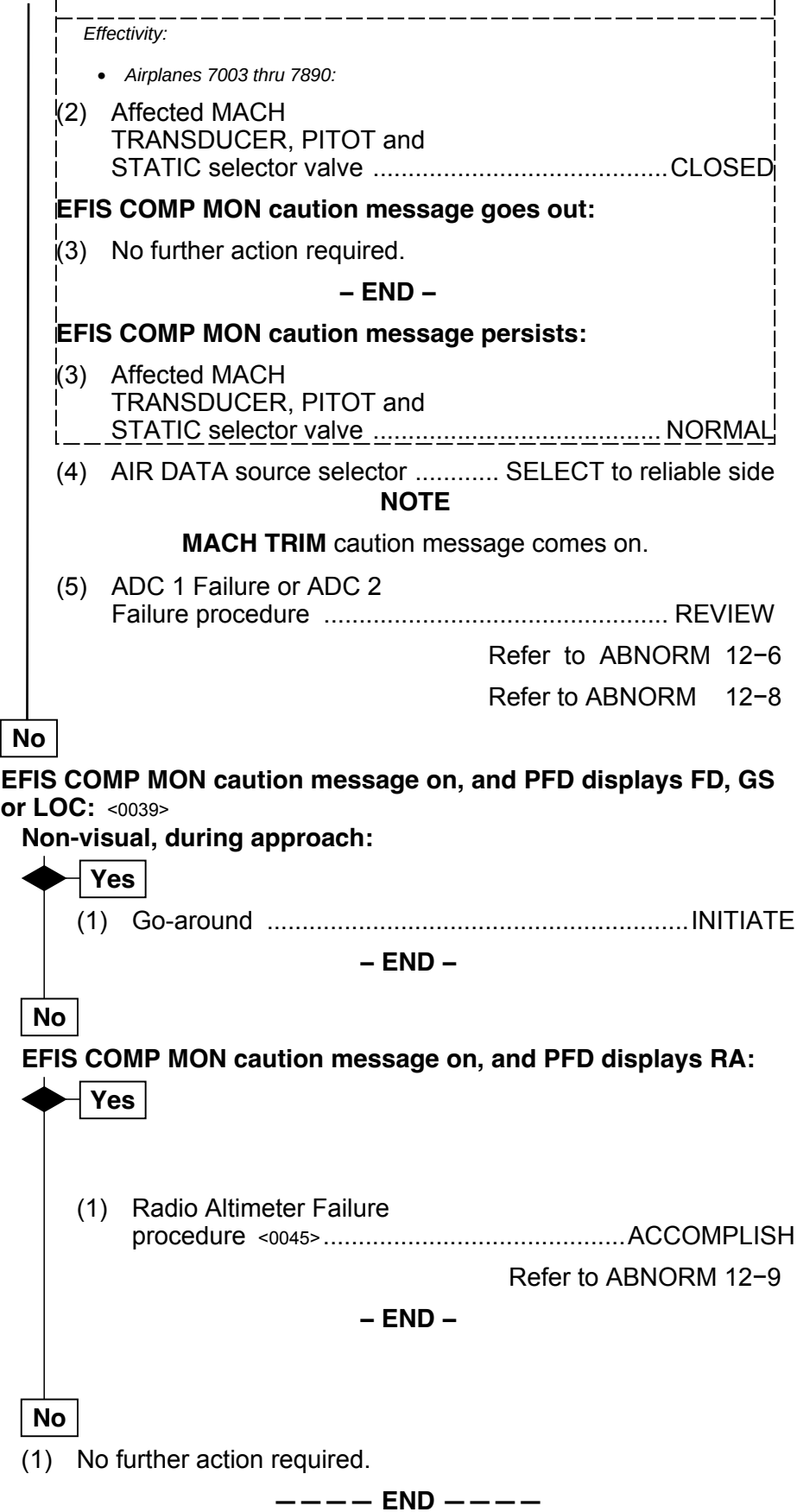
EMER

IMM

ABNORM

INDEX

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



EFIS COMP INOP

- (1) Affected flight instrumentsMONITOR
- (2) Standby flight instrumentsCROSS-CHECK

NOTE
Use **Unreliable Airspeed In-flight procedure:**
Refer to EMER 17-2

- If pitch attitude, thrust setting or external noise not consistent with indicated airspeed,
- If no reliable airspeed side can be determined.

----- **END** -----
▼

Primary Flight Display Failure

- (1) Affected display reversionary panel selector SELECT PFD

NOTE
Applicable MFD defaults to primary flight display.

----- **END** -----

DISPLAY TEMP Annunciation

- On any two displays:**
- (1) Land at the nearest suitable airport.

----- **END** -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



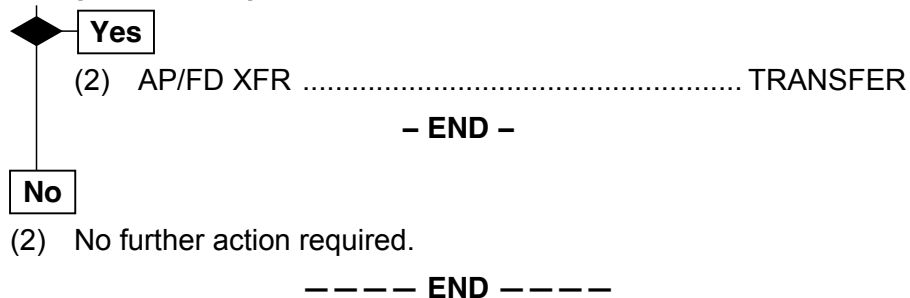
Display Control Panel Failure

(1) DSPL CONT source selector SELECT operative side

NOTE

HDG knob remains inoperative until operable display control panel is selected.

Autopilot/FD coupled to affected side:



ADC 1 Failure

NOTE

- 1. Category II and IIIa operations may be affected. Review the requirements.
- 2. RVSM operations are affected. Review the requirements. <0030>

(1) AIR DATA source selector2

NOTE

- 1. Use **Unreliable Airspeed In-flight procedure:**
Refer to EMER 17-2
 - If pitch attitude, thrust setting or external noise not consistent with indicated airspeed,
 - If no reliable airspeed side can be determined.
- 2. **MACH TRIM** caution message comes on and MACH TRIM INOP light on.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

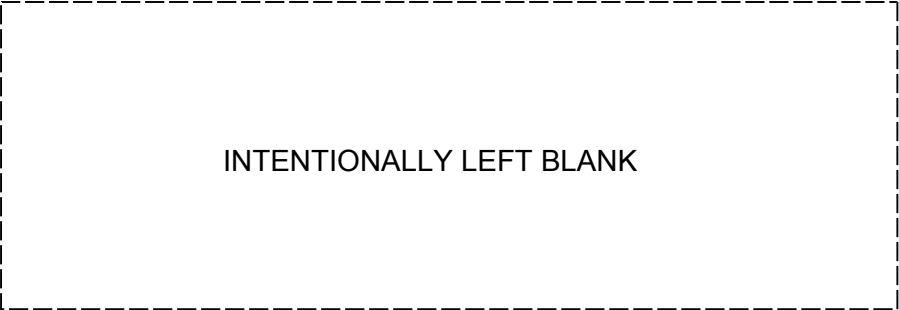
EICAS

EMER

IMM

ABNORM

INDEX

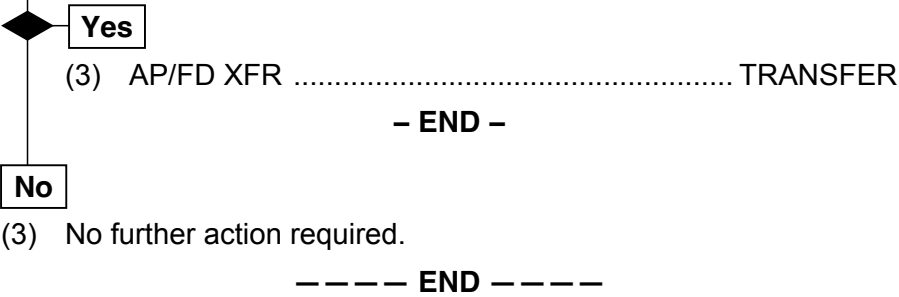


Airplane with EGPWS installed: <0040>

NOTE

Windshear guidance is operative on PFD 2 only. <0040>

Autopilot/FD coupled to affected side:



SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

ADC 2 Failure

NOTE

- 1. Category II and IIIa operations may be affected. Review the requirements.
- 2. RVSM operations are affected. Review the requirements. <0030>

(1) AIR DATA source selector1

NOTE

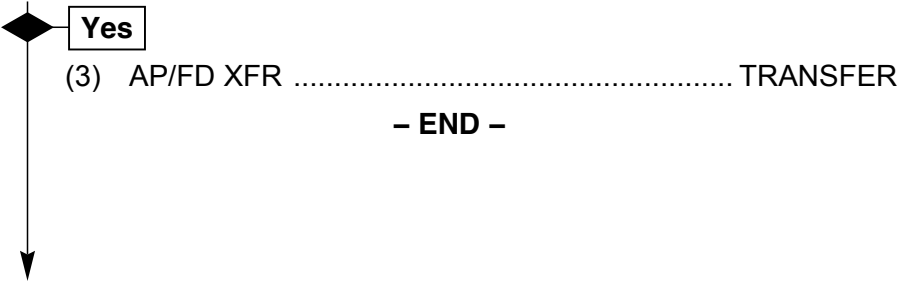
- 1. Use **Unreliable Airspeed In-flight procedure:**
Refer to EMER 17-2
 - If pitch attitude, thrust setting or external noise not consistent with indicated airspeed,
 - If no reliable airspeed side can be determined.
- 2. **MACH TRIM** caution message comes on and MACH TRIM INOP light on.

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NOTE

Windshear guidance is operative on PFD 1 only.<0040>

Autopilot/FD coupled to affected side:



SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



No

(3) No further action required.

----- END -----

Radio Altimeter Failure <0045>

Both RAs have failed:

Yes

NOTE

1. Traffic alert and collision avoidance system is inoperative. **TCAS FAIL** PFD/MFD status message comes on.
2. Enhanced ground proximity warning system is inoperative. **GPWS FAIL** status message comes on. <0040>
3. Windshear detection and guidance is inoperative. **WINDSHEAR FAIL** status message comes on.

(1) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.15 (15%)	1.20 (20%)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



	Runway Surface	
	Dry	Wet or Contaminated
Without Thrust Reverser(s)	1.20 (20%)	1.20 (20%)

After touchdown:

- (2) FLIGHT SPOILER lever SELECT MAX deployment

NOTE

FLT SPOILER DEPLOY and GND SPLR DEPLOY caution messages may come on upon touchdown for approximately 4 seconds.

– END –

No

One RA has failed

- (1) No action required.

NOTE

- 1. If RA 1 has failed, windshear guidance is operative on PFD 2 only.
- 2. If RA 2 has failed, windshear guidance is operative on PFD 1 only.
- 3. With one radio altimeter failed, both TCAS and GPWS remain operational.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Inertial Reference System Failure <0025>

NOTE

1. If an IRS is in ATT mode, a heading drift rate of up to 15 degrees per hour can occur. The IRS CONTROL page data on the FMS should be updated periodically from the best available alternate navigation source.
2. If an IRS is in ATT mode, windshear guidance is inoperative on the affected side.

IRS 1 has failed:

Yes

NOTE

1. Autopilot and affected yaw damper will disconnect.
2. Flight director 1 is inoperative, **YD 1 INOP** and **FD 1 FAIL** status message come on.
3. Windshear guidance is operative on PFD 2 only. <0040>

(1) ATTD HDG source selector2

NOTE

PFD 1 indicates ATT 2 and MAG 2 selected.

Autopilot/FD coupled to affected side:

Yes

(2) AP/FD XFR TRANSFER

– END –

No

(2) No further action required.

– END –

No

IRS 2 has failed:

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

NOTE

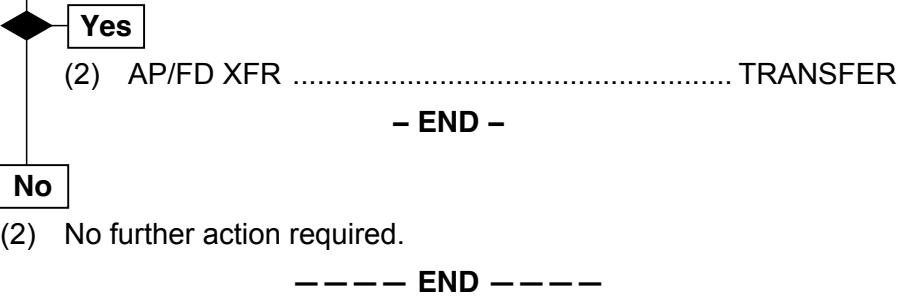
- 1. Autopilot and affected yaw damper will disconnect.
- 2. Flight director 2 is inoperative, **YD 2 INOP** and **FD 2 FAIL** status message come on.
- 3. Windshear detection and guidance is inoperative, **WINDSHEAR FAIL** status messages come on.
- 4. Windshear guidance is operative on PFD 1 only. <0040>

(1) ATTD HDG source selector1

NOTE

PFD 2 indicates ATT 1 and MAG 1 selected.

Autopilot/FD coupled to affected side:



SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

	ABNORM 12-13
	Rev. 70, Jul 13/2018

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SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Radio Tuning Unit Failure

(1) Affected RTU INHIBSELECT

(2) Operable RTU USE 1/2 switch to tune cross-side radio.

----- END -----

Air Traffic Control Transponder Failure

NOTE

1. RVSM operations are affected. Review the requirements. <0030>

2. SSR Mode S Enhanced Surveillance operations are affected. Review the applicable limitations. <0096>

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Altitude Alerting System Failure <0030>

NOTE

RVSM operations are affected. Review the requirements.

----- END -----

Position Information Unreliable <0040>

FMS DR message on FMS CDU, or insufficient sensors available on the MFD FMS POSITION SUMMARY page.

IRS ONLY message on FMS CDU, or insufficient sensors available on the MFD FMS POSITION SUMMARY page. <0025>

- (1) GRND PROX, TERRAIN OFF

----- END -----

Uncommanded True Heading Indication <0025>

White **TRU**, **TRU1** or **TRU2** indicators on PFD above HSI lubber line and on MFD near digital heading indication. Indicators will continuously flash below 10000 feet.

NOTE

This procedure is not applicable if operating in the regions where true heading is used intentionally.

- (1) EFIS and standby compass CROSS-CHECK
if difference is more than 3 degrees, use standby heading.

- (2) Air Traffic Control NOTIFY as required

When flight conditions permit, on FMS CDU:

- (3) INDEX function key SELECT
to access INDEX menu page.

- (4) FMS CTL line select key SELECT
to access FMS CONTROL page.

NOTE

On DISPLAY MODE line **MAG** is displayed in white and **TRUE** is displayed in green.

- (5) MAG/TRUE line select key SELECT

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

NOTE

Expect **REQ PENDING** message to be displayed for a few seconds.

- If **MAG** is displayed in green and **TRUE** is displayed in white, proceed to step (6).
- If **MAG** is displayed in white and **TRUE** is displayed in green, proceed to step (8).

MAG is displayed in green and TRUE is displayed in white:

(6) Confirm that PFD and MFD no longer display white TRU indicator.

(7) No further action required.

MAG is displayed in white and TRUE is displayed in green:

(8) EFIS and standby compassCONTINUE to CROSS-CHECK

if difference is more than 3 degrees, use standby heading.

(9) Air Traffic Control NOTIFY as required

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

I

	ABNORM 13-1
	Rev. 76, Jul 02/2021

CONTENTS	PAGE
----------	------

BRAKES

- **A/SKID INBD** [ABNORM 13-2](#)
- **A/SKID OUTBD** [ABNORM 13-3](#)
- **A/SKID INBD and A/SKID OUTBD** [ABNORM 13-3](#)
- **IB BRAKE PRESS or OB BRAKE PRESS** [ABNORM 13-4](#)
- **PARK BRAKE SOV** [ABNORM 13-5](#)

NOSE WHEEL STEERING

- **STEERING INOP** [ABNORM 13-6](#)

LANDING GEAR

- **MLG OVHT FAIL** [ABNORM 13-7](#)
- **Landing Gear Manual Extension** [ABNORM 13-7](#)

WEIGHT-ON-WHEELS

- **PROX SYS CHAN** [ABNORM 13-9](#)
- **WOW INPUT** [ABNORM 13-13](#)
- **WOW OUTPUT** [ABNORM 13-13](#)

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

A/SKID INBD

NOTE

1.

Anti-skid braking is lost on 1 or 2 inboard wheels or parking brake SOV has failed.
2.

The landing distance factors that follow are based upon both anti-skid systems inoperative.

(1) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.75 (75%)	1.95 (95%)
Without Thrust Reverser(s)	1.95 (95%)	1.95 (95%)



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

A/SKID OUTBD

NOTE

1.

Anti-skid braking is lost on 1 or 2 outboard wheels.
2.

The landing distance factors that follow are based upon both anti-skid systems inoperative.

(1) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.75 (75%)	1.95 (95%)
Without Thrust Reverser(s)	1.95 (95%)	1.95 (95%)



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

----- END -----

A/SKID INBD and A/SKID OUTBD

NOTE

Anti-skid braking is lost on 2 or more wheels.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

- (1) ANTI-SKID OFF
- (2) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.75 (75%)	1.95 (95%)
Without Thrust Reverser(s)	1.95 (95%)	1.95 (95%)



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

----- END -----

IB BRAKE PRESS or OB BRAKE PRESS

- (1) Hydraulic pressure and fluid quantityCHECK affected system
- (2) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.50 (50%)	1.60 (60%)
Without Thrust Reverser(s)	1.60 (60%)	1.60 (60%)

NOTE

Apply brakes carefully and maximize the use of reverse thrust.

----- END -----

PARK BRAKE SOV

NOTE

The parking brake SOV is failed or is closed with the parking brake not set, possibly affecting the inboard anti-skid system performance.

- (1) PARKING BRAKE handle CHECK released
- (2) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.75 (75%)	1.95 (95%)

	Runway Surface	
	Dry	Wet or Contaminated
Without Thrust Reverser(s)	1.95 (95%)	1.95 (95%)



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

After landing:

- (3) Wheel chocksINSTALL before turning off hydraulic systems 2 and 3

NOTE

The parking brake system will deplete more rapidly than normal with the parking brake set and hydraulic systems 2 and 3 selected off.

----- END -----

STEERING INOP

- (1) N/W STRG OFF

NOTE

If the **STEERING INOP** caution message is displayed with the N/W STRG switch OFF, but not displayed with the N/W STRG switch ARMED, the steering shut-off valve has failed (stuck open). The N/W STRG switch must be selected to ARMED to enable nosewheel steering normal operation.

- (2) N/W STRGARMED

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



STEERING INOP caution message persists:

Yes

- (3) N/W STRG OFF
- (4) Use differential braking, rudder and engine thrust as required to assist in directional control.

NOTE

1. Select the longest runway available with minimum cross-wind and turbulence.
2. In high cross-wind conditions, rudder effectiveness may be limited after landing with maximum reverse thrust selected.

– END –

No

- (3) No further action required.

----- END -----

MLG OVHT FAIL

- (1) No action required. The MLG bay overheat detection system is inoperative.
- (2) BTMSMONITOR

----- END -----

Landing Gear Manual Extension

- (1) AirspeedNot more than 250 KIAS
- (2) HYDRAULIC 2 and 3B pump ON

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE

The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 lbs. The flight crew may choose to reposition the seat prior to manual release handle deployment. In safe flight conditions, the flight crew may also choose to stand up to pull the manual release handle.

- (3) LANDING GEAR MANUAL RELEASE handlePULL to full extension



Nosewheel steering may not be available upon landing, even if the **STEERING INOP caution message is not displayed.**

NOTE

1. The **NOSE DOOR OPEN** warning message is displayed when the LANDING GEAR MANUAL RELEASE handle is pulled.

2. With the landing gear manual release handle pulled, the **HYD 3 HI TEMP** caution message may eventually be posted.

- (4) LDG GEAR lever DN

- If any landing gear fails to down-and-lock following manual release attempt, proceed to step (5).

• If landing gear is down-and-locked, proceed to step (6).

Any landing gear fails to down-and-lock following manual release attempt:

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(5) LANDING GEAR MANUAL
RELEASE handle STOW and PULL to full
extension

NOTE

Step (5) may be repeated as many times as
desired.

- If landing gear is down-and-locked, proceed to step (6).
- If any landing gear still fails to down-and-lock, proceed to
step (8).

Landing gear is down-and-locked:

(6) LANDING GEAR MANUAL
RELEASE handleSTOW

NOTE

In some cases, stowing the landing gear manual
release handle may cause the main or nose gear
to retract. If this occurs, the landing gear manual
release handle must again be pulled to full
extension and left in this position until the landing
gear locking pins have been installed.

(7) No further action required.

Any landing gear still fails to down-and-lock:

(8) Land at the nearest suitable airport.

(9) Landing Gear Up / Unsafe
Landing procedureACCOMPLISH

Refer to EMER 13-9

PROX SYS CHAN <0039>

(1) Cabin crewAlert and brief that PASS
SIGNS are inoperative.

NOTE

TCAS and WINDSHEAR alerts may be
unreliable.

Prior to landing:

(2) AirspeedNot more than 250 KIAS

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(3) HYDRAULIC 2 and 3B pump ON



Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE

The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 lbs. The flight crew may choose to reposition the seat prior to manual release handle deployment. In safe flight conditions, the flight crew may also choose to stand up to pull the manual release handle.

(4) LANDING GEAR MANUAL RELEASE handlePULL to full extension

NOTE

1.

The **NOSE DOOR OPEN** warning message is displayed when the LANDING GEAR MANUAL RELEASE handle is pulled.
2.

The outboard anti-skid channel will be inoperative and the **A/SKID OUTBD** caution message will be displayed after the landing gear is fully extended.

(5) LDG GEAR lever DN

(6) N/W STRG switch OFF

- If landing gear is down-and-locked, proceed to step (7).
- If any landing gear fails to down-and-lock following manual release attempt, proceed to step (9).

Landing gear is down-and-locked:

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

(7) LANDING GEAR MANUAL
RELEASE handleSTOW

NOTE

1. In some cases, stowing the landing gear manual release handle may cause the main or nose gear to retract. If this occurs, the landing gear manual release handle must again be pulled to full extension and left in this position until the landing gear locking pins have been installed.

(8) Proceed to step (14).

Any landing gear fails to down-and-lock following manual release attempt:

(9) HYDRAULIC 3A and 3B pump OFF

(10) HYDRAULIC pageSELECT and VERIFY
system 3 pressure
decreasing

When hydraulic system 3 pressure is less than 200 psi:

(11) LANDING GEAR MANUAL
RELEASE handleSTOW and pull to full
extension

NOTE

Step (11) may be repeated as many times as
desired.

(12) LANDING GEAR MANUAL
RELEASE handleSTOW

(13) HYDRAULIC 3A and 3B pump ON

- If landing gear is down-and-locked, proceed to step (14).
- If any landing gear still fails to down-and-lock, proceed to step (16).

Landing gear is down-and-locked:

(14) GRND PROX. FLAP <0040> OVRD

(15) Actual landing distance INCREASE

NOTE

Apply the following factors to FLAP 45 actual landing distances for either Dry surfaces in PERF-04 or to Wet or Contaminated surfaces in PERF-06. If dispatched with Anti-skid inoperative, apply to dry or wet actual landing distances found in PERF-07.

	Runway Surface	
	Dry	Wet or Contaminated
With Two Thrust Reversers	1.75 (75%)	1.95 (95%)
Without Thrust Reverser(s)	1.95 (95%)	1.95 (95%)



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

NOTE

1. Select the longest runway available with minimum cross-wind and turbulence.
2. Use differential braking, rudder and engine thrust as required to assist in directional control.
3. In high cross-wind conditions, rudder effectiveness may be limited after landing with maximum reverse thrust selected.

Any landing gear still fails to down-and-lock:

(16) Landing Gear Up / Unsafe
Landing ProcedureACCOMPLISH

Refer to EMER 13-9

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



WOW INPUT

NOTE

Nosewheel steering may be inoperative upon landing.

END

WOW OUTPUT

On the ground:

Yes

- (1) Do not take-off
- (2) ADG DEPLOY CONT AUTO circuit breaker (2N6) OPEN
- (3) APU (if operating) SHUTDOWN
- (4) ANTI-ICE, LH and RH PROBES OFF
- (5) CRT displaysMONITOR for high temperature
- (6) ANTI-SKID OFF

END

No

In flight:

- (1) Manual Pressurization Control ProcedureACCOMPLISH
Refer to ABNORM 2-11
- (2) BEACON OFF then ON to maintain flight data recorder operation.
- (3) GND LIFT DUMPINGMAN DISARM
- (4) N/W STRG OFF

After touchdown:

- (5) GND LIFT DUMPINGMAN ARM
- (6) Flight spoilers lever SELECT MAX deployment

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

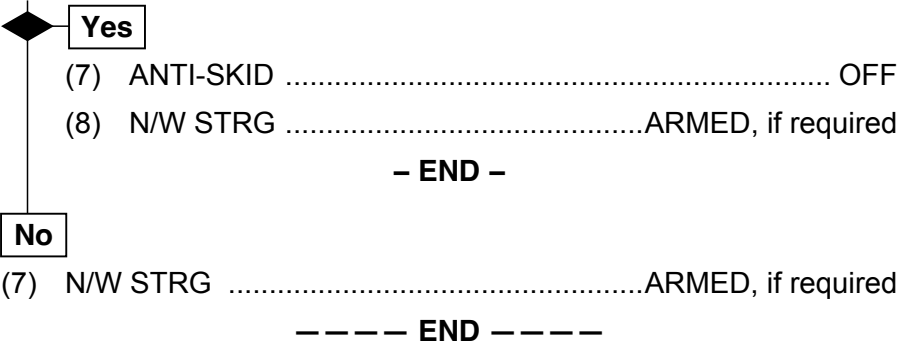
ABNORM

INDEX



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

Loss of braking occurs below 35 kts:



SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

CONTENTS	PAGE
----------	------

MISCELLANEOUS SYSTEMS

- **ELT ON** ABNORM 14-2
- **EMER LTS OFF** ABNORM 14-2
- **OXY LO PRESS** ABNORM 14-2
- **PASS OXY ON** ABNORM 14-2
- Passenger Oxygen, Auto-deploy Failure ABNORM 14-2
- **SMOKE TOILET** (Caution) ABNORM 14-3
- Pilot Incapacitation ABNORM 14-3

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

ELT ON

(1) ELT <Type Spec> or <0092> ON, then ARM/RESET.
Check that the ELT is not transmitting.

----- END -----

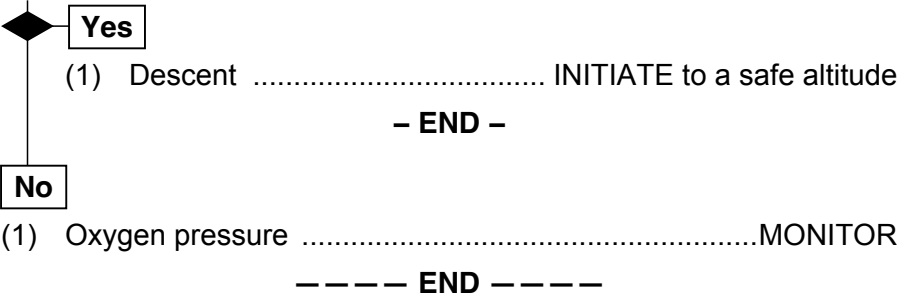
EMER LTS OFF

(1) EMER LTS CHECK status of EMER LTS

----- END -----

OXY LO PRESS

Oxygen is in use or leak is suspected:



PASS OXY ON

(1) Passenger oxygen CHECK status

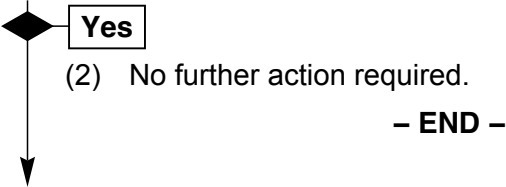
----- END -----

Passenger Oxygen, Auto-deploy Failure

Cabin altitude is above 14250 (±750) feet and oxygen masks have not deployed automatically:

(1) PASS OXY ON

Masks deploy:



No

(2) Flight attendant(s)ADVISE to open oxygen mask compartments manually.

----- END -----

SMOKE TOILET (Caution)

(1) SMOKE TOILET ProcedureACCOMPLISH
Refer to EMER 2-14

Pilot Incapacitation

In case of pilot incapacitation, the following are general guidelines:

- Ensure autopilot is engaged.
- Call flight attendant to flight compartment.
- Check that incapacitated pilot does not interfere with flight controls. It is preferable to have the incapacitated pilot removed from the flight compartment.
- If unable to remove the incapacitated pilot, instruct flight attendant to lock shoulder harness of incapacitated pilot and move seat rearwards.
- If an immediate landing is imperative, obtain advice on most suitable airport where medical assistance can be readily rendered.
- Check on possibility of obtaining assistance from pilots who may be traveling as passengers on board the airplane.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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CONTENTS	PAGE
----------	------

AURAL / VISUAL WARNING SYSTEM

- EICAS Primary Display Failure ABNORM 15-2
- EICAS Secondary Display Failure ABNORM 15-2
- EICAS Control Panel Failure ABNORM 15-2
- **EICAS COMP INOP** ABNORM 15-3
- TCAS System Failure (TCAS FAIL Flag on PFD/MFD) ABNORM 15-3
- TCAS Resolution Advisory Failure (TCAS RA FAIL Flag on PFD) ABNORM 15-3
- TCAS Traffic Display Failure (TD FAIL Flag on MFD) ABNORM 15-4
- TCAS Traffic Display Failure (TCAS DISPLAY FAIL Flag on MFD) <0039> ABNORM 15-4
- **DCU 1 INOP or DCU 2 INOP or DCU 1 AURAL INOP or DCU 2 AURAL INOP** (Status Message) ABNORM 15-4
- **GPWS FAIL** (Status Message) <0040>..... ABNORM 15-4
- **TERRAIN FAIL** (Status Message) <0040> ABNORM 15-4

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

EICAS Primary Display Failure

- (1) EICAS source selectorED 2
- (2) Applicable display reversionary panelEICAS

NOTE

1. Applicable MFD defaults to EICAS STAT (status) page.
2. All page functions are then selectable on the EICAS control panel.

----- END -----

EICAS Secondary Display Failure

- (1) Applicable display reversionary panelEICAS

NOTE

1. Applicable MFD defaults to EICAS STAT (status) page.
2. All page functions are selectable on the EICAS control panel.

----- END -----

EICAS Control Panel Failure

To display a synoptic page (ECS, HYDRAULIC, AC ELECTRICAL or DC ELECTRICAL, FUEL, FLIGHT CONTROLS, ANTI-ICE, or DOORS):

- (1) STEPSELECT as required

NOTE

1. PRI, STAT, CAS and STEP buttons are operable.
2. ECS, HYD, ELEC, FUEL, F/CTL, A/ICE, DOORS, MENU, SEL, UP and DN buttons are inoperative.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

EICAS COMP INOP

NOTE

Navigation data may be lost/unreliable.

Two (2) Data Concentrator Units (DCUs) remain operative:

Yes

(1) EICAS displays (both)PRI PAGE

(2) Engine ITT, N₁ and N₂ indicationsCROSS-CHECK

(3) EICAS PRI pageUSE reliable side

(4) Affected engine and/or systemMAINTAIN within limits

— END —

No

Only one (1) Data Concentrator Unit (DCU) remains operative:

(1) Engine ITT, N₁ and N₂ indicationsMONITOR

----- END -----

TCAS System Failure (TCAS FAIL Flag on PFD/MFD)

(1) Radio tuning unit, TCAS main display page, MODE line select keySTBY MODE

----- END -----

TCAS Resolution Advisory Failure (TCAS RA FAIL Flag on PFD)

(1) Radio tuning unit, TCAS main display page, MODE line select keyTA MODE ONLY

NOTE

Pilot with functional RA display will be required to carry out any subsequent avoidance maneuvers.

----- END -----

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

TCAS Traffic Display Failure (TD FAIL Flag on MFD)

TCAS DISPLAY FAIL caution message displayed on MFD:

(1) Functional MFD Use for TCAS data

----- END -----

TCAS Traffic Display Failure (TCAS DISPLAY FAIL Flag on MFD) <0039>

TCAS DISPLAY FAIL caution message displayed on one MFD:

(1) Functional MFD Use for TCAS data

----- END -----

DCU 1 INOP or DCU 2 INOP or DCU 1 AURAL INOP or DCU 2 AURAL INOP (Status Message)

(1) Affected AUDIO WARNING DISABLE

(2) Operative AUDIO WARNING NORMAL

----- END -----

GPWS FAIL (Status Message) <0040>

(1) No action required. Assume loss of GPWS mode 1 thru 6 aural.

----- END -----

TERRAIN FAIL (Status Message) <0040>

(1) No action required. **TERRAIN FAIL** status message is pilot advisory of Terrain Awareness Alerting and Display and Terrain Clearance Floor System failure.

NOTE

- 1. Reversion of both MFDs will result in a loss of the Terrain / Obstacle Awareness Display (**TERRAIN FAIL** status message comes on).
- 2. If full functionality of EGPWS is desired, revert one display back to MFD format.

----- END -----

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

ADVISORY

The following are system response or acknowledgement – type messages.

Engine	
L REV ARMED R REV ARMED	L or R REVERSER switch set to ARM.
APR ARM	APR system is armed.
APR TEST 1 OK APR TEST 2 OK	APR test successful at DCU 1 (DCU 2).
IGNITION A IGNITION B	Ignition A or B system is on.
IGNITION A/B	Ignition system A and B are on during continuous ignition conditions. (One igniter is inoperative).

Bleed Air	
DUCT TEST OK	Bleed leak detection test successful.

APU	
APU AVAIL Light	APU may be used to supply bleed air and/or electrical power.
APU SOV CLSD	APU fuel shutoff valve is confirmed closed during an APU fire condition.

Fire Protection	
APU SQUIB 1 APU SQUIB 2	APU firex bottle squib 1 or squib 2 test successful.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



ADVISORY-2

Rev. 67, Jan 13/2017

Fire Protection

ENG SQUIB 1 L ENG SQUIB 1 R ENG SQUIB 2 L ENG SQUIB 2 R	Applicable engine firex bottle squib 1 or squib 2 test successful.
CARGO SQUIB 1 CARGO SQUIB 2	Cargo bay firex bottle squib 1 or squib 2 test successful.

Flight Controls

PLT ROLL CMD CPLT ROLL CMD	Pilot (copilot) roll authority selected.
FLT SPLR DEPLOY <JAA>	Flight spoiler deployed above 800 feet AGL.
GND SPLR DEPLOY	Any ground spoilers deployed (on ground or below 10 feet AGL).
GLD MAN ARM	Ground lift dumping, manually armed.
T/O CONFIG OK	Airplane configured for take-off.

Fuel

GRAV XFLOW OPEN	Gravity cross-flow valve confirmed open.
L ENG SOV CLSD R ENG SOV CLSD	L or R engine fuel firewall SOV confirmed closed.
L FUEL PUMP ON R FUEL PUMP ON	L or R fuel pump operating.

Hydraulics

HYD SOV 1 CLSD HYD SOV 2 CLSD	Hydraulic SOV 1 or 2 closed.
----------------------------------	------------------------------

QUICK REFERENCE
HANDBOOK
CSP A-022-PRS

ADVISORY MESSAGES

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Ice	
WING A/ICE OK	Wing anti-ice test successful.
ICE <0039>	Ice detected and system operational.
L COWL A/I ON <0039>	Left cowl anti-ice selected on.
R COWL A/I ON <0039>	Right cowl anti-ice selected on.
COWL A/I ON <0039>	L and R cowl anti-ice selected on.
WING A/I ON <0039>	Wing anti-ice switch selected on.
WING/COWL A/I ON <0039>	Wing and cowl anti-ice switches selected on.

Miscellaneous	
FDR EVENT	Flight data recorder event switch pressed.

Landing Gear	
PARKING BRAKE ON	Parking brake set with one or both engines less than 70% N ₁ .

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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STATUS

Engine	
L ENG ECU FAIL R ENG ECU FAIL	APR system failure. Failure at applicable engine control unit (APR output not triggered by the applicable ECU).
L APR ECU FAIL <0039>	APR system failure. Failure at applicable engine control unit (APR output not triggered by the applicable ECU).
R APR ECU FAIL <0039>	APR system failure. Failure at applicable engine control unit (APR output not triggered by the applicable ECU).
L ENGINE START R ENGINE START	Applicable engine start in progress.
CONT IGNITION	Continuous ignition (A and B igniters) selected for both engines.
DCU 1 APR FAIL DCU 2 APR FAIL	APR system failure. Failure at applicable DCU (APR relay not energized by DCU when under test).
IGNITION A/B <0039>	Ignition system A and B are on during continuous ignition conditions. (One igniter is inoperative).
ENG TYPE MISCOMP <0005>	DCU configuration problem. Do not start the APU or the engines.

Air-conditioning and Pressurization	
AUTO PRESS 1 FAIL AUTO PRESS 2 FAIL	Applicable automatic pressurization system controller failed.
CABIN (CKPT) TEMP MAN	Cabin (cockpit) temperature in manual control.
CABIN PRESS MAN	Pressurization system under manual control.
CARGO FAN FAIL	Cargo bay fan failed.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Air-conditioning and Pressurization	
CARGO SOV FAIL	Cargo bay SOV failed.
COOL EXHAUST FAIL	Avionics exhaust fan failed, or low flow from cooling exhaust.
CPAM FAIL	CPCP indication fail. Airplane altitude max FL300.
L PACK OFF R PACK OFF	Applicable pack switched off. Airplane altitude not above 25000 feet.
RAM AIR OPEN	Ram air SOV selected open.
CKPT COOL FAIL	Cockpit display cooling SOV failed or SOV not in the correct position, (closed on the ground, open in the air).
OVBD COOL FAIL	Overboard avionics cooling SOV failed closed. Refer to ABNORM 2–9
INBD COOL FAIL	Inboard avionics cooling SOV failed closed. SOV not open when passenger door closed.

Automatic Flight Control System	
FD 1 FAIL FD 2 FAIL	L or R flight director failed.
YD 1 INOP YD 2 INOP	Yaw damper 1 is off or failed. Yaw damper 2 is off or failed.

APU	
APU IN BITE	APU under BITE prior to start (approximately 5 to 8 seconds).
APU LCV OPEN	APU load control valve open.
APU SOV OPEN	APU SOV confirmed open and no APU fire present.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

APU	
APU START	APU start in progress.
APU ECU FAIL	APU ECU is inoperative and APU will not start.

APU Door Messages	
DOOR CLSD	Normal position when APU is shut down.
DOOR MID	Normal position when above 13000 feet or more than 260 KIAS.
DOOR OPEN	Normal position when below 13000 feet and less than 260 KIAS.
DOOR INHIB/CLSD DOOR INHIB/MID DOOR INHIB/OPEN	Door inhibited in the applicable position.
DOOR (with dashes --)	Door position unknown. Refer to ABNORM 4–5 Limited to 300 KIAS or the APU must be operating.
DOOR INHIB (with dashes --)	Door actuator or position sensor failure. The door will remain in the last known position. There is a possibility of an APU hung start if the APU is started in flight.

Bleed Air	
L 10TH ARM OPEN <0039>	Left 10TH stage switch is selected open but SOV is closed. Review Single PACK dispatch requirements.
R 10TH ARM OPEN <0039>	Right 10TH stage switch is selected open but SOV is closed. Review Single PACK dispatch requirements.
L 14TH ARM CLSD <0039>	Left 14TH stage SOV selected CLSD.
R 14TH ARM CLSD <0039>	Right 14TH stage SOV selected CLSD.
10TH ISOL OPEN	10TH stage bleed air isolation valve selected open.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Bleed Air	
14TH ISOL OPEN	14TH stage bleed air isolation valve selected open.
DUCT MON LOOP A DUCT MON LOOP B	Loop A or Loop B under test.
L 10TH SOV CLSD R 10TH SOV CLSD	Applicable 10TH stage SOV selected closed. Starter-assisted relight not possible on affected engine.
L 14TH SOV CLSD R 14TH SOV CLSD	Applicable 14TH stage SOV selected closed.

Electrical	
IDG 1 DISC IDG 2 DISC	Applicable IDG disconnected with the engine running.
AC ESS ALTN	Essential transfer contactor closed, AC ESS bus supplied by AC BUS 2 either automatically or by manual selection.
AC UTIL 1 OFF AC UTIL 2 OFF	Applicable AC utility bus shed. Associated battery charger is not available.
AC 1 AUTOXFER OFF AC 2 AUTOXFER OFF	Applicable auto transfer inhibited.
APU BATT CHGR <0039>	APU battery charger not charging.
DC ESS TIE CLSD	DC essential tie contactor energized.
DC TIE 1 CLSD DC TIE 2 CLSD	Associated DC tie contactor energized. Associated DC utility bus, cabin reading lights, is not available.
MAIN BATT CHGR <0039>	Main battery charger not charging.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

<i>Effectivity:</i> - Airplanes 7901 and subsequent; and - Airplanes 7003 to 7067, 7069 to 7900 incorporating the following Service Bulletin: - SB 601R-27-116, <i>Flight Controls – Flaps Skew Detection System (SDS) – Installation and Activation of the Skew Detection System.</i>	
Flight Controls	
FLAPS DEGRADED	Displayed for all skew detection sensor faults with the exception of the DC power supply fault.

Flight Controls	
FLAPS HALFSPEED	Flaps channel 1 or 2 failed. Flaps operate at half speed.
FLT SPLRS FAULT	Loss of flight spoiler redundancy: - One PCU on one side, or - One SECU module. The flight spoilers operate normally.
FLUTTER DAMP FAIL	Hydraulic fluid level in the affected flutter damper is approximately 20% of normal level. Flight may continue but maintenance on the affected flutter damper must be carried out within one flight day.
FLUTTER DAMPER <0039>	Hydraulic fluid level in the affected flutter damper is approximately 20% of normal level. Flight may continue but maintenance on the affected flutter damper must be carried out within one flight day.
GLD MAN DISARM	Ground lift dumping selected to MAN DISARM.
IB GND SPLR FAULT OB GND SPLR FAULT	Loss of redundancy of the applicable ground spoiler control. Affected ground spoiler operates normally.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Flight Controls	
SPOILERONS FAULT	Spoilerons fault detected: - One PCU on one side, or - One SECU module. Spoilerons operate normally.
STAB CH1 INOP STAB CH2 INOP	Single channel pitch trim failure. Pitch trim automatically transfers to other channel. Refer to ABNORM 8–9

Fuel	
AUTO XFLOW INHIB	Automatic cross-flow inhibited.
FUEL CH 1 FAIL FUEL CH 2 FAIL	Applicable fuel quantity computer channel failed.
GRAV XFLOW FAIL	Gravity cross-flow not in confirmed position (open or closed).
L AUTO XFLOW ON R AUTO XFLOW ON	Applicable cross-flow SOV open in auto mode.
L XFLOW ON R XFLOW ON	Applicable cross-flow SOV open.
MAN XFLOW	Manual cross-flow selected.

Ice Detection	
ICE	Ice detected and system operational.
ICE DET 1 FAIL ICE DET 2 FAIL	Ice detector 1 or 2 failed and Ice detector 2 or 1 not failed. - Refer to ABNORM 11–8. During icing conditions or when icing conditions are anticipated: - Select ANTI-ICE, WING switch to NORM, and - Select ANTI-ICE, COWL switch to ON.

SUPPLIMITSPERFMISCPLANINTERMIXEICASEMERIMMABNORMINDEX

Instruments	
IRS 1 IN ATT <0025>	IRS 1 in attitude mode.
IRS 2 IN ATT <0025>	IRS 2 in attitude mode.
IRS 1 ON BATT <0025>	IRS 1 on battery backup power.
IRS 2 ON BATT <0025>	IRS 2 on battery backup power.
IRS 1 DC FAIL <0025>	IRS 1 DC power supply failed.
IRS 2 DC FAIL <0025>	IRS 2 DC power supply failed.
IRS 1 OVERTEMP <0025>	IRS 1 DC over-temperature detected.
IRS 2 OVERTEMP <0025>	IRS 2 DC over-temperature detected.
SPEED REFS INDEP	Pilot and copilot V-speed selection are not synchronized. Each pilot must set V-speed using their on-side ARP.

Landing Gear	
BTMU FAIL	Brake temperature monitoring unit fault detected. Brake temperature readouts indicate ---- (dashed).
HORN MUTED	Landing gear horn is muted.
PROX SYS FAULT	The PSEU has detected a sensor or internal failure.
WOW OUTPUT FAIL	Weight-on-wheels output fault detected (non-critical systems). • System has failed to the in-air mode if the airplane is on the ground, or • System has failed to the on-ground mode if the airplane is in the air.

Miscellaneous	
NO SMOKING	No smoking signs selected on.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Miscellaneous	
SEAT BELTS	Seat belts signs selected on.
EMER LTS ON	Emergency lights system selected on.
FDR ACCEL FAIL	Failure of FDR accelerometer.
FDR FAIL	FDR data recorder failed.
IAPS DEGRADED	IAPS bus fault. One of the four IAPS quadrants is inoperative.
IAPS OVERTEMP	IAPS power supply overtemperature detected: - One of the two cooling fans is inoperative. - One-half of IAPS will shut down. There is a possible loss of significant communication and navigation systems.

Aural/Visual Warning System	
CAS MISCOMP	An EICAS warning, caution or aural message miscompare situation has been detected.
DCU 1 AURAL INOP DCU 2 AURAL INOP	Self detected internal aural fault or aural inoperative manually disabled. Refer to ABNORM 15-4.
DCU 1 INOP DCU 2 INOP	Self detected internal fault or fault detected by other DCUs. Refer to ABNORM 15-4.
GPWS FAIL	GPWS failed (mode 1 thru 6).
GS CANCEL	Glideslope cancel mode selected.
TERRAIN FAIL <0040>	EGPWS terrain map failure, not available for display.
TERRAIN OFF <0040>	EGPWS Terrain map manually deselected.
TERRAIN NOT AVAIL <0040>	EGPWS terrain map not available for display due to position inaccuracy.
WINDSHEAR FAIL	Windshear detection system in GPWS failed, or windshear guidance failed.

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

A

A/SKID INBDABNORM 13-2

A/SKID INBD and A/SKID OUTBDABNORM 13-3

A/SKID OUTBDABNORM 13-3

Abnormal Increase of Centre Tank QuantityABNORM 9-11

AC 1 AUTOXFERABNORM 6-2

AC 1 AUTOXFER OFF STATUS-4

AC 2 AUTOXFERABNORM 6-2

AC 2 AUTOXFER OFF STATUS-4

AC BUS 1ABNORM 6-3

AC BUS 2ABNORM 6-4

AC ESS ALTN STATUS-4

AC ESS BUSABNORM 6-5

AC SERV BUSABNORM 6-6

AC UTIL 1 OFF STATUS-4

AC UTIL 2 OFF STATUS-4

ADG Inadvertent Deployment.....ABNORM 6-8

AFCS MSGS FAIL EMER 4-2

ADC 1 FailureABNORM 12-6

ADC 2 FailureABNORM 12-8

Air Traffic Control Transponder FailureABNORM 12-14

Aileron PCU RunawayABNORM 8-3

Aileron System Jammed IMM ACTION ITEMS-8

Aileron or Rudder Trim RunawayABNORM 8-8

Altitude Alerting System Failure <0030>ABNORM 12-15

ANTI-ICE DUCT EMER 12-3

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



AP PITCH TRIMABNORM 3-3

Effectivity:

- Airplanes that comply with Transport Canada Airworthiness Directive CF-2002-35R2, Aileron Stiffness, Part C – Termination of Control Checks, or paragraph (i) of FAA Airworthiness Directive 2005-18-06:

AP TRIM IS LWDABNORM 3-2

AP TRIM IS NDABNORM 3-2

AP TRIM IS NUABNORM 3-2

AP TRIM IS RWDABNORM 3-2

APR ARMADVISORY-1

APR CMD SETABNORM 1-14

APR INOPABNORM 1-14

APR TEST 1 OKADVISORY-1

APR TEST 2 OKADVISORY-1

APU AVAIL

Light.....ADVISORY-1

APU BATT CHGR <0039> STATUS-4

Effectivity:

- Airplanes **not incorporating** the following Service Bulletin:
 - SB 601R-31-022, Indicating and Recording – Engine Indication and Crew Alert System (EICAS) – EICAS 2000 Avionics System Upgrade (CSUs, DCPs, DCUs, EFDs, IOCs, FMC, and MDC).

APU BATT CHGRABNORM 6-6

APU BATT OFFABNORM 6-7

APU BLEED ONABNORM 4-2

APU BTL LOABNORM 7-3

APU Door FailureABNORM 4-5

APU ECU FAIL STATUS-3

APU FAULTABNORM 4-3

APU FIRE IMM ACTION ITEMS-7

APU FIRE FAILABNORM 7-2

APU GEN OFFABNORM 6-7

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

APU GEN OVLD	ABNORM 6-8
APU IN BITE	STATUS-2
APU LCV FAIL	ABNORM 4-2
APU LCV OPEN	STATUS-2
APU OIL PRESS	ABNORM 4-3
APU OIL TEMP	ABNORM 4-4
APU OVERSPEED	EMER 5-2
APU OVERTEMP	EMER 5-2
APU SOV CLSD	ADVISORY-1
APU SOV FAIL	ABNORM 4-6
APU SOV OPEN	ABNORM 4-6
APU SOV OPEN	STATUS-2
APU SQUIB 1	ADVISORY-1
APU SQUIB 2	ADVISORY-1
APU START	STATUS-3
Windshield or Window Cracking, Shattering, Arcing or Delamination	ABNORM 11-17
ARINC COOL	ABNORM 2-7
Autopilot Failure	EMER 4-2
AUTO PRESS	ABNORM 2-10
AUTO PRESS 1 FAIL	STATUS-1
AUTO PRESS 2 FAIL	STATUS-1
AUTO XFLOW INHIB	STATUS-6
AV BAY DOOR	ABNORM 5-7

B

BATTERY BUS	ABNORM 6-9
BLEED MISCONFIG <0039>	ABNORM 2-10
BRAKE OVHT	IMM ACTION ITEMS-10
BTMU FAIL	STATUS-7
BULK FUEL TEMP	ABNORM 9-8

C

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX



INDEX-4

Rev. 75, Jan 04/2021

CABIN ALTABNORM 2-11

CABIN ALT
(Warning Message) EMER 3-2

CABIN ALT
(Warning Message) IMM ACTION ITEMS-7

CABIN PRESS MAN STATUS-1

CABIN (CKPT) TEMP MAN STATUS-1

CARGO BTL LO <0013> or
<0034> or <0043> or <0053> or <0059>
or <0074>ABNORM 7-4

CARGO DOORABNORM 5-7

CARGO FAN FAIL STATUS-1

CARGO OVHT <0013> or
<0034> or <0074>ABNORM 2-13

CARGO SOV FAIL STATUS-2

CARGO SQUIB 1ADVISORY-2

CARGO SQUIB 2ADVISORY-2

CAS MISCOMP STATUS-8

CKPT COOL FAIL STATUS-2

Configuration Warning IMM ACTION ITEMS-11

CONT IGNITION STATUS-1

COOL EXHAUST FAIL STATUS-2

COWL A/I ON <0039>ADVISORY-3

CPAM FAIL STATUS-2

CPLT ROLL CMDADVISORY-2

Crew Escape Hatch UnsafeABNORM 5-7

D

DC BUS 1ABNORM 6-10

DC BUS 2ABNORM 6-11

DC EMER BUSABNORM 6-13

DC ESS BUSABNORM 6-13

DC ESS TIE CLSD STATUS-4

DC SERV BUSABNORM 6-14

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

DC TIE 1 CLSD STATUS-4

DC TIE 2 CLSD STATUS-4

DCU 1 APR FAIL STATUS-1

DCU 1 AURAL INOP STATUS-8

DCU 1 AURAL INOP
(Status Messages).....ABNORM 15-4

DCU 1 INOP STATUS-8

DCU 1 INOP
(Status Message)ABNORM 15-4

DCU 2 APR FAIL STATUS-1

DCU 2 AURAL INOP
(Status Messages).....ABNORM 15-4

DCU 2 AURAL INOP STATUS-8

DCU 2 INOP STATUS-8

DCU 2 INOP
(Status Message)ABNORM 15-4

DIFF PRESS EMER 3-4

Display Control Panel FailureABNORM 12-6

DISPLAY COOLABNORM 2-8

Ditching or Forced Landing
Imminent IMM ACTION ITEMS-7

DOOR (with dashes – –) STATUS-3

DOOR CLSD STATUS-3

DOOR INHIB (with dashes– –) STATUS-3

DOOR INHIB/CLSD STATUS-3

DOOR INHIB/MID STATUS-3

DOOR INHIB/OPEN STATUS-3

DOOR MID STATUS-3

DOOR OPEN STATUS-3

Double Engine Failure IMM ACTION ITEMS-6

DUCT MON LOOP A STATUS-4

DUCT MON LOOP B STATUS-4

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

DUCT TEST OKADVISORY-1

During Landing - Excessive
Asymmetry EMER 13-12

During Landing - Loss of
Braking EMER 13-12

E

EFIS COMP INOPABNORM 12-5

EFIS COMP MON <0025>ABNORM 12-2

EICAS COMP INOPABNORM 15-3

EICAS Control Panel FailureABNORM 15-2

EICAS Primary Display
FailureABNORM 15-2

EICAS Secondary Display
FailureABNORM 15-2

Elevator System Jammed IMM ACTION ITEMS-9

ELT ONABNORM 14-2

EMER LTS OFFABNORM 14-2

EMER LTS ON STATUS-8

EMER PWR ONLY EMER 9-2

Emergency Descent
Procedure EMER 3-2

Emergency Descent
Procedure IMM ACTION ITEMS-7

ENG BTL 1 LOABNORM 7-4

ENG BTL 2 LOABNORM 7-4

ENG SQUIB 1 LADVISORY-2

ENG SQUIB 1 RADVISORY-2

ENG SQUIB 2 LADVISORY-2

ENG SQUIB 2 RADVISORY-2

Engine Failure During
ApproachABNORM 1-3

Engine Failure in Climb
During ALTS CAPABNORM 1-4

Engine Hot StartABNORM 1-21

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Engine Oil Level Indication
System FailureABNORM 1-18

ENGINE OVERSPD IMM ACTION ITEMS-6

ENG TYPE MISCOMP <0005>..... STATUS-1

Engine Starter Fails to
CutoutABNORM 1-19

ESS TRU 1ABNORM 6-14

ESS TRU 2ABNORM 6-15

F

FD 1 FAIL STATUS-2

FD 2 FAIL STATUS-2

FDR ACCEL FAIL STATUS-8

FDR EVENTADVISORY-3

FDR FAIL STATUS-8

FLAPS DEGRADED STATUS-5

FLAPS FAILABNORM 8-13

FLAPS HALFSPEED STATUS-5

Flap Twist (In Flight)ABNORM 8-9

Flight Control Computer
FailureABNORM 3-5

Flight Director Guidance
FailedABNORM 3-5

FLIGHT SPOILER Lever
Jammed (Spoilers Deployed)ABNORM 8-22

FLT SPLR DEPLOY <JAA>ABNORM 8-17

FLT SPLR DEPLOY <JAA>ADVISORY-2

FLT SPLRSABNORM 8-18

FLT SPLRS FAULT STATUS-5

Fluctuating Engine Oil
PressureABNORM 1-18

FLUTTER DAMP FAIL STATUS-5

FLUTTER DAMPER <0039> STATUS-5

Forced Landing EMER 7-5

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

FUEL CH 1 FAIL

STATUS-6

FUEL CH 1/2 FAIL <0039>

ABNORM 9-2

FUEL CH 2 FAIL

STATUS-6

FUEL IMBALANCE

ABNORM 9-2

Fuel Leak Procedure

ABNORM 9-13

G

GEAR DISAGREE

EMER 13-3

GEN 1 OFF

ABNORM 6-15

GEN 1 OVLD

ABNORM 6-16

GEN 2 OFF

ABNORM 6-15

GEN 2 OVLD

ABNORM 6-16

GLD MAN ARM

ADVISORY-2

GLD MAN DISARM

STATUS-5

GLD NOT ARMED

ABNORM 8-27

GLD UNSAFE

ABNORM 8-26

GND SPLR DEPLOY

ADVISORY-2

GND SPLR DEPLOY

ABNORM 8-23

GPWS FAIL <0040>

(Status message)

ABNORM 15-4

GPWS FAIL

STATUS-8

GRAV XFLOW FAIL

STATUS-6

GRAV XFLOW OPEN

ADVISORY-2

Gravity Cross-feed

Procedure

ABNORM 9-10

GS CANCEL

STATUS-8

H

I

High Oil Temperature

ABNORM 1-24

HORN MUTED

STATUS-7

Horizontal Stabilizer Trim

Malfunction

ABNORM 8-8

HYD 1 HI TEMP

ABNORM 10-9

HYD 1 LO PRESS

ABNORM 10-2

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

HYD 1 LO PRESS and HYD
2 LO PRESS

ABNORM 10-17

HYD 1 LO PRESS AND
HYD 3 LO PRESS

ABNORM 10-20

HYD 2 HI TEMP

ABNORM 10-12

HYD 2 LO PRESS

ABNORM 10-3

HYD 2 LO PRESS and HYD
3 LO PRESS

ABNORM 10-24

HYD 3 HI TEMP

ABNORM 10-14

HYD 3 LO PRESS

ABNORM 10-5

HYD EDP 1A

ABNORM 10-15

HYD EDP 2A

ABNORM 10-15

HYD PUMP 1B

ABNORM 10-16

HYD PUMP 2B

ABNORM 10-16

HYD PUMP 3A

ABNORM 10-17

HYD PUMP 3B

ABNORM 10-17

HYD SOV 1 CLSD

ADVISORY-2

HYD SOV 1 OPEN

ABNORM 10-17

HYD SOV 2 CLSD

ADVISORY-2

HYD SOV 2 OPEN

ABNORM 10-17

I

IAPS DEGRADED

STATUS-8

IAPS OVERTEMP

STATUS-8

IB BRAKE PRESS

ABNORM 13-4

IB GND SPLR FAULT

STATUS-5

IB GND SPLRS

ABNORM 8-24

ICE

ABNORM 11-8

ICE

STATUS-6

ICE <0039>

ADVISORY-3

ICE DET 1 FAIL
(Status Message)

ABNORM 11-8

ICE DET 1 FAIL

STATUS-6

ICE DET 2 FAIL
(Status Message)

ABNORM 11-8

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

ICE DET 2 FAIL STATUS-6

ICE DETECT FAIL
(Caution Message)ABNORM 11-8

Ice Dispersal ProcedureABNORM 11-2

IDG 1ABNORM 6-17

IDG 1 DISC STATUS-4

IDG 2ABNORM 6-17

IDG 2 DISC STATUS-4

IGNITION A/BADVISORY-1

IGNITION A/B <0039> STATUS-1

IGNITION AADVISORY-1

IGNITION BADVISORY-1

INBD COOL FAIL STATUS-2

In-flight Engine ShutdownABNORM 1-4

Inertial Reference System
Failure <0025>ABNORM 12-11

Interrupted StartABNORM 1-20

IRS 1 DC FAIL <0025> STATUS-7

IRS 1 IN ATT <0025> STATUS-7

IRS 1 ON BATT <0025> STATUS-7

IRS 1 OVERTEMP <0025> STATUS-7

IRS 2 DC FAIL <0025> STATUS-7

IRS 2 IN ATT <0025> STATUS-7

IRS 2 ON BATT <0025> STATUS-7

IRS 2 OVERTEMP <0025> STATUS-7

L

L 10TH DUCT EMER 6-2

L 10TH SOV CLSD STATUS-4

L 10TH ARM OPEN <0039> STATUS-3

L 14TH ARM CLSD <0039> STATUS-3

L 14TH DUCT EMER 6-4

L 14TH SOV CLSD STATUS-4

L AOA HEATABNORM 11-9

L AOA HEAT and R AOA
HEATABNORM 11-9

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

L APR ECU FAIL <0039>.....

STATUS-1

L AUTO XFLOW ON

STATUS-6

L COWL A/ICE

ABNORM 11-3

L COWL A/ ON <0039>.....

ADVISORY-3

L EMER DOOR

ABNORM 5-7

L ENG ECU FAIL

STATUS-1

L ENG FIRE

(In Flight)

IMM ACTION ITEMS-4

L ENG FIRE

(On Ground)

IMM ACTION ITEMS-5

L ENG MISCOMP

ABNORM 1-16

L ENG OIL PRESS

EMER 1-16

L ENG SOV CLSD

ADVISORY-2

L ENG SOV CLSD

ABNORM 9-4

L ENG SOV FAIL

ABNORM 9-4

L ENG SOV OPEN

ABNORM 9-4

L ENGINE START

STATUS-1

L FIRE FAIL

ABNORM 7-2

L FLT SPLR

ABNORM 8-19

L FUEL FILTER

ABNORM 9-8

L FUEL LO PRESS

ABNORM 9-5

L FUEL LO TEMP

ABNORM 9-8

L FUEL PUMP

ABNORM 9-6

L FUEL PUMP ON

ADVISORY-2

L JET OVHT FAIL

ABNORM 7-3

L JETPIPE OVHT

EMER 1-4

L JETPIPE OVHT

IMM ACTION ITEMS-5

L MAIN EJECTOR

ABNORM 9-5

L MAIN EJECTOR and L

SCAV EJECTOR

ABNORM 9-2

L PACK HI PRESS

ABNORM 2-2

L PACK HI TEMP

ABNORM 2-5

L PACK OFF

STATUS-2

L PITOT HEAT

ABNORM 11-11

L REV ARMED

ADVISORY-1

L REV UNLOCKED

IMM ACTION ITEMS-6

L REV UNSAFE

ABNORM 1-13

L SCAV EJECTOR

ABNORM 9-6

- SUPP
- LIMITS
- PERF
- MISC
- PLAN
- INTERMIX
- EICAS
- EMER
- IMM
- ABNORM
- INDEX

L SPOILERONABNORM 8-21

L STATIC HEATABNORM 11-14

L WINDOW HEATABNORM 11-16

L WING A/ICEABNORM 11-4

L WSHLD HEATABNORM 11-16

L XFER SOVABNORM 9-6

L XFLOW ON STATUS-6

L XFLOW SOVABNORM 9-7

LDG GEAR Lever Jammed
in the UP Position EMER 13-7

Landing Gear Manual
ExtensionABNORM 13-7

Landing Gear Up / Unsafe
Landing Procedure EMER 13-9

Loss of all AC Power IMM ACTION ITEMS-8

Low Engine Oil Pressure
Indication EMER 1-16

Low Fuel EMER 11-2

Low IdlingABNORM 1-21

M

MACH TRIMABNORM 8-5

Effectivity:

Airplanes **not incorporating** the following Service Bulletin:

SB 601R-31-022, Indicating and Recording – Engine Indication and Crew Alert System (EICAS) – EICAS 2000 Avionics System Upgrade (CSUs, DCPs, DCUs, EFDs, IOCs, FMC, and MDC).

MAIN BATT CHGRABNORM 6-18

MAIN BATT CHGR <0039> STATUS-4

MAIN BATT OFFABNORM 6-18

MAN XFLOW STATUS-6

Manual Pressurization
Control ProcedureABNORM 2-11

MLG BAY OVHT IMM ACTION ITEMS-10

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

MLG OVHT FAILABNORM 13-7

N

N₁ Fan VibrationABNORM 1-21
N₂ Core VibrationABNORM 1-24
N₂ StagnationABNORM 1-21
No LightoffABNORM 1-20
NO SMOKING STATUS-7
NO STRTR CUTOUTABNORM 1-19
NOSE DOOR OPEN EMER 13-12

O

OB BRAKE PRESSABNORM 13-4
OB GND SPLR FAULT STATUS-5
OB GND SPLRSABNORM 8-25
OVBD COOLABNORM 2-9
OVBD COOL FAIL
Status Message (On the
Ground only).....ABNORM 2-9
OVBD COOL FAIL STATUS-2
OXY LO PRESSABNORM 14-2

P

PARK BRAKE SOVABNORM 13-5
PARKING BRAKE EMER 13-11
PARKING BRAKE ONADVISORY-3
PASSENGER DOOR EMER 14-2
Passenger EvacuationIMM ACTION ITEMS-8
Passenger Oxygen,
Auto-deploy FailureABNORM 14-2
PASS OXY ONABNORM 14-2

- SUPP
- LIMITS
- PERF
- MISC
- PLAN
- INTERMIX
- EICAS
- EMER
- IMM
- ABNORM
- INDEX



PAX DR LATCHABNORM 5-5

Effectivity:

- Airplanes 7390, 7396, 7409, 7412, 7416, 7423, 7432, 7444, 7452, 7455, 7457 and subsequent:

PAX DR OUT HNDLABNORM 5-2

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Pilot IncapacitationABNORM 14-3

PITOT BASE HEATABNORM 11-13

Planned Ditching EMER 7-3

PLT ROLL CMDADVISORY-2

Post-Shutdown Engine Tail

Pipe FireIMM ACTION ITEMS-5

Primary Flight Display Failure.....ABNORM 12-5

PROX SYS CHAN <0039>ABNORM 13-9

PROX SYS FAULT STATUS-7

R

R 10TH DUCTEMER 6-3

R 10TH SOV CLSD STATUS-4

R 10TH ARM OPEN <0039> STATUS-3

R 14TH ARM CLSD <0039> STATUS-3

R 14TH DUCTEMER 6-4

R 14TH SOV CLSD STATUS-4

Radio Altimeter Failure

<0045>ABNORM 12-9

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

Radio Tuning Unit FailureABNORM 12-14

RAM AIR OPEN STATUS-2

Rejected Take-off Before
Achieving V₁ IMM ACTION ITEMS-4

Rejected Take-off Before
Achieving V₁ EMER 1-2

R AOA HEATABNORM 11-9

R APR ECU FAIL <0039> STATUS-1

R AUTO XFLOW ON STATUS-6

R COWL A/ICE ONABNORM 11-3

R COWL A/ICE ON <0039>ADVISORY-3

R EMER DOORABNORM 5-7

R ENG FIRE
(In Flight) IMM ACTION ITEMS-4

R ENG FIRE
(On Ground) IMM ACTION ITEMS-5

R ENG MISCOMPABNORM 1-16

R ENG OIL PRESS EMER 1-16

R ENG SOV CLSDABNORM 9-4

R ENG SOV CLSDADVISORY-2

R ENG SOV FAILABNORM 9-4

R ENG SOV OPENABNORM 9-4

R ENGINE START STATUS-1

R FIRE FAILABNORM 7-2

R FLT SPLRABNORM 8-19

R FUEL FILTERABNORM 9-8

R FUEL LO PRESSABNORM 9-5

R FUEL LO TEMPABNORM 9-8

R FUEL PUMPABNORM 9-6

R FUEL PUMP ONADVISORY-2

R JET OVHT FAILABNORM 7-3

R JETPIPE OVHT EMER 1-4

R JETPIPE OVHT IMM ACTION ITEMS-5

SUPP
LIMITS
PERF
MISC
PLAN
INTERMIX
EICAS
EMER
IMM
ABNORM
INDEX

R MAIN EJECTORABNORM 9-5

R MAIN EJECTOR and R
SCAV EJECTORABNORM 9-2

R PACK HI PRESSABNORM 2-2

R PACK HI TEMPABNORM 2-5

R PACK OFF STATUS-2

R PITOT HEATABNORM 11-12

R REV ARMEDADVISORY-1

R REV UNLOCKED IMM ACTION ITEMS-6

R REV UNSAFEABNORM 1-13

RAM AIR OPEN STATUS-2

Recommended Configuration EMER 7-2

R SCAV EJECTORABNORM 9-6

R SPOILERONABNORM 8-21

R STATIC HEATABNORM 11-14

Rudder System Jammed IMM ACTION ITEMS-9

R WINDOW HEATABNORM 11-16

R WING A/ICEABNORM 11-4

R WSHLD HEATABNORM 11-16

R XFER SOVABNORM 9-6

R XFLOW ON STATUS-6

R XFLOW SOVABNORM 9-7

S

SEAT BELTS STATUS-8

SERVICE DOORABNORM 5-7

Severe Engine Damage (In
Flight) IMM ACTION ITEMS-4

Severe Engine Damage (On
Ground) IMM ACTION ITEMS-5

Single Engine Approach and
LandingABNORM 1-12

I SMOKE CARGO (In Flight)
<0013> or <0034> or <0043> or
<0053> or <0059> or <0074> EMER 2-15

I SMOKE CARGO (On
Ground) <0013> or <0034> or
<0043> or <0053> or <0059> or <0074> EMER 2-16

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

SMOKE CARGO (On Ground) <0013> or <0034> or <0043> or <0053> or <0059> or <0074> IMM ACTION ITEMS-6

SMOKE TOILET EMER 2-14

Smoke/Fire/Fumes Procedure EMER 2-2

Smoke or Fumes Removal Procedure EMER 2-13

SPEED REFS INDEP STATUS-7

SPOILERONS ABNORM 8-21

SPOILERONS FAULT STATUS-6

SPOILERONS ROLL ABNORM 8-20

STAB CH1 INOP STATUS-6

STAB CH2 INOP STATUS-6

STAB TRIM ABNORM 8-5

Stabilizer Trim Runaway IMM ACTION ITEMS-9

STALL FAIL ABNORM 8-2

Stall Recovery IMM ACTION ITEMS-10

Starter-assisted Cross Bleed Relight ABNORM 1-6

STBY PITOT HEAT ABNORM 11-15

STEERING INOP ABNORM 13-6

Suspected Leak into Centre Tank ABNORM 9-11

T

T/O CONFIG OK ADVISORY-2

TAT PROBE HEAT ABNORM 11-15

TCAS Resolution Advisory Failure (TCAS RA FAIL FLAG ON PFD) ABNORM 15-3

TCAS System Failure (TCAS FAIL Flag on PFD/MFD) ABNORM 15-3

TCAS Traffic Display Failure (TCAS DISPLAY FAIL Flag on MFD) <0039> ABNORM 15-4

TCAS Traffic Display Failure (TD FAIL Flag on MFD) ABNORM 15-4

TERRAIN FAIL <0040> (Status Message) ABNORM 15-4

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX



TERRAIN FAIL <0040> STATUS-8

TERRAIN NOT AVAIL <0040>..... STATUS-8

TERRAIN OFF <0040> STATUS-8

TIE CLOSED Light(s) OnABNORM 6-20

Thrust Lever Jammed or
InoperativeABNORM 1-16

U

Uncommanded AccelerationIMM ACTION ITEMS-6

Uncommanded Deceleration,
Abnormal Engine Oscillation
/ FluctuationABNORM 1-15

Unpressurized Flight
Procedure (PACKs off)ABNORM 2-12

Unpressurized Flight
Procedure (PACKs on)ABNORM 2-13

Uncommanded True
Heading Indication <0025>ABNORM 12-15

Uncommanded Yaw MotionIMM ACTION ITEMS-7

Uncommanded Yaw Motion EMER 4-2

Unreliable Airspeed In-flight EMER 17-2

Unreliable Airspeed In-flightIMM ACTION ITEMS-11

V

W

Windmilling RelightABNORM 1-10

WINDSHEAR FAIL STATUS-8

WING A/ICE OKADVISORY-3

WING A/I ON <0039>ADVISORY-3

INTENTIONALLY LEFT BLANK

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

INTENTIONALLY LEFT BLANK

WING A/ICE SNSRABNORM 11-6

WING OVHT EMER 12-2

WING/COWL A/I ON <0039>ADVISORY-3

WOW INPUTABNORM 13-13

WOW OUTPUTABNORM 13-13

WOW OUTPUT FAIL STATUS-7

X

XFLOW APU/PUMPABNORM 9-9

Y

YAW DAMPERABNORM 3-3

YD 1 INOP STATUS-2

YD 2 INOP STATUS-2

NUMERIC

10TH ISOL OPEN STATUS-3

14TH ISOL OPEN STATUS-4

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

	INDEX-20
	Rev. 70, Jul 13/2018

SUPP

LIMITS

PERF

MISC

PLAN

INTERMIX

EICAS

EMER

IMM

ABNORM

INDEX

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